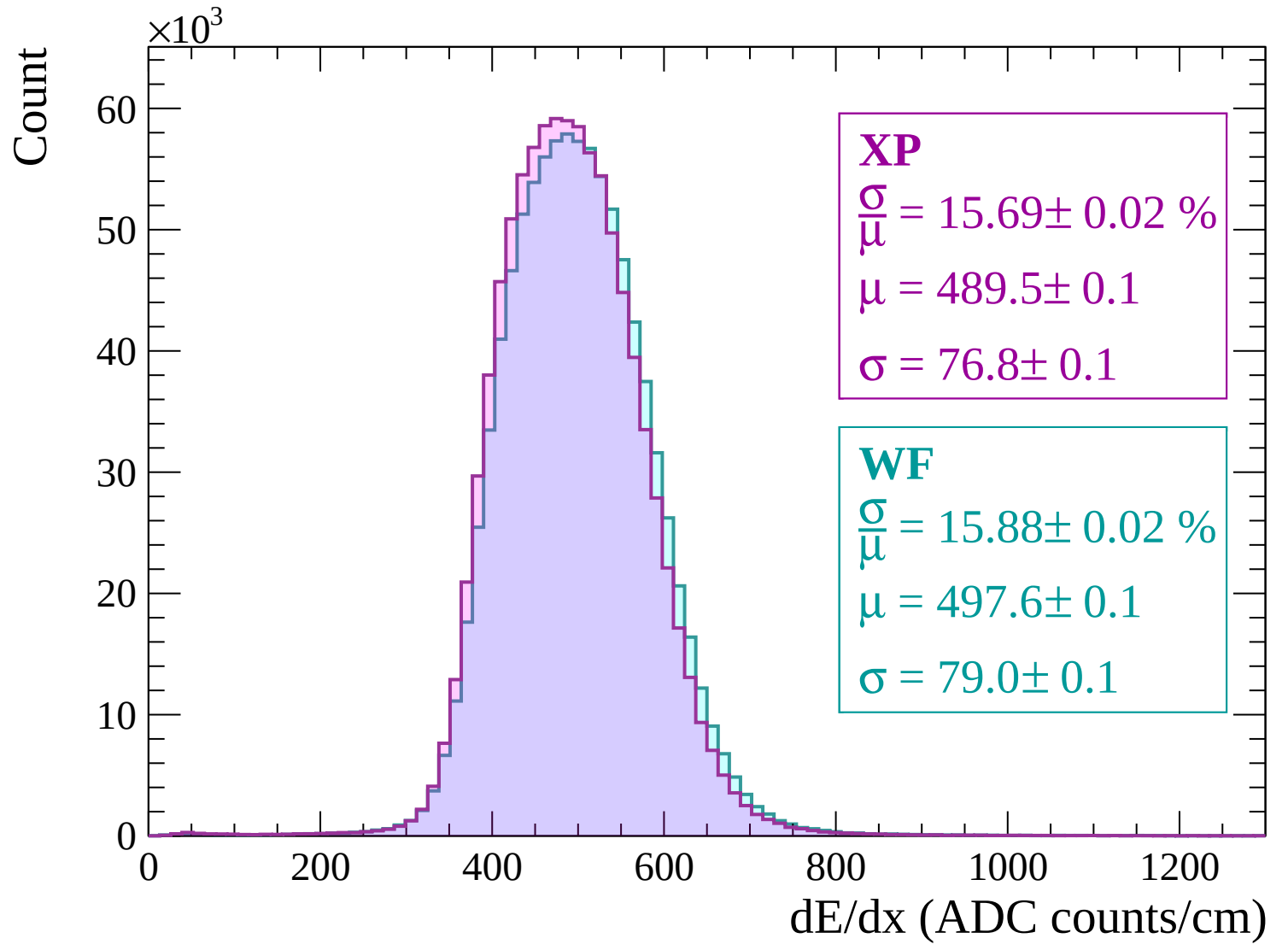
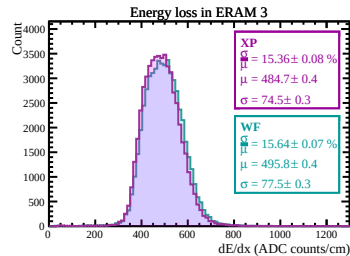
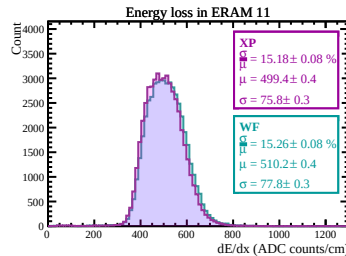
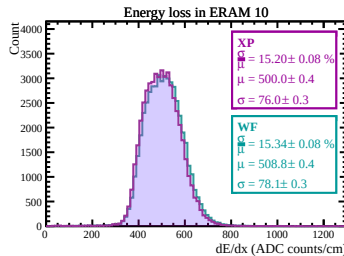
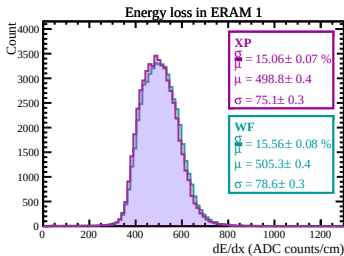
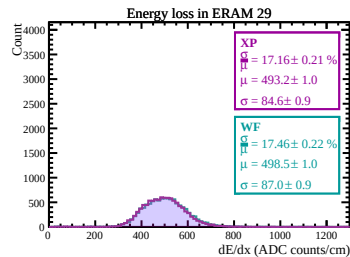
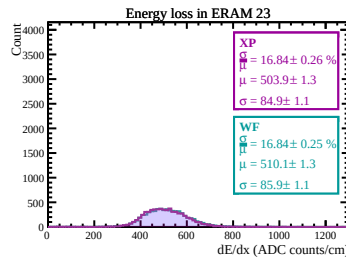
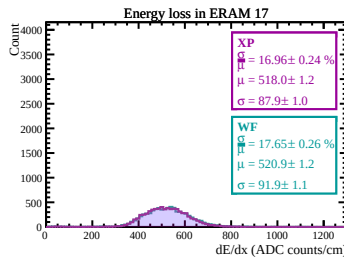
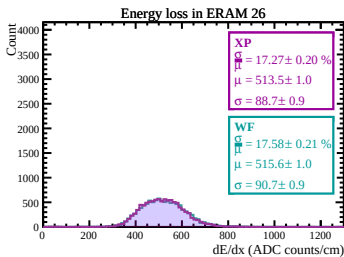
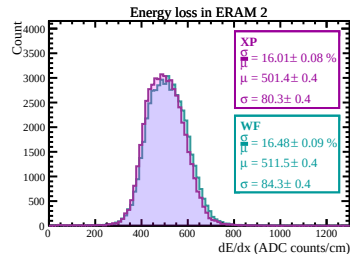
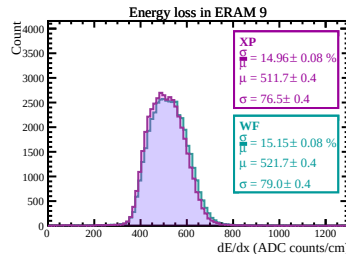
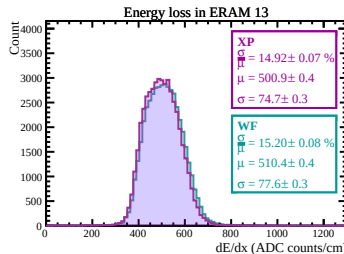
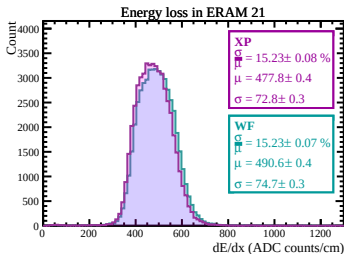
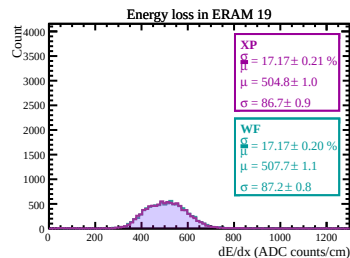
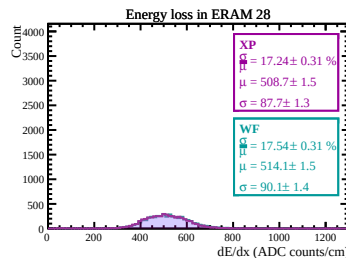
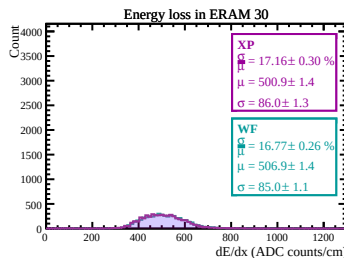
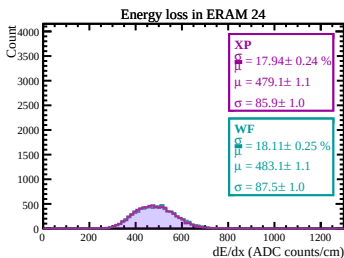
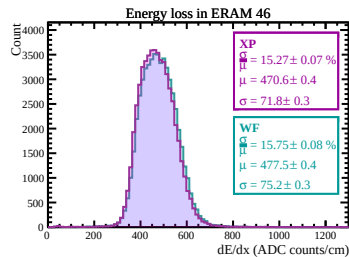
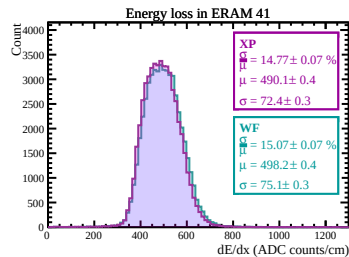
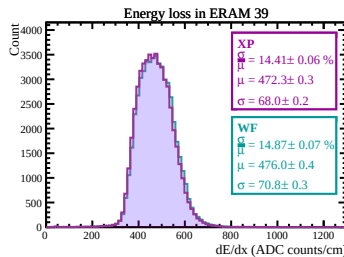
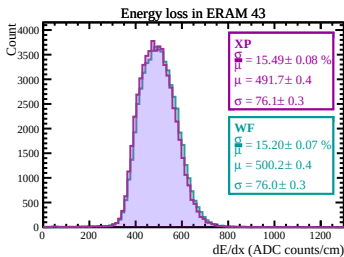
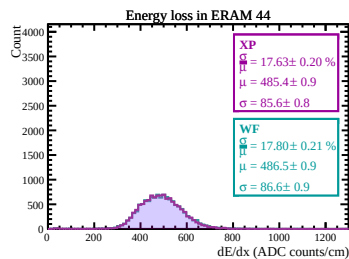
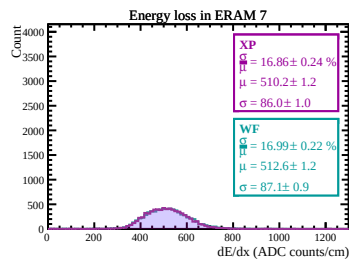
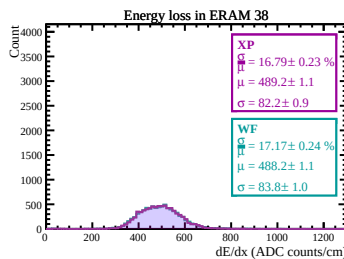
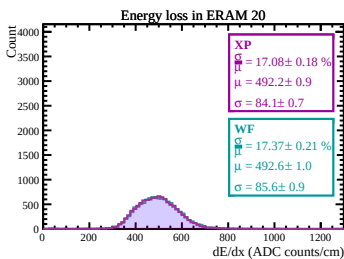
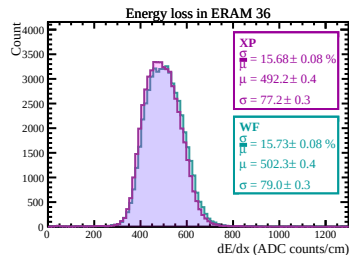
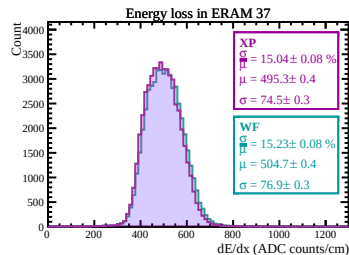
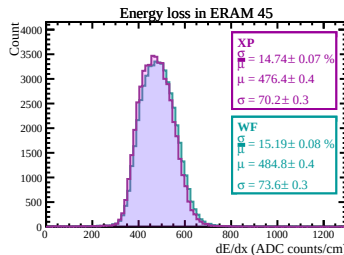
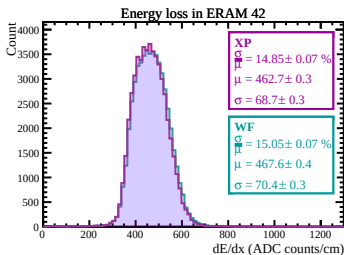
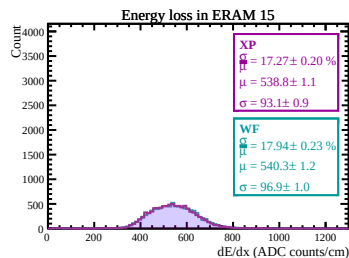
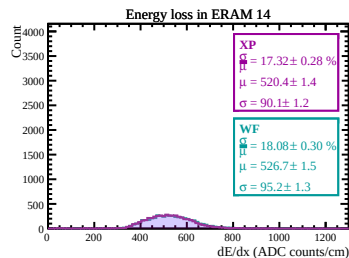
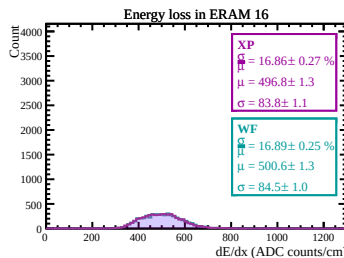
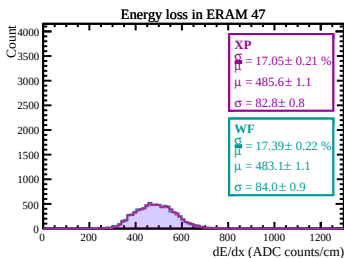
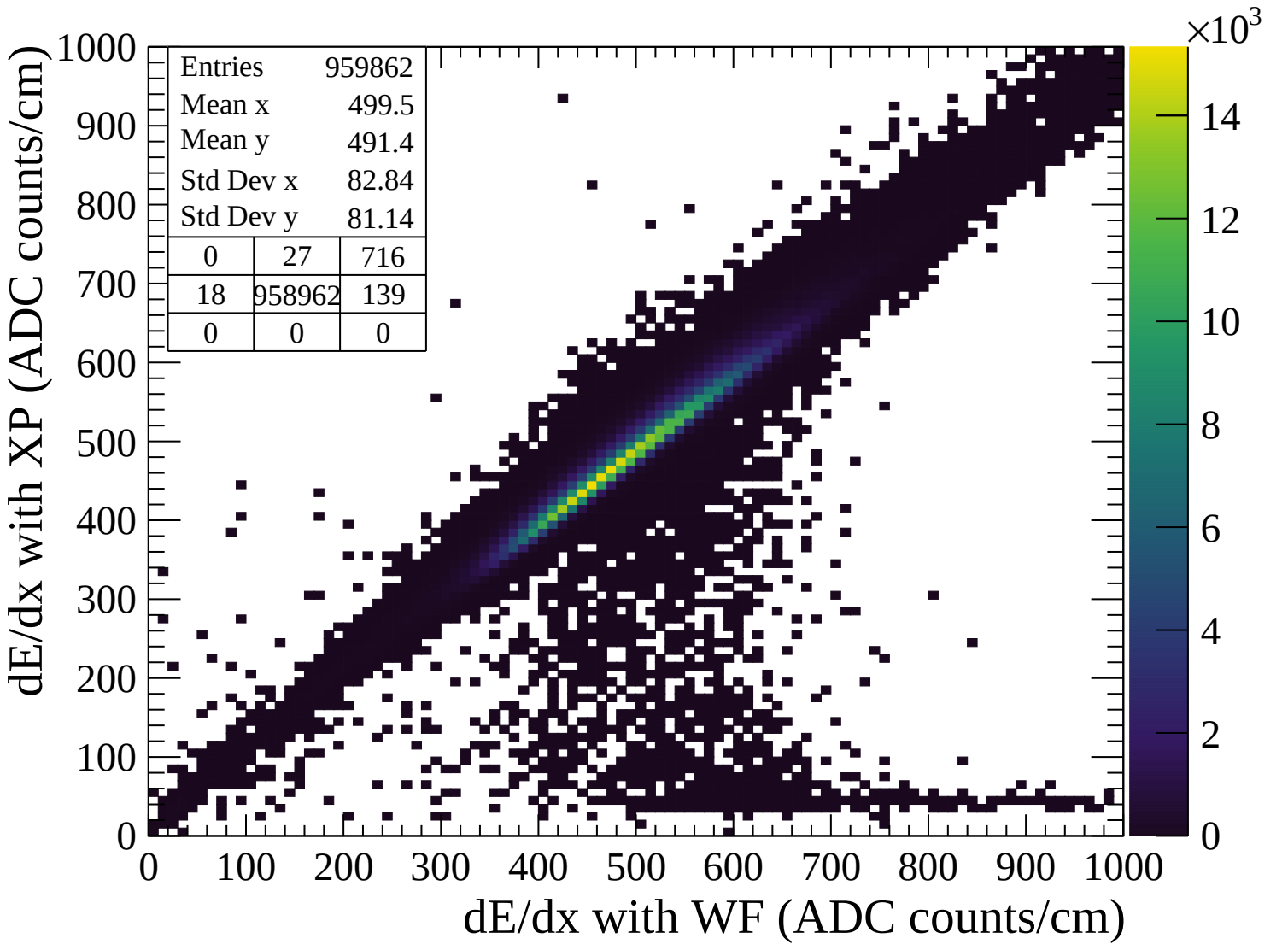
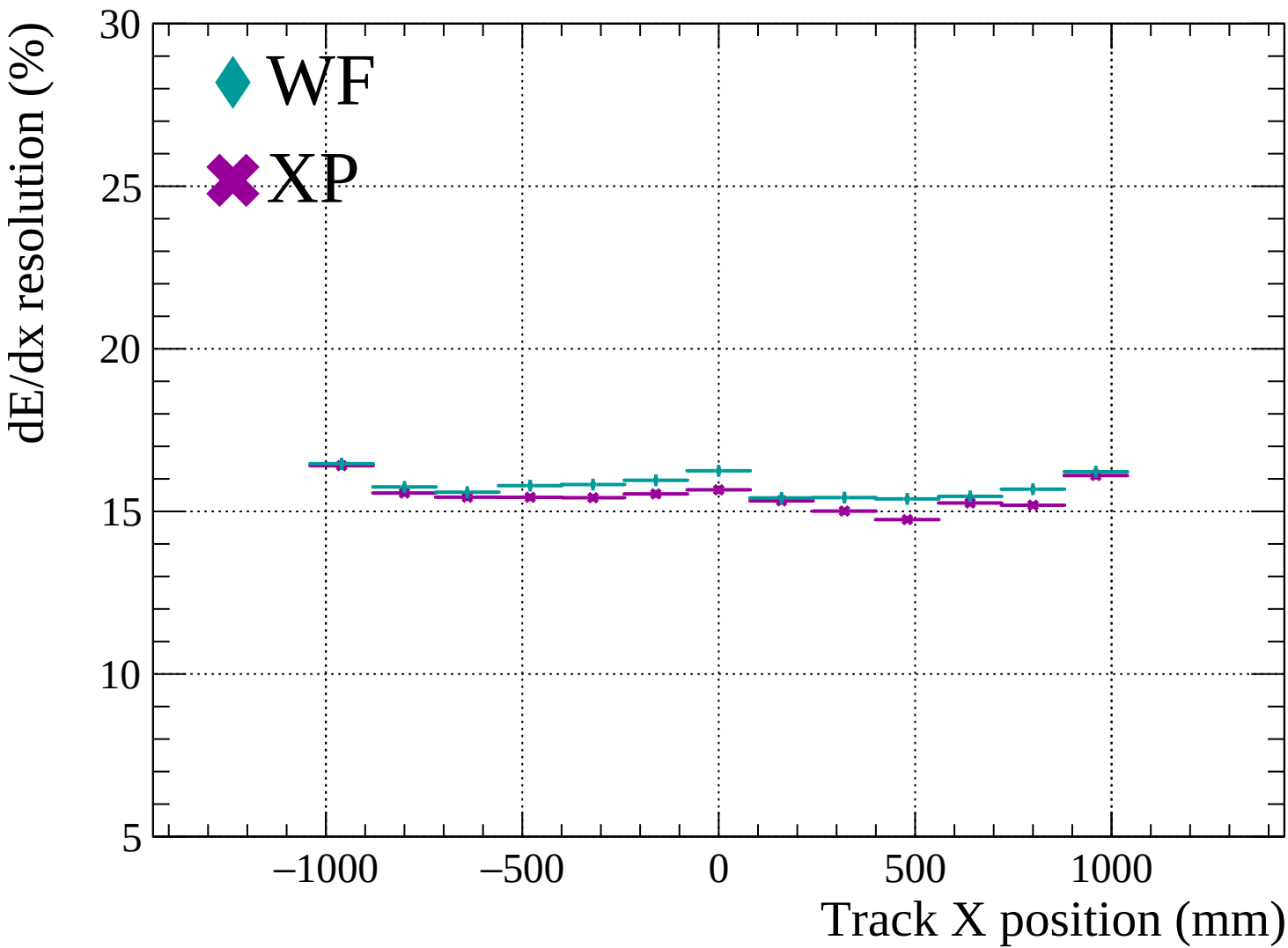


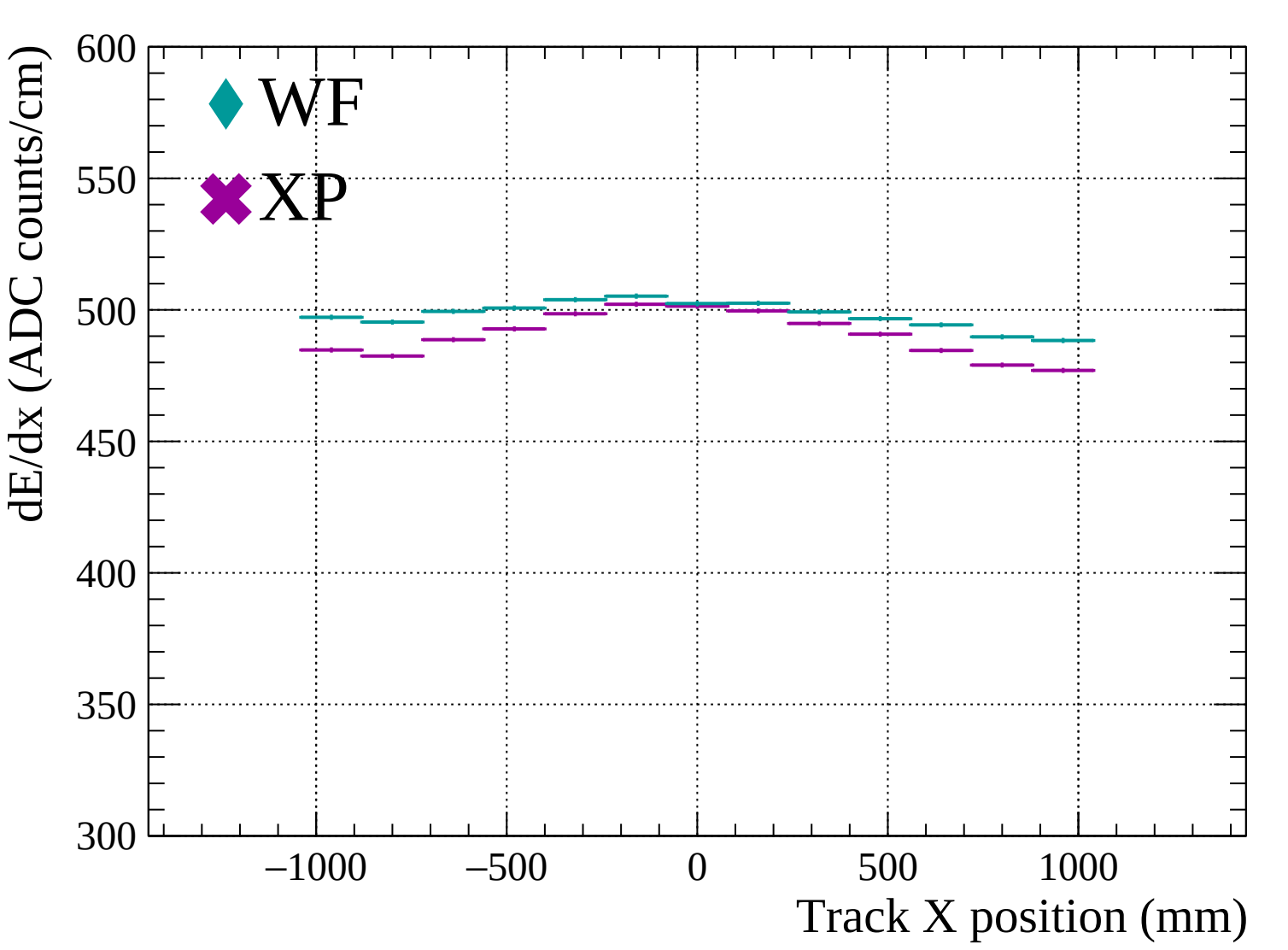


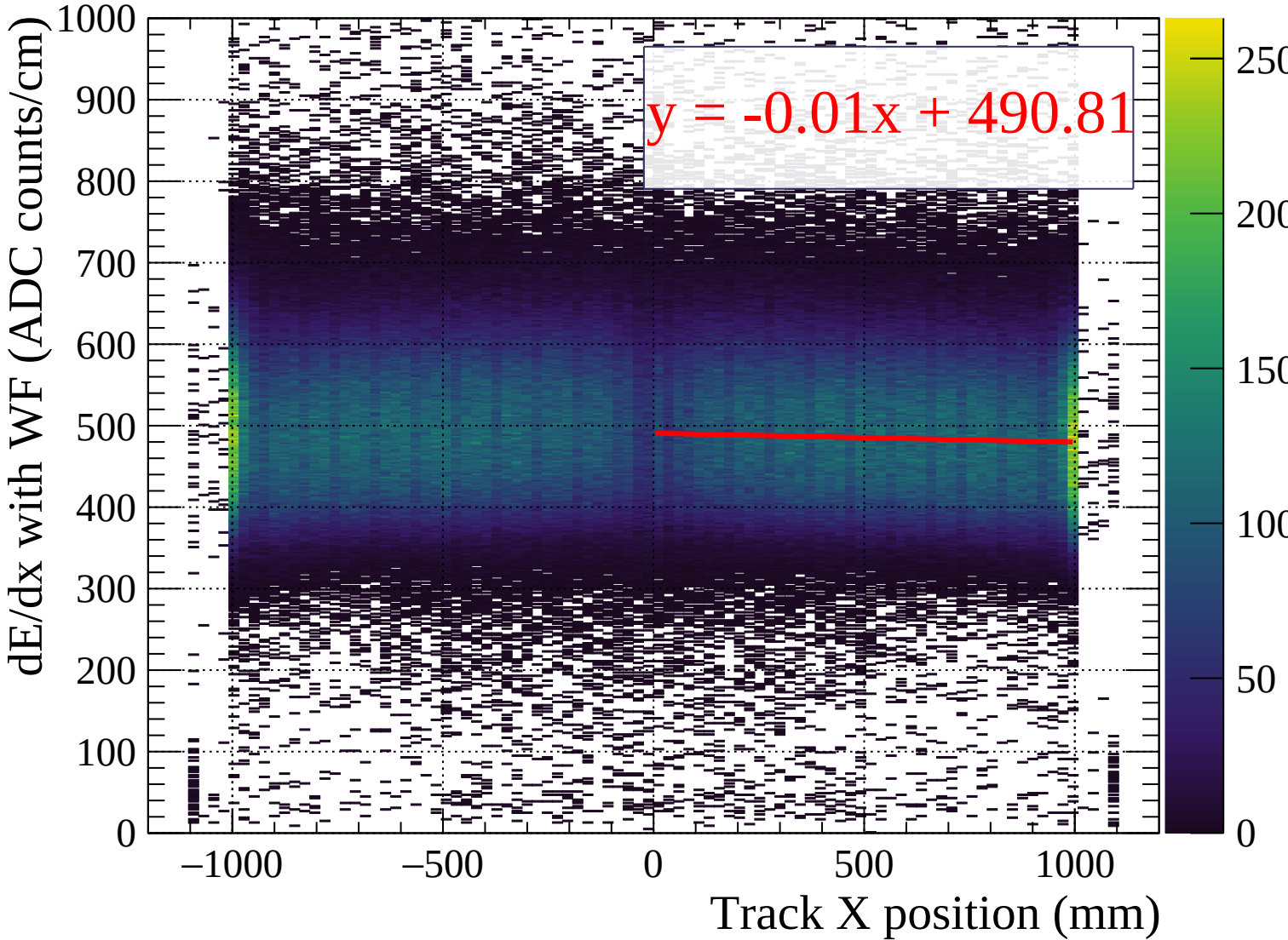


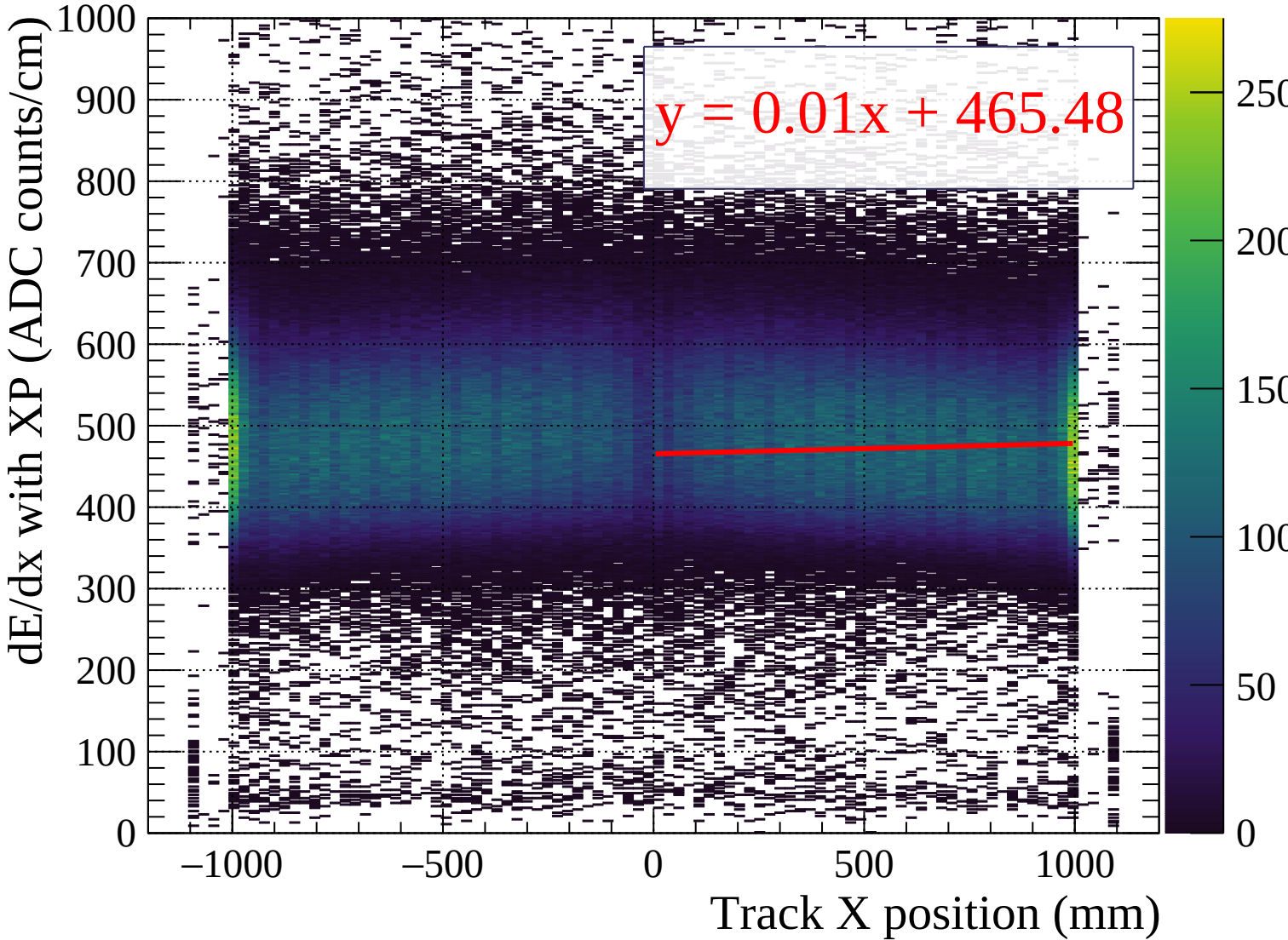


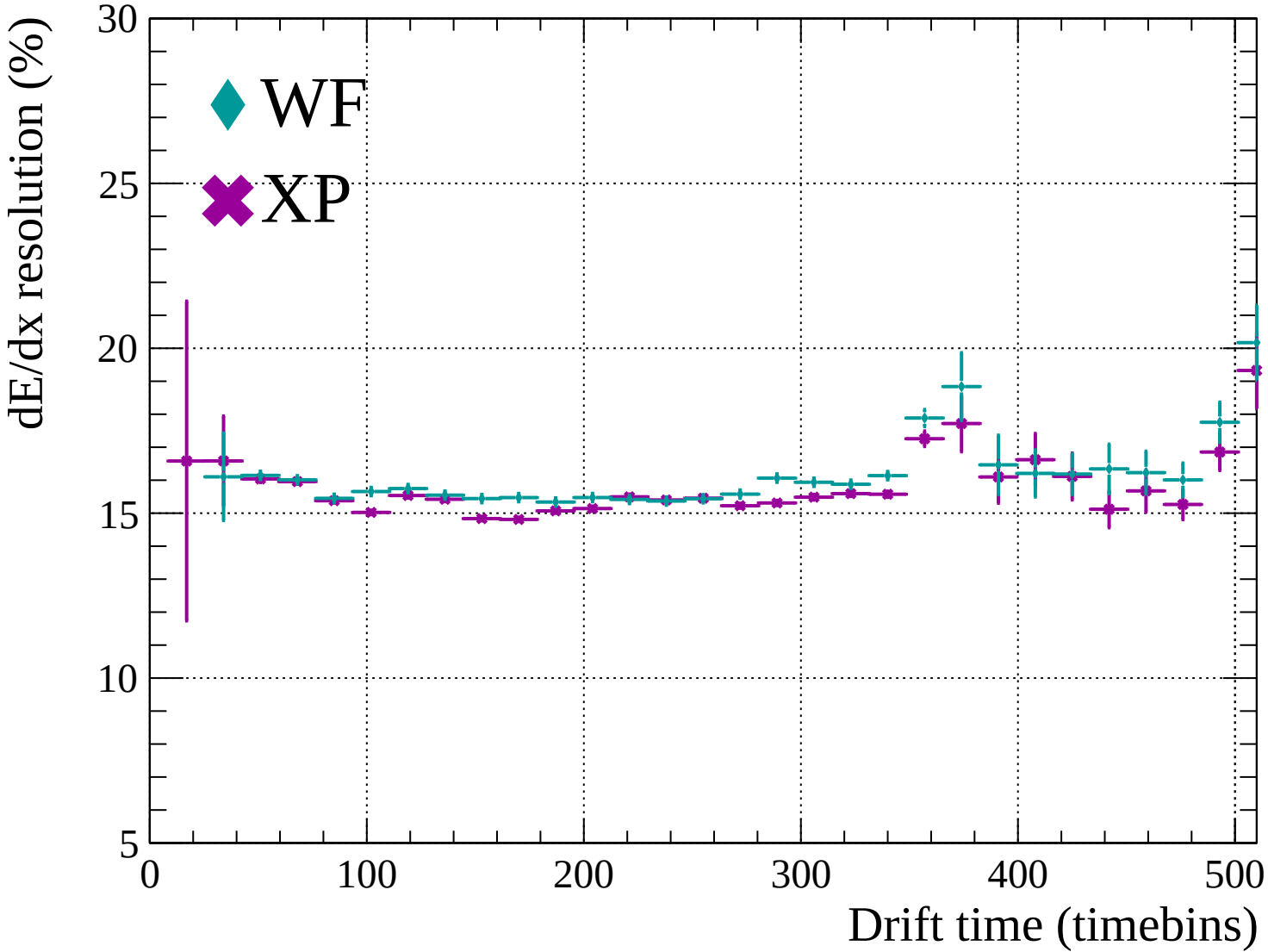


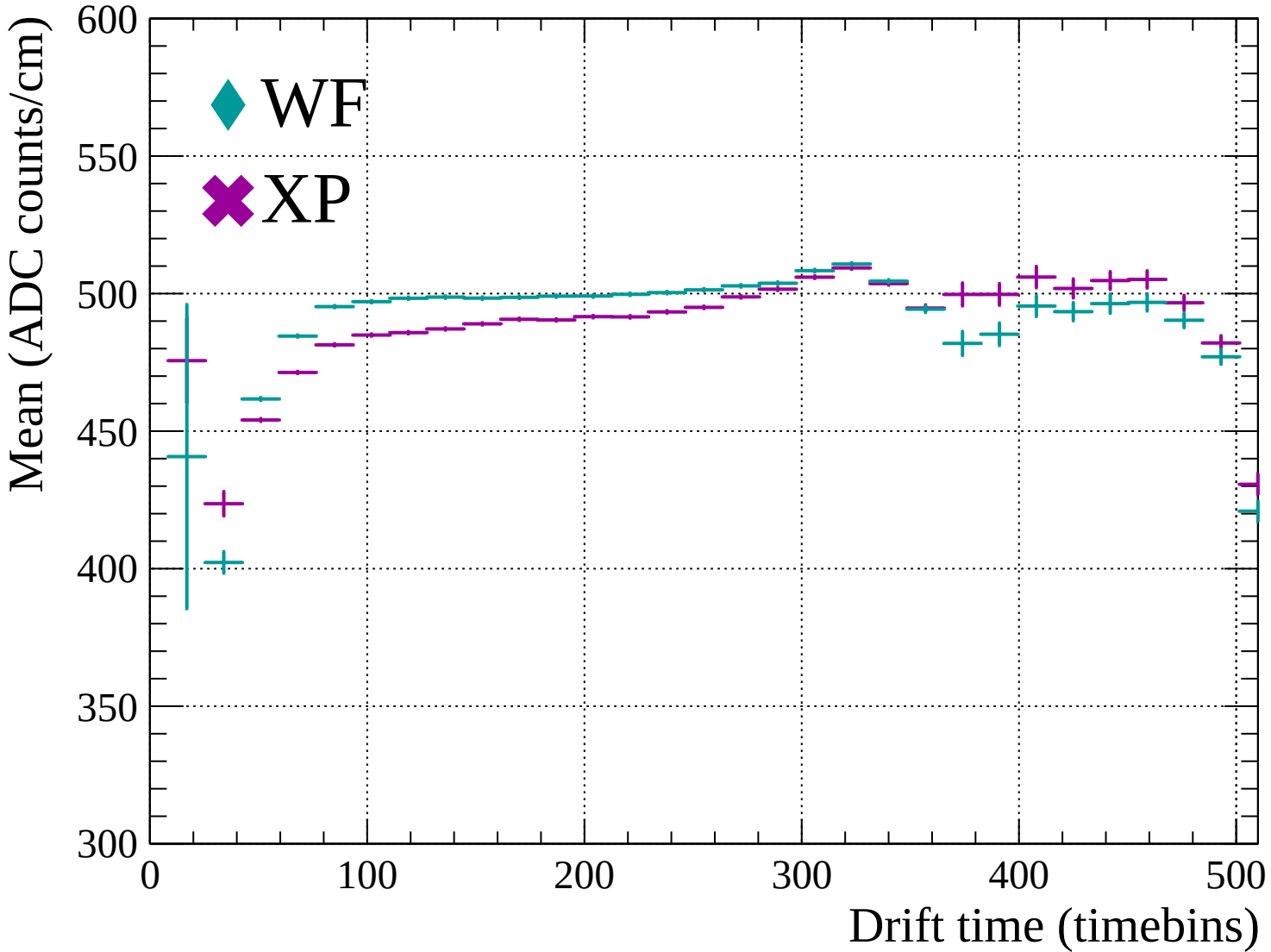


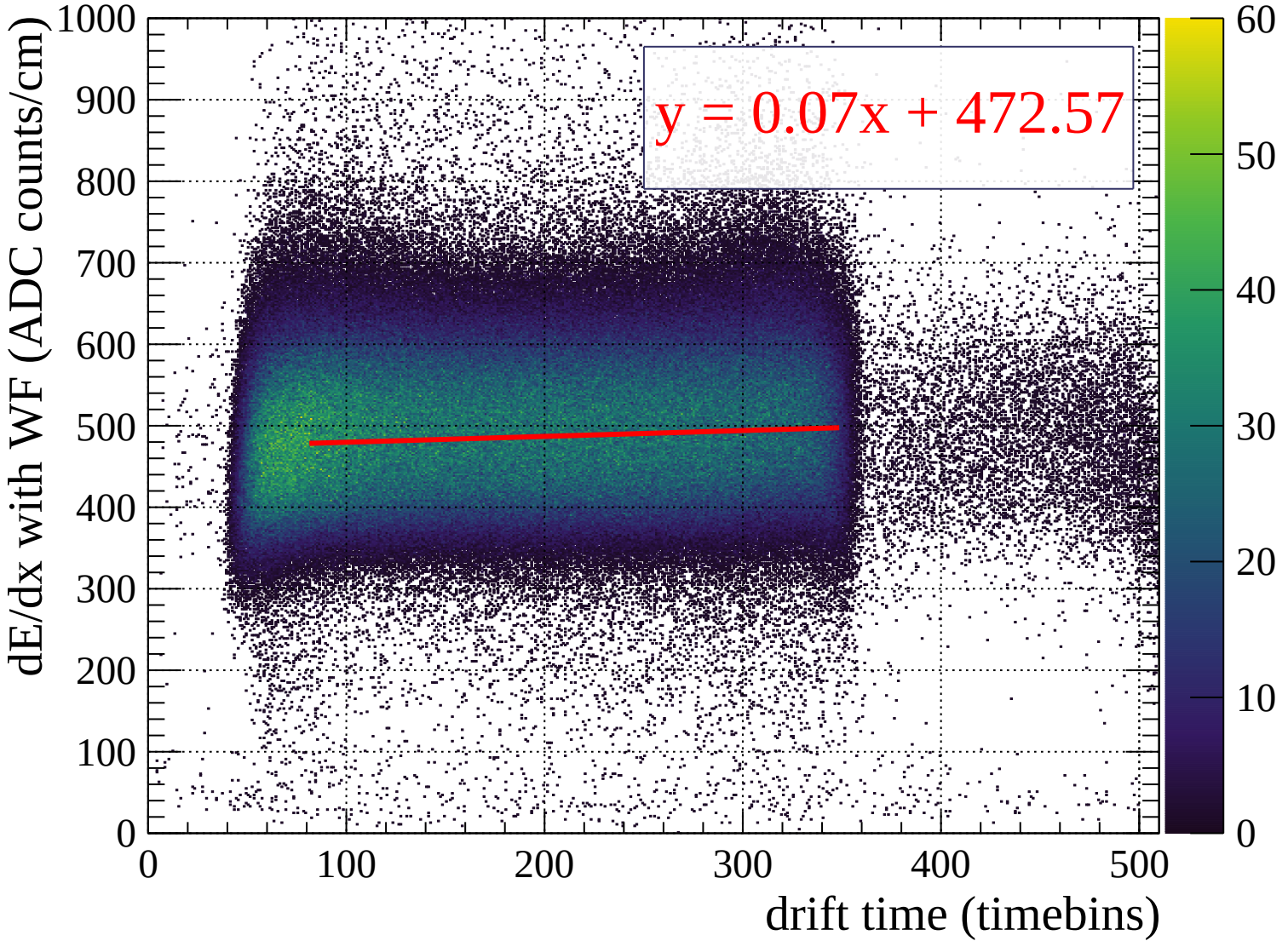


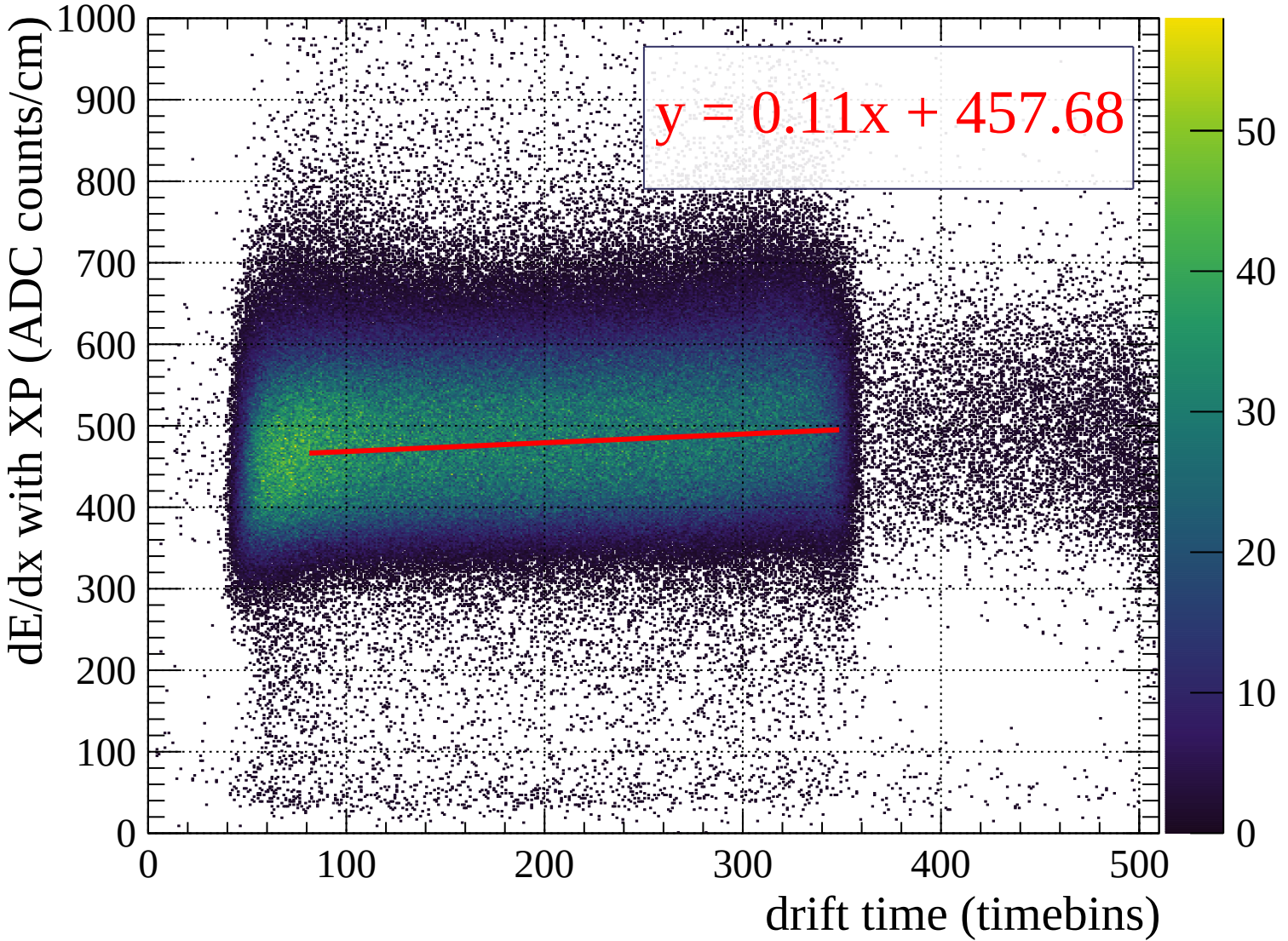


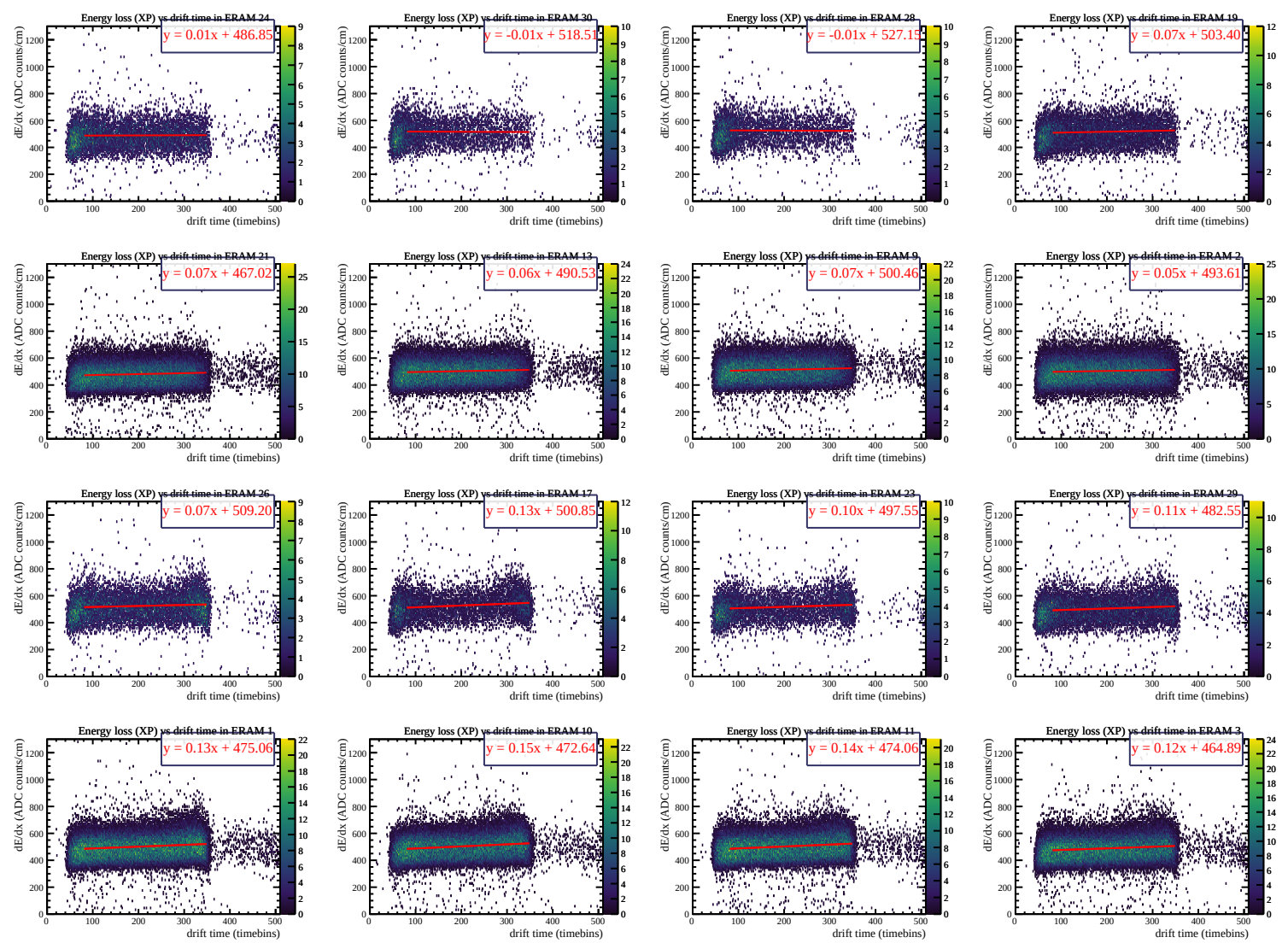


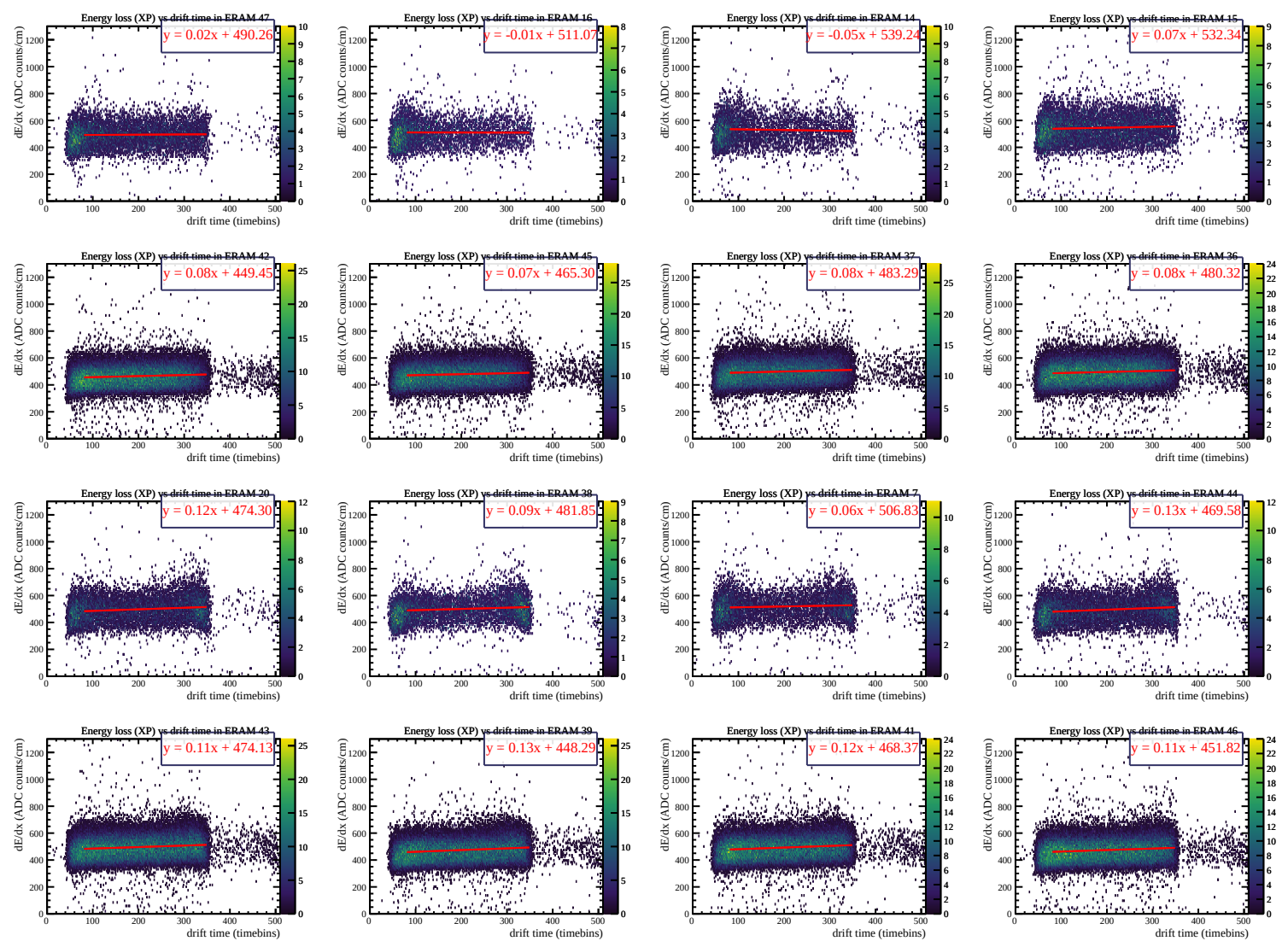


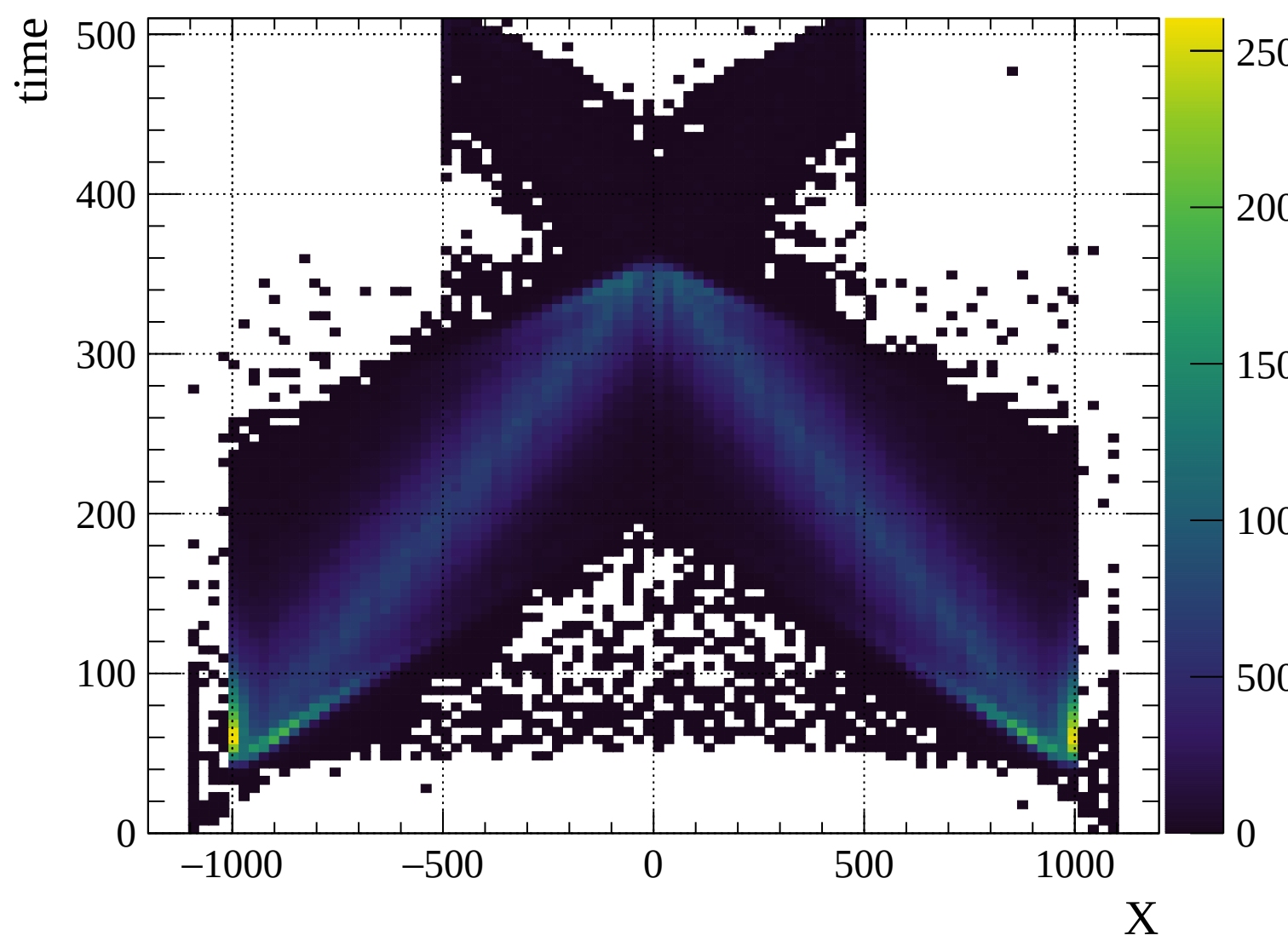


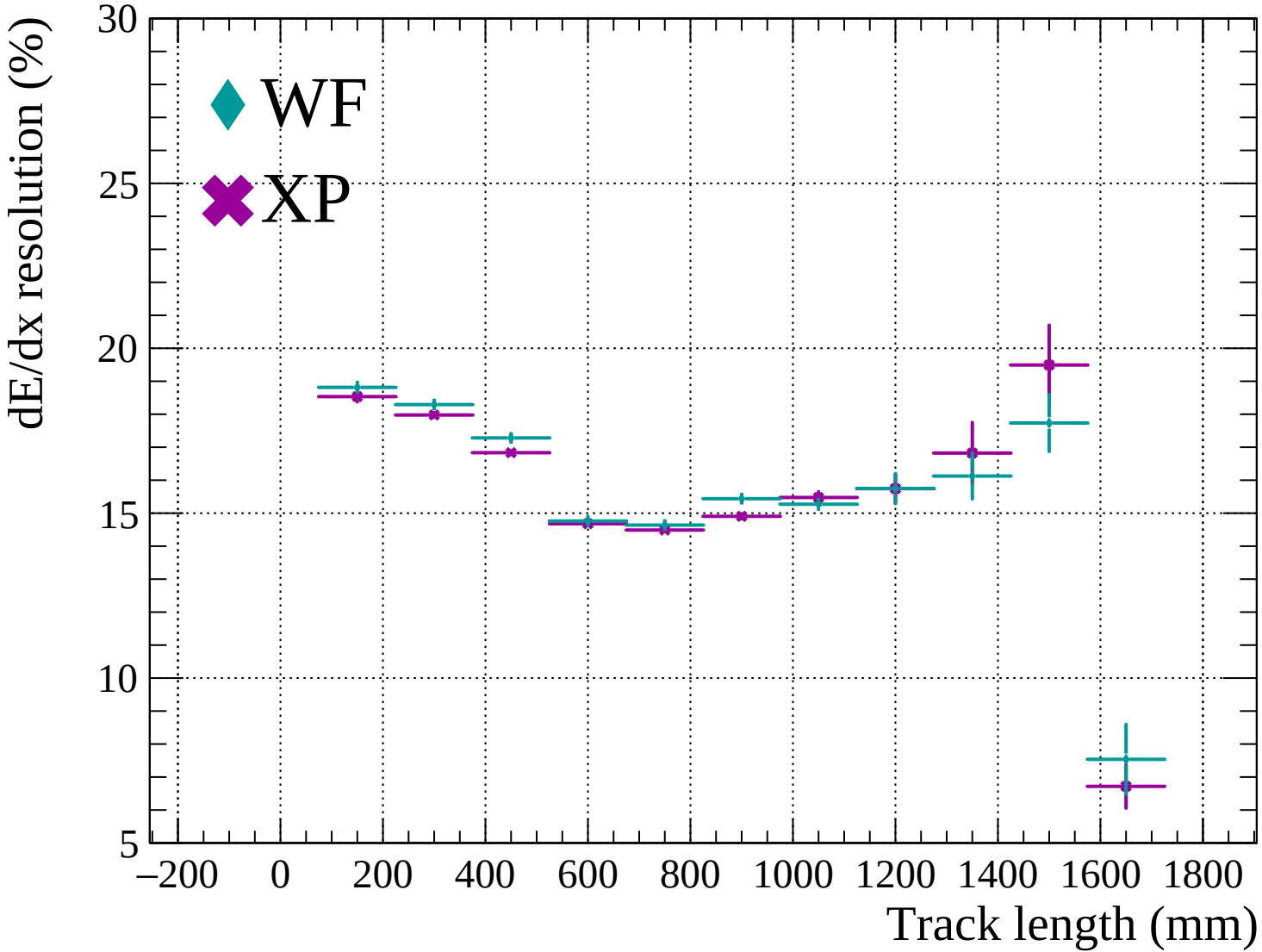


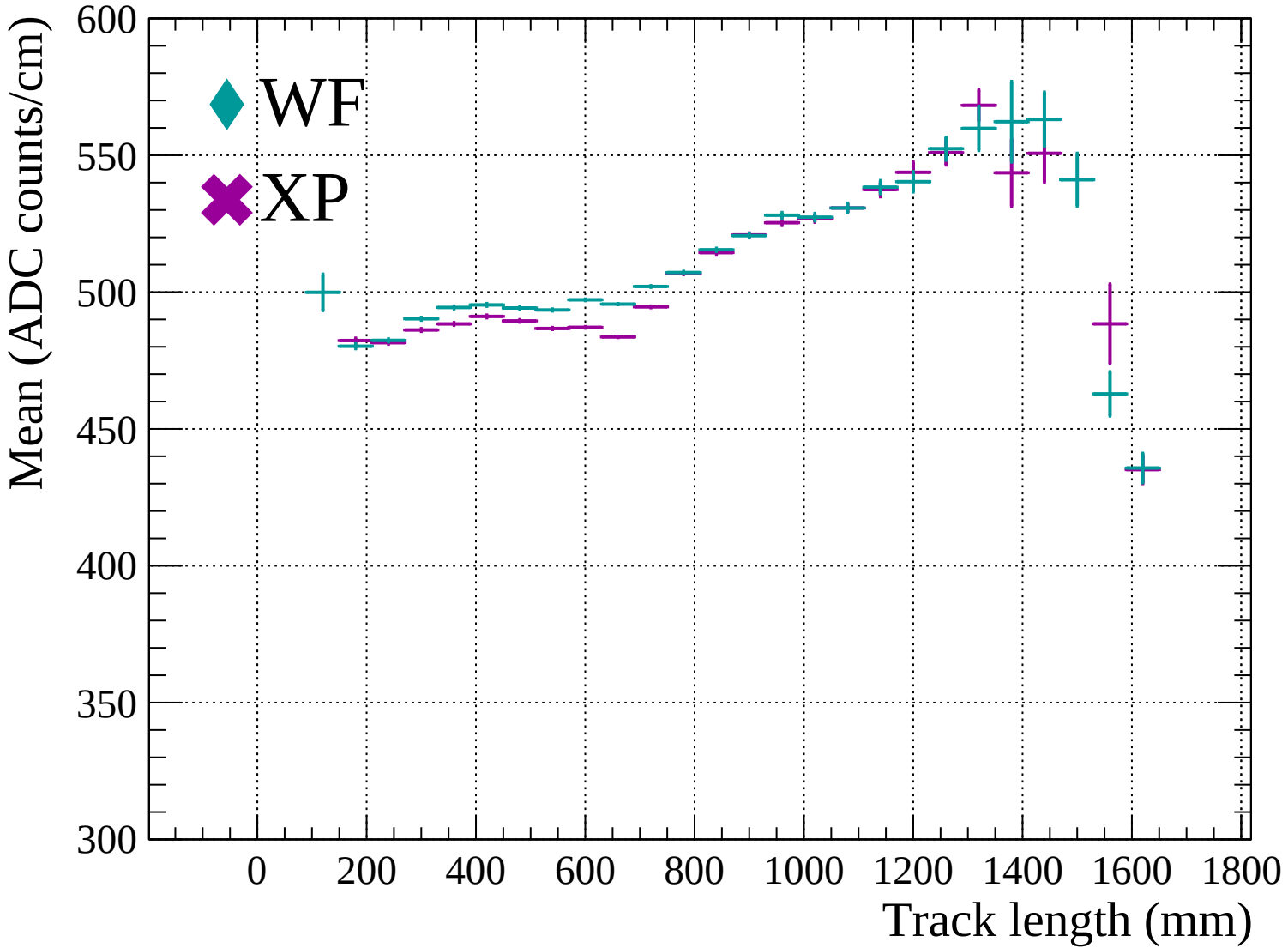


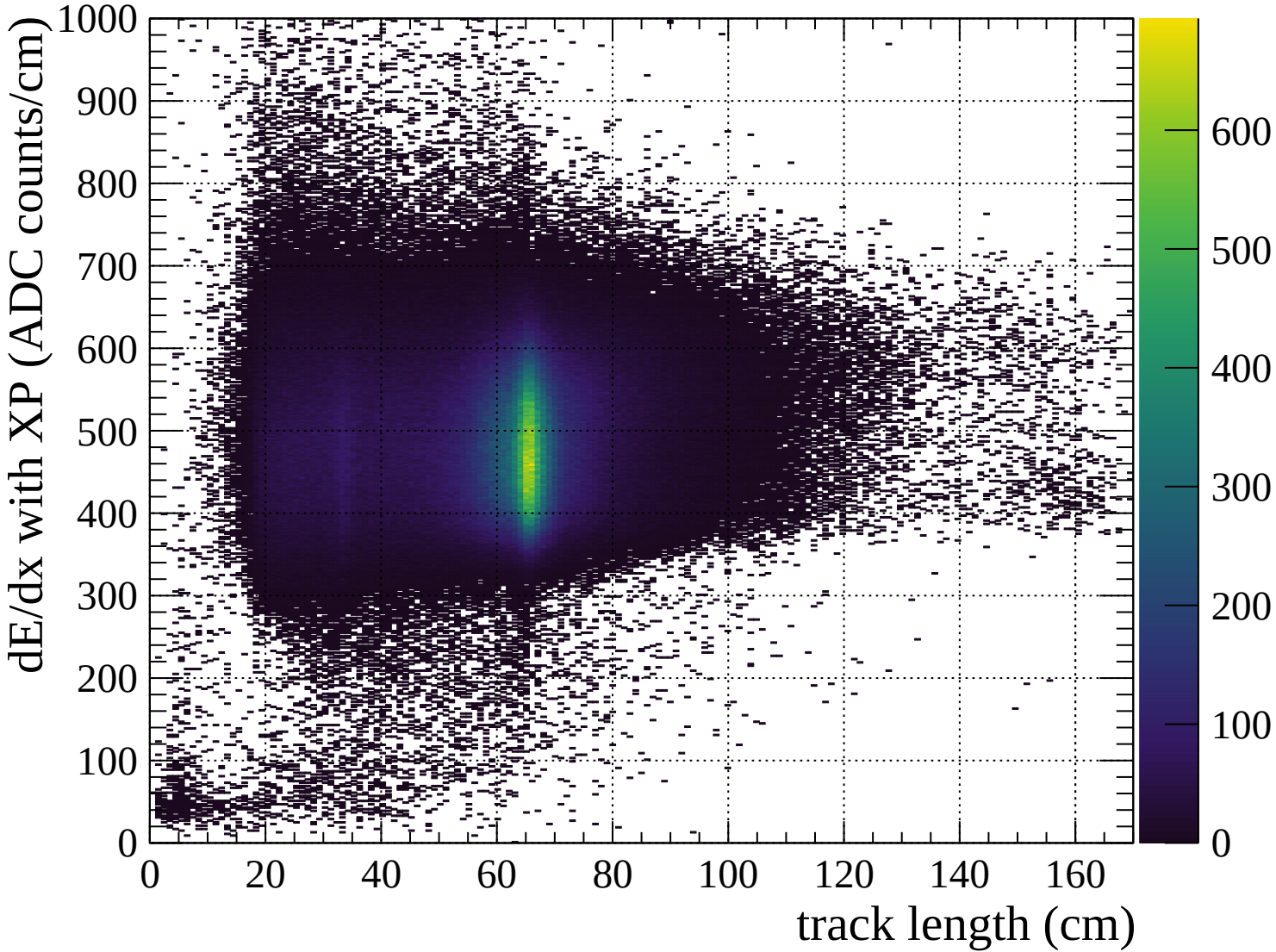


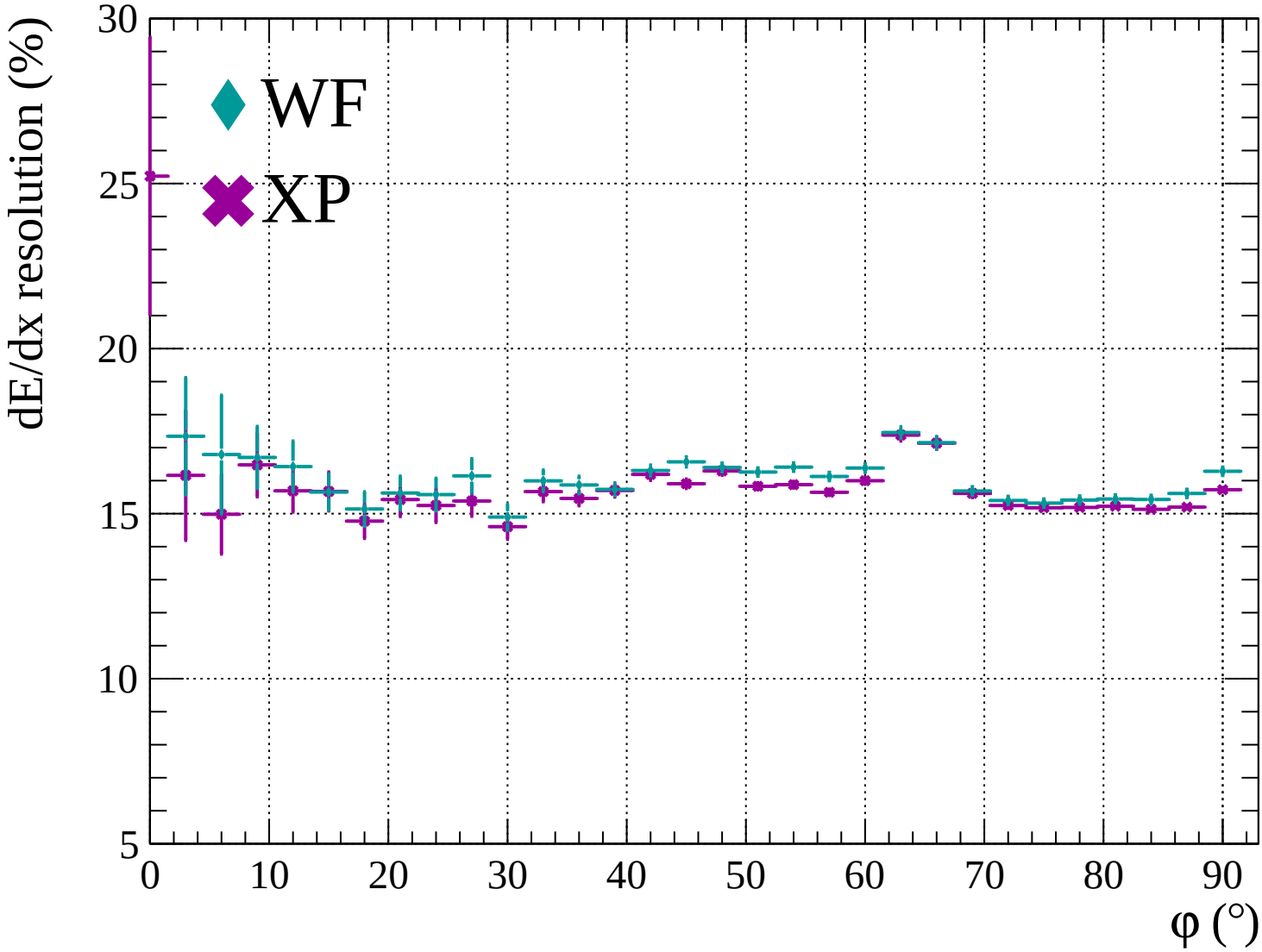


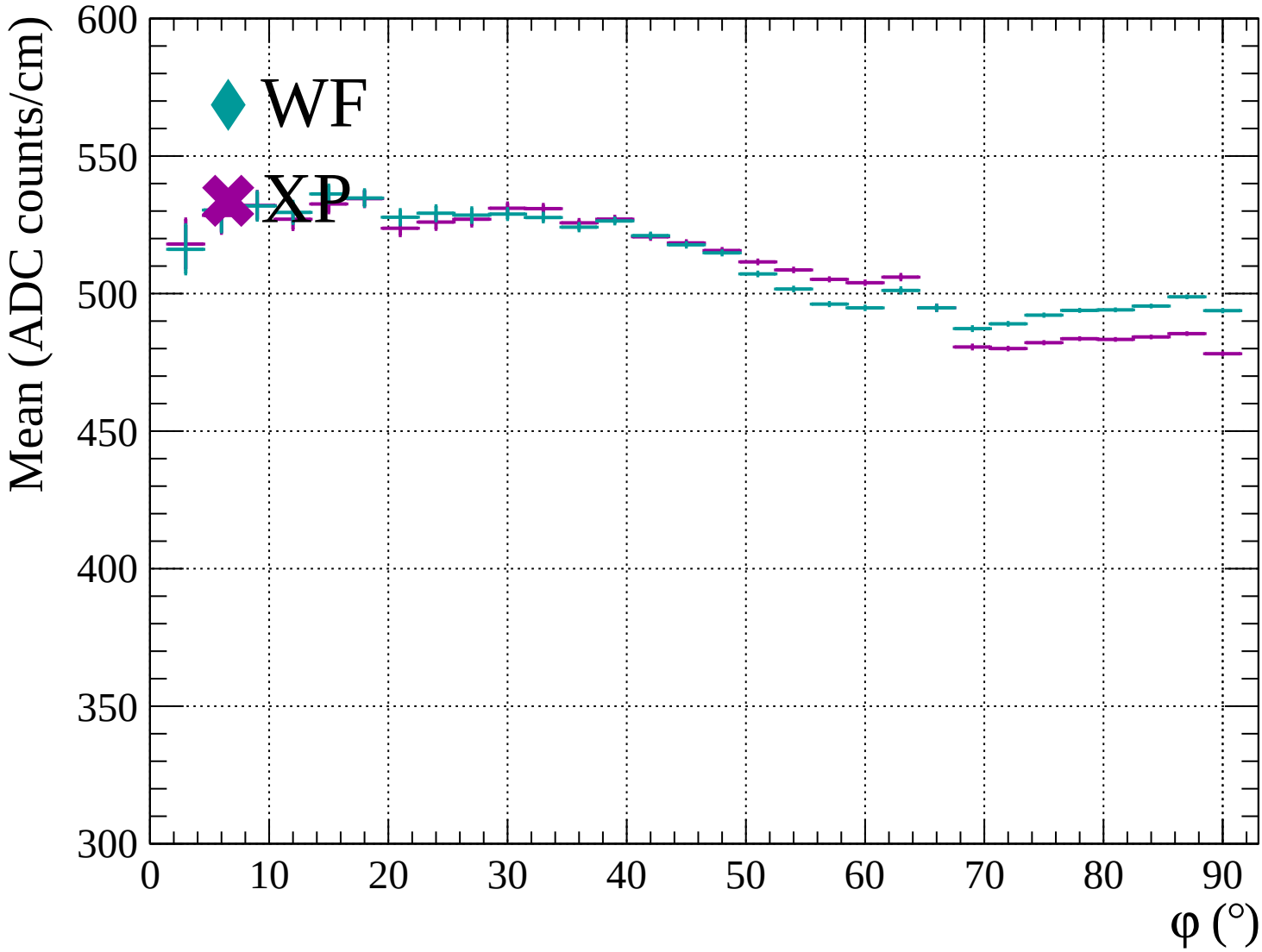


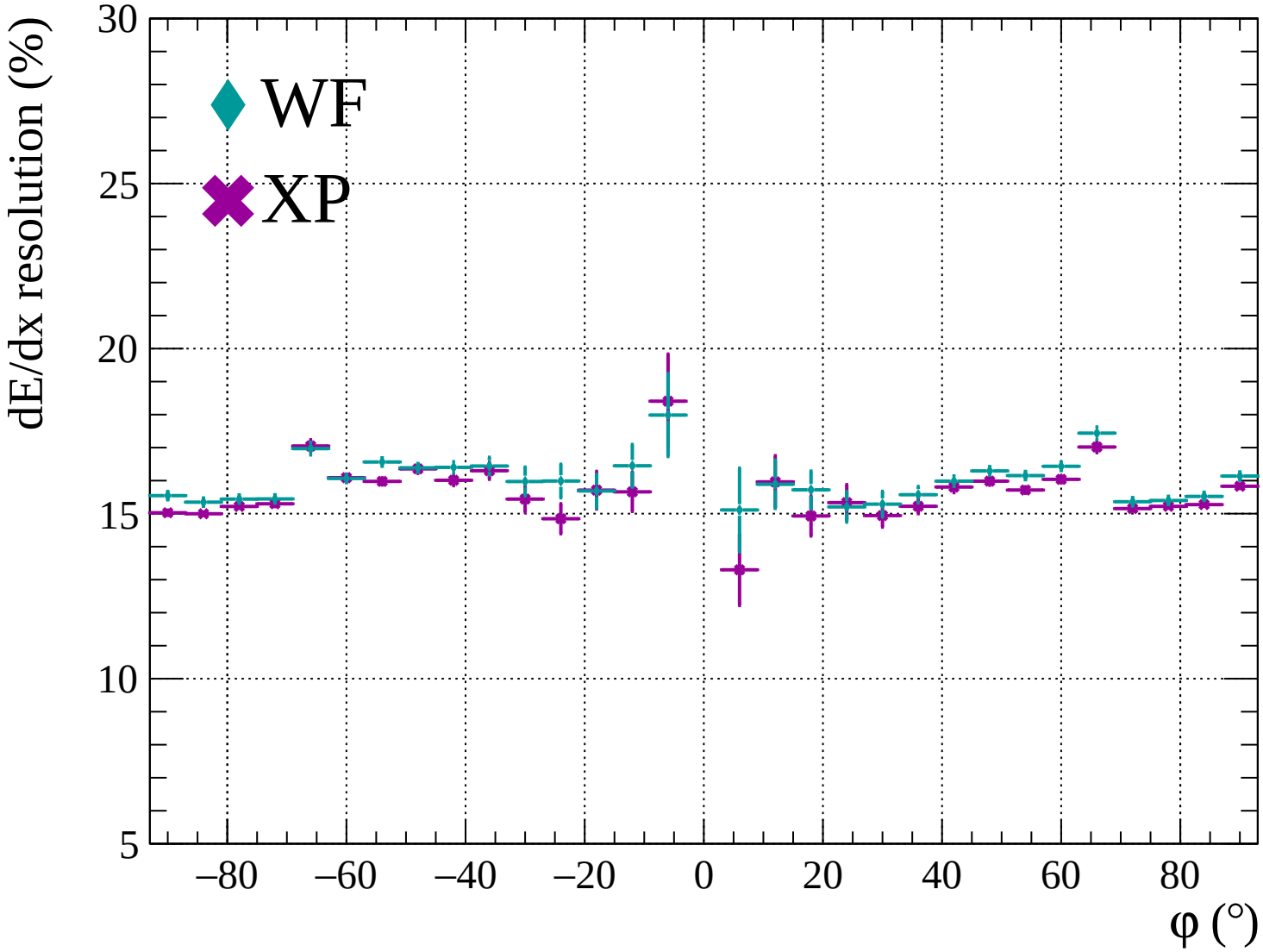


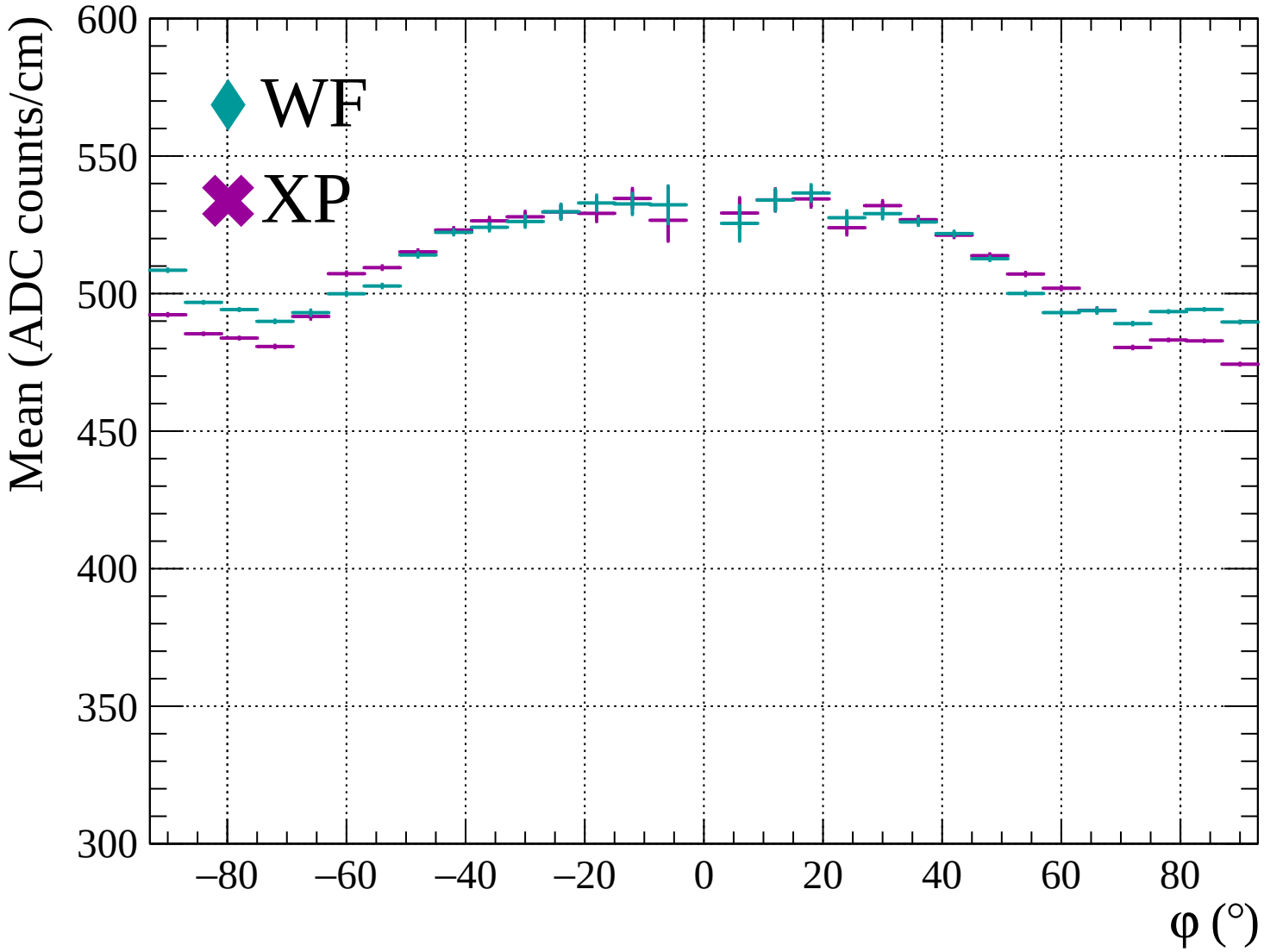


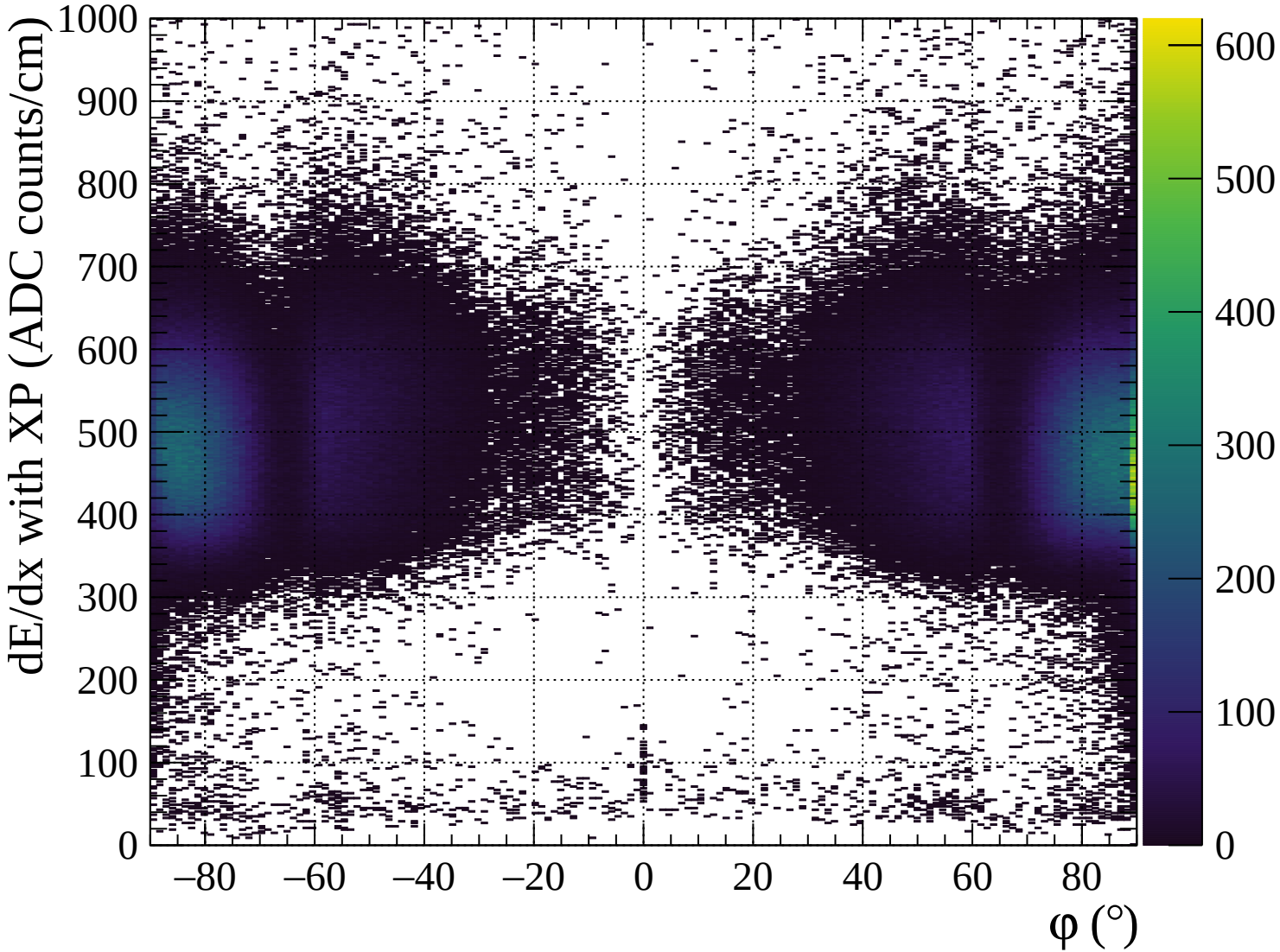


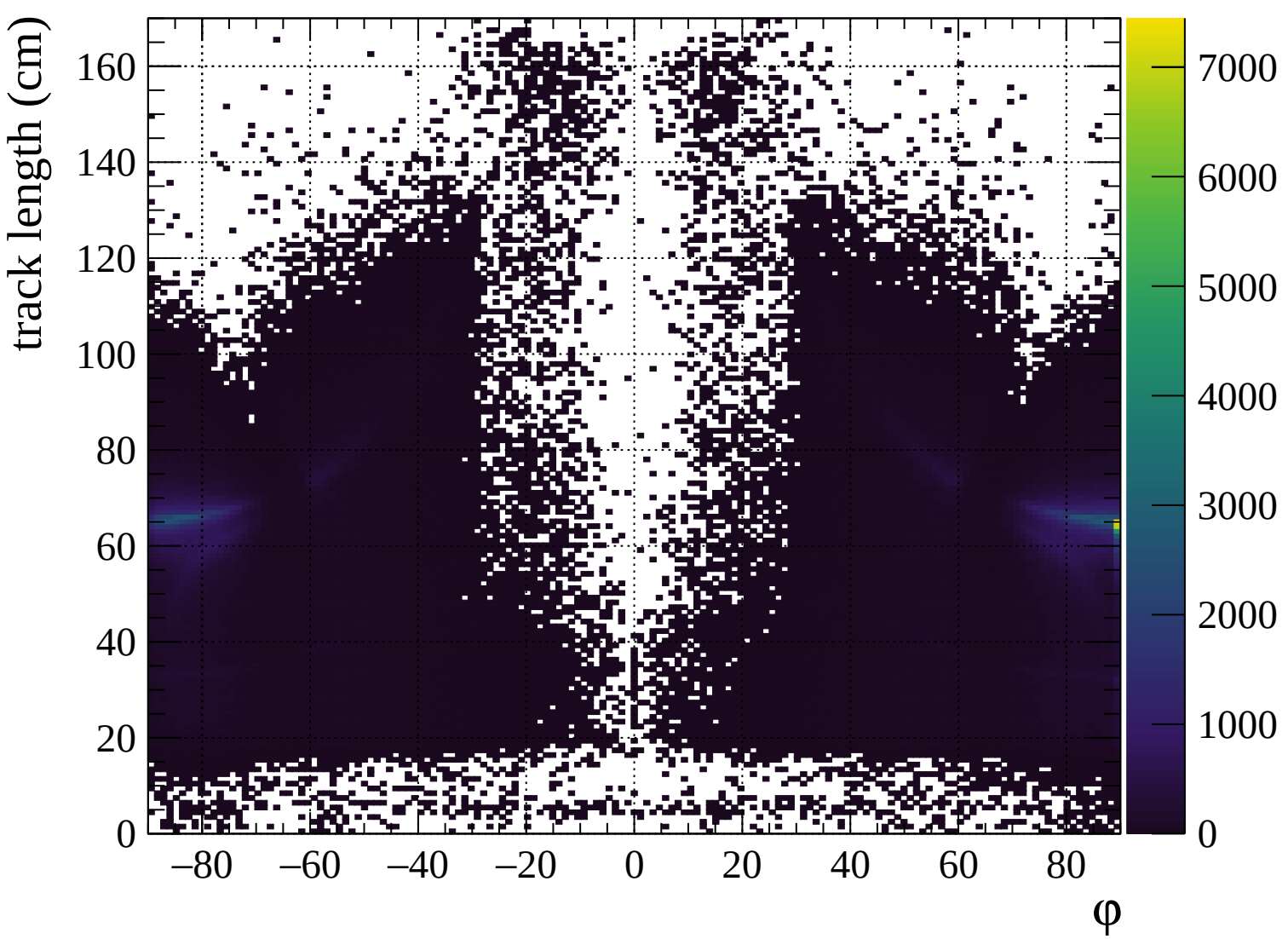


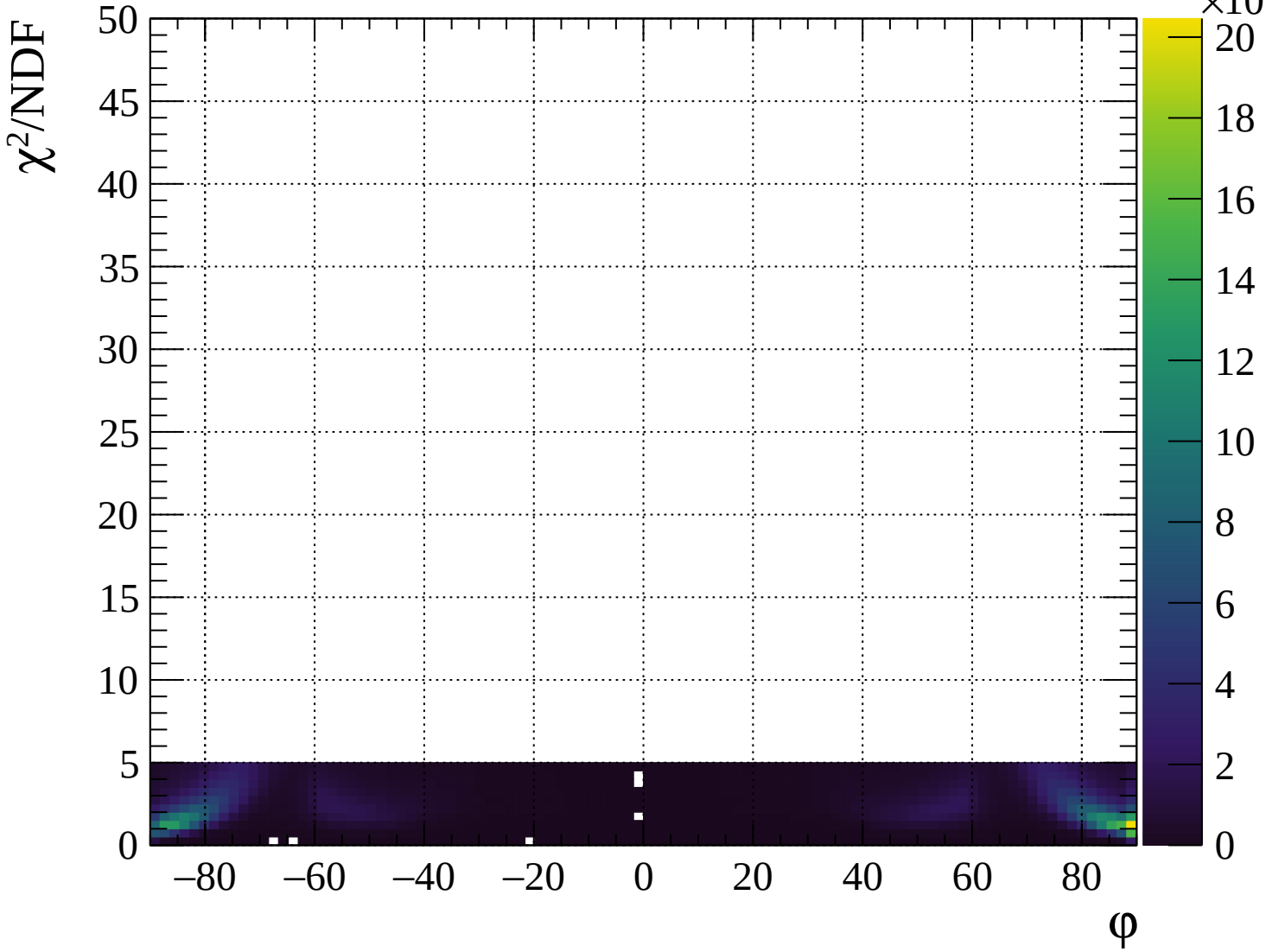


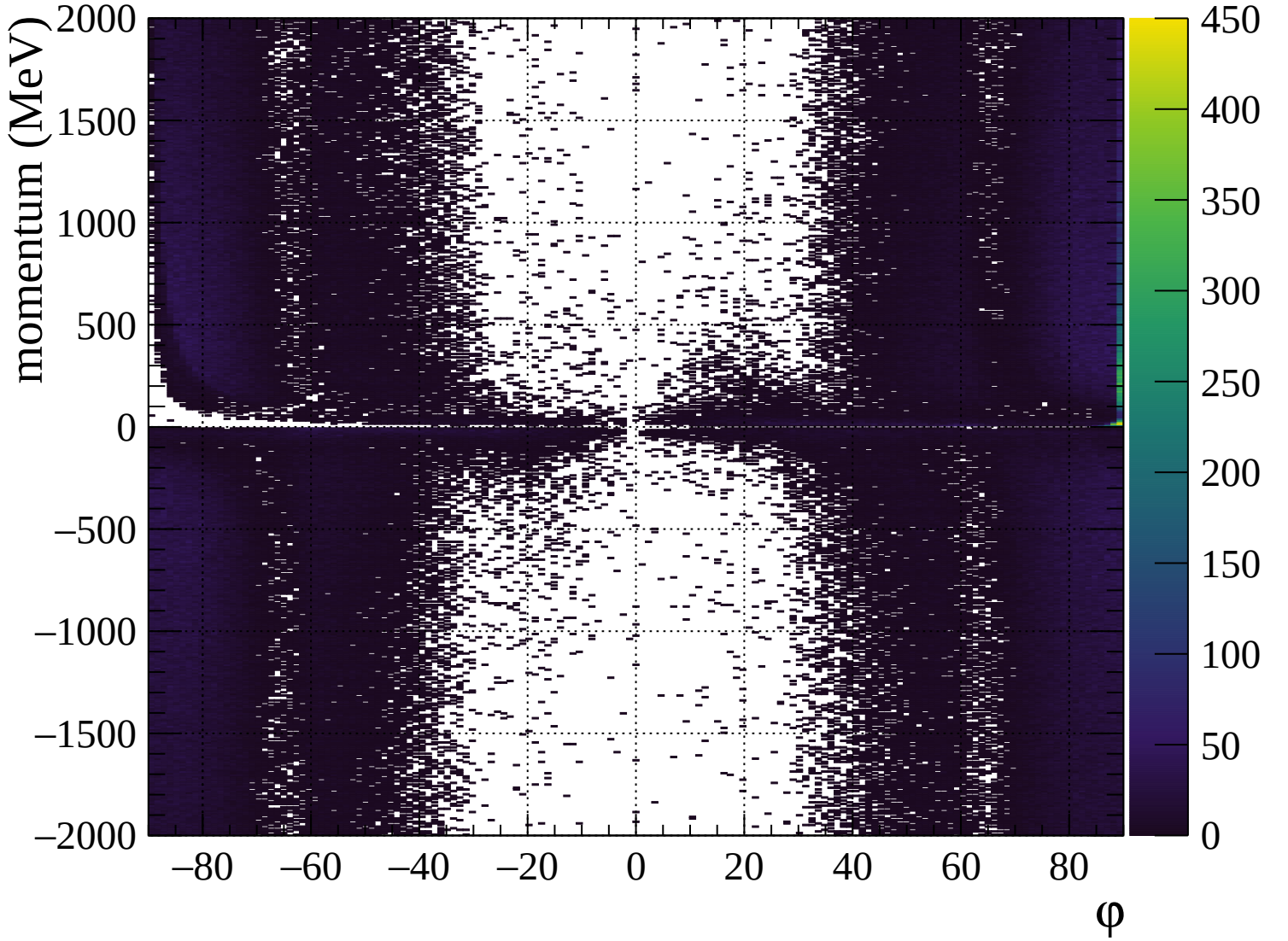


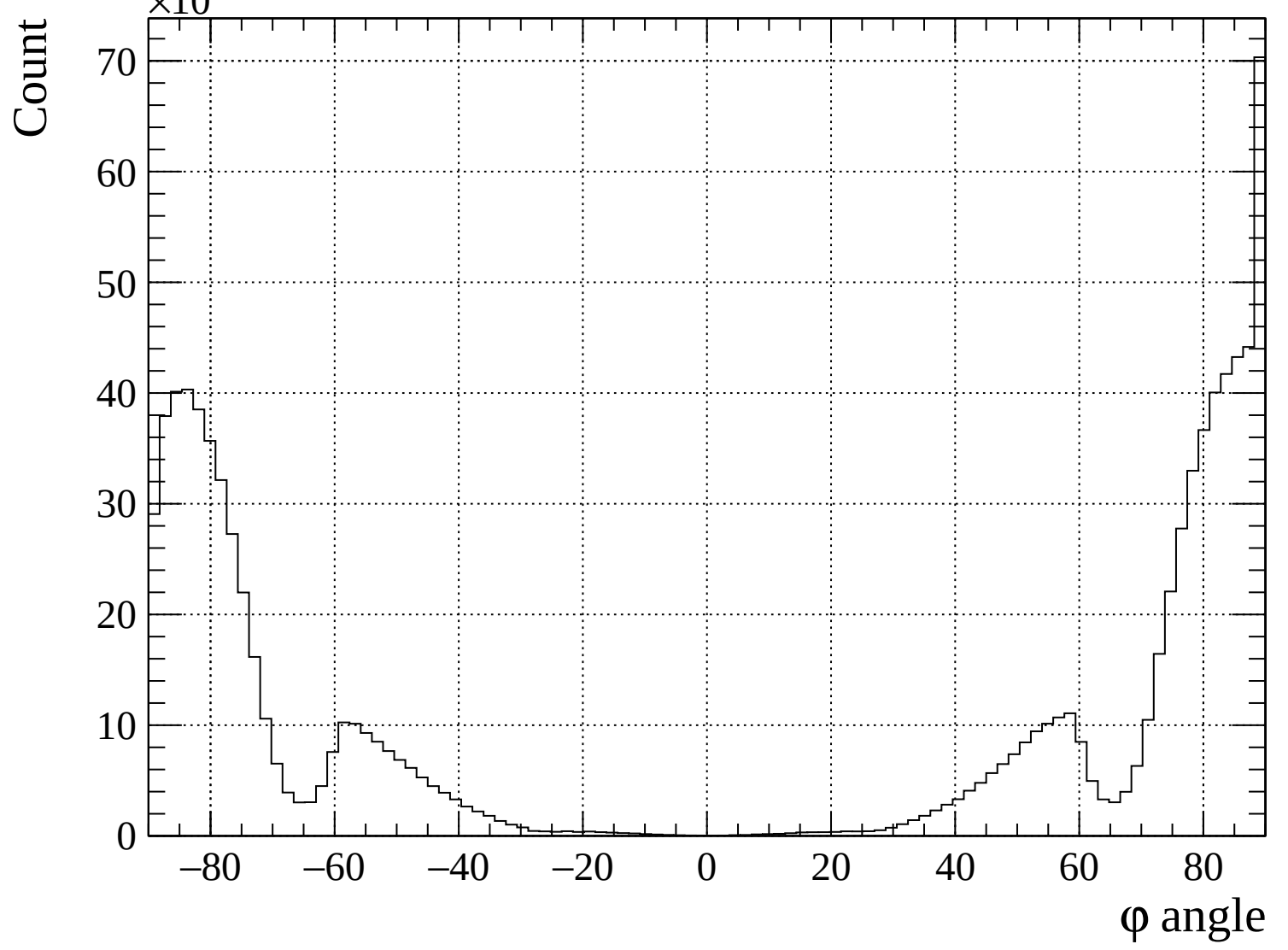


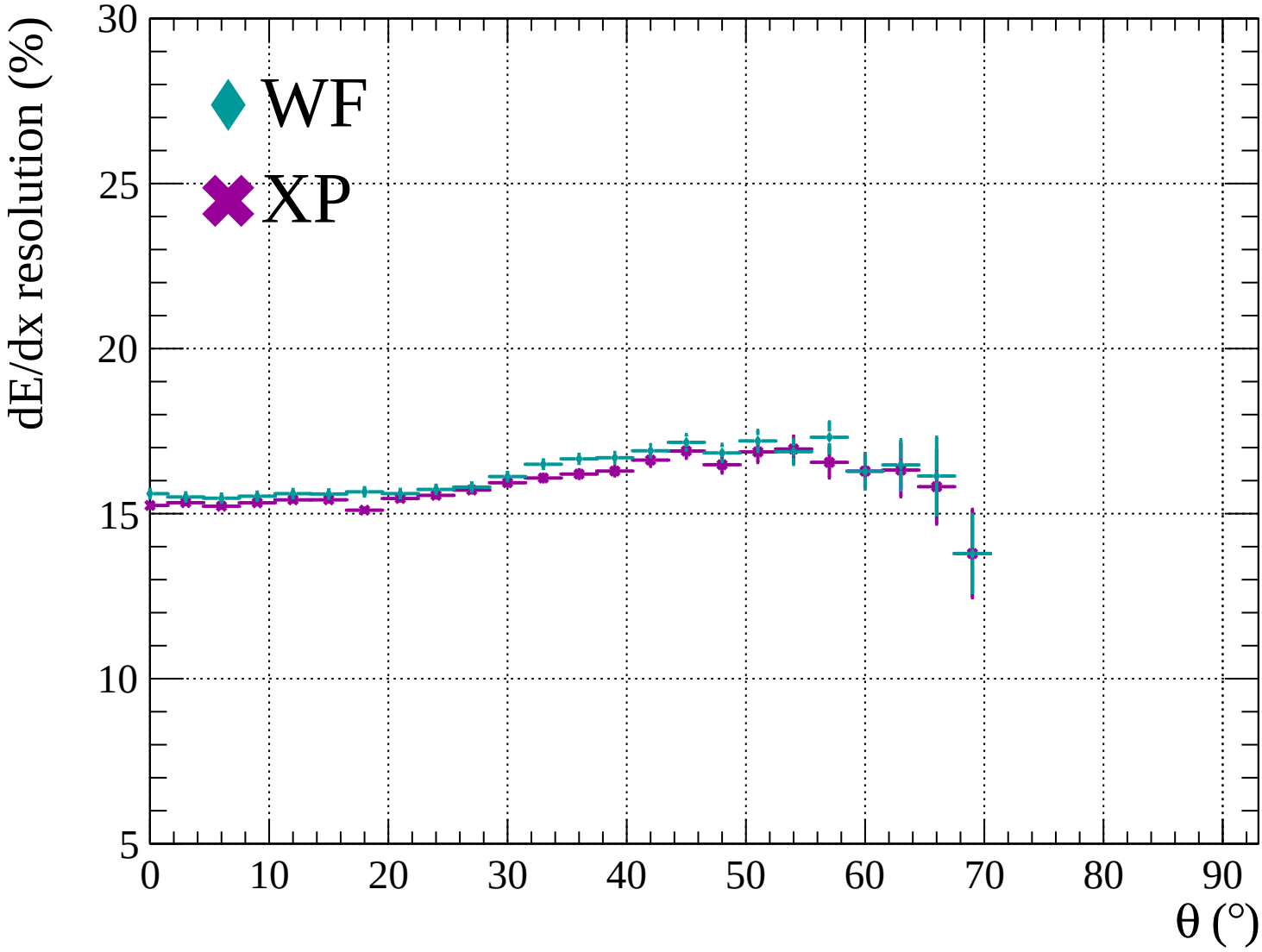


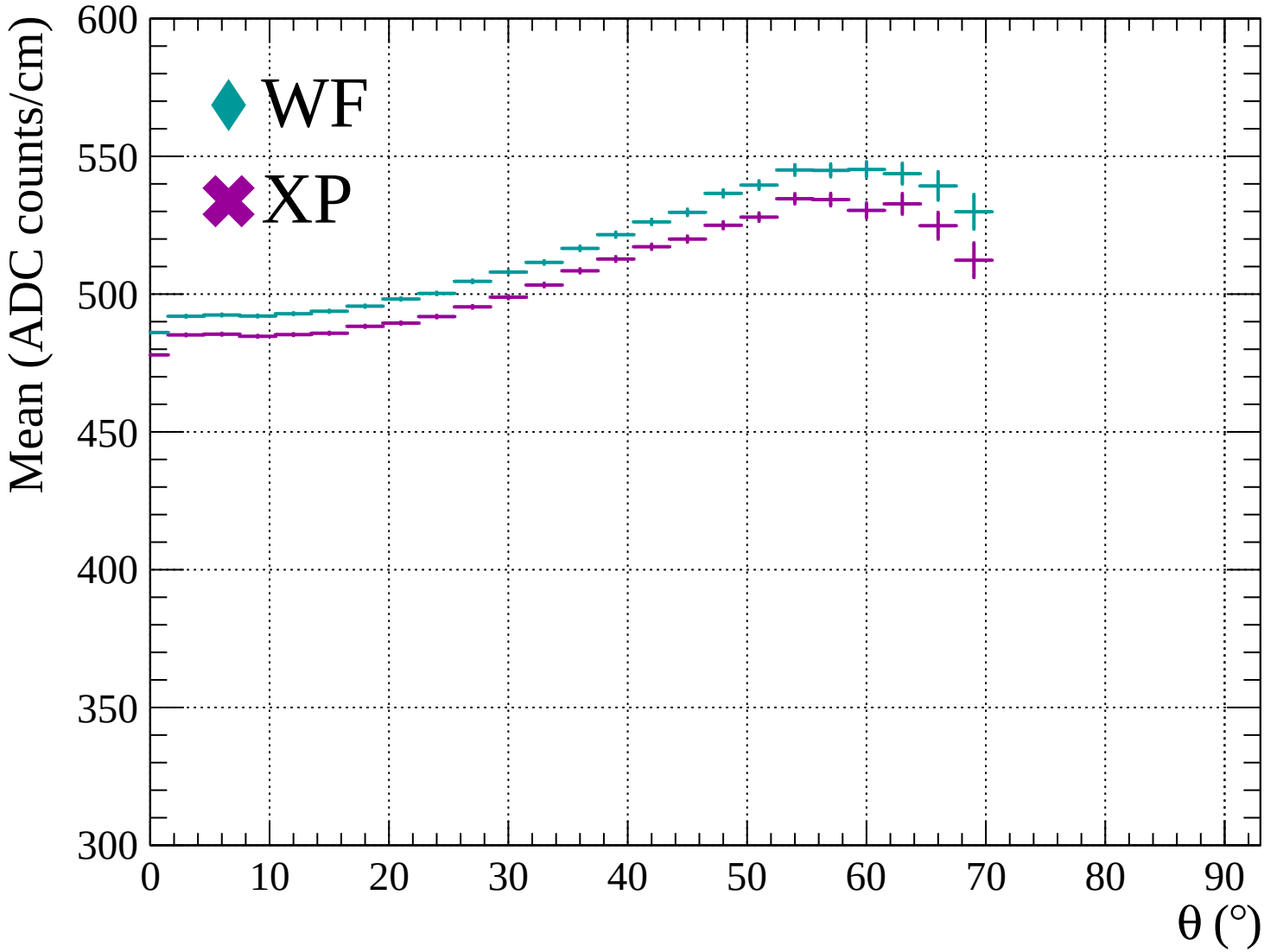


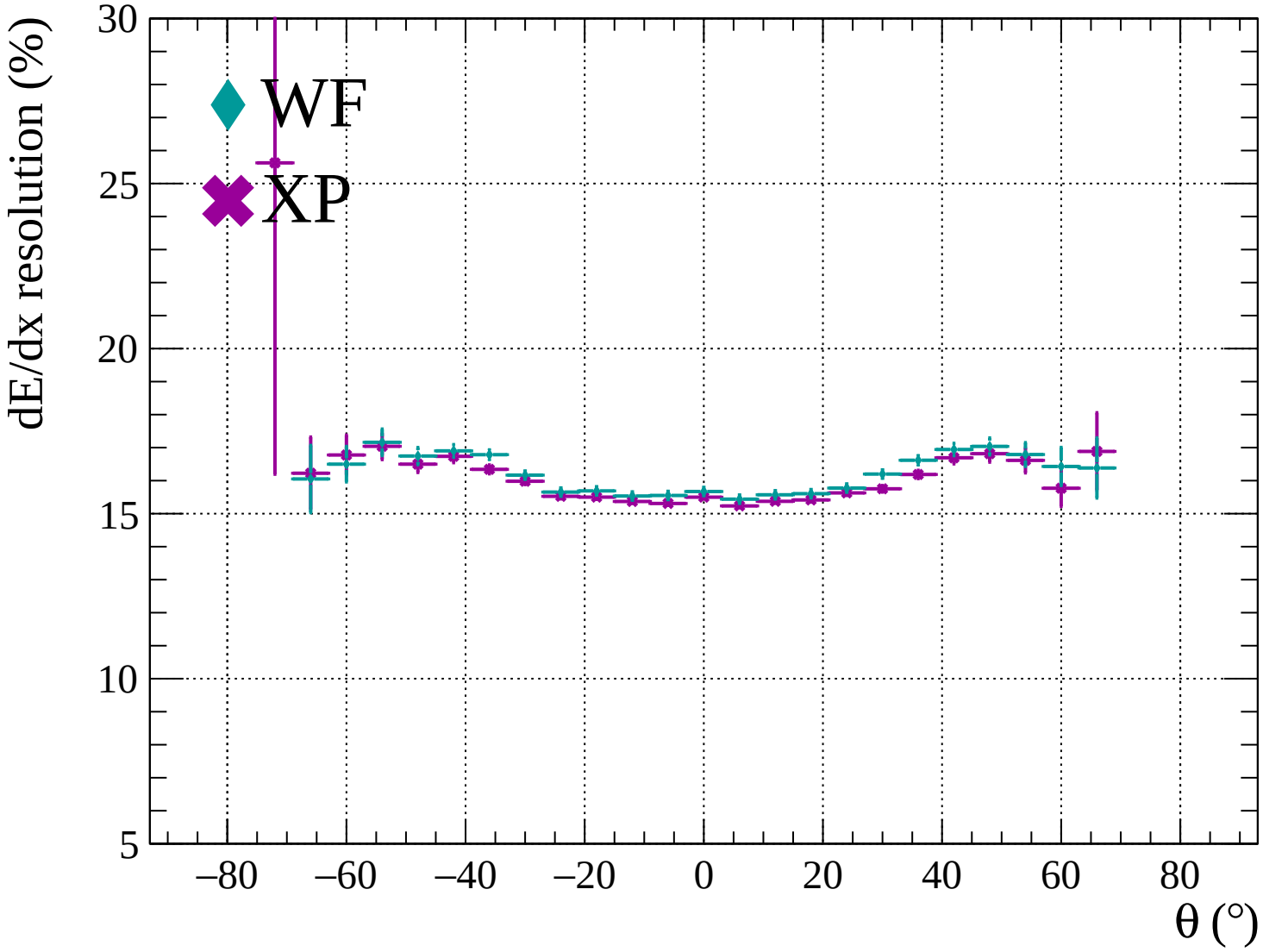


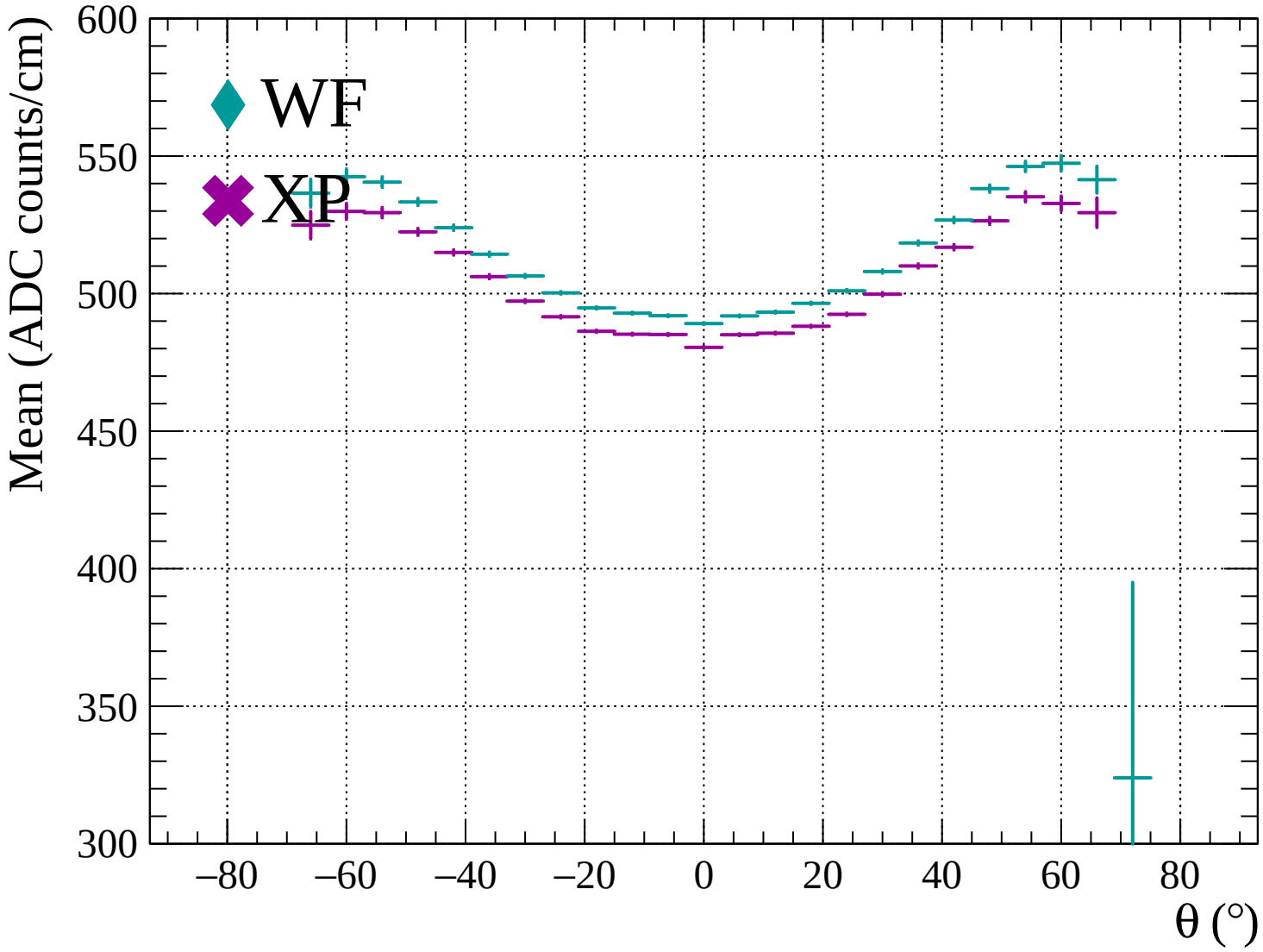


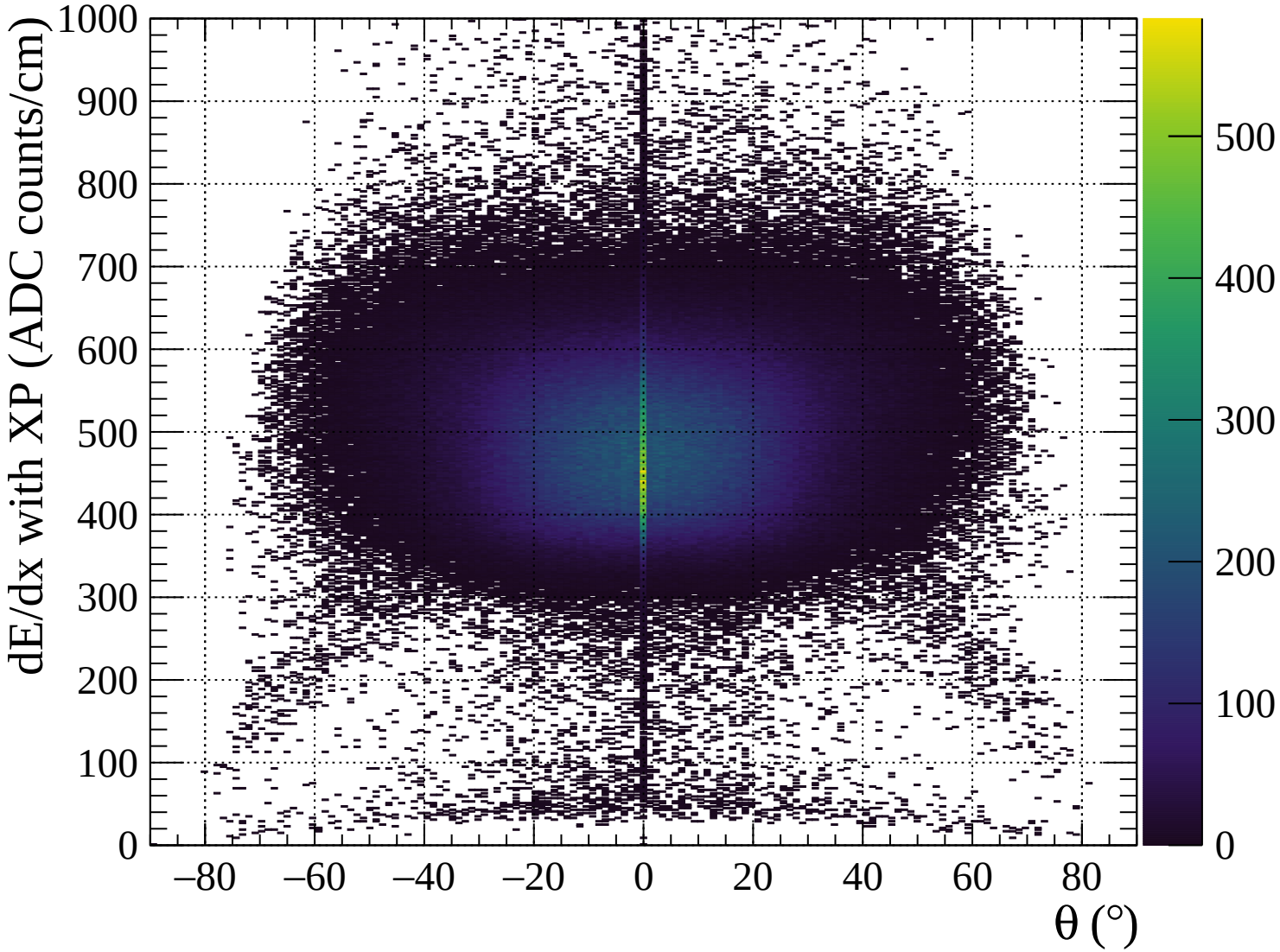


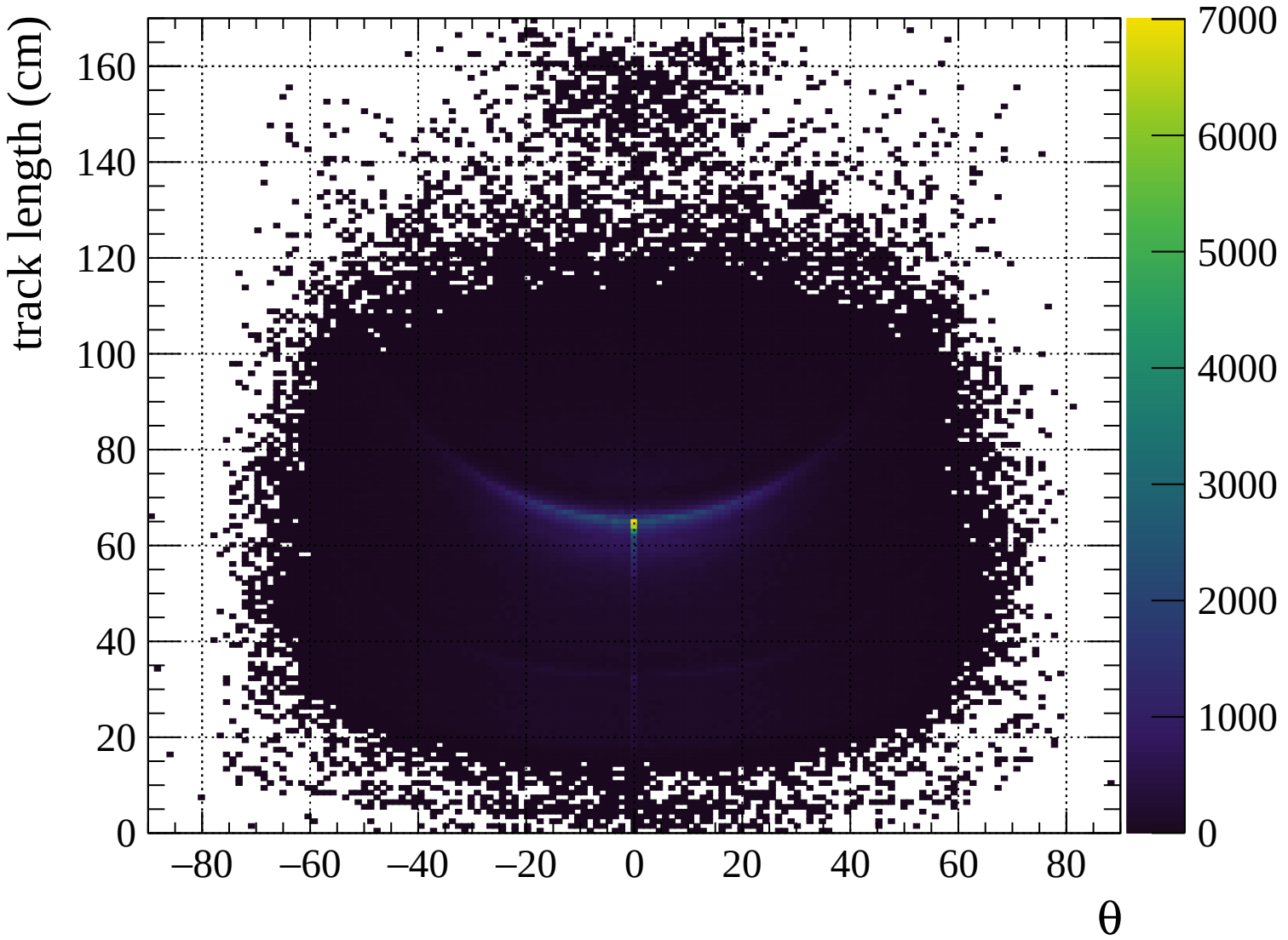


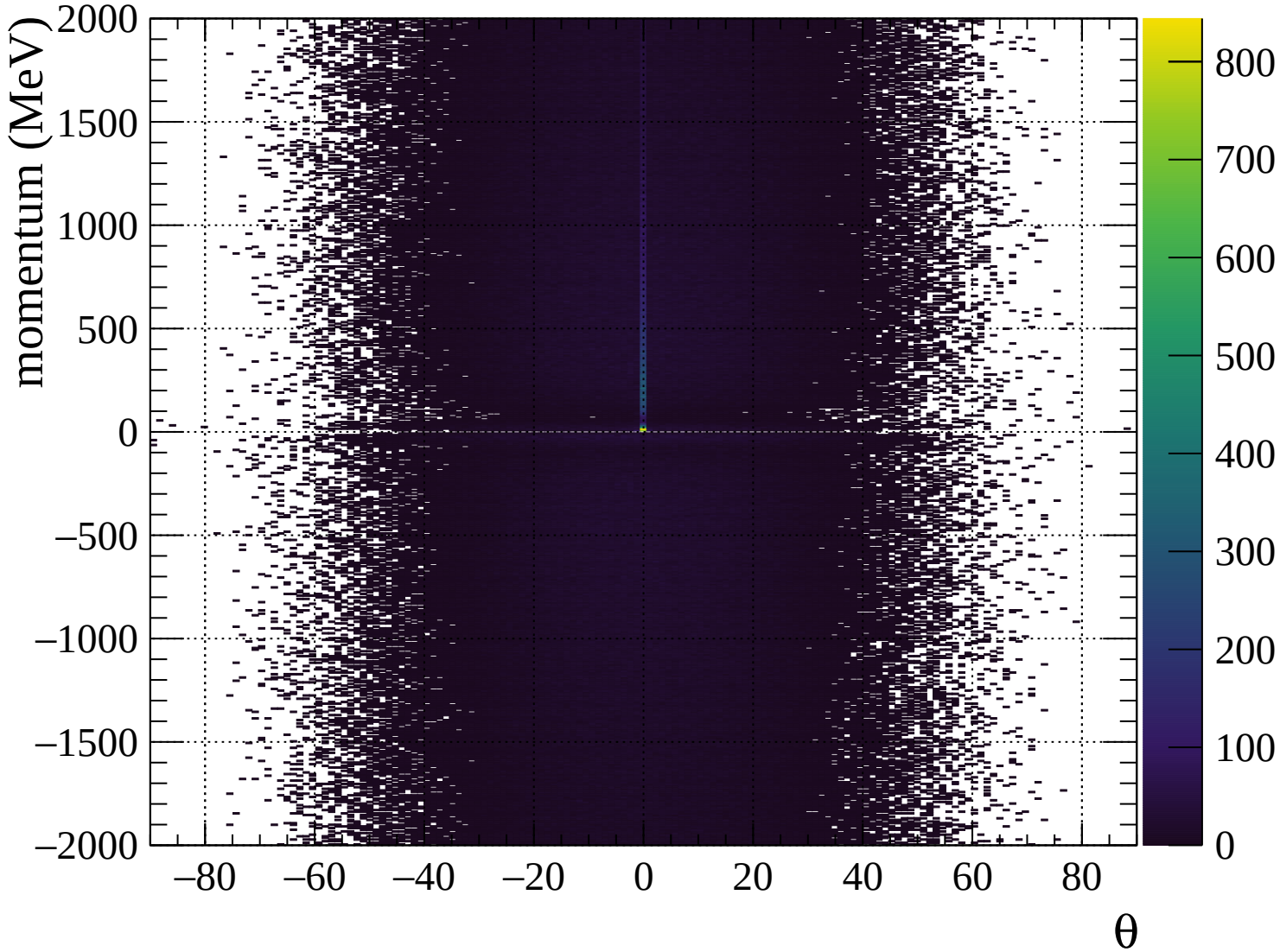


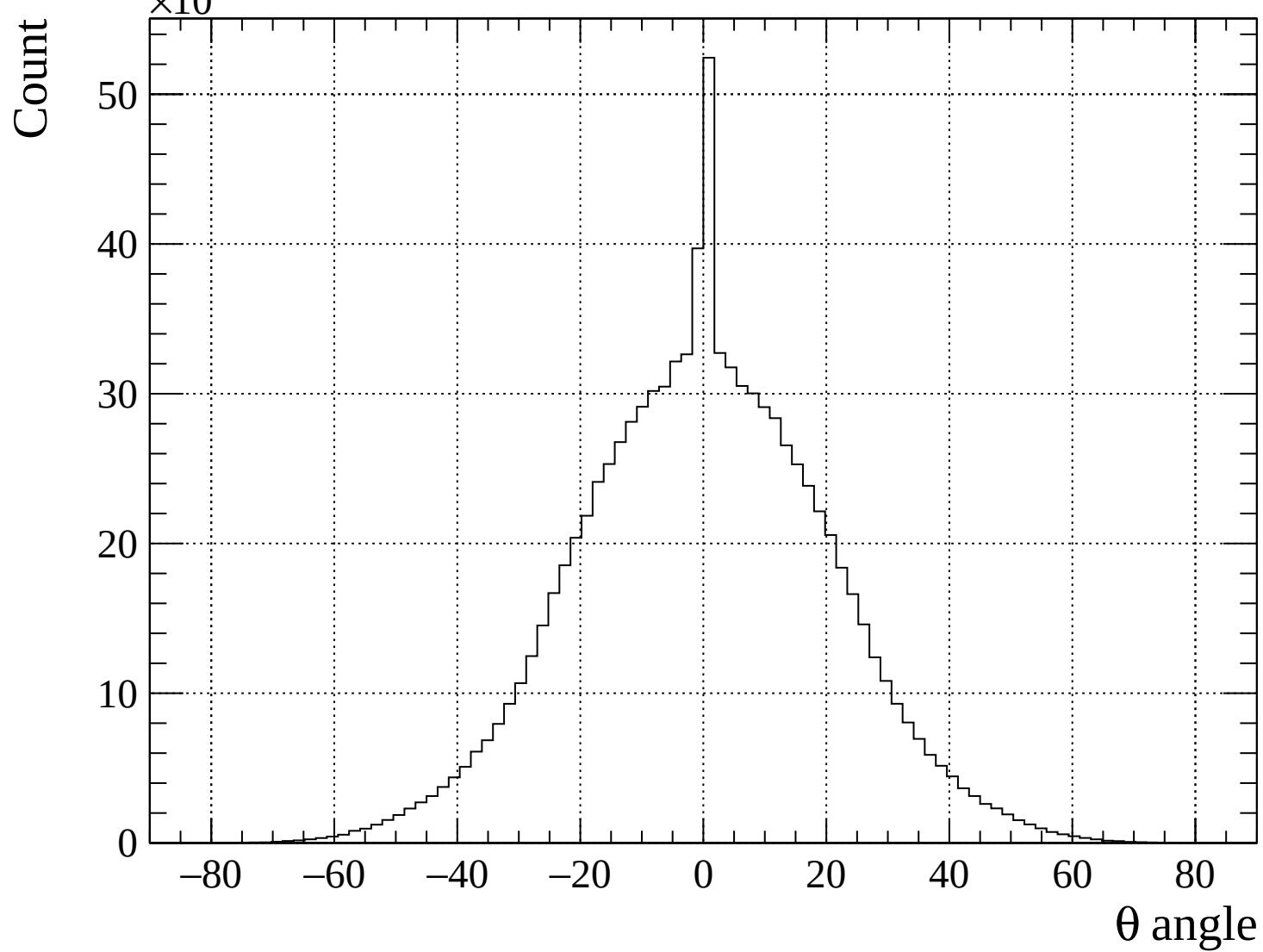


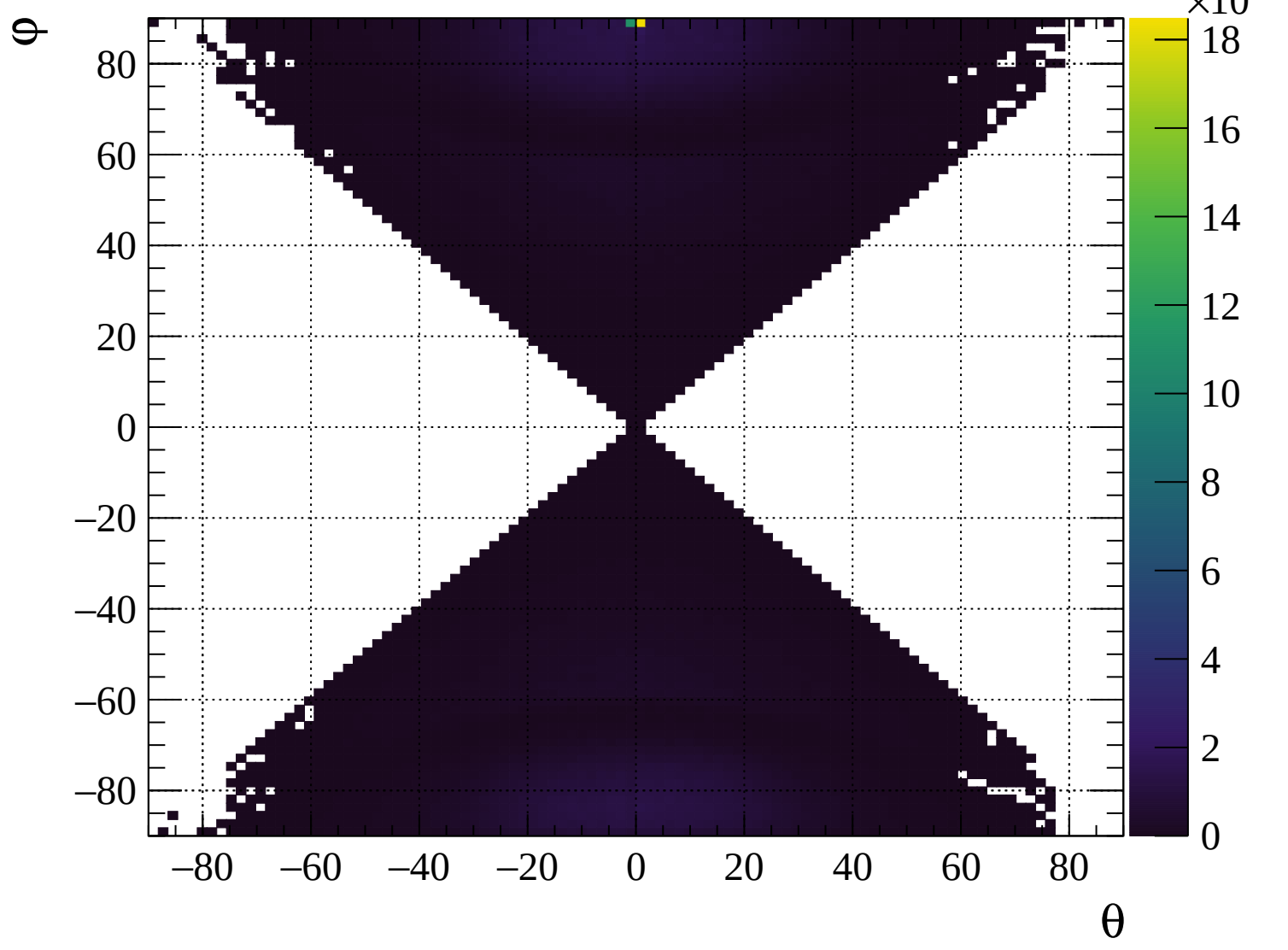


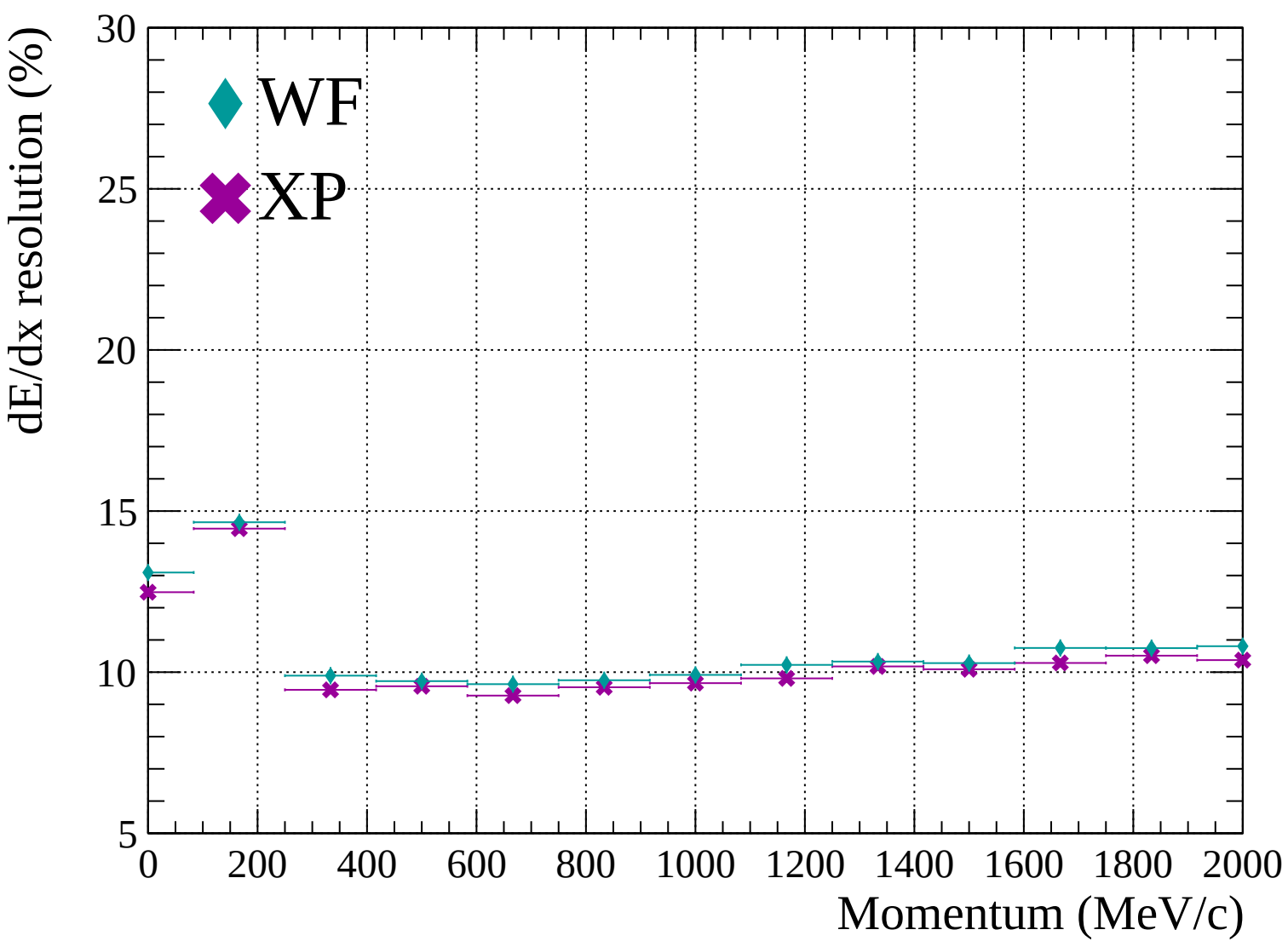


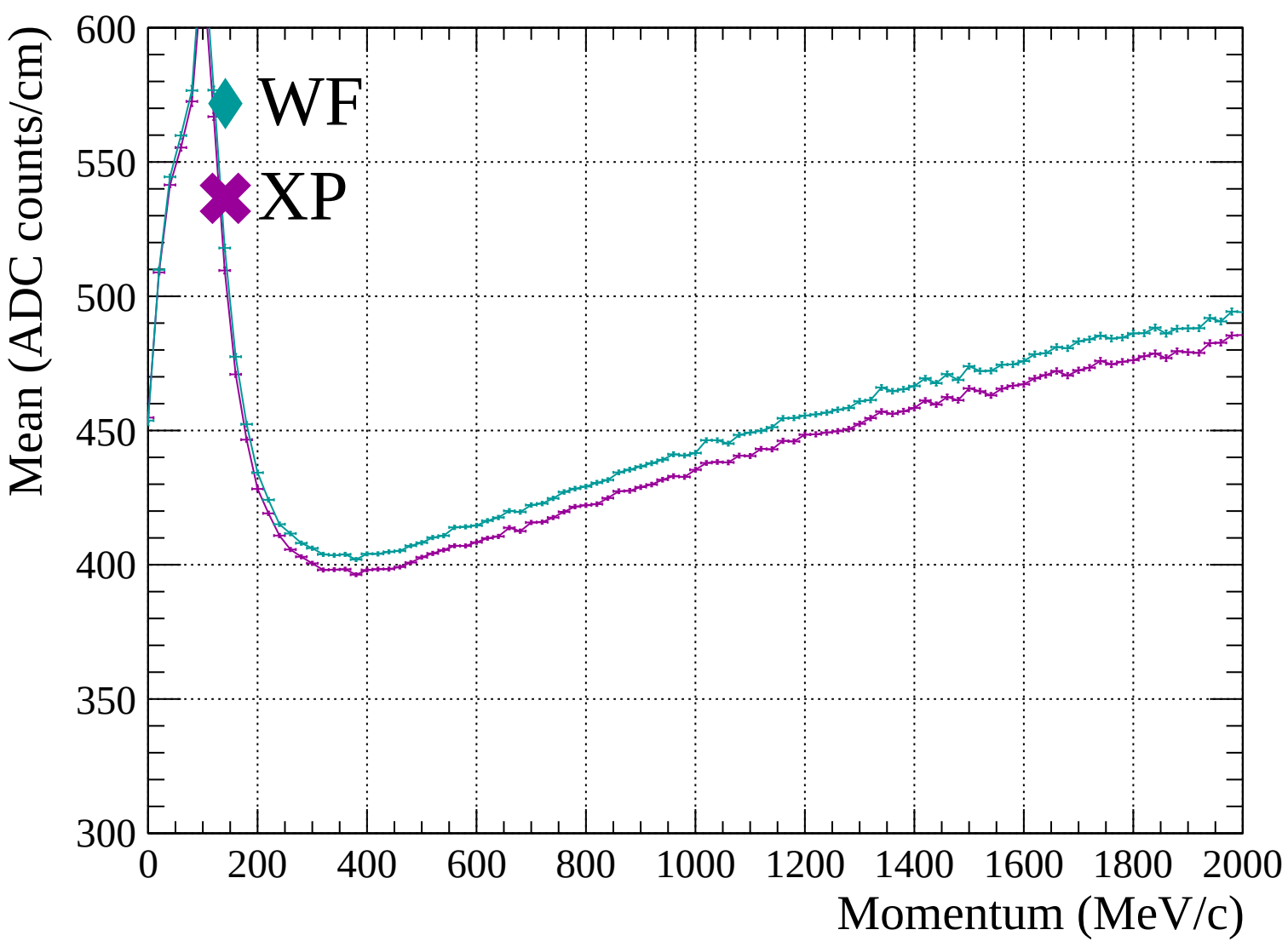


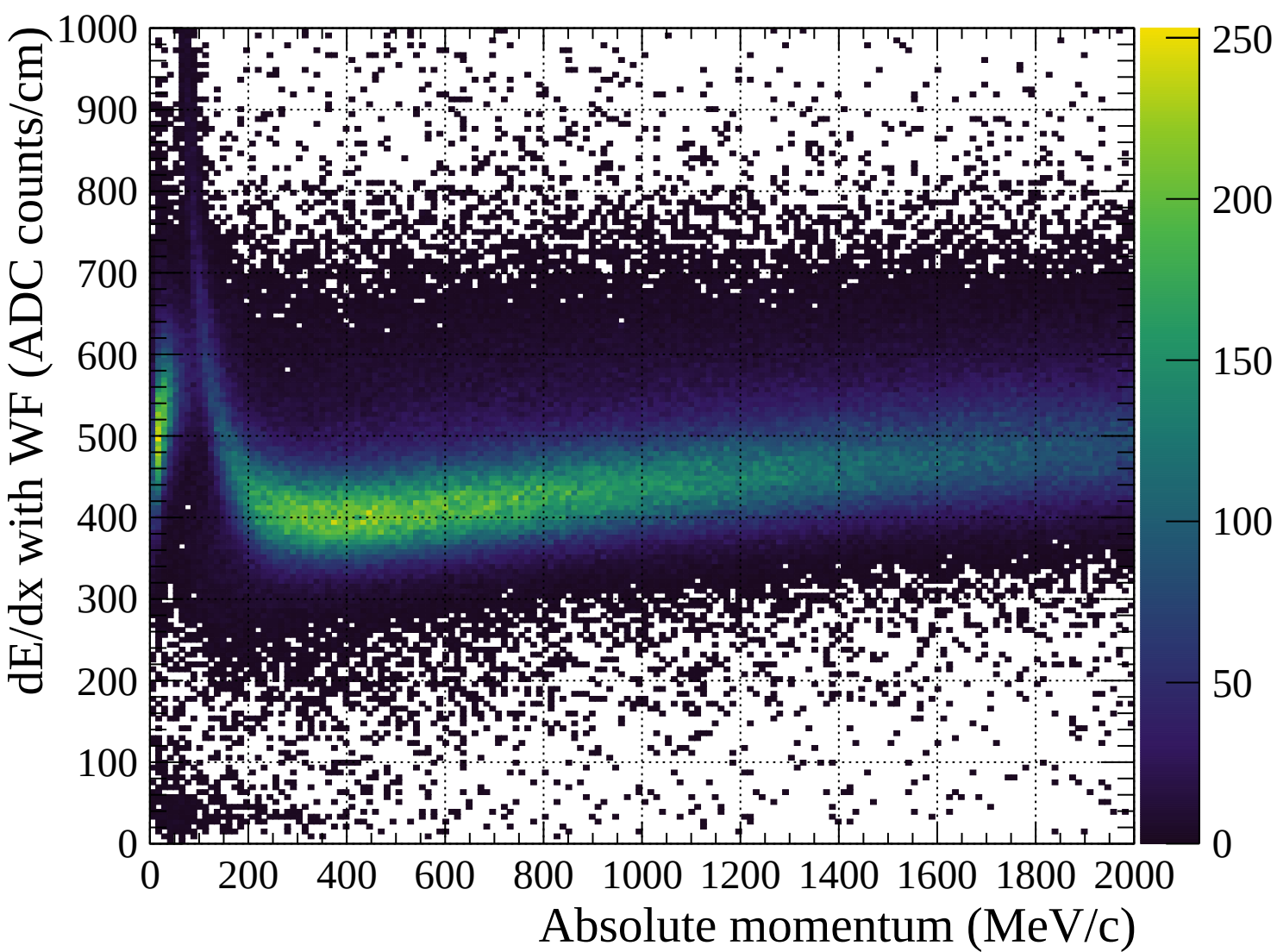


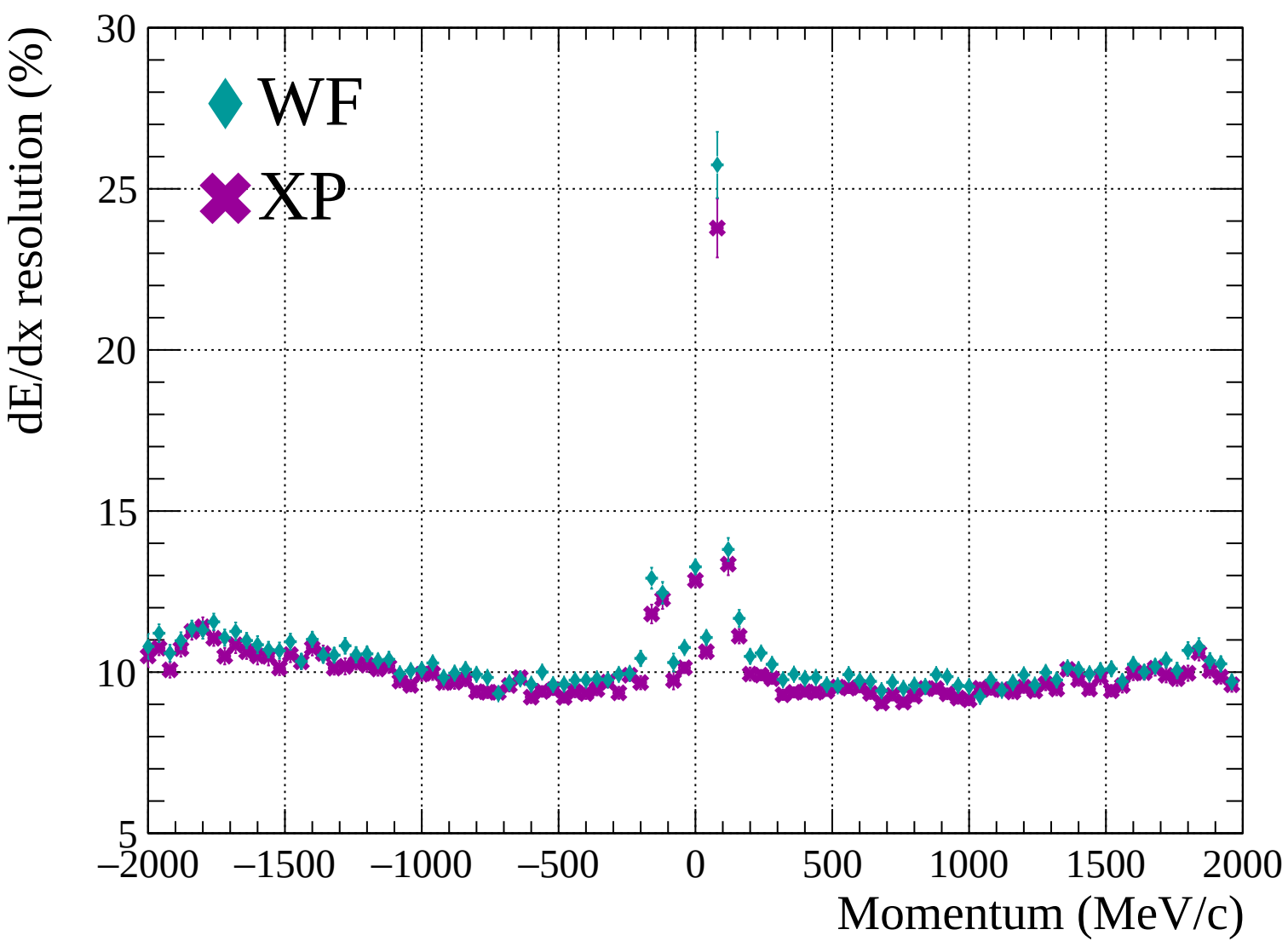


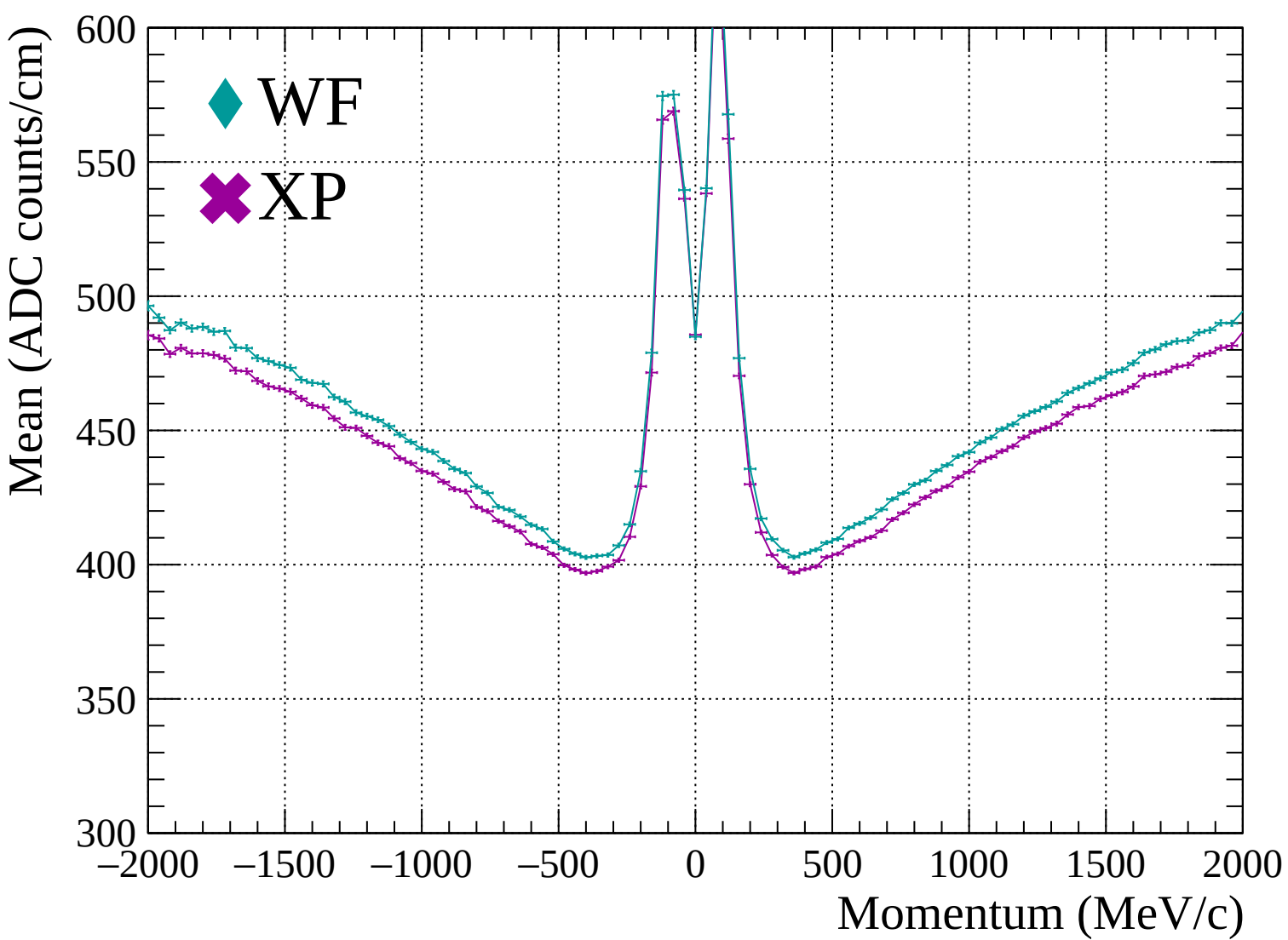


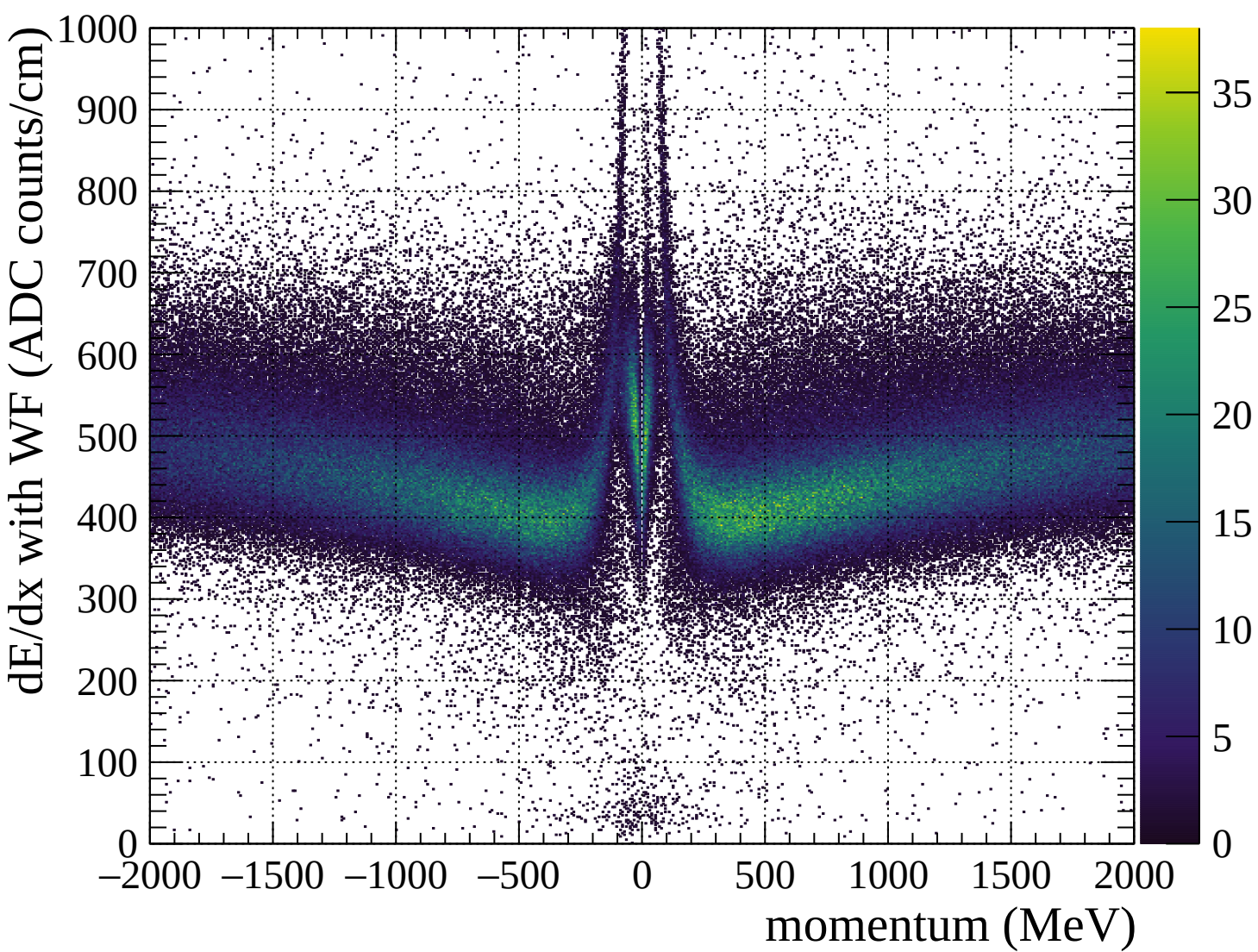


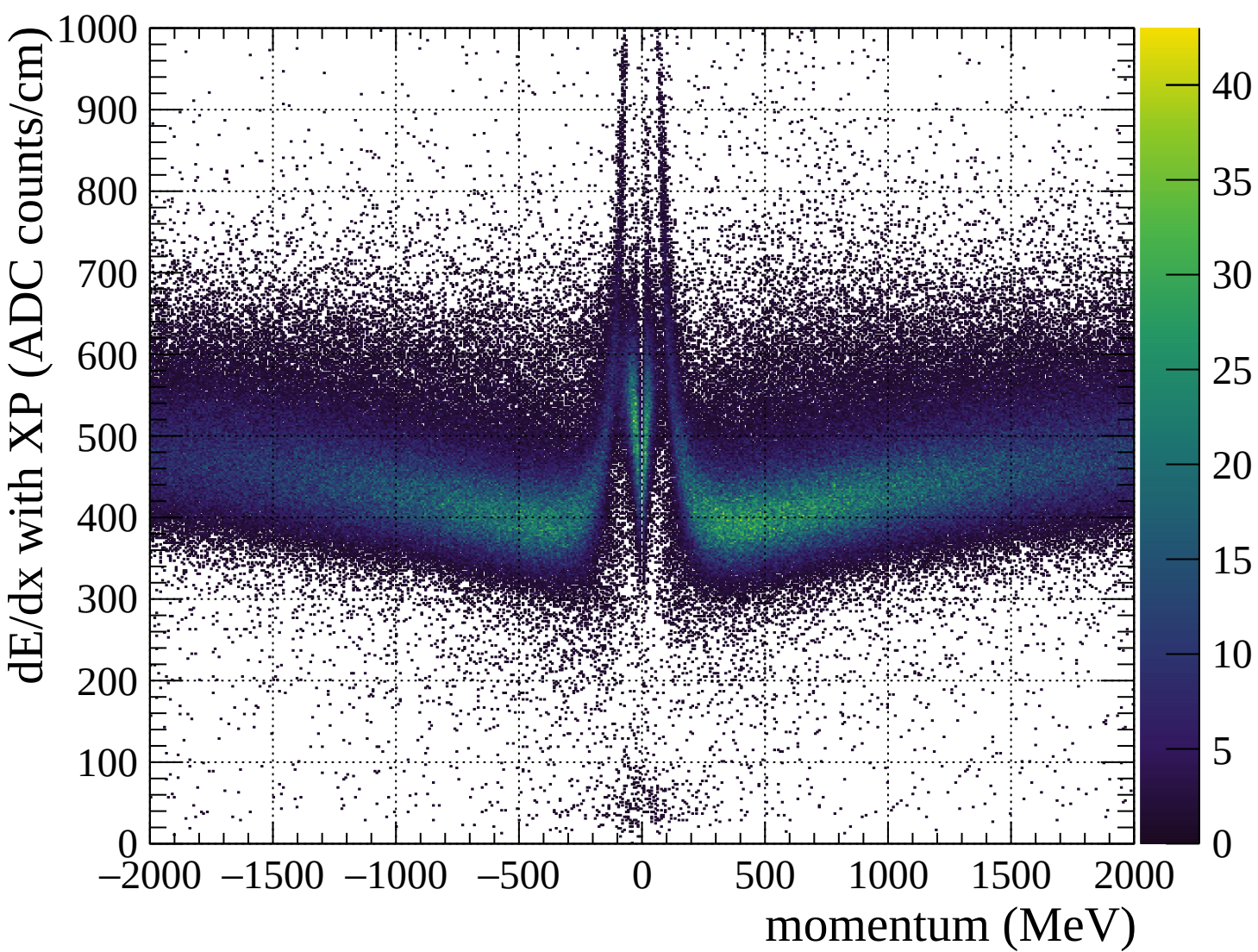


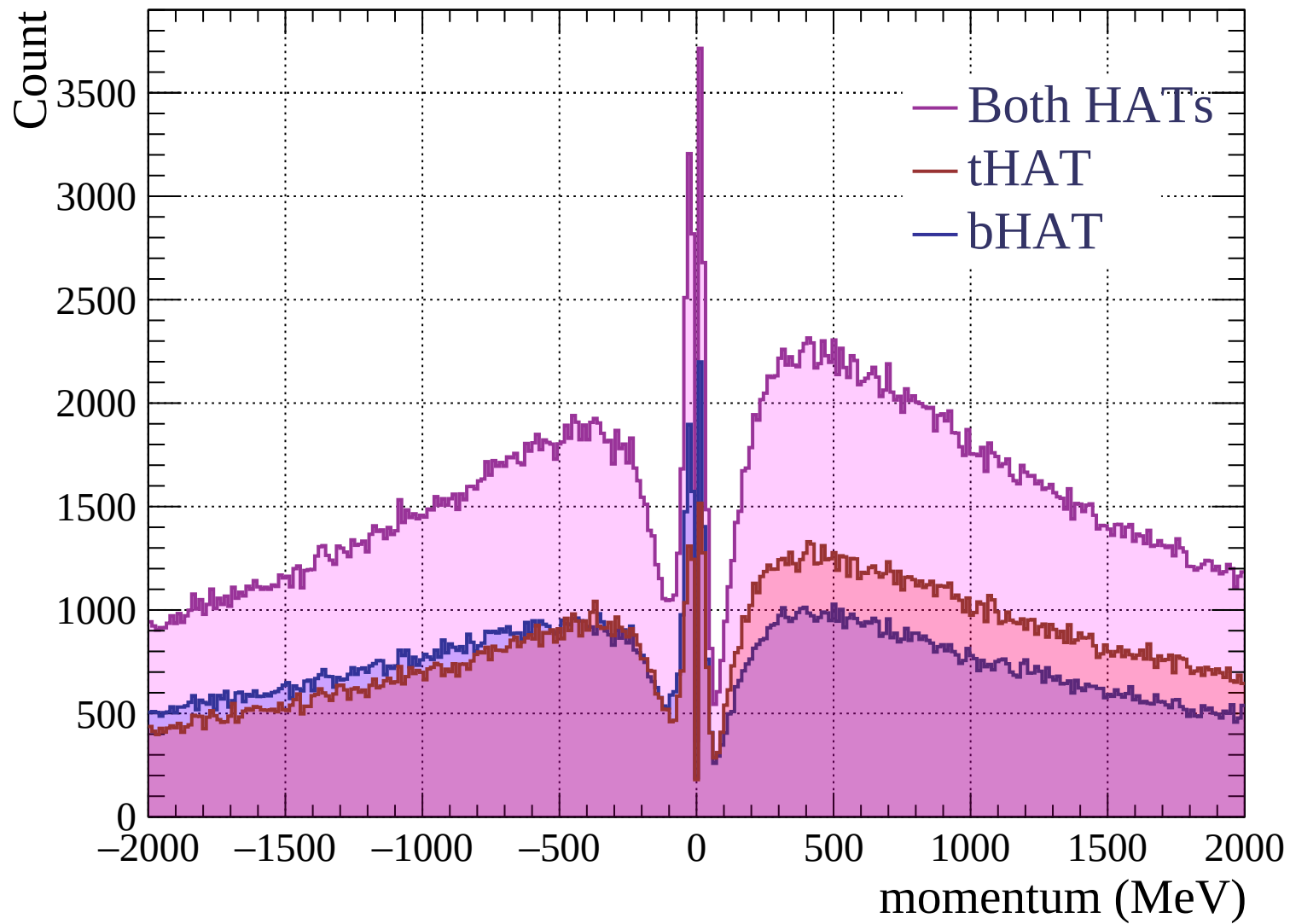


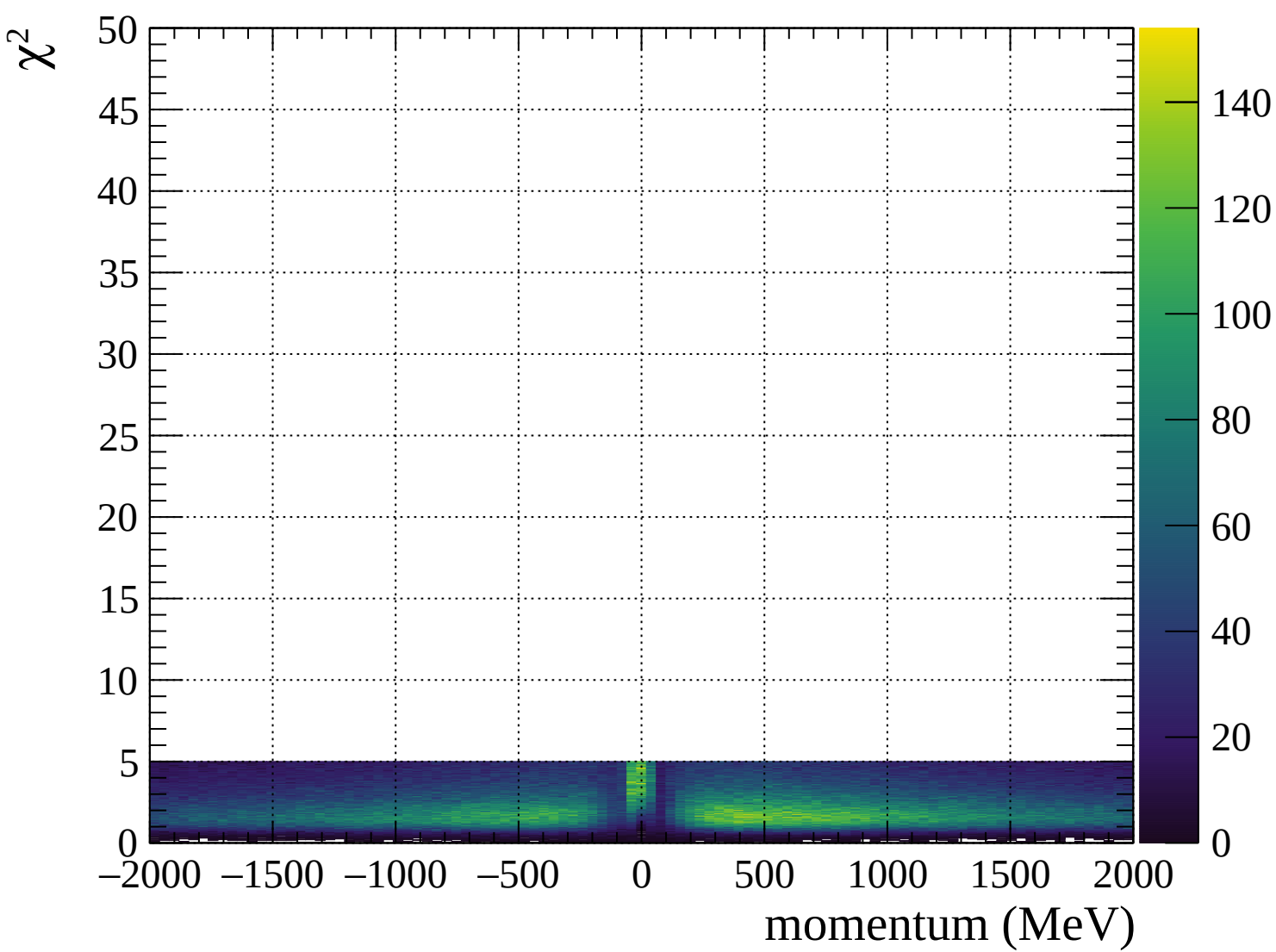




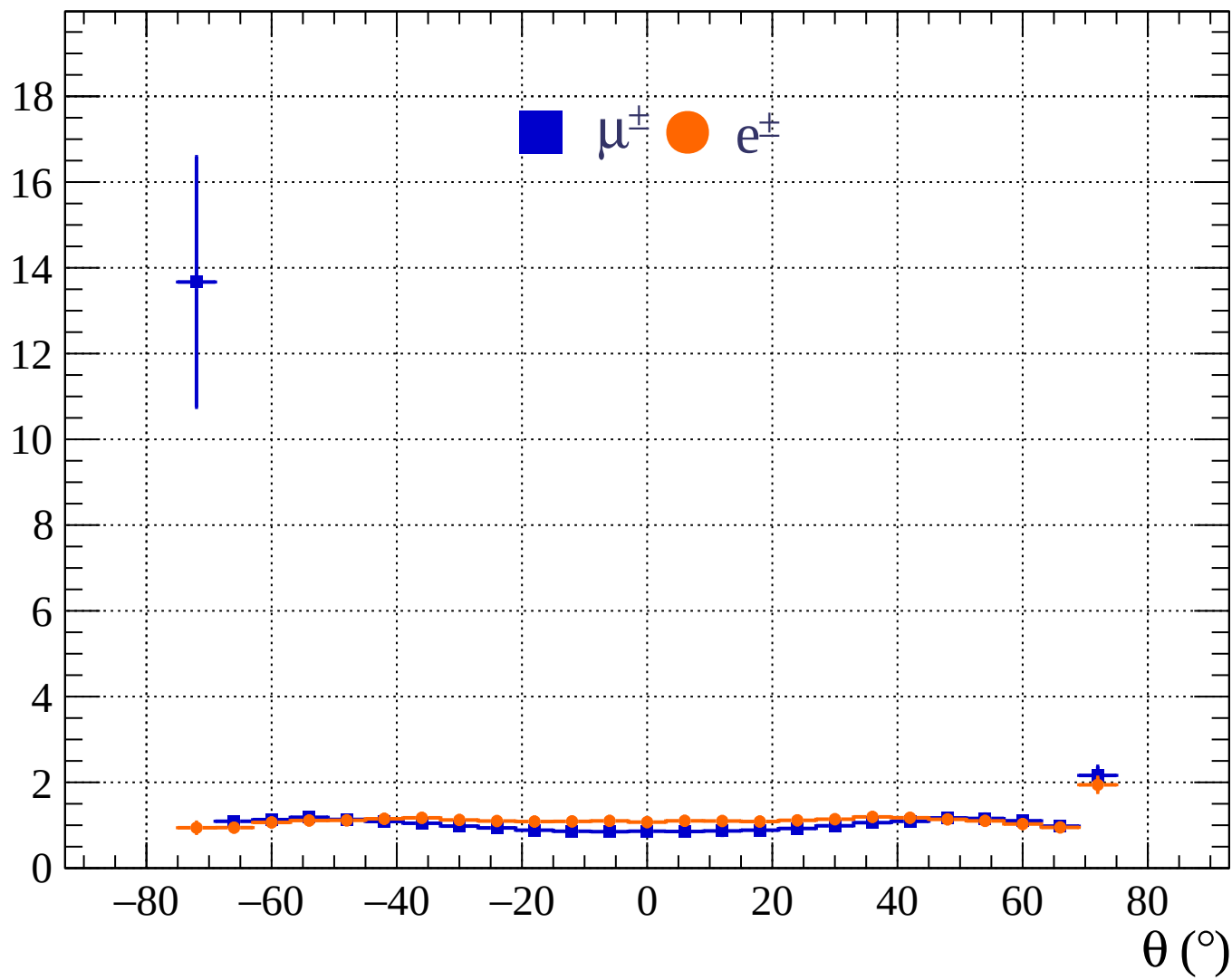


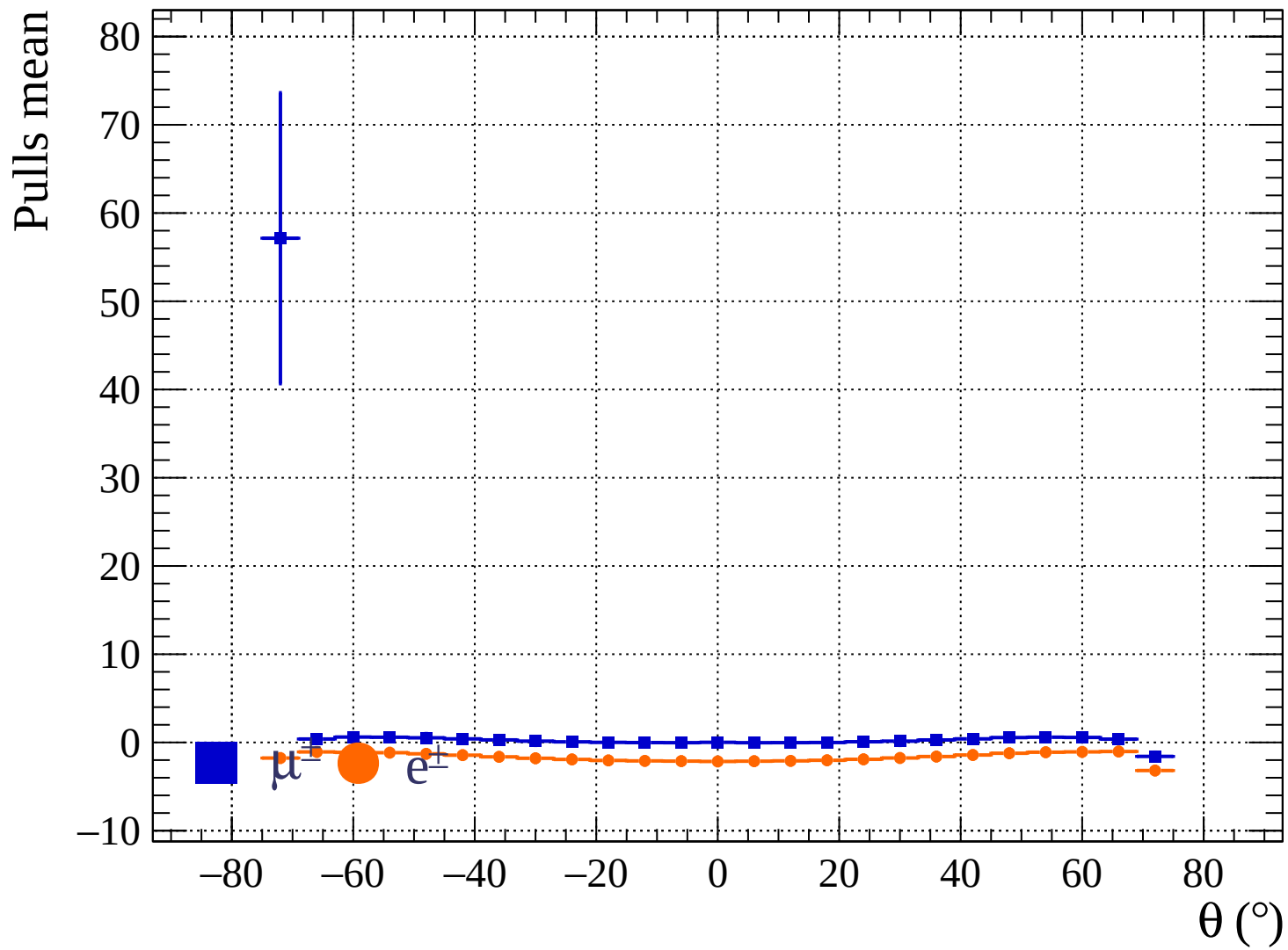




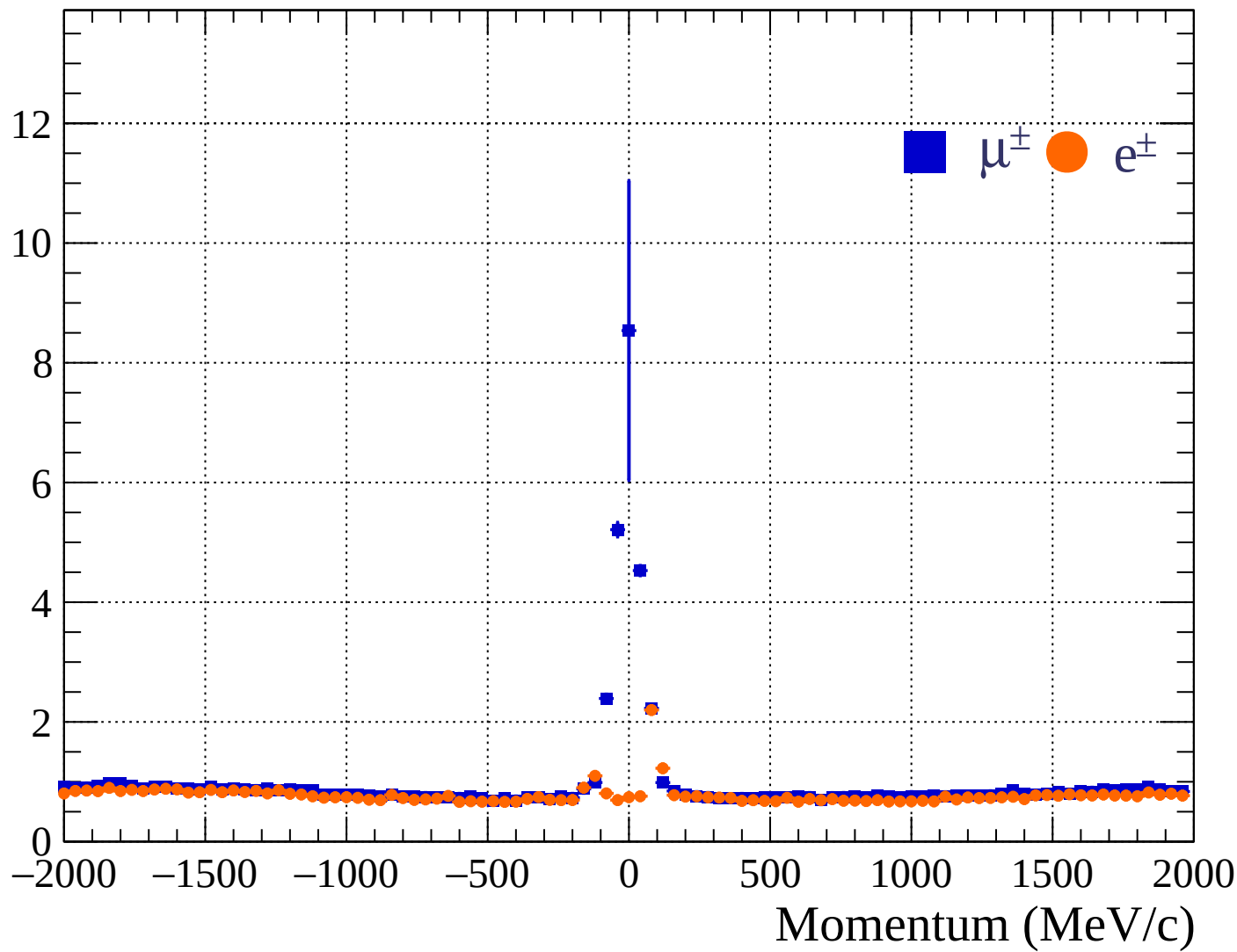


Pulls standard deviation

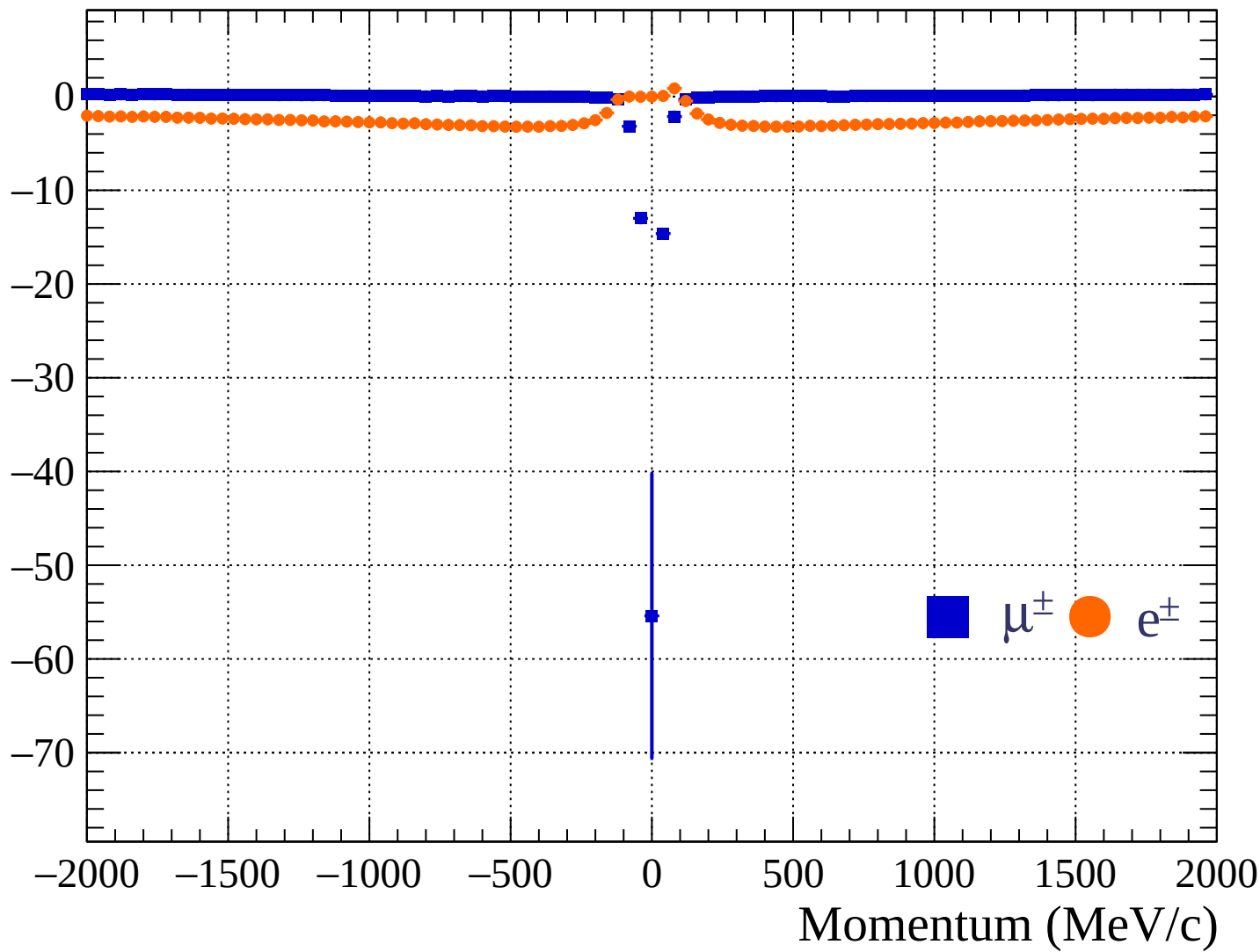


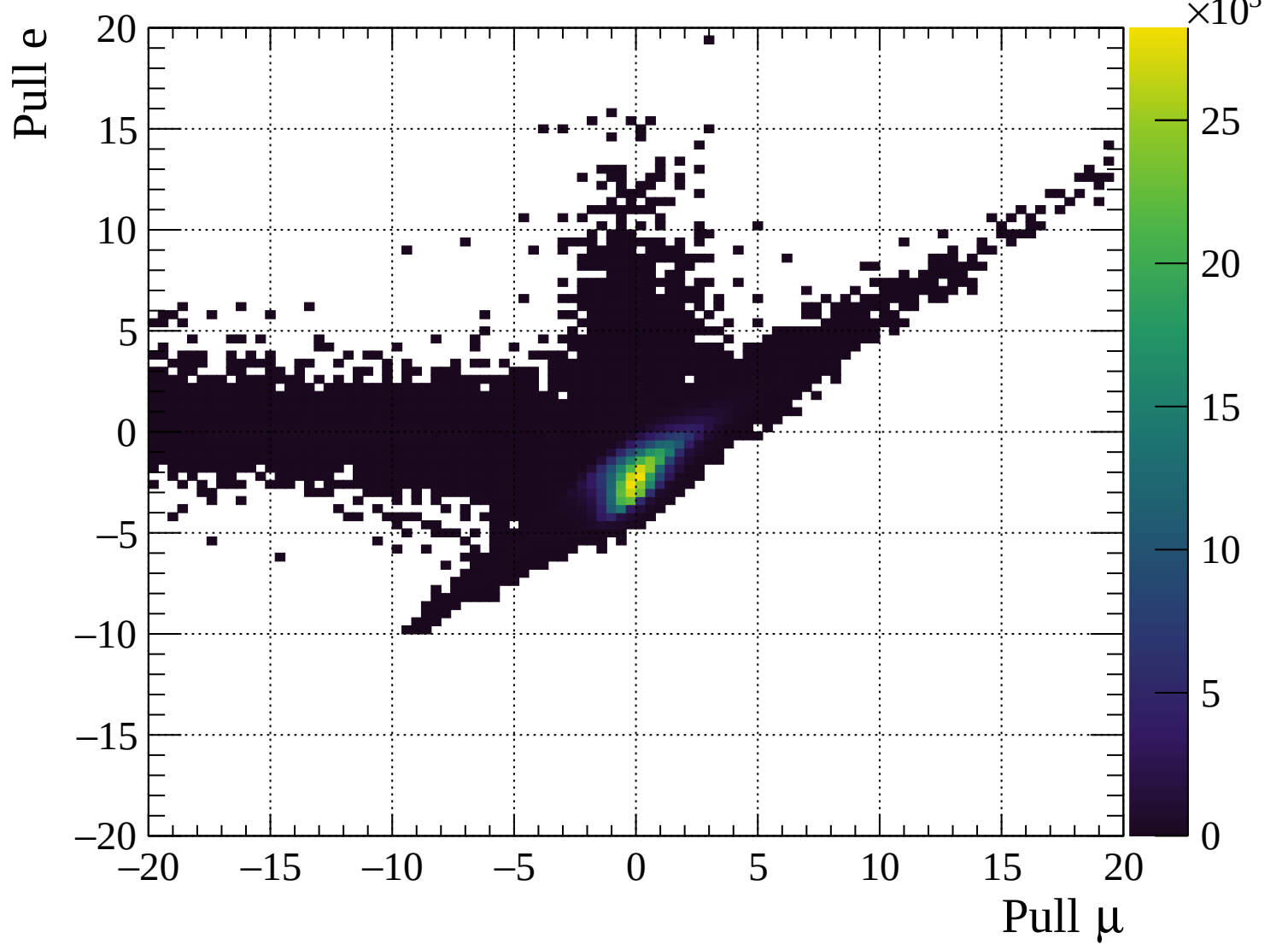


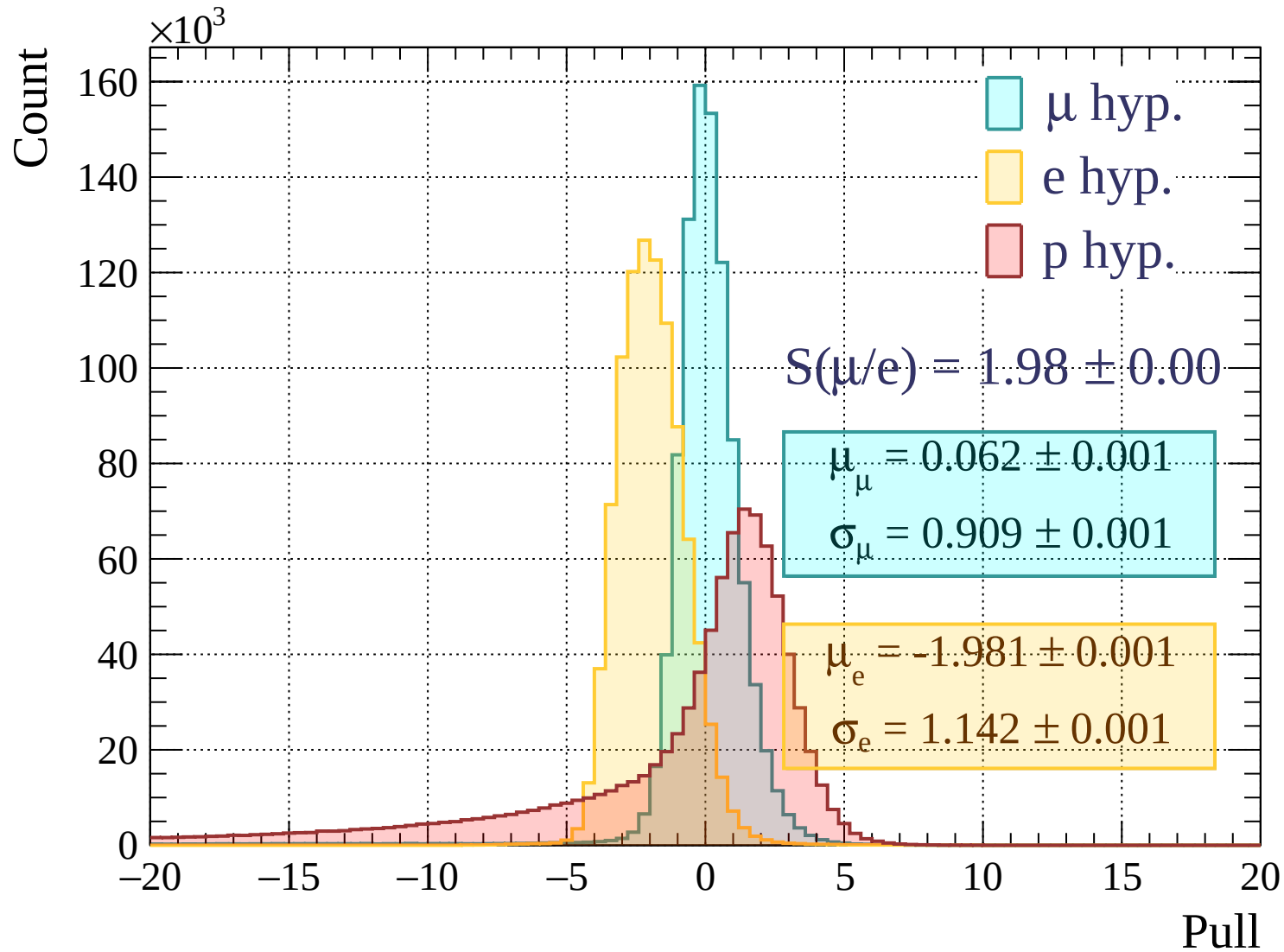
Pulls standard deviation

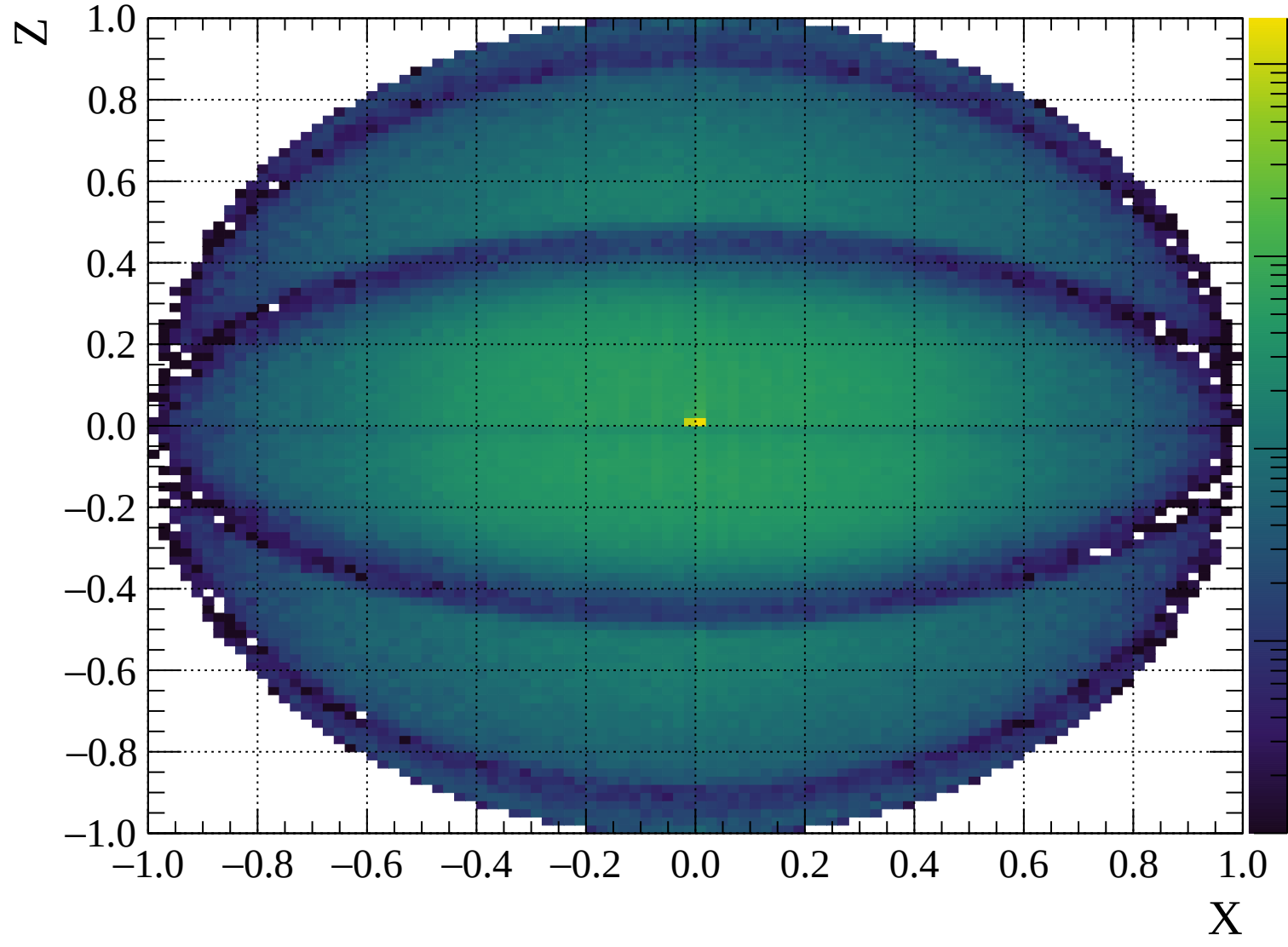


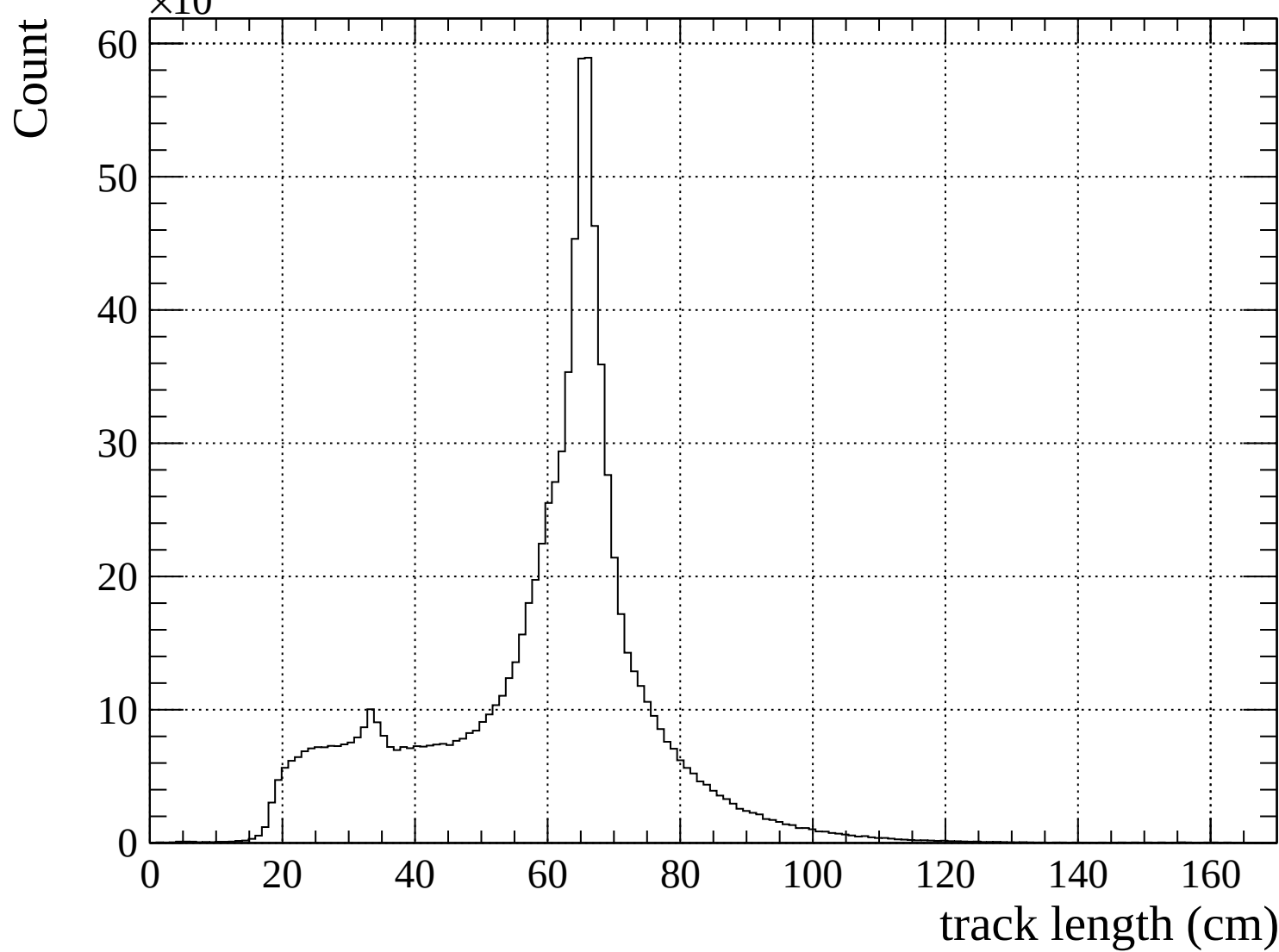
Pulls mean

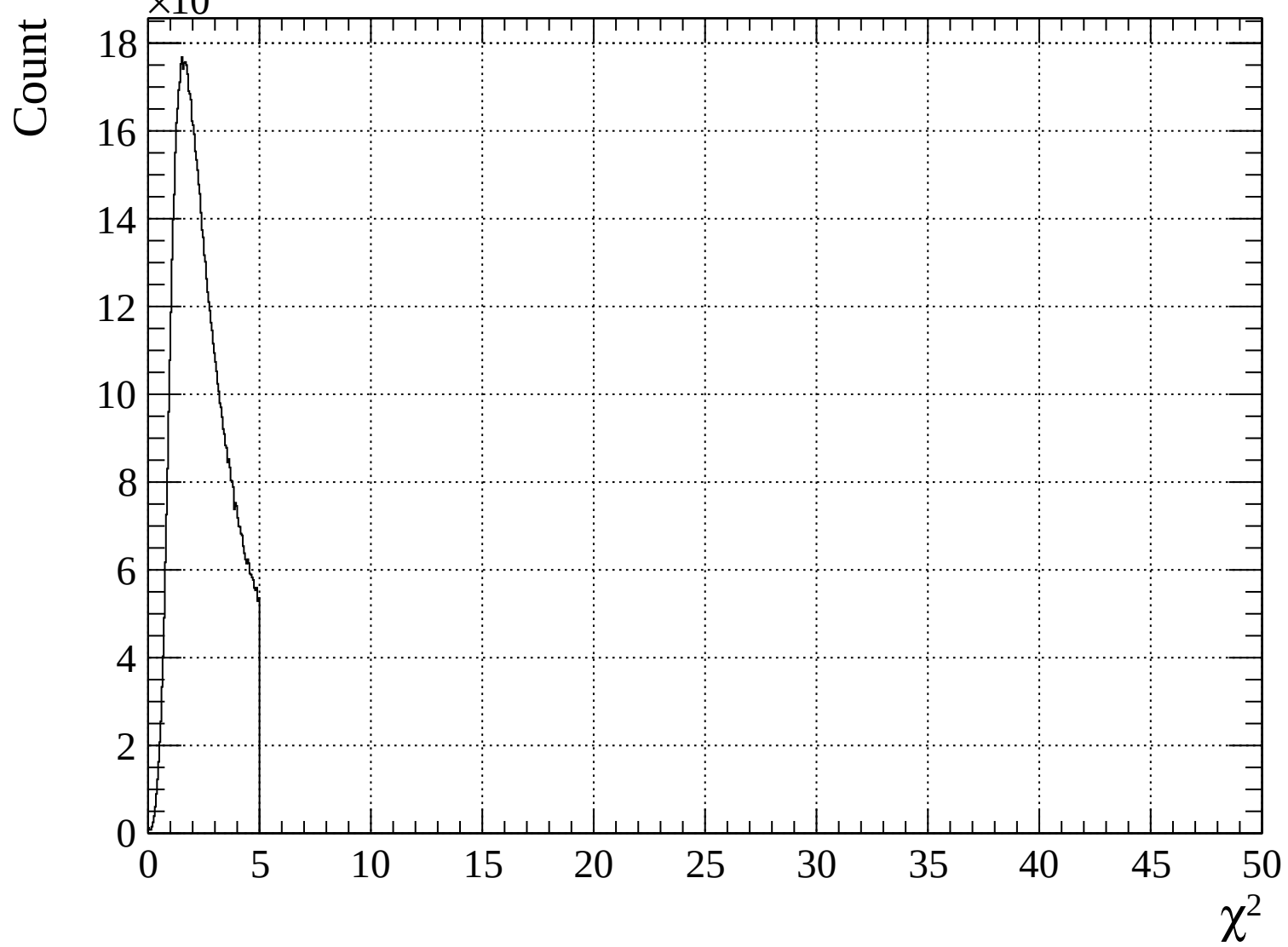


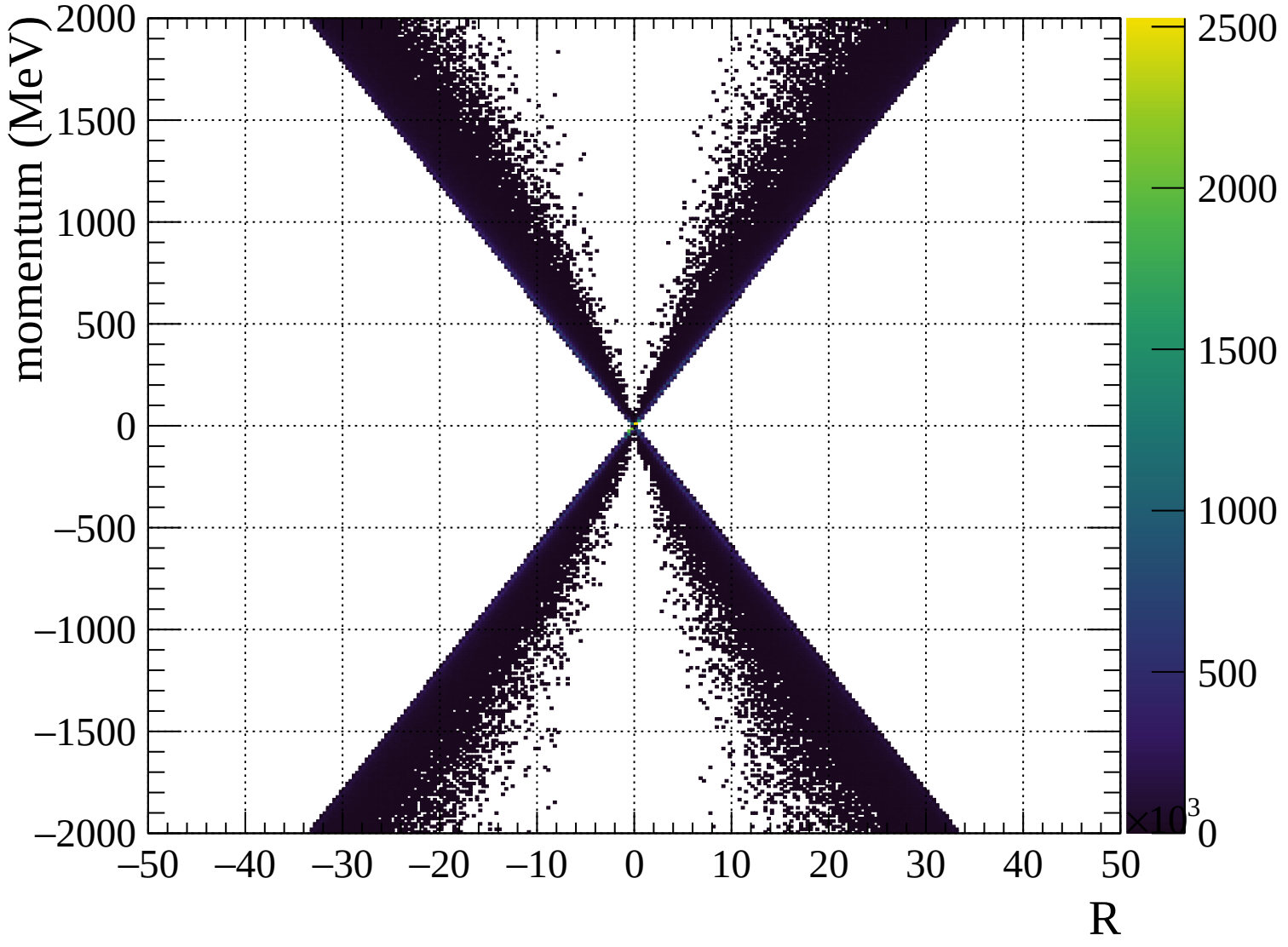


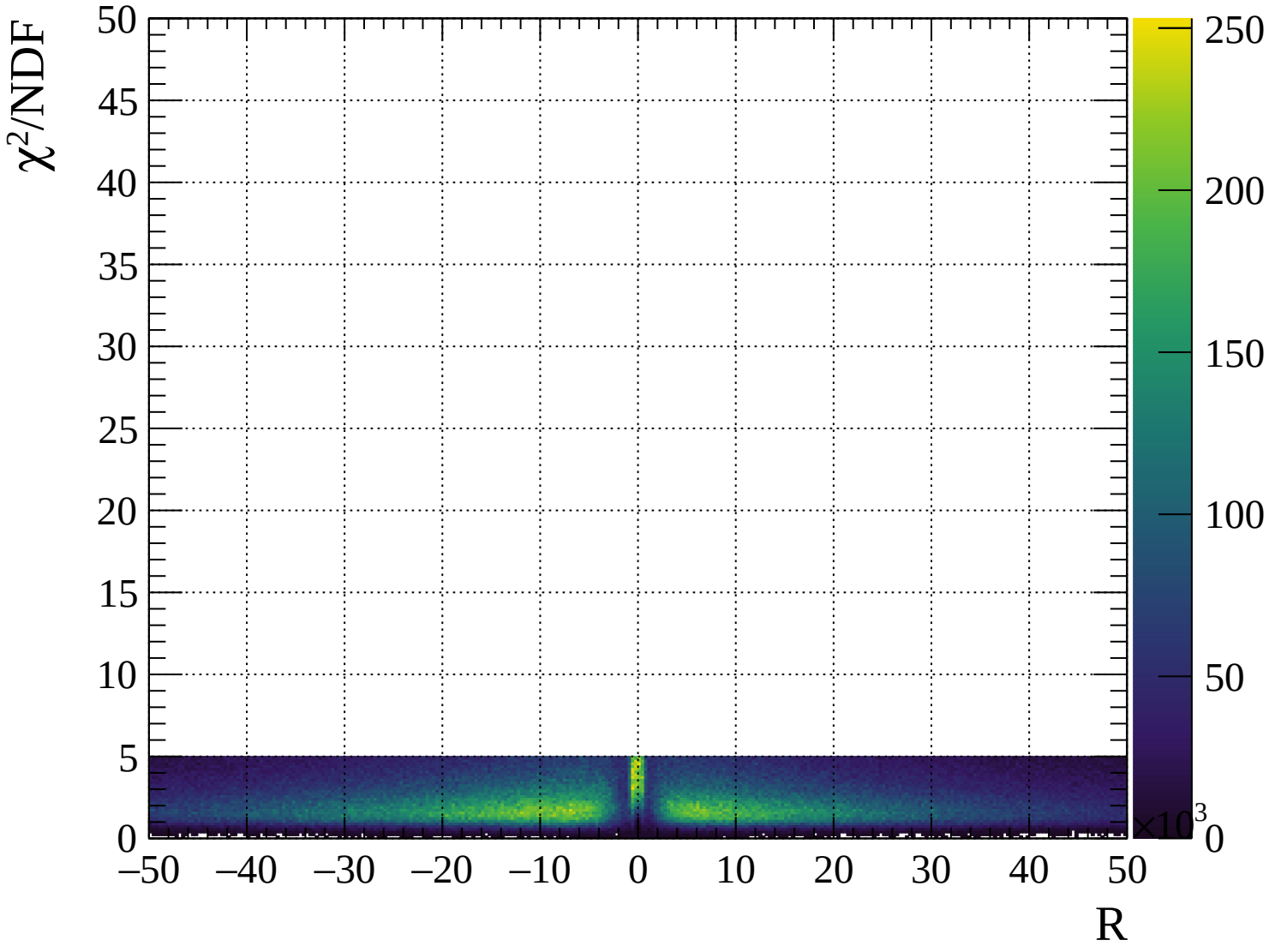






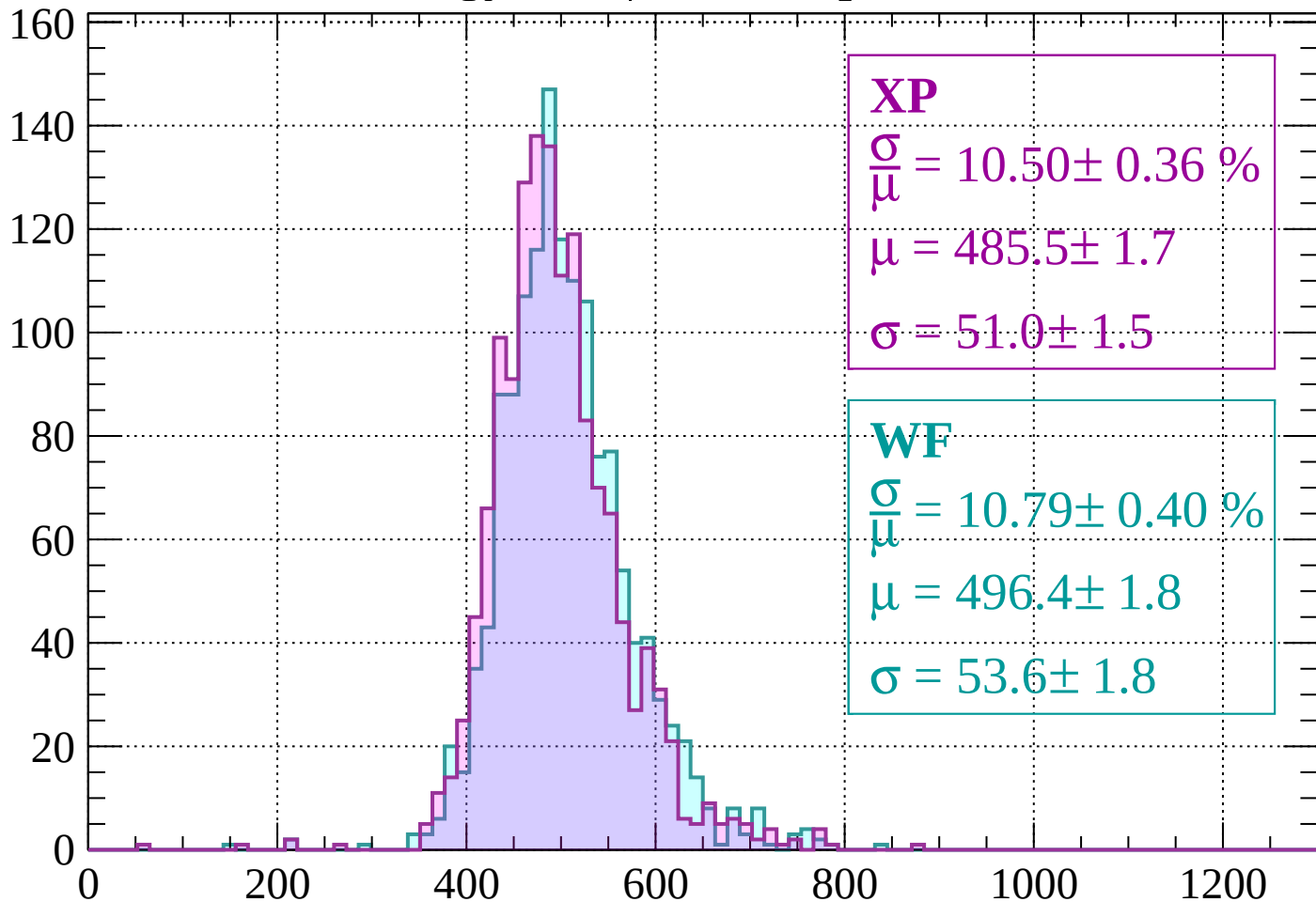






Energy loss | -2000 < p < -1960

Count



XP

$$\frac{\sigma}{\mu} = 10.50 \pm 0.36 \%$$

$$\mu = 485.5 \pm 1.7$$

$$\sigma = 51.0 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 10.79 \pm 0.40 \%$$

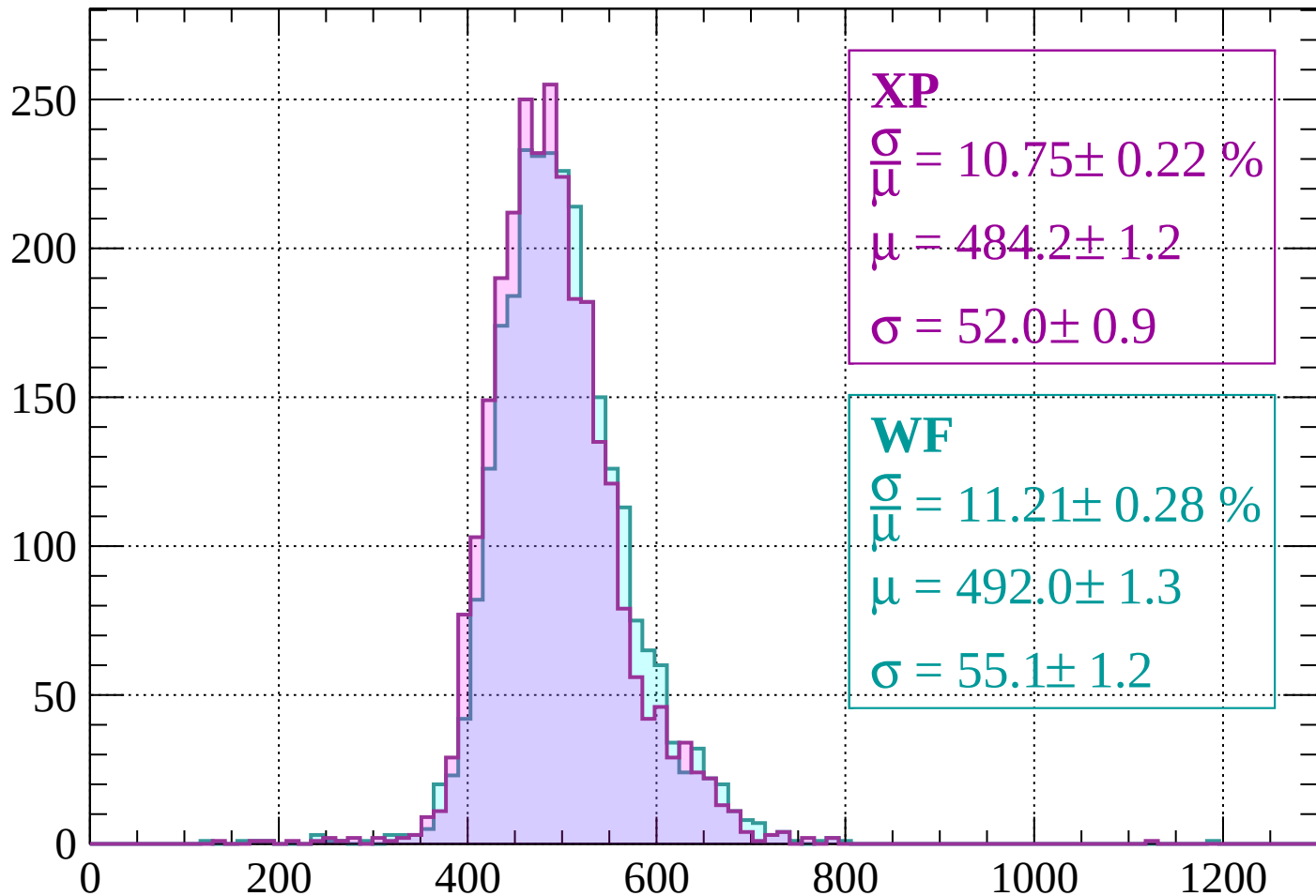
$$\mu = 496.4 \pm 1.8$$

$$\sigma = 53.6 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1960 < p < -1920$

Count



XP

$$\frac{\sigma}{\mu} = 10.75 \pm 0.22 \%$$

$$\mu = 484.2 \pm 1.2$$

$$\sigma = 52.0 \pm 0.9$$

WF

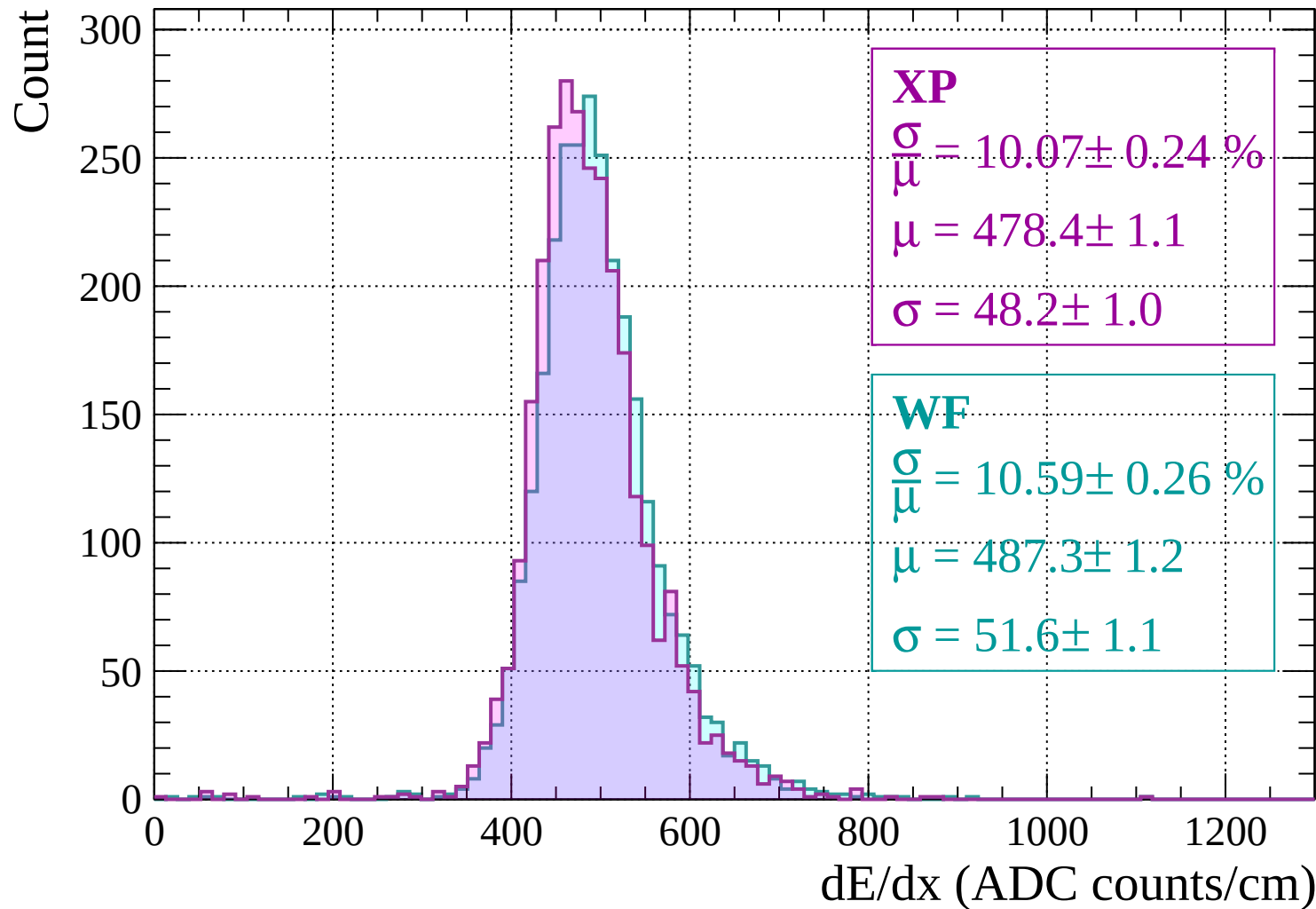
$$\frac{\sigma}{\mu} = 11.21 \pm 0.28 \%$$

$$\mu = 492.0 \pm 1.3$$

$$\sigma = 55.1 \pm 1.2$$

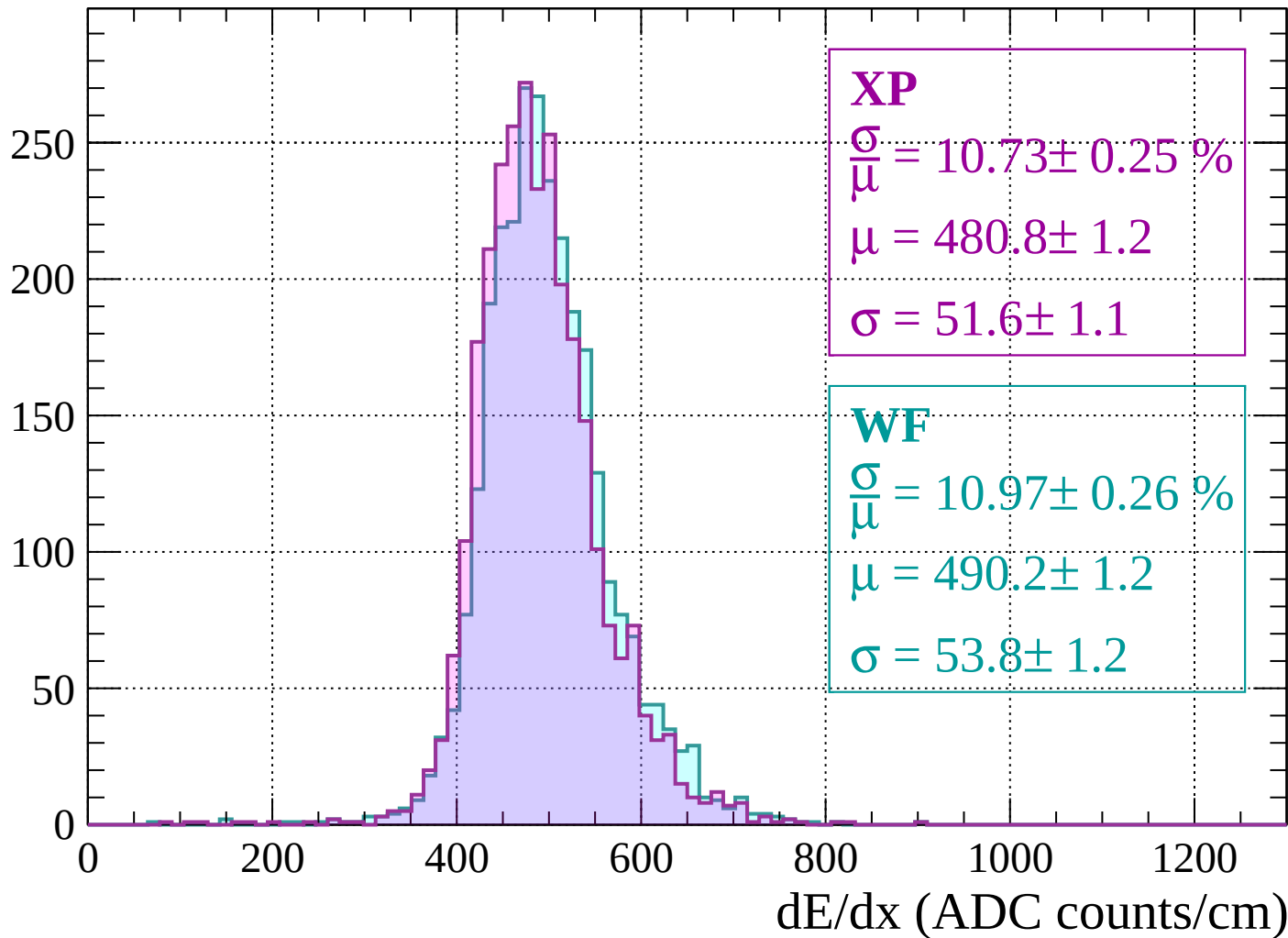
dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880



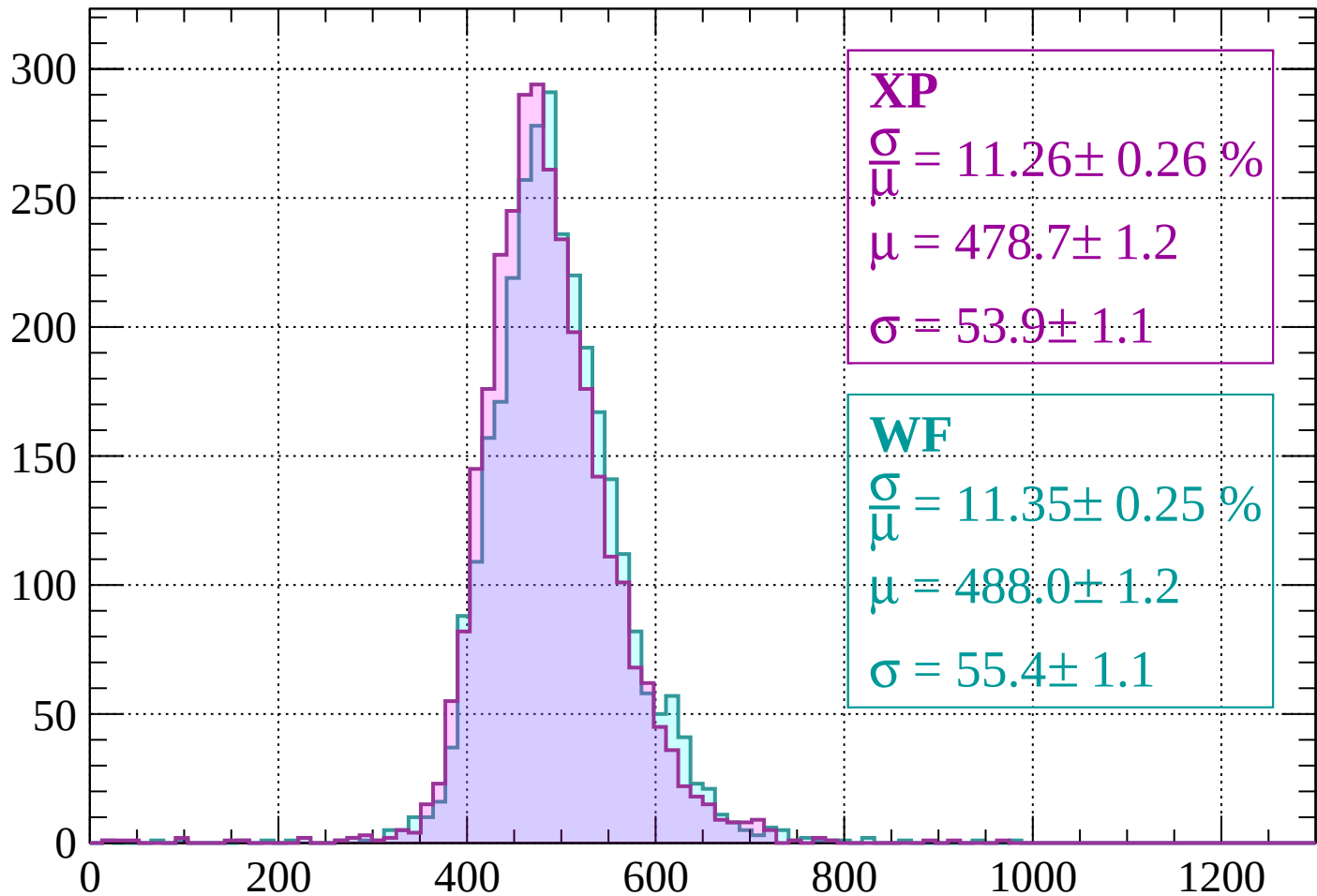
Energy loss | -1880 < p < -1840

Count



Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 11.26 \pm 0.26 \%$$

$$\mu = 478.7 \pm 1.2$$

$$\sigma = 53.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.35 \pm 0.25 \%$$

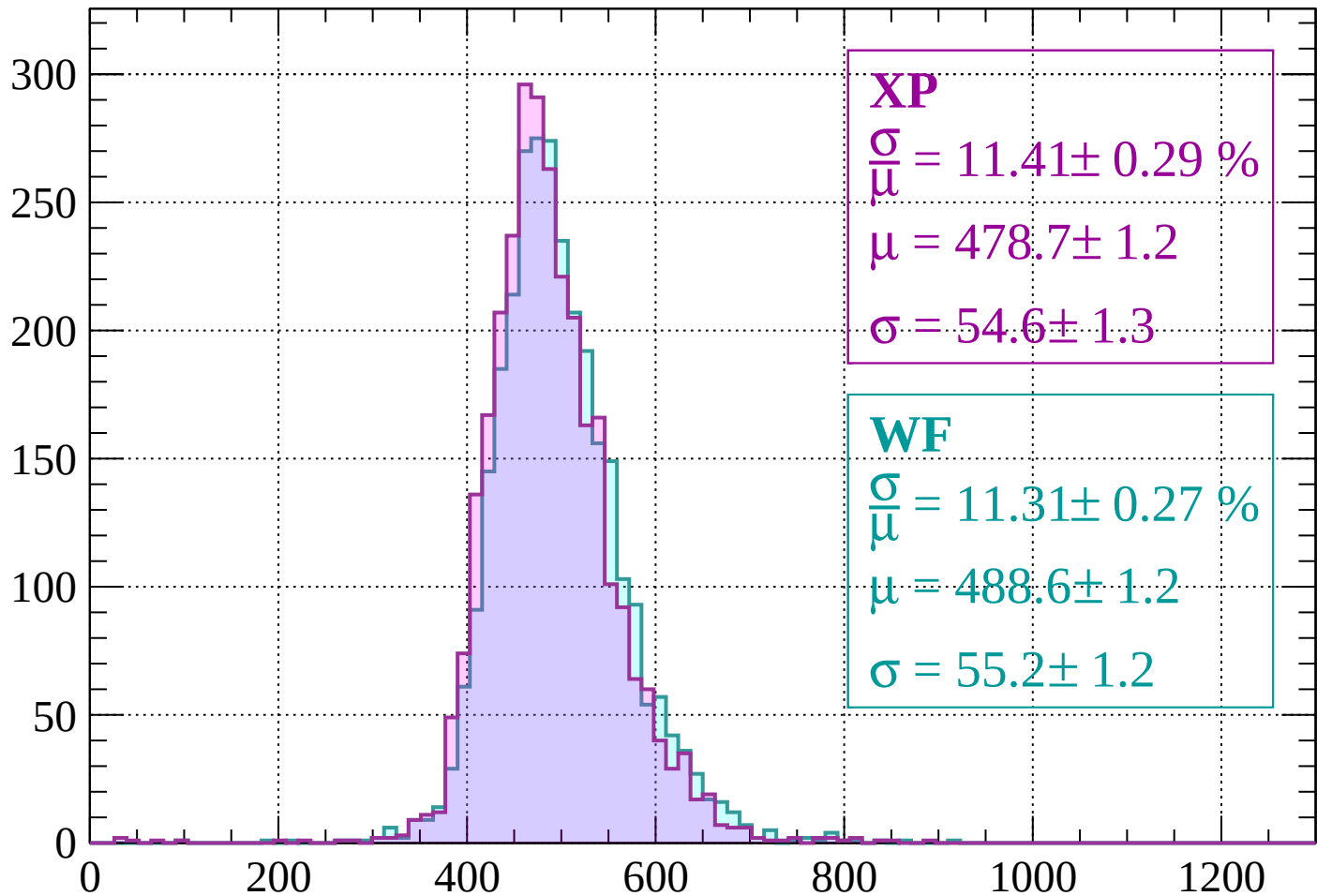
$$\mu = 488.0 \pm 1.2$$

$$\sigma = 55.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1800 < p < -1760

Count



XP

$$\frac{\sigma}{\mu} = 11.41 \pm 0.29 \%$$

$$\mu = 478.7 \pm 1.2$$

$$\sigma = 54.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.31 \pm 0.27 \%$$

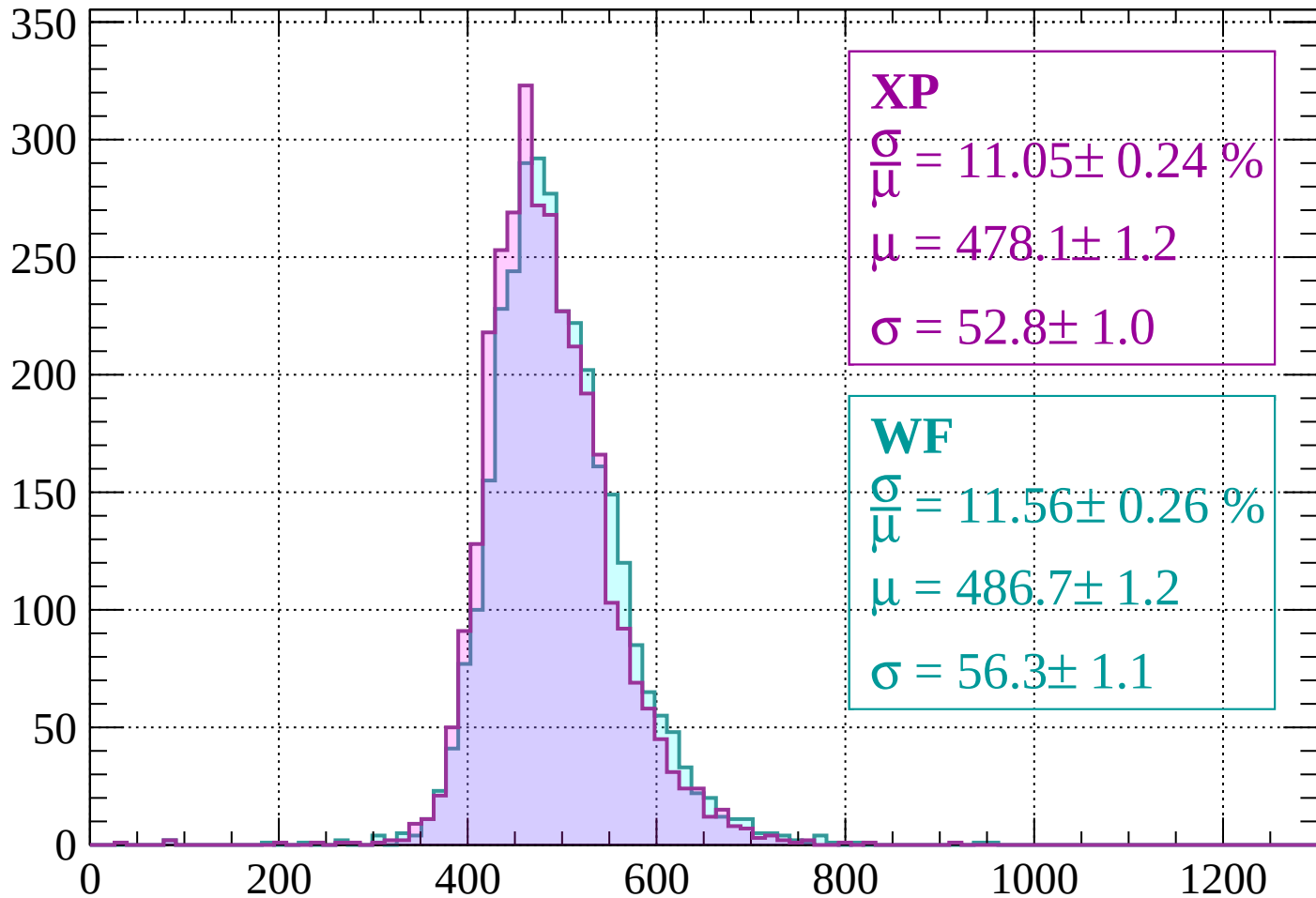
$$\mu = 488.6 \pm 1.2$$

$$\sigma = 55.2 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$

Count



XP

$$\frac{\sigma}{\mu} = 11.05 \pm 0.24 \%$$

$$\mu = 478.1 \pm 1.2$$

$$\sigma = 52.8 \pm 1.0$$

WF

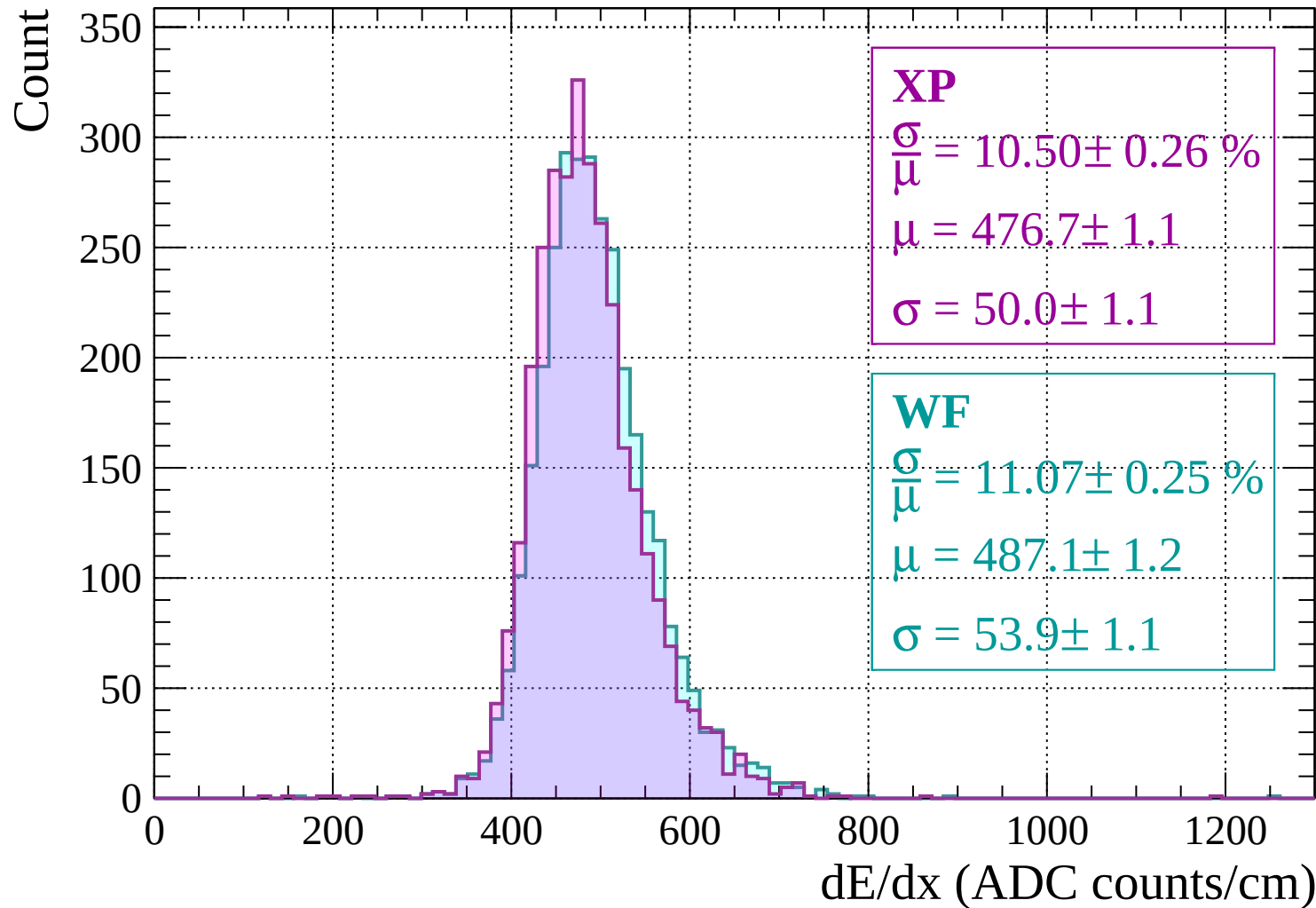
$$\frac{\sigma}{\mu} = 11.56 \pm 0.26 \%$$

$$\mu = 486.7 \pm 1.2$$

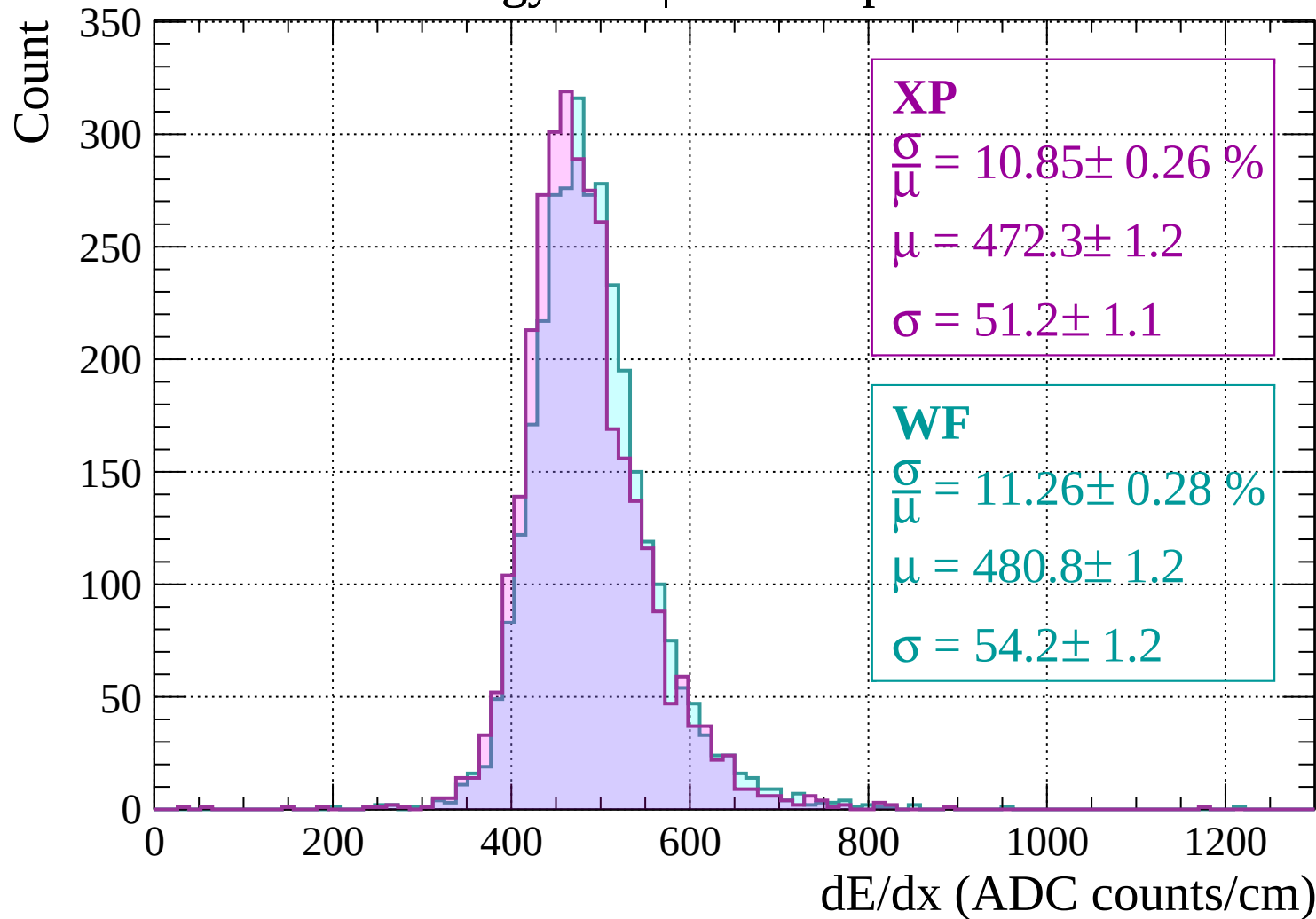
$$\sigma = 56.3 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1720 < p < -1680

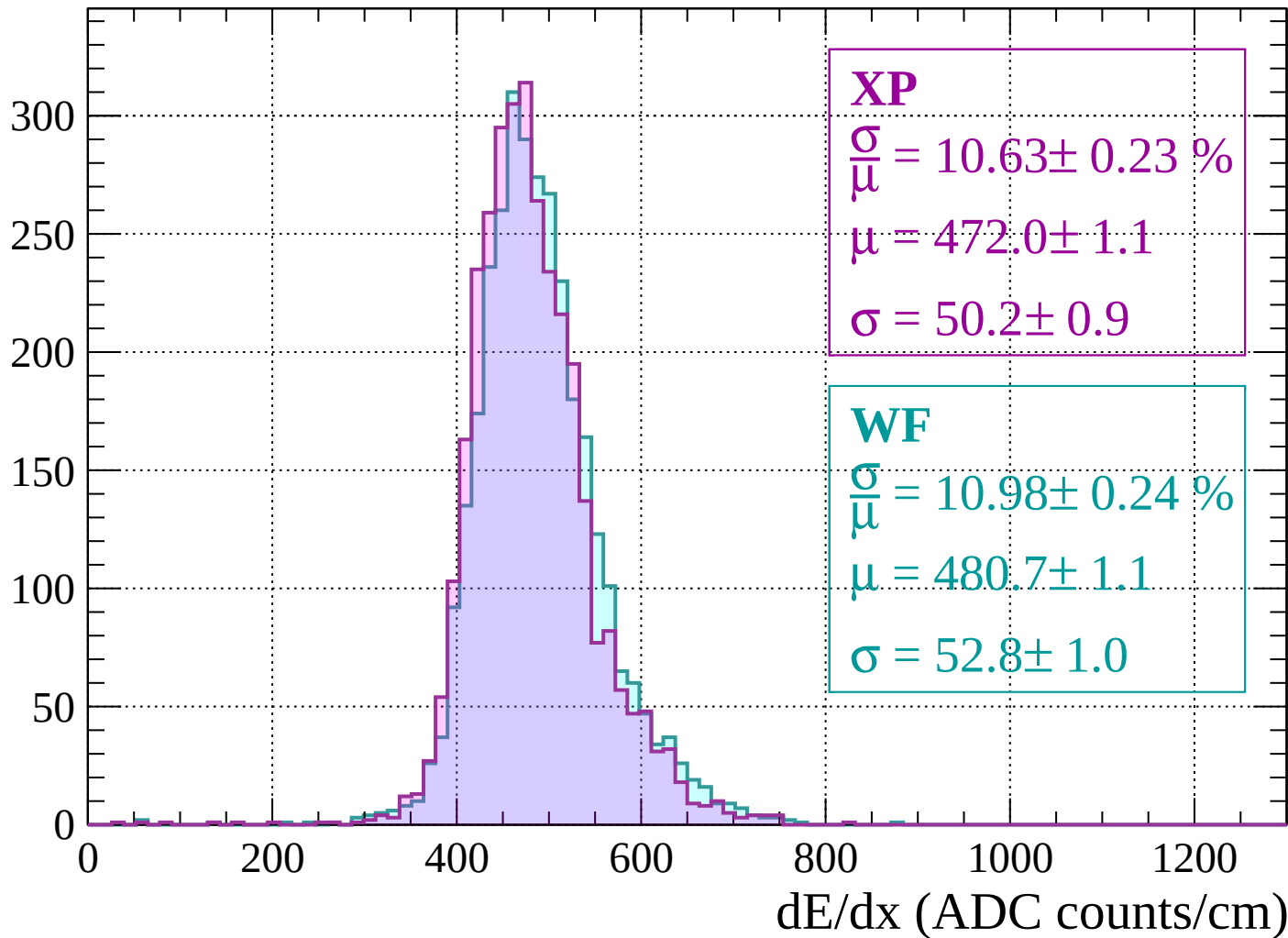


Energy loss | $-1680 < p < -1640$



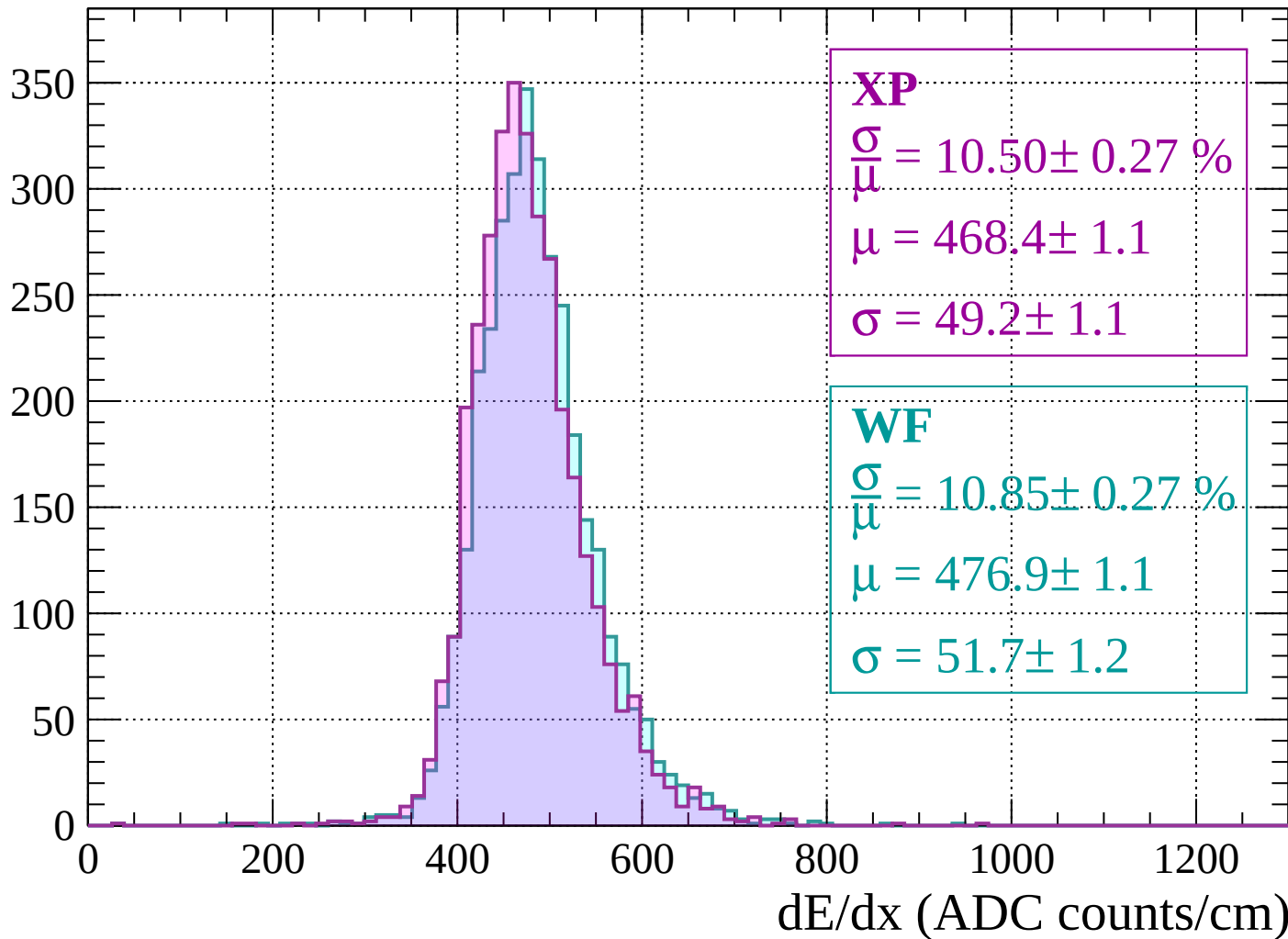
Energy loss | -1640 < p < -1600

Count



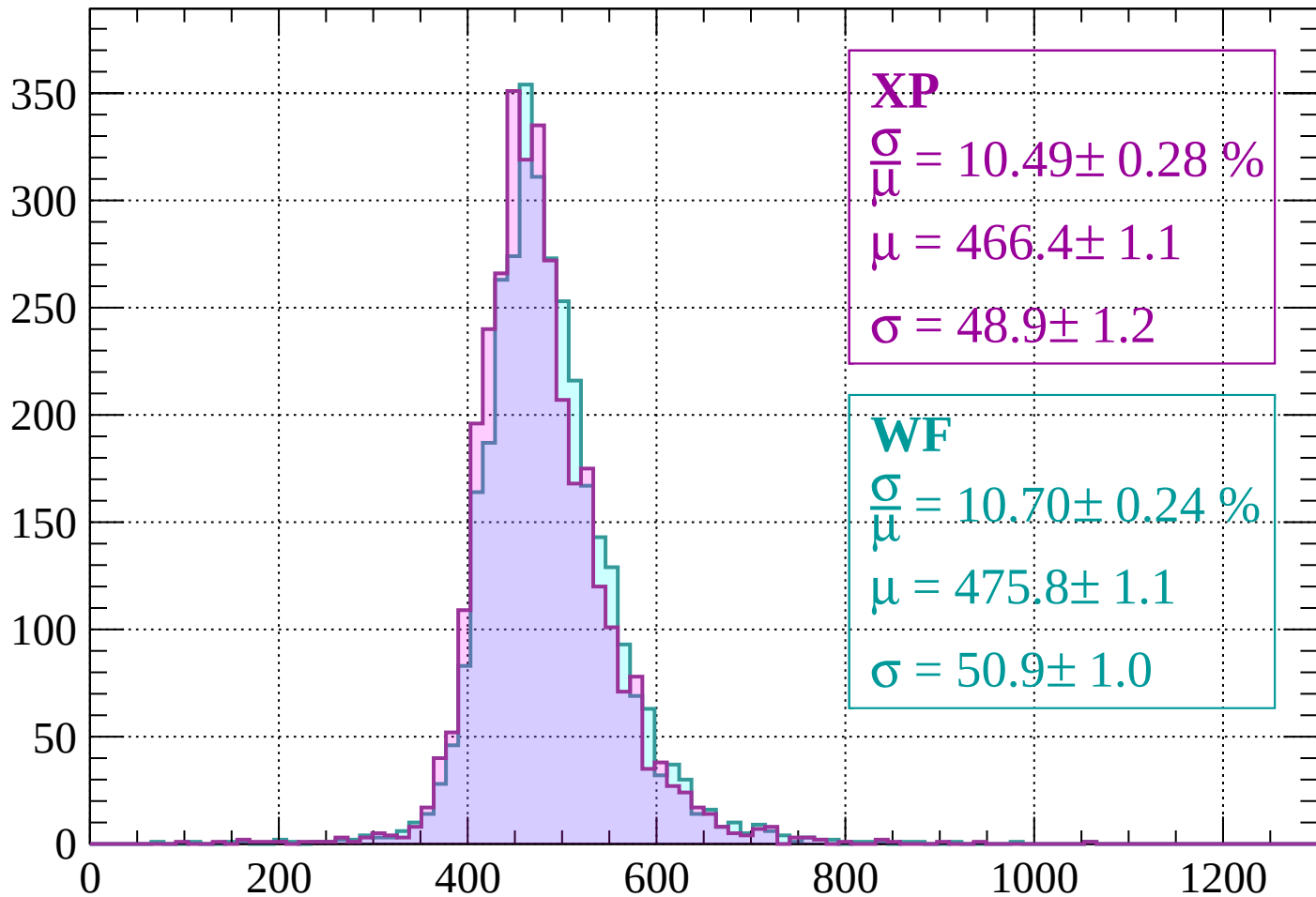
Energy loss | $-1600 < p < -1560$

Count



Energy loss | $-1560 < p < -1520$

Count



XP

$$\frac{\sigma}{\mu} = 10.49 \pm 0.28 \%$$

$$\mu = 466.4 \pm 1.1$$

$$\sigma = 48.9 \pm 1.2$$

WF

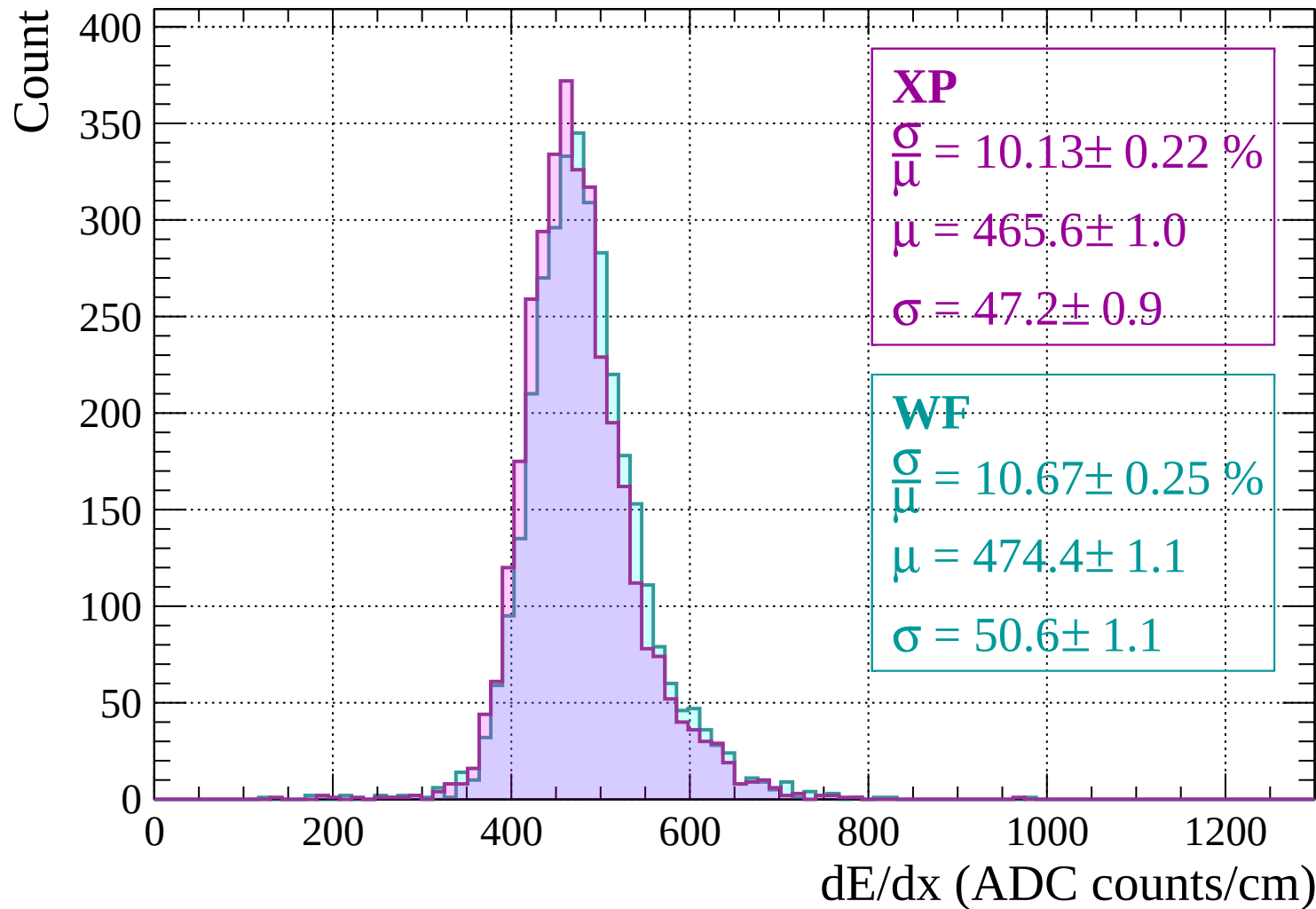
$$\frac{\sigma}{\mu} = 10.70 \pm 0.24 \%$$

$$\mu = 475.8 \pm 1.1$$

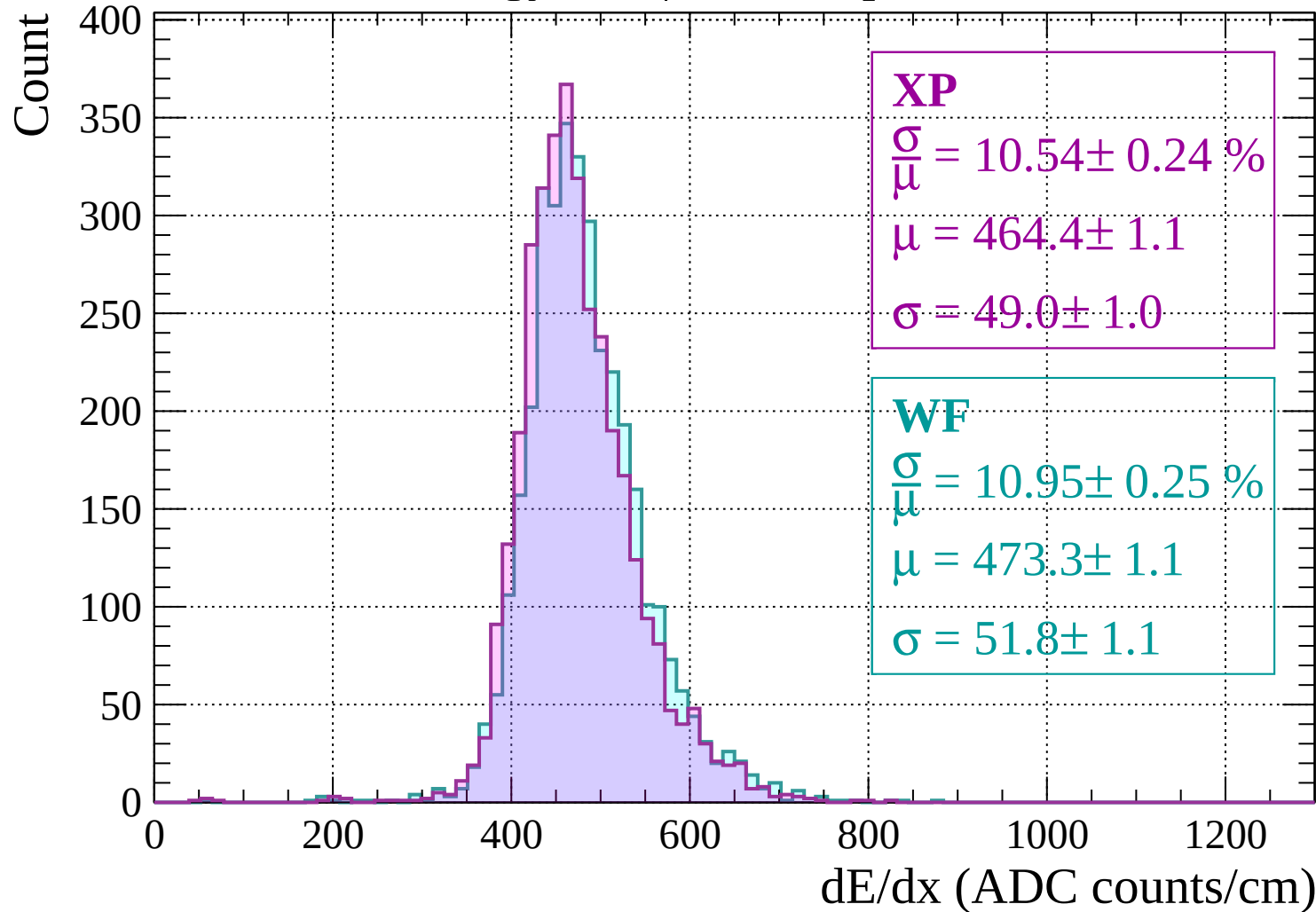
$$\sigma = 50.9 \pm 1.0$$

dE/dx (ADC counts/cm)

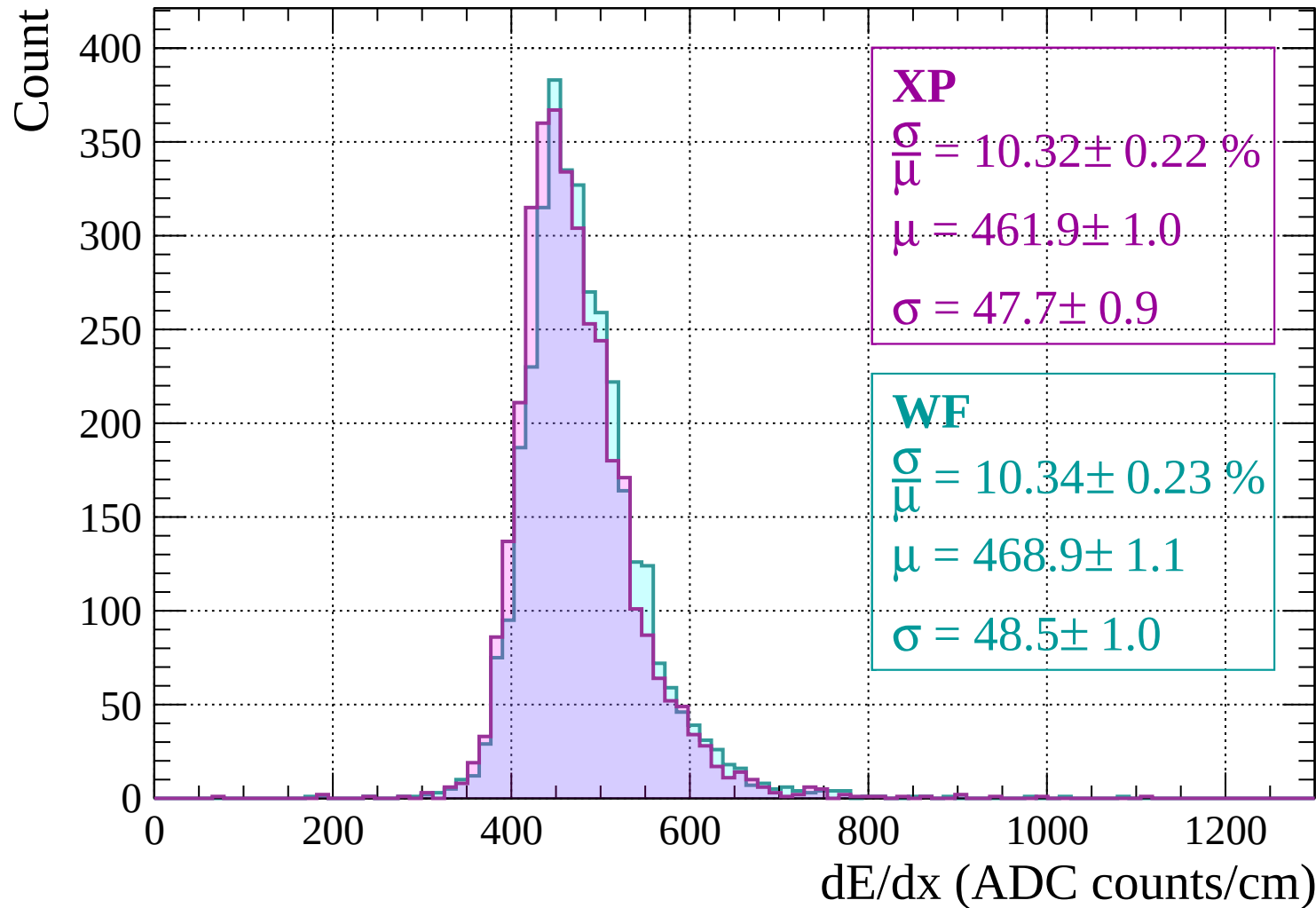
Energy loss | -1520 < p < -1480



Energy loss | -1480 < p < -1440

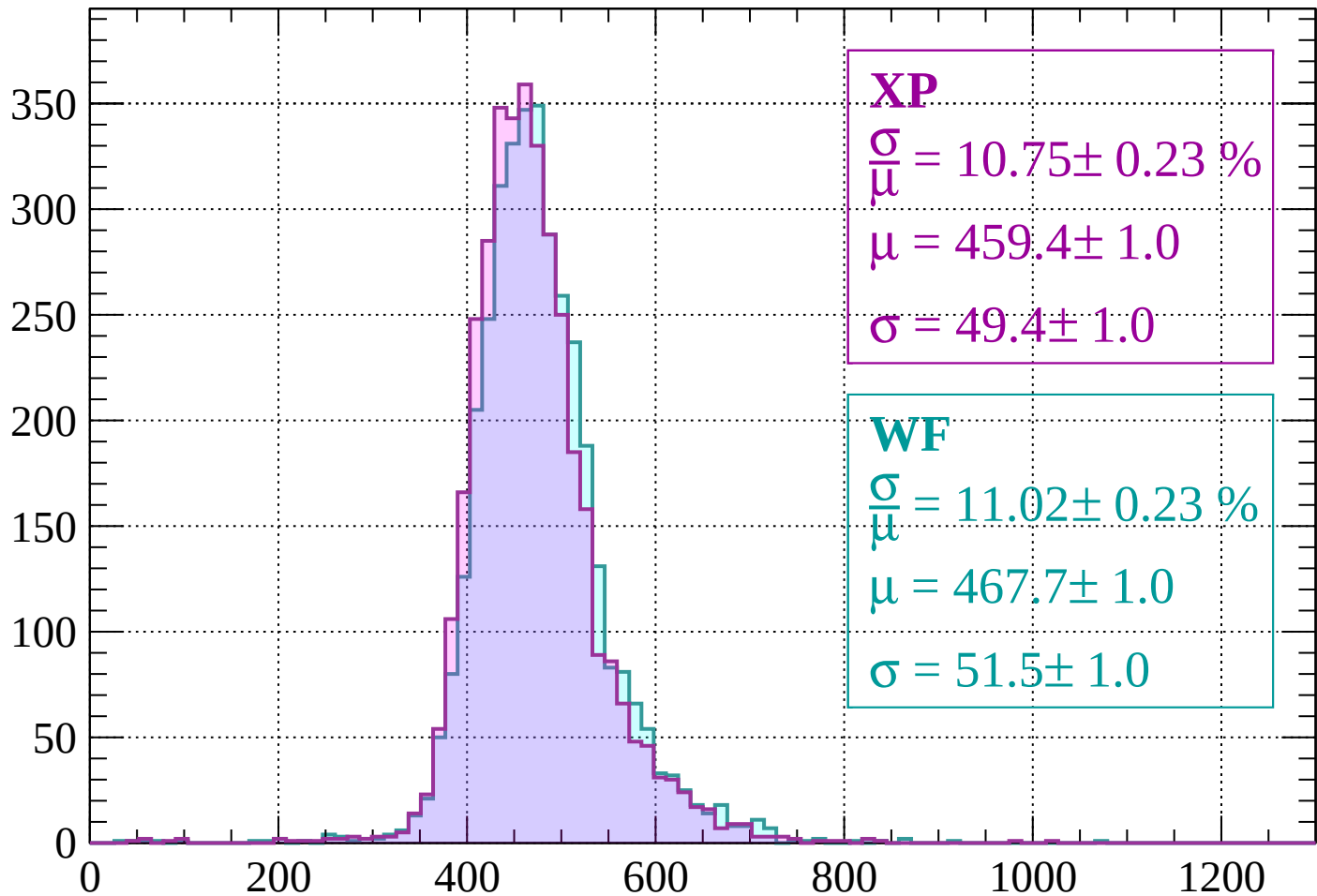


Energy loss | -1440 < p < -1400



Energy loss | -1400 < p < -1360

Count



XP

$$\frac{\sigma}{\mu} = 10.75 \pm 0.23 \%$$

$$\mu = 459.4 \pm 1.0$$

$$\sigma = 49.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.02 \pm 0.23 \%$$

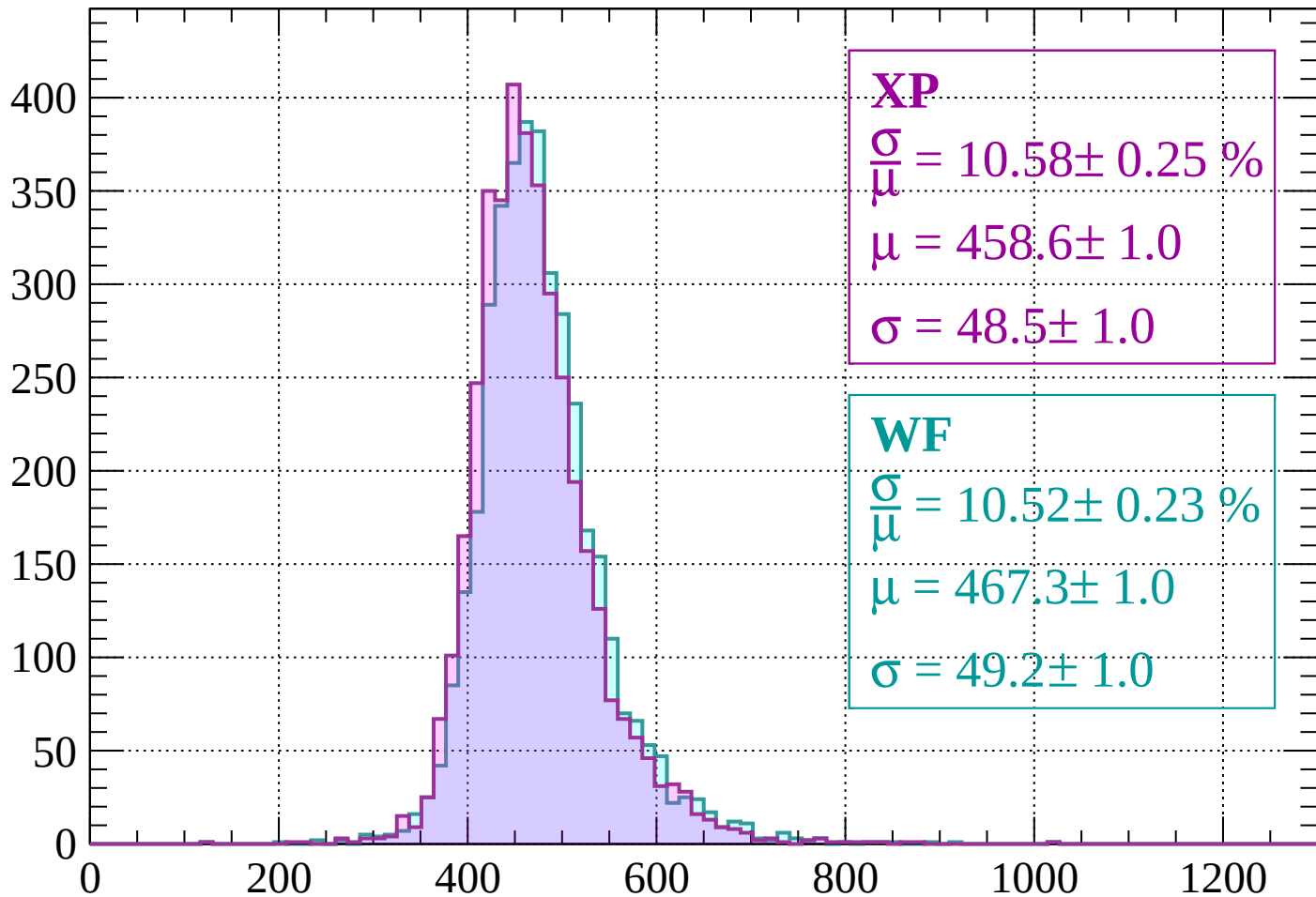
$$\mu = 467.7 \pm 1.0$$

$$\sigma = 51.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 10.58 \pm 0.25 \%$$

$$\mu = 458.6 \pm 1.0$$

$$\sigma = 48.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.52 \pm 0.23 \%$$

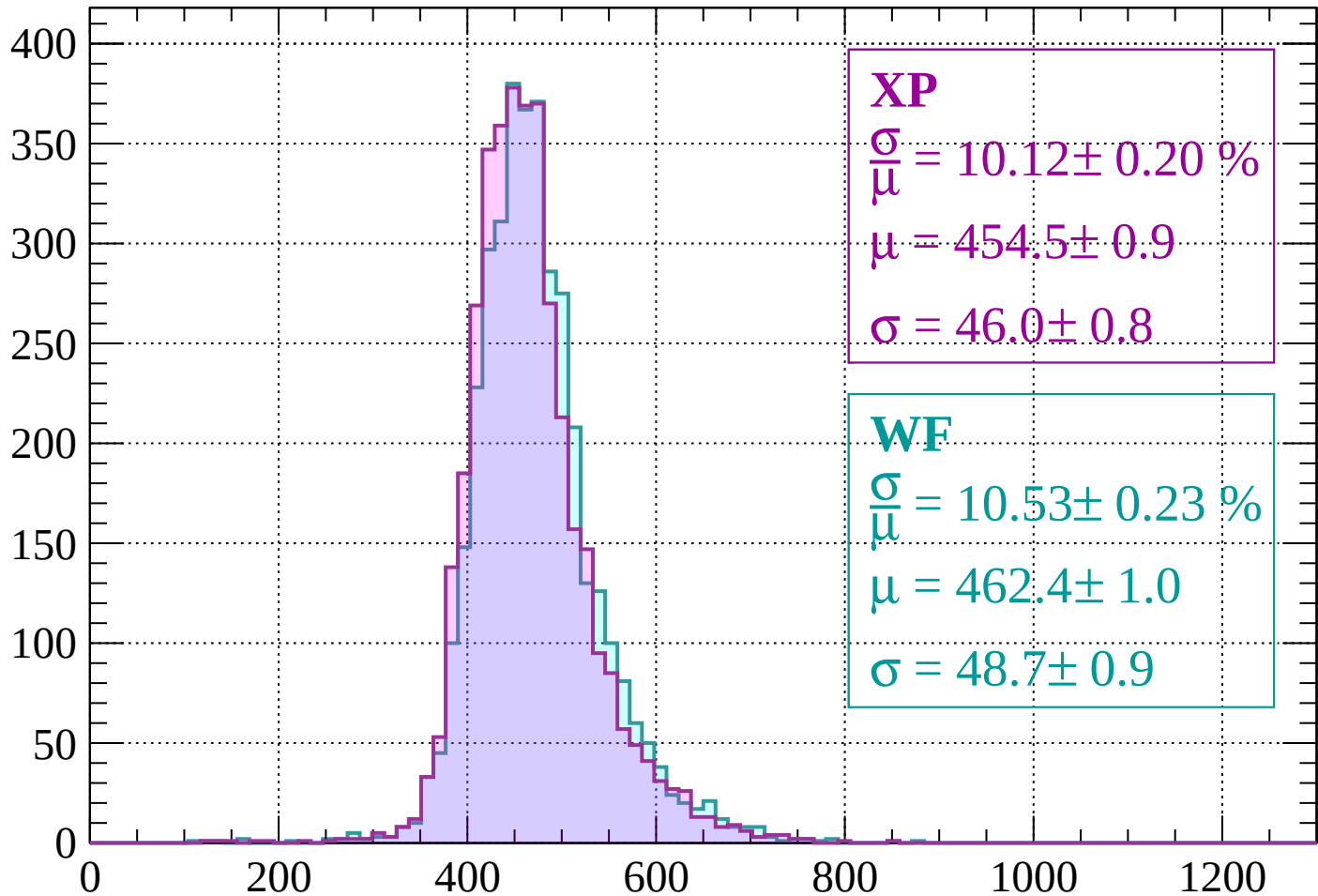
$$\mu = 467.3 \pm 1.0$$

$$\sigma = 49.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 10.12 \pm 0.20 \%$$

$$\mu = 454.5 \pm 0.9$$

$$\sigma = 46.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.53 \pm 0.23 \%$$

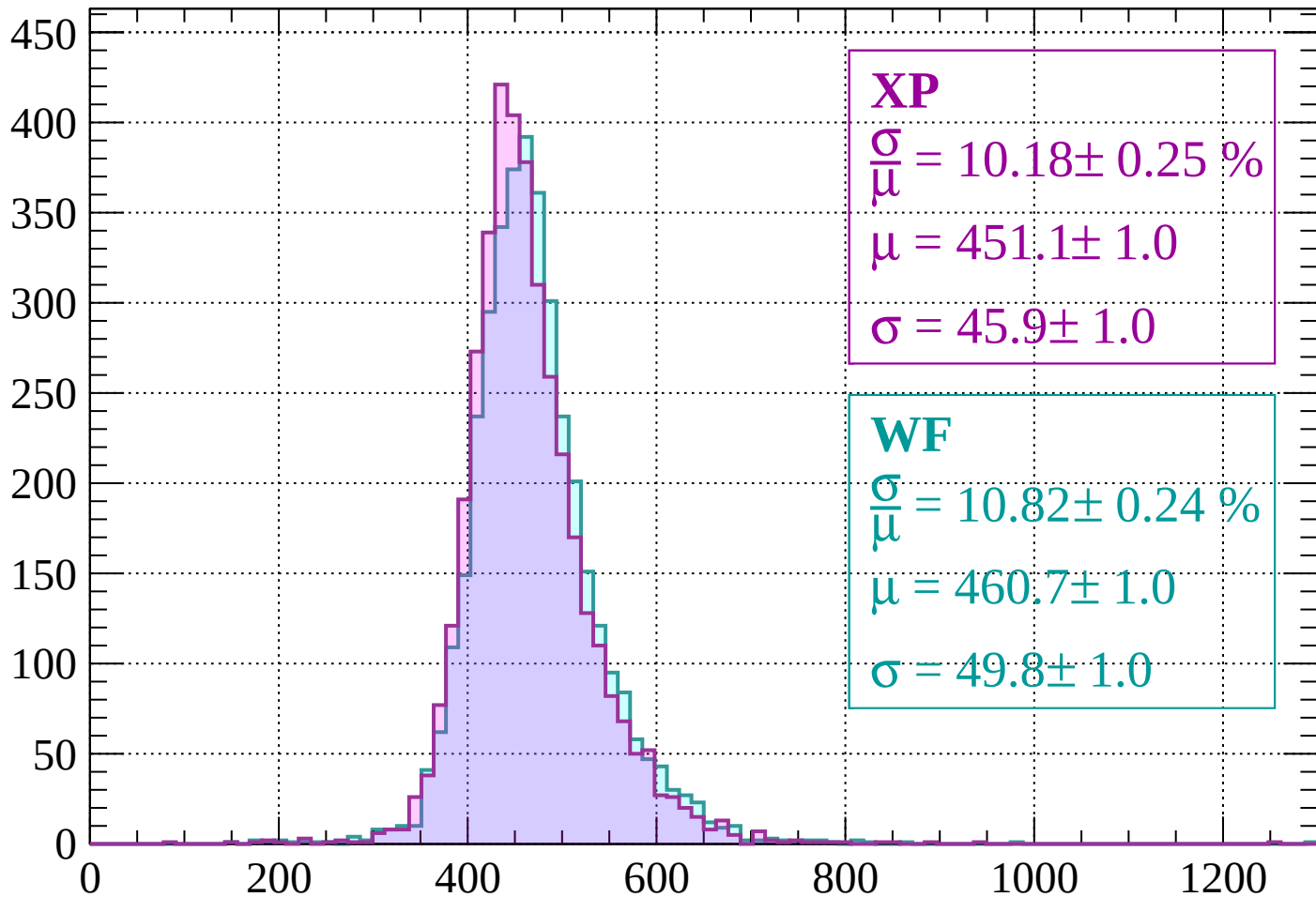
$$\mu = 462.4 \pm 1.0$$

$$\sigma = 48.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 10.18 \pm 0.25 \%$$

$$\mu = 451.1 \pm 1.0$$

$$\sigma = 45.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.82 \pm 0.24 \%$$

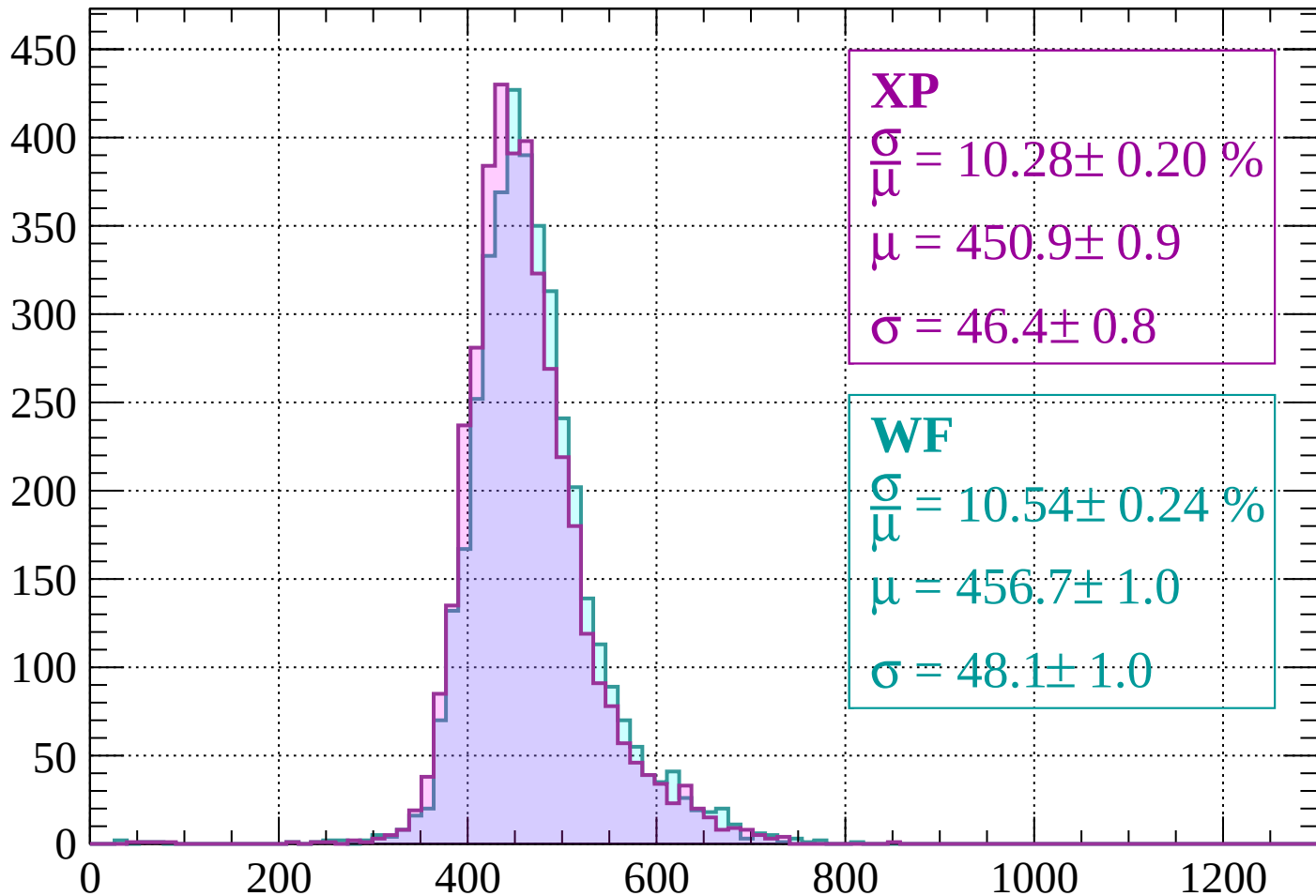
$$\mu = 460.7 \pm 1.0$$

$$\sigma = 49.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1240 < p < -1200

Count



XP

$$\frac{\sigma}{\mu} = 10.28 \pm 0.20 \%$$

$$\mu = 450.9 \pm 0.9$$

$$\sigma = 46.4 \pm 0.8$$

WF

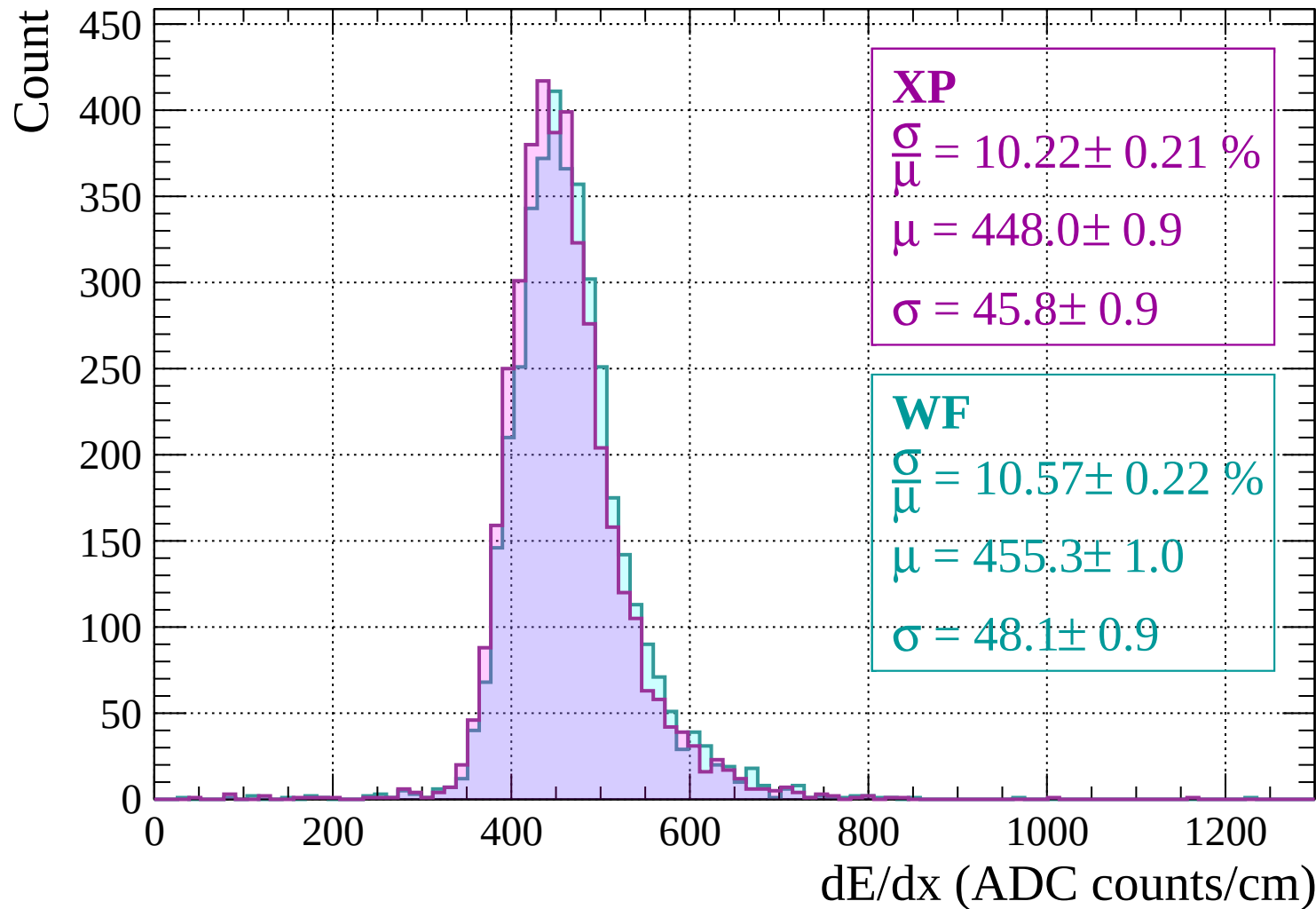
$$\frac{\sigma}{\mu} = 10.54 \pm 0.24 \%$$

$$\mu = 456.7 \pm 1.0$$

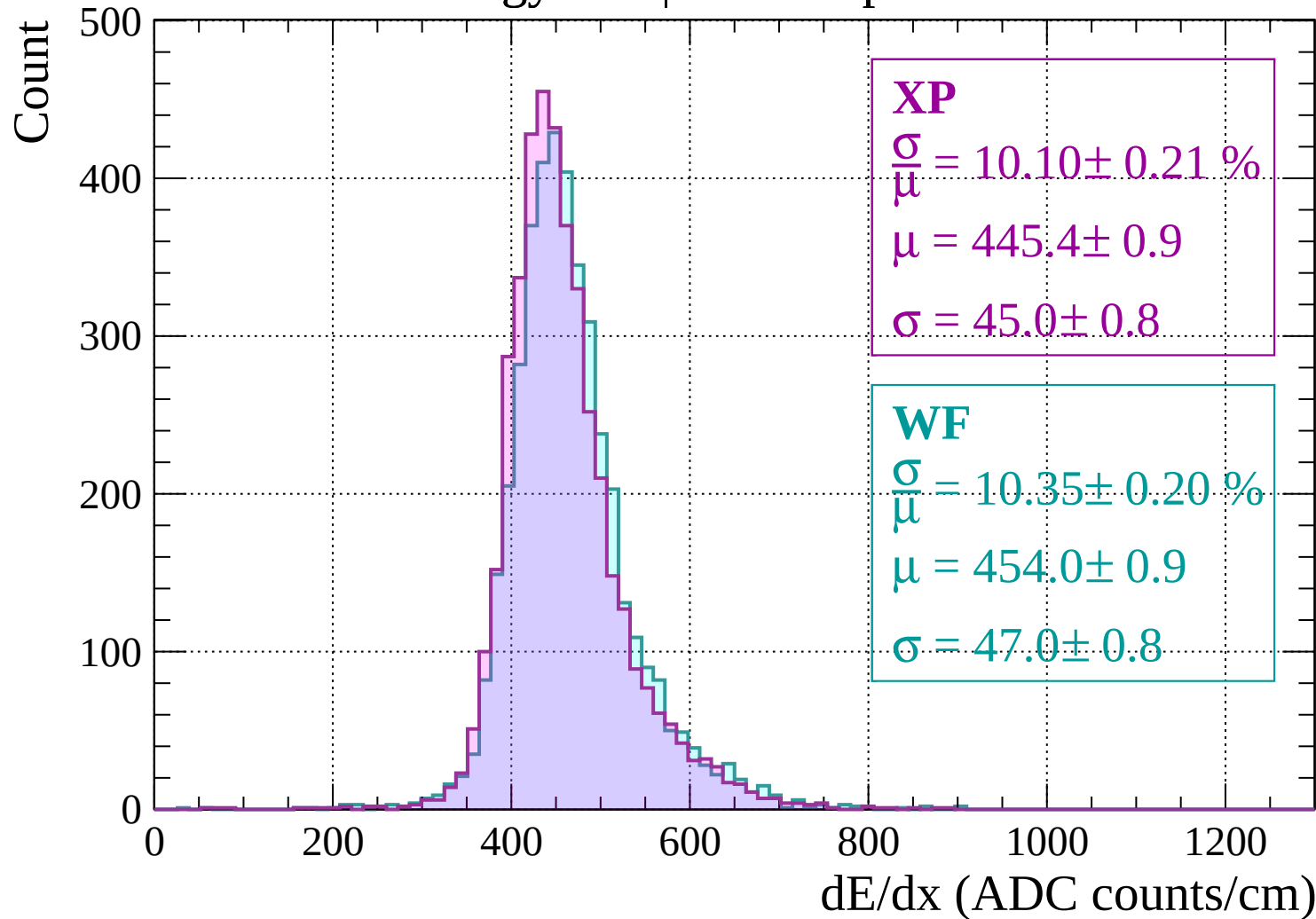
$$\sigma = 48.1 \pm 1.0$$

dE/dx (ADC counts/cm)

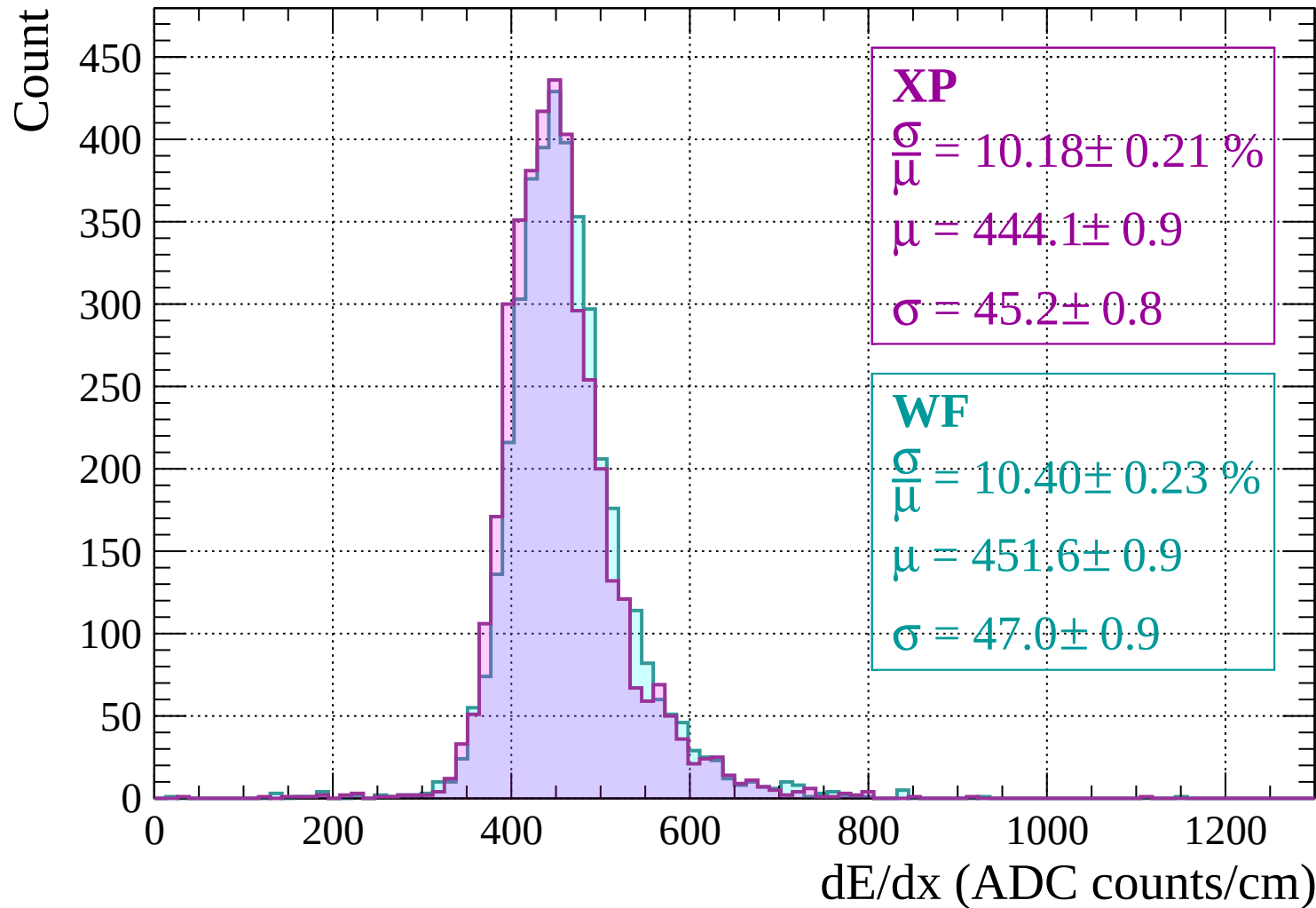
Energy loss | -1200 < p < -1160



Energy loss | $-1160 < p < -1120$

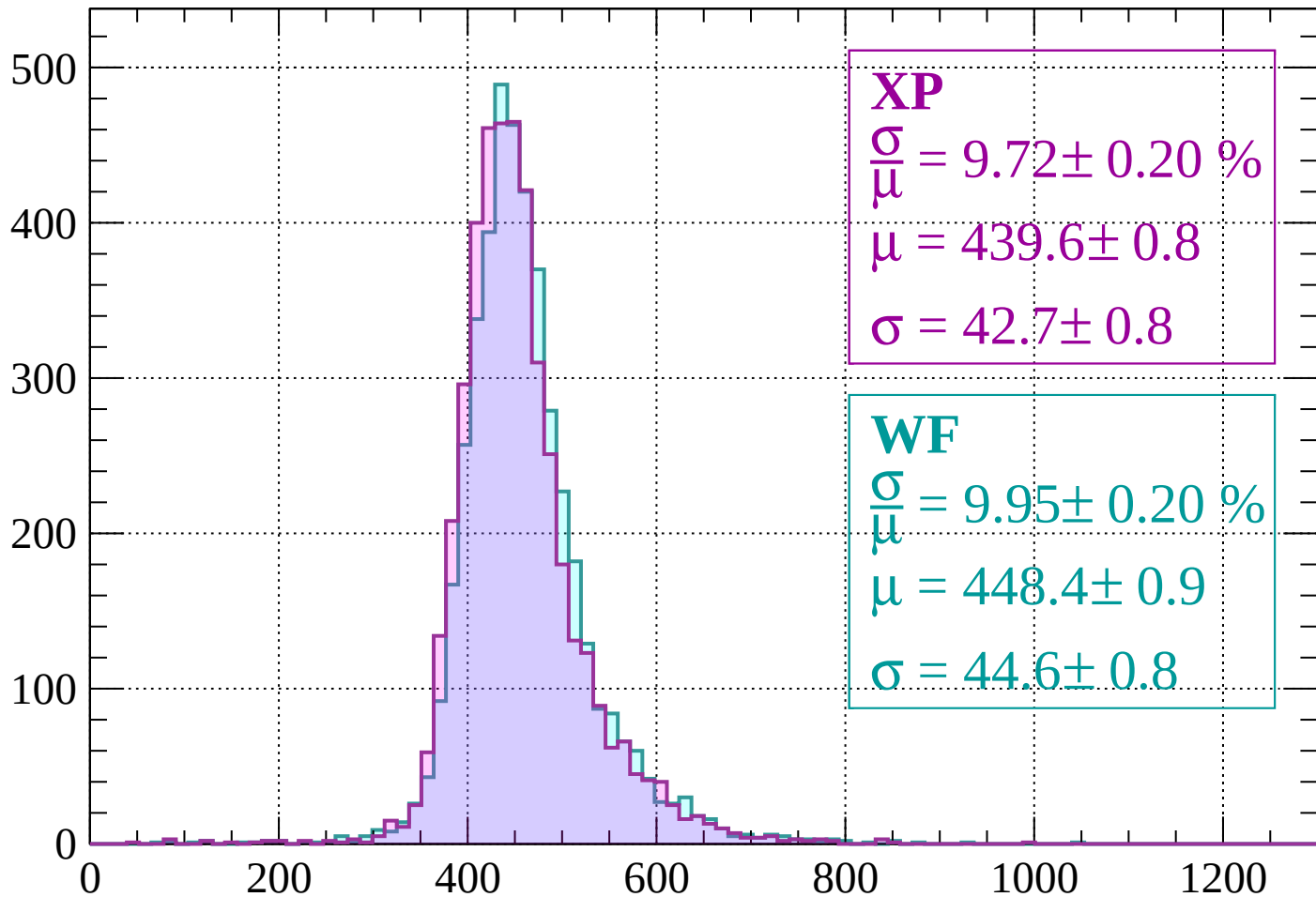


Energy loss | -1120 < p < -1080



Energy loss | $-1080 < p < -1040$

Count



XP

$$\frac{\sigma}{\mu} = 9.72 \pm 0.20 \%$$

$$\mu = 439.6 \pm 0.8$$

$$\sigma = 42.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.95 \pm 0.20 \%$$

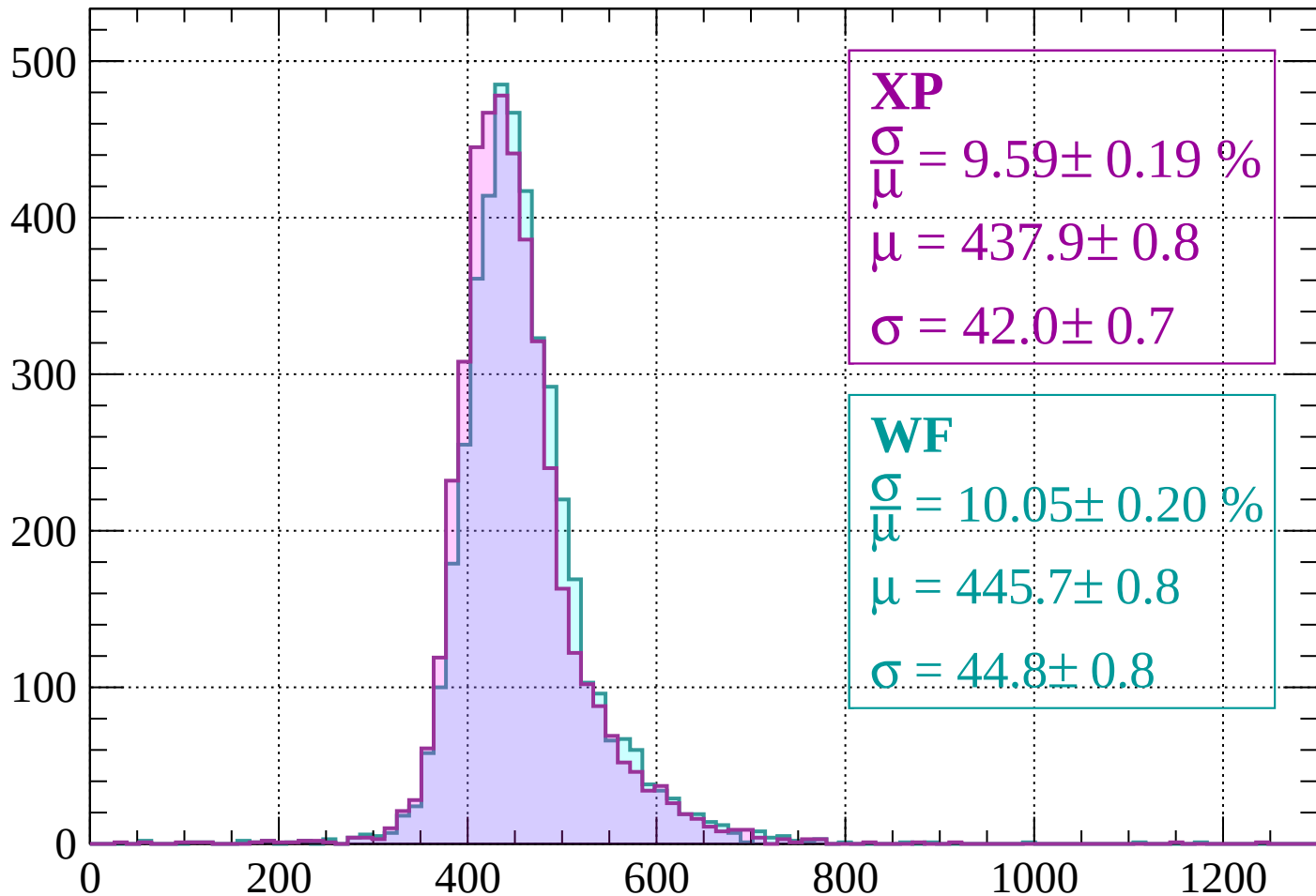
$$\mu = 448.4 \pm 0.9$$

$$\sigma = 44.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 9.59 \pm 0.19 \%$$

$$\mu = 437.9 \pm 0.8$$

$$\sigma = 42.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.05 \pm 0.20 \%$$

$$\mu = 445.7 \pm 0.8$$

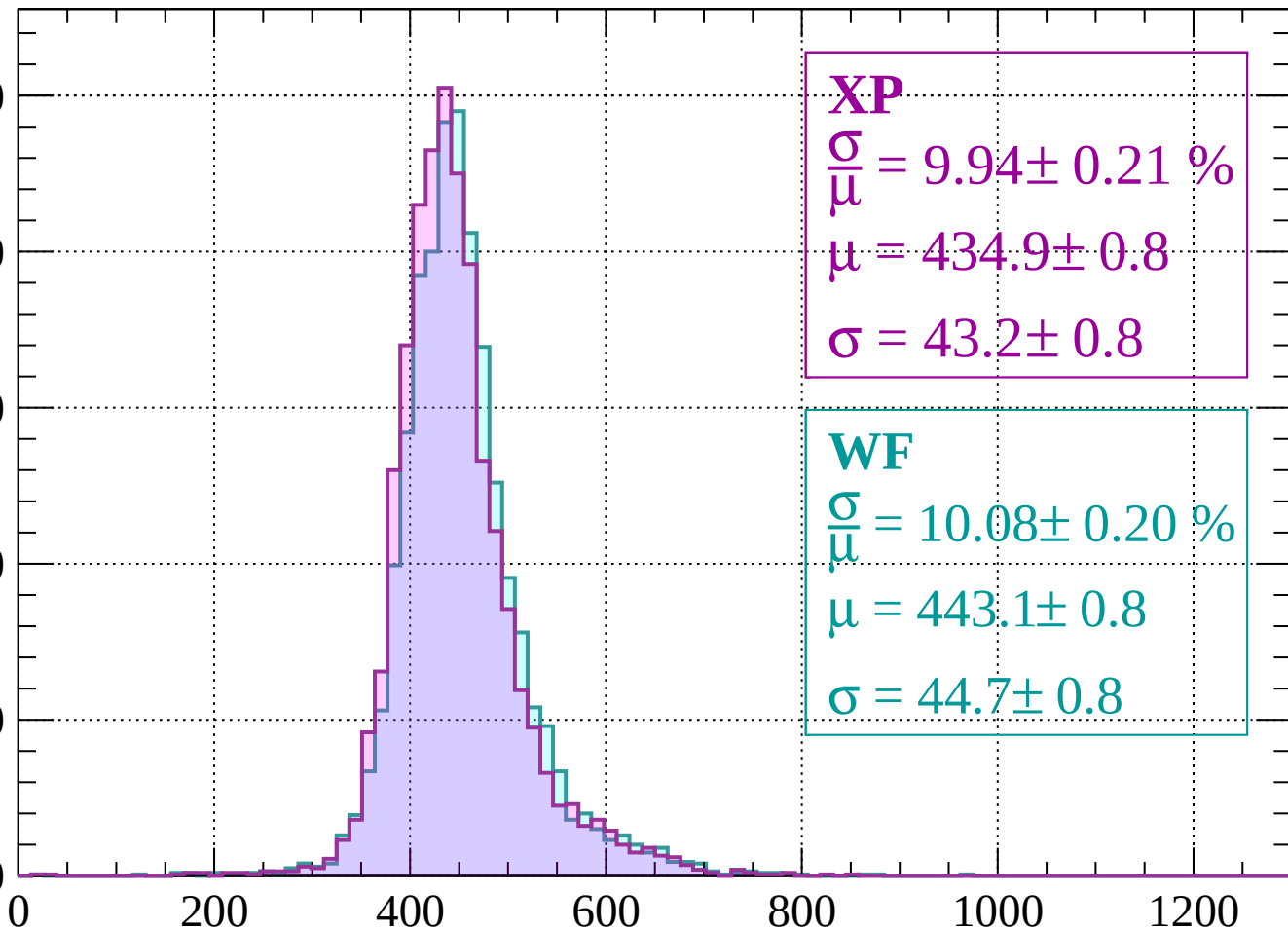
$$\sigma = 44.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\bar{\mu}} = 9.94 \pm 0.21 \%$$

$$\mu = 434.9 \pm 0.8$$

$$\sigma = 43.2 \pm 0.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.08 \pm 0.20 \%$$

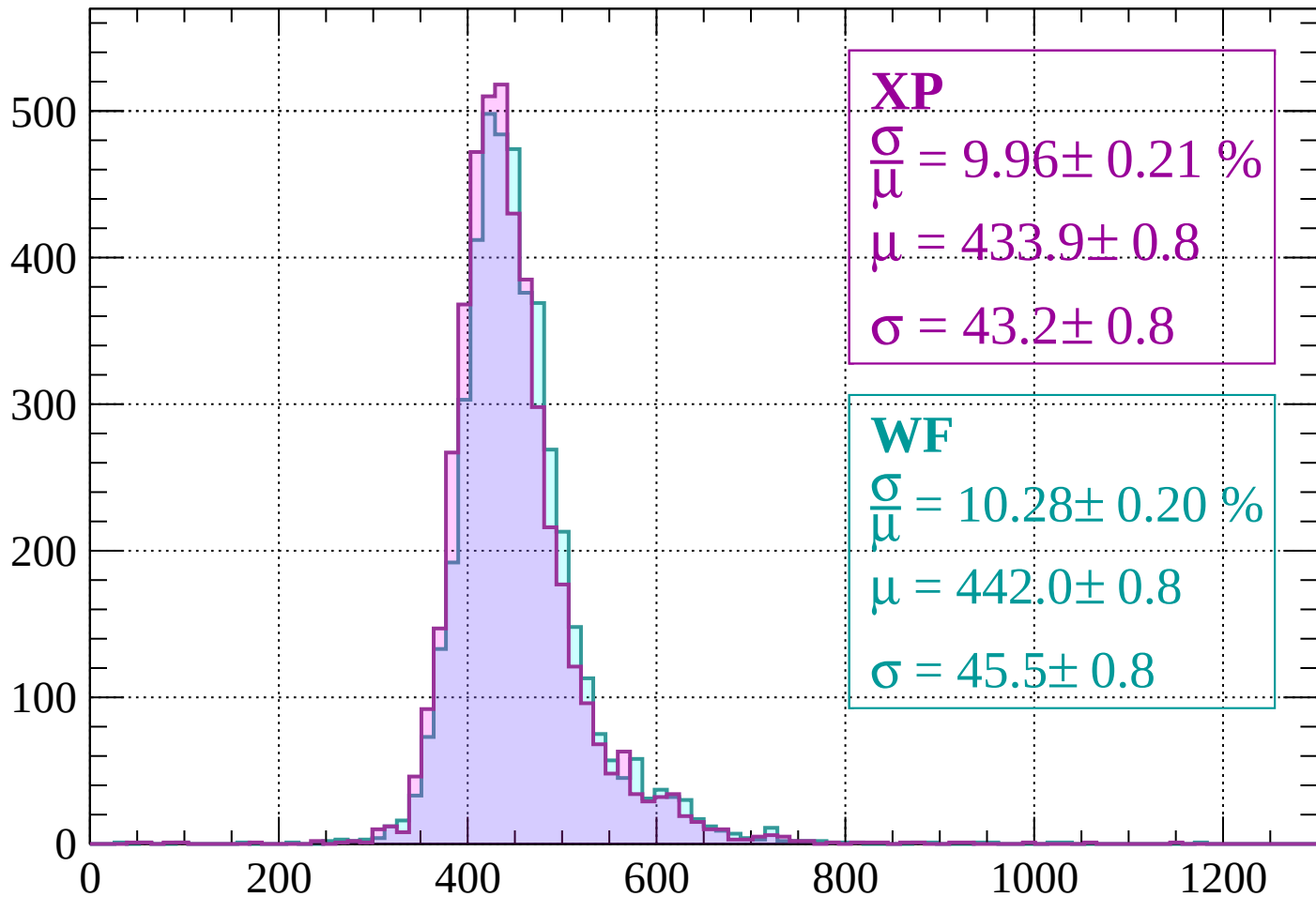
$$\mu = 443.1 \pm 0.8$$

$$\sigma = 44.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 9.96 \pm 0.21 \%$$

$$\mu = 433.9 \pm 0.8$$

$$\sigma = 43.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.28 \pm 0.20 \%$$

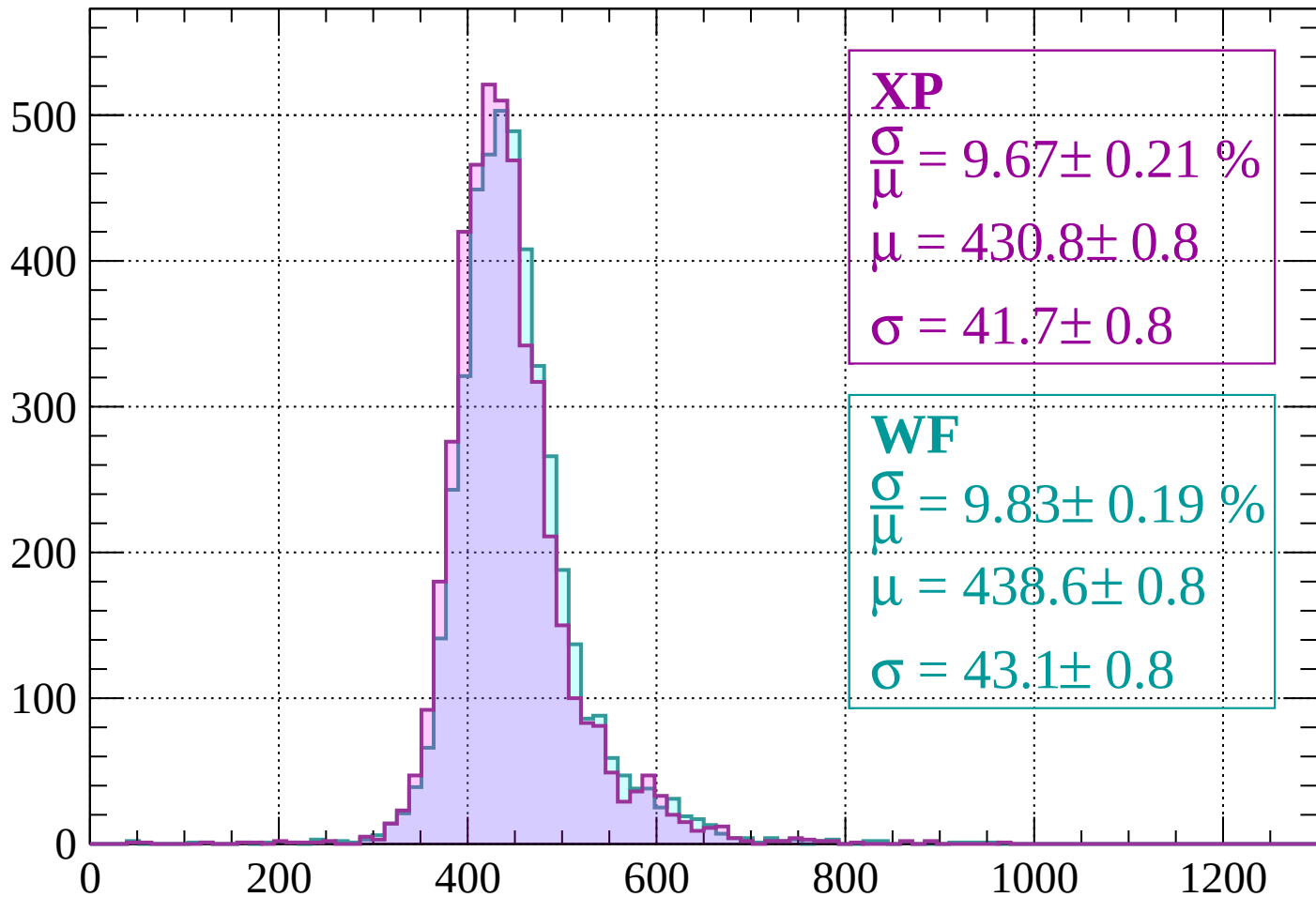
$$\mu = 442.0 \pm 0.8$$

$$\sigma = 45.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 9.67 \pm 0.21 \%$$

$$\mu = 430.8 \pm 0.8$$

$$\sigma = 41.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.83 \pm 0.19 \%$$

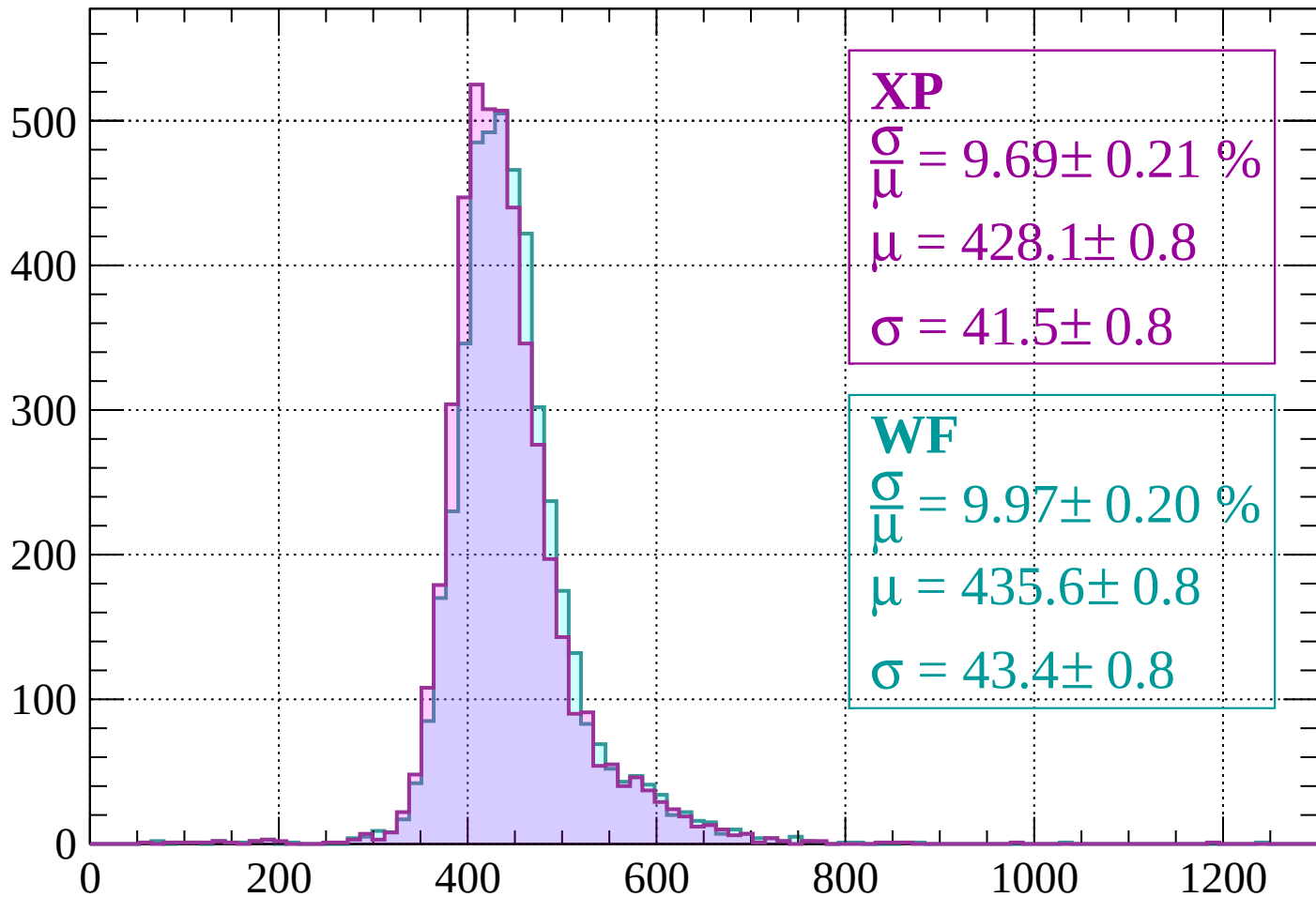
$$\mu = 438.6 \pm 0.8$$

$$\sigma = 43.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.69 \pm 0.21 \%$$

$$\mu = 428.1 \pm 0.8$$

$$\sigma = 41.5 \pm 0.8$$

WF

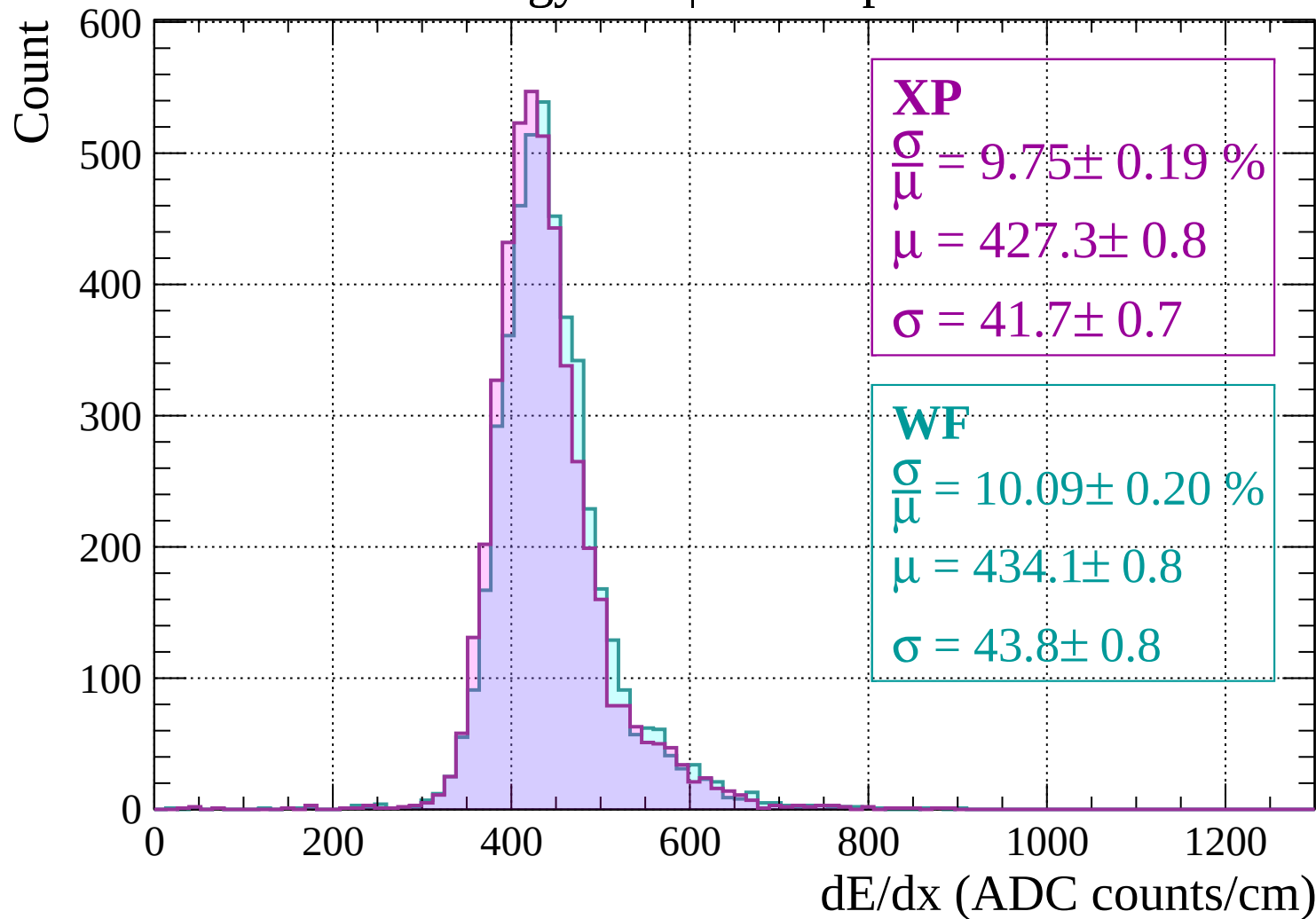
$$\frac{\sigma}{\mu} = 9.97 \pm 0.20 \%$$

$$\mu = 435.6 \pm 0.8$$

$$\sigma = 43.4 \pm 0.8$$

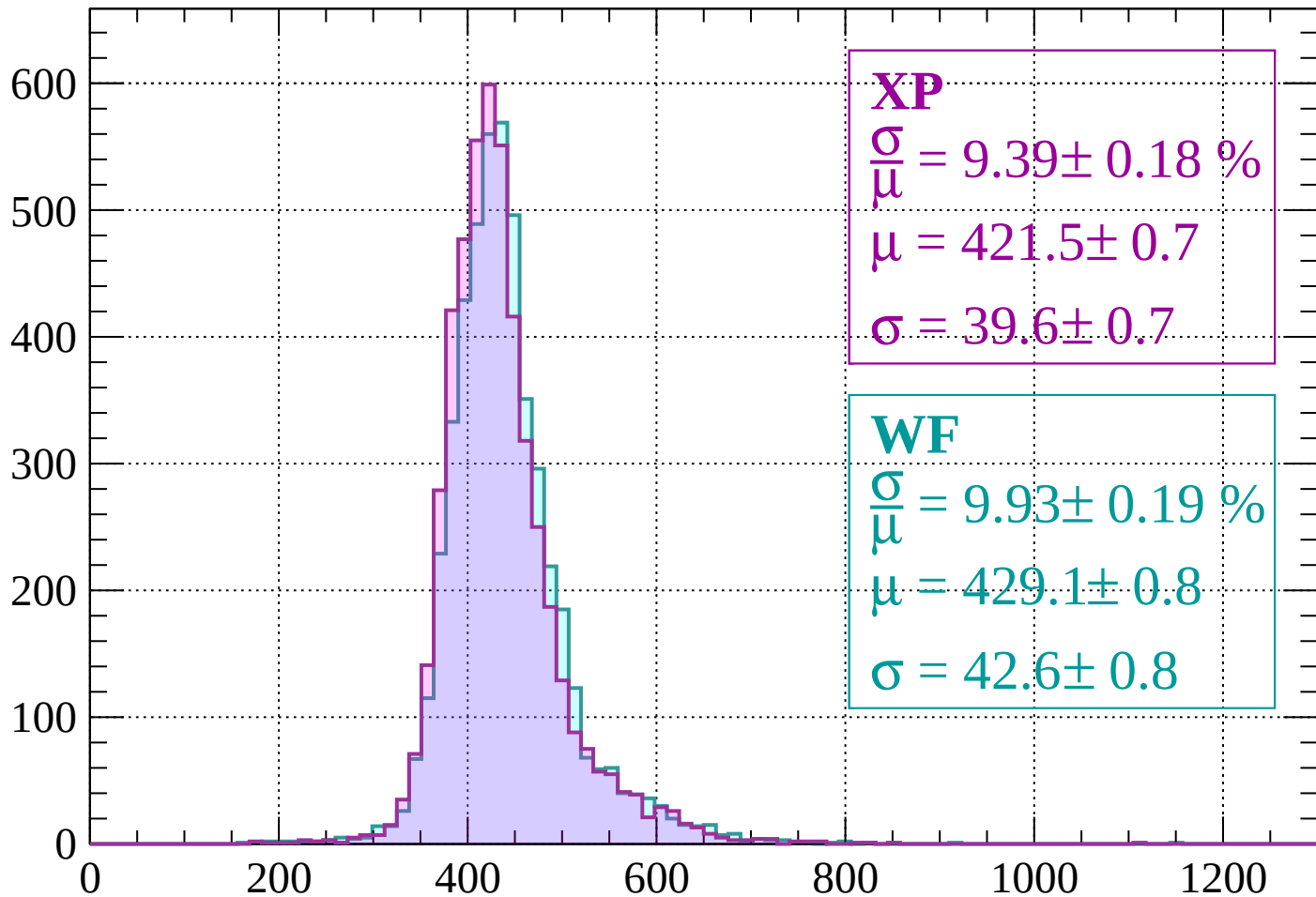
dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$



Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.18 \%$$

$$\mu = 421.5 \pm 0.7$$

$$\sigma = 39.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.93 \pm 0.19 \%$$

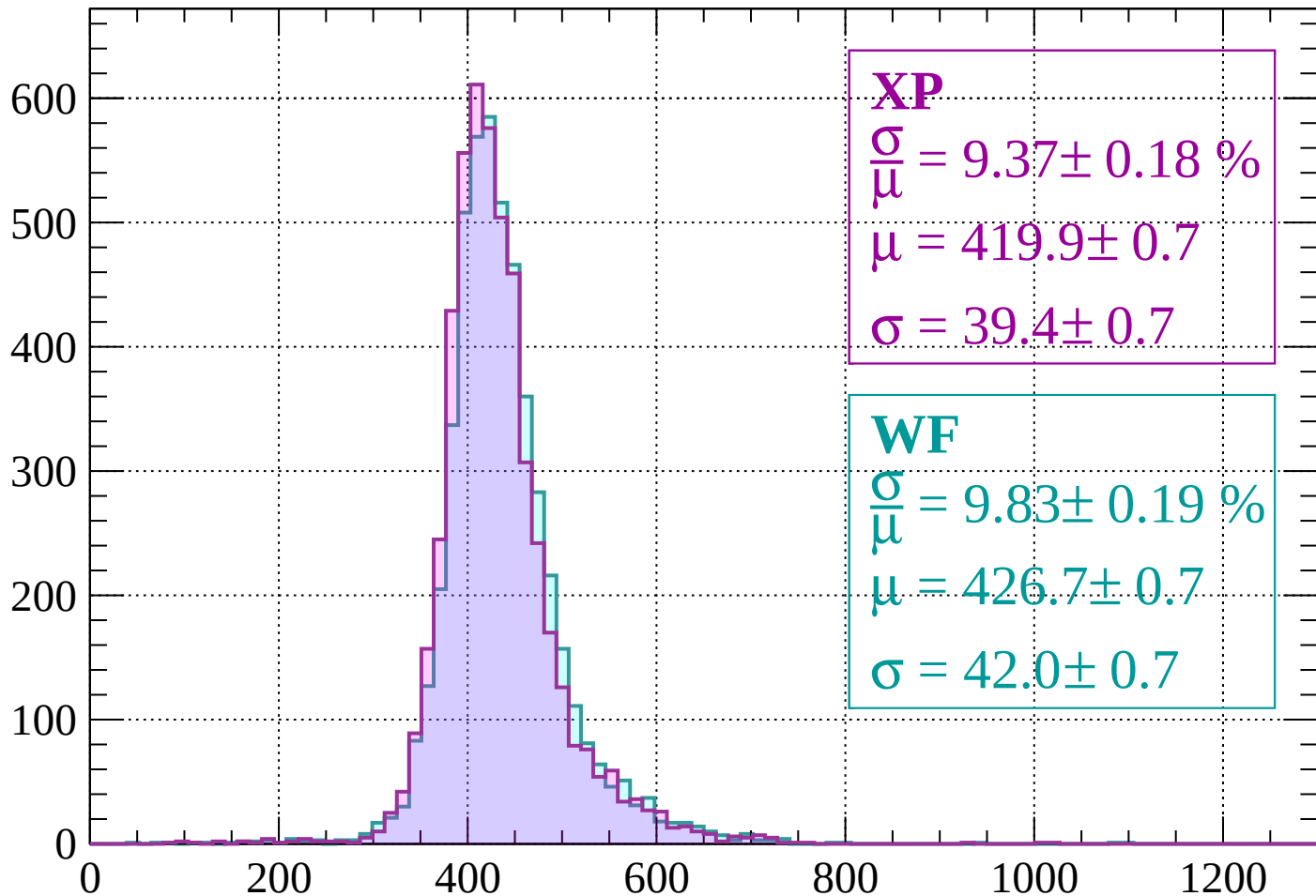
$$\mu = 429.1 \pm 0.8$$

$$\sigma = 42.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.37 \pm 0.18 \%$$

$$\mu = 419.9 \pm 0.7$$

$$\sigma = 39.4 \pm 0.7$$

WF

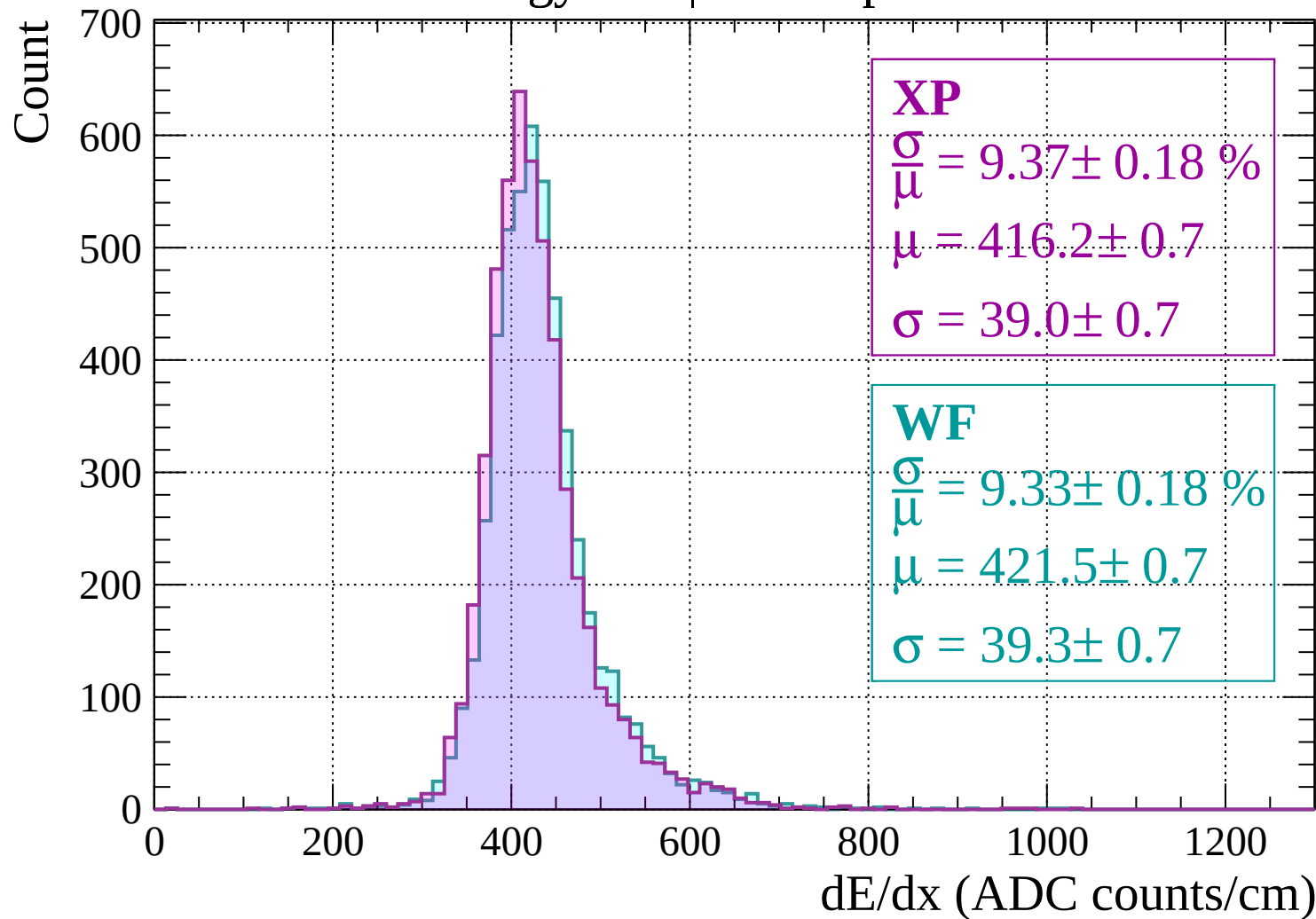
$$\frac{\sigma}{\mu} = 9.83 \pm 0.19 \%$$

$$\mu = 426.7 \pm 0.7$$

$$\sigma = 42.0 \pm 0.7$$

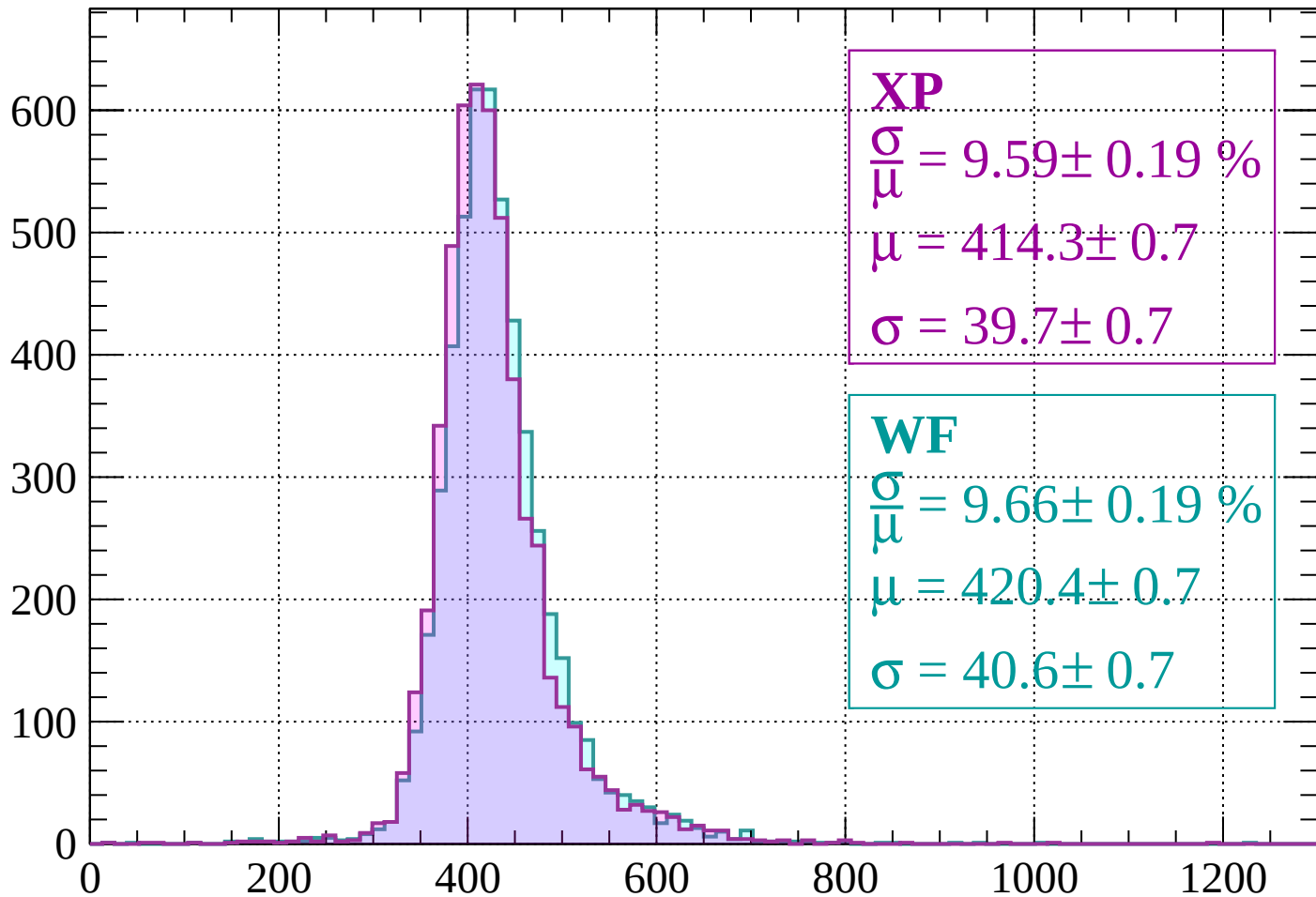
dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$



Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 9.59 \pm 0.19 \%$$

$$\mu = 414.3 \pm 0.7$$

$$\sigma = 39.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.66 \pm 0.19 \%$$

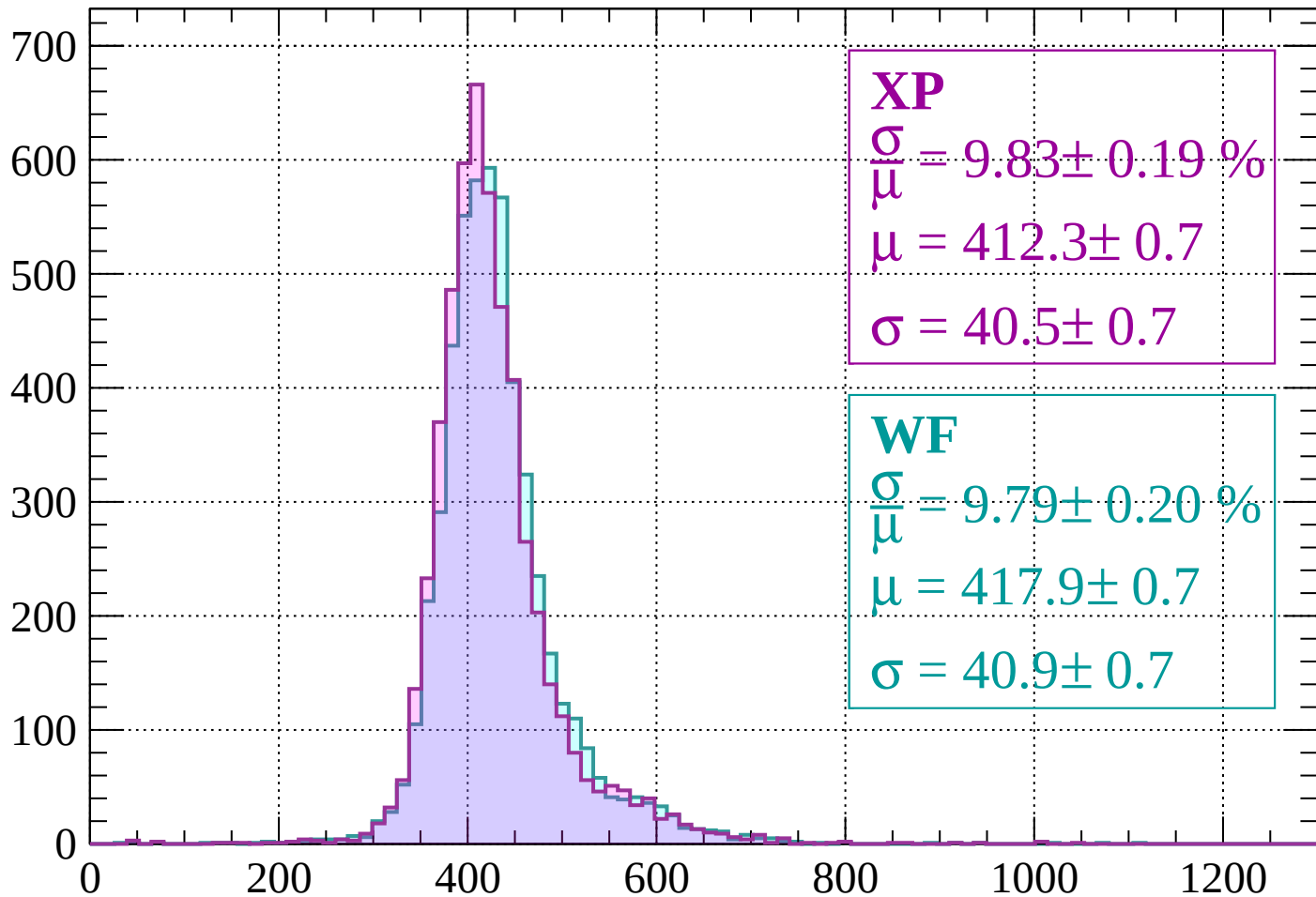
$$\mu = 420.4 \pm 0.7$$

$$\sigma = 40.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 9.83 \pm 0.19 \%$$

$$\mu = 412.3 \pm 0.7$$

$$\sigma = 40.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.20 \%$$

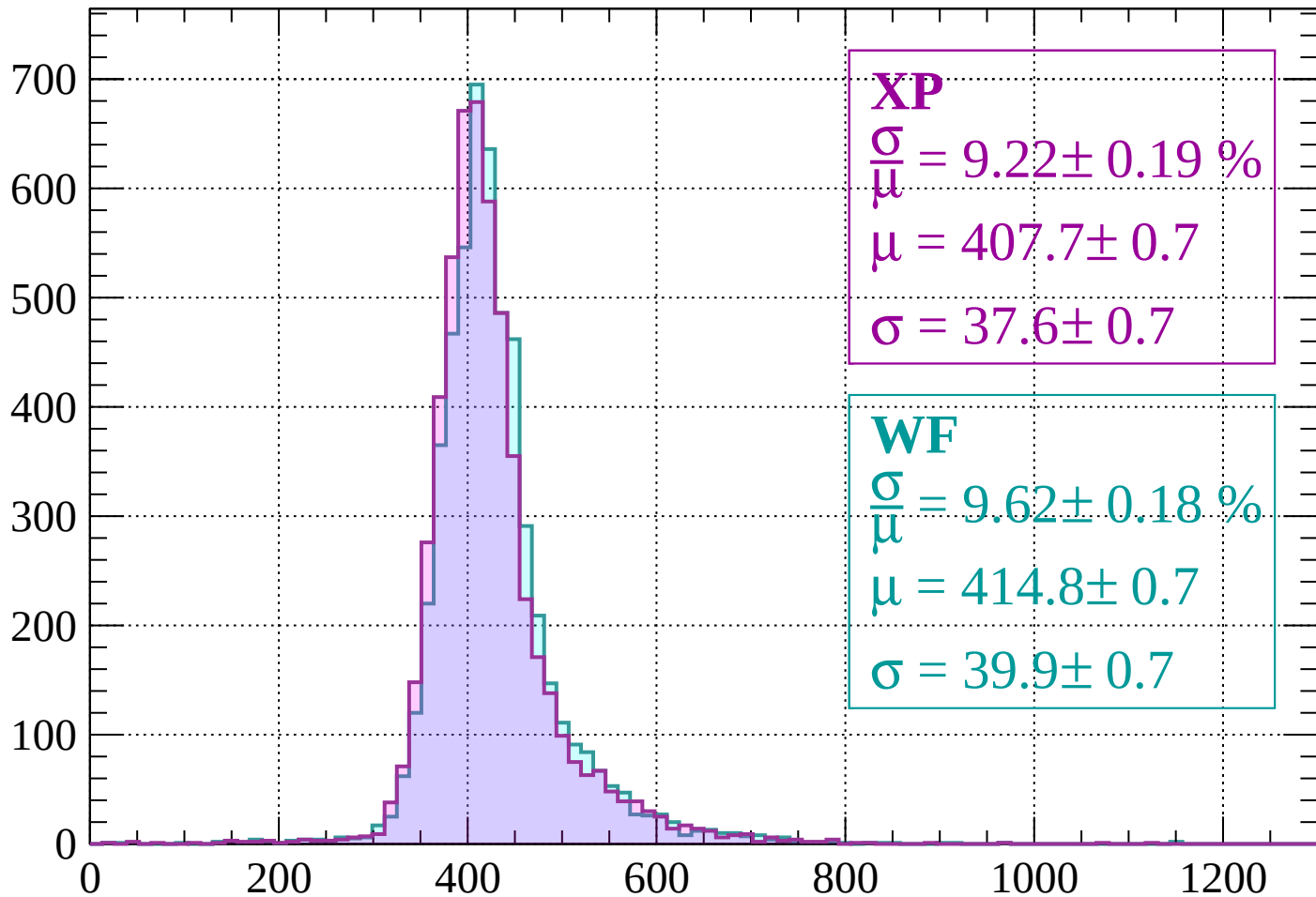
$$\mu = 417.9 \pm 0.7$$

$$\sigma = 40.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 9.22 \pm 0.19 \%$$

$$\mu = 407.7 \pm 0.7$$

$$\sigma = 37.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.62 \pm 0.18 \%$$

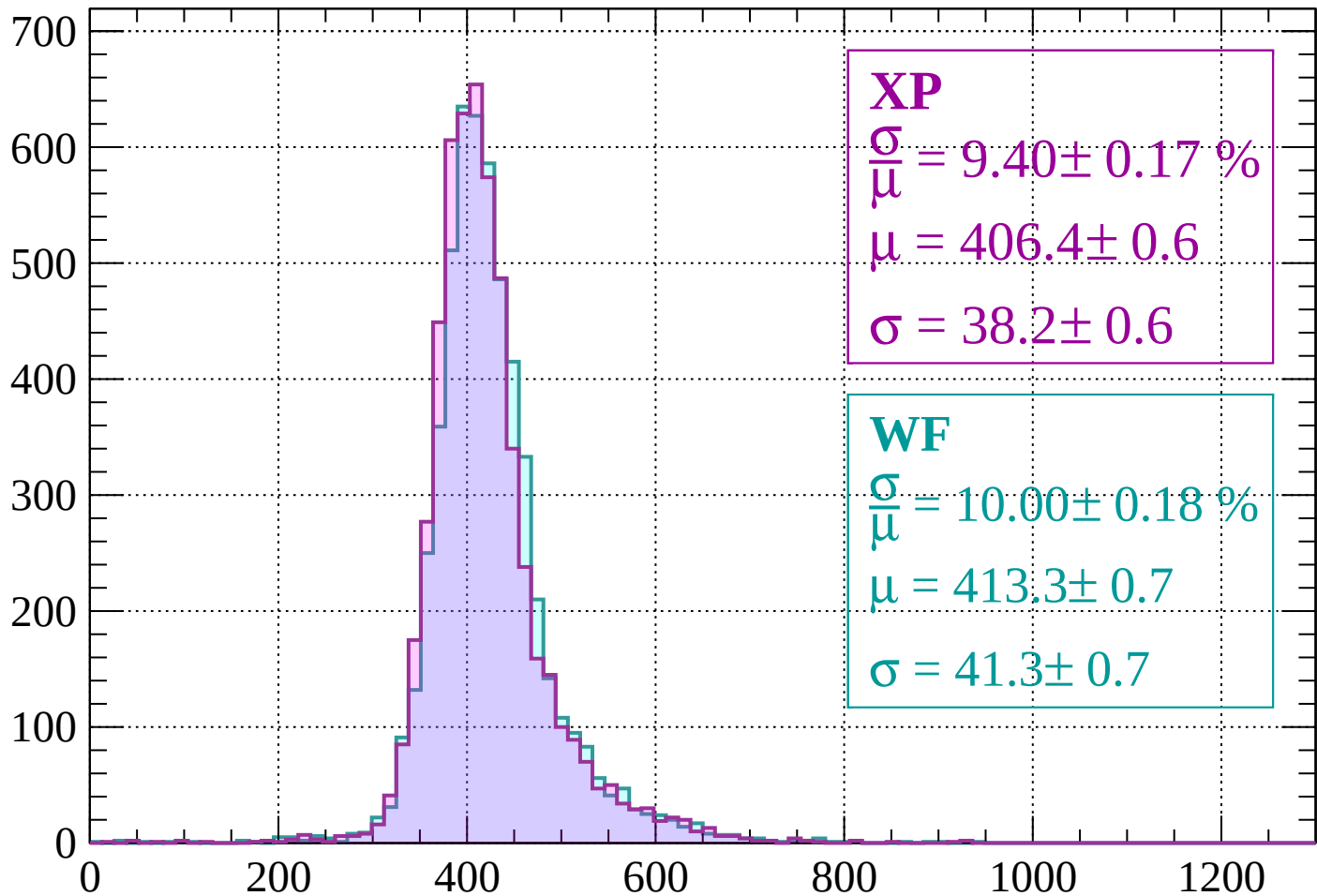
$$\mu = 414.8 \pm 0.7$$

$$\sigma = 39.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.17 \%$$

$$\mu = 406.4 \pm 0.6$$

$$\sigma = 38.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.00 \pm 0.18 \%$$

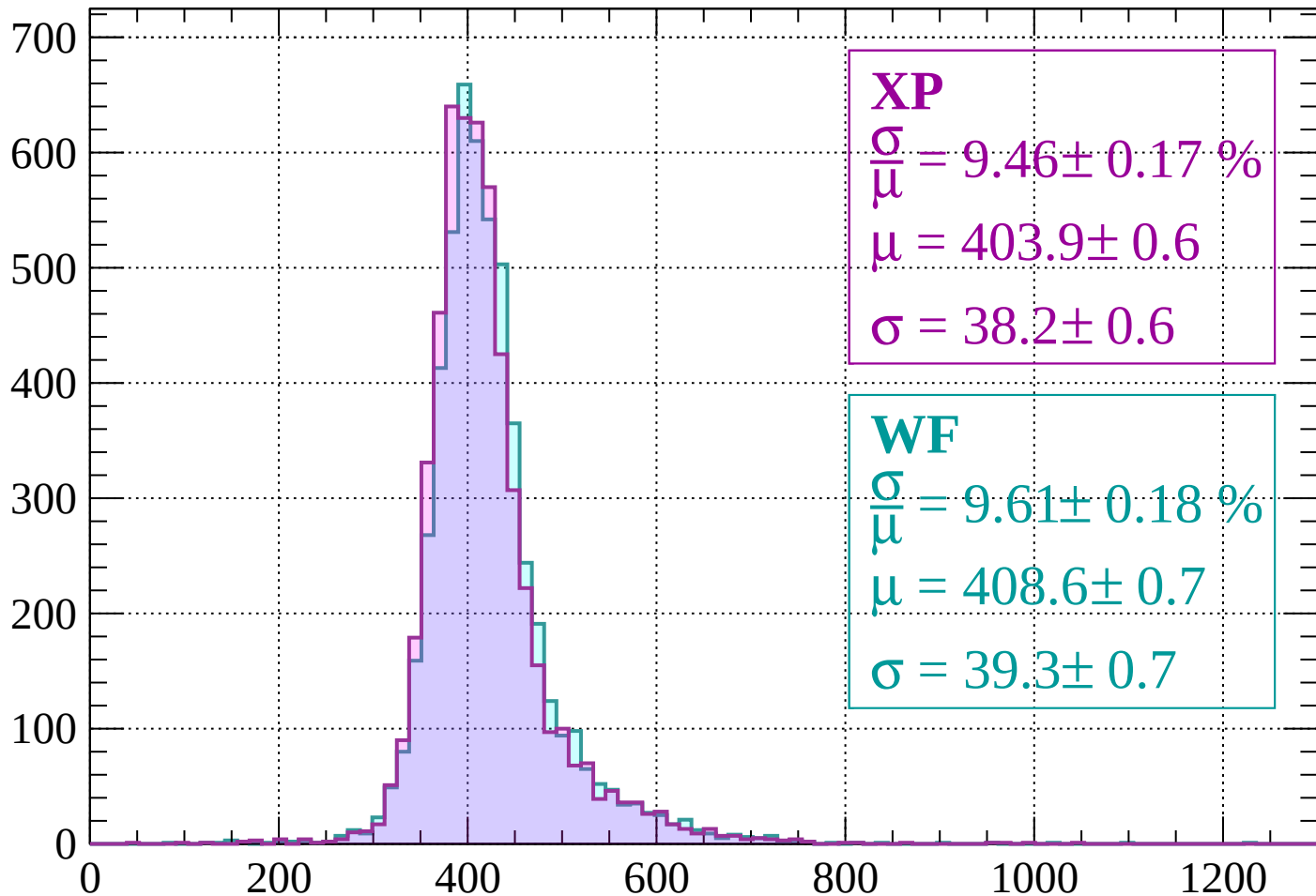
$$\mu = 413.3 \pm 0.7$$

$$\sigma = 41.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.46 \pm 0.17 \%$$

$$\mu = 403.9 \pm 0.6$$

$$\sigma = 38.2 \pm 0.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.61 \pm 0.18 \%$$

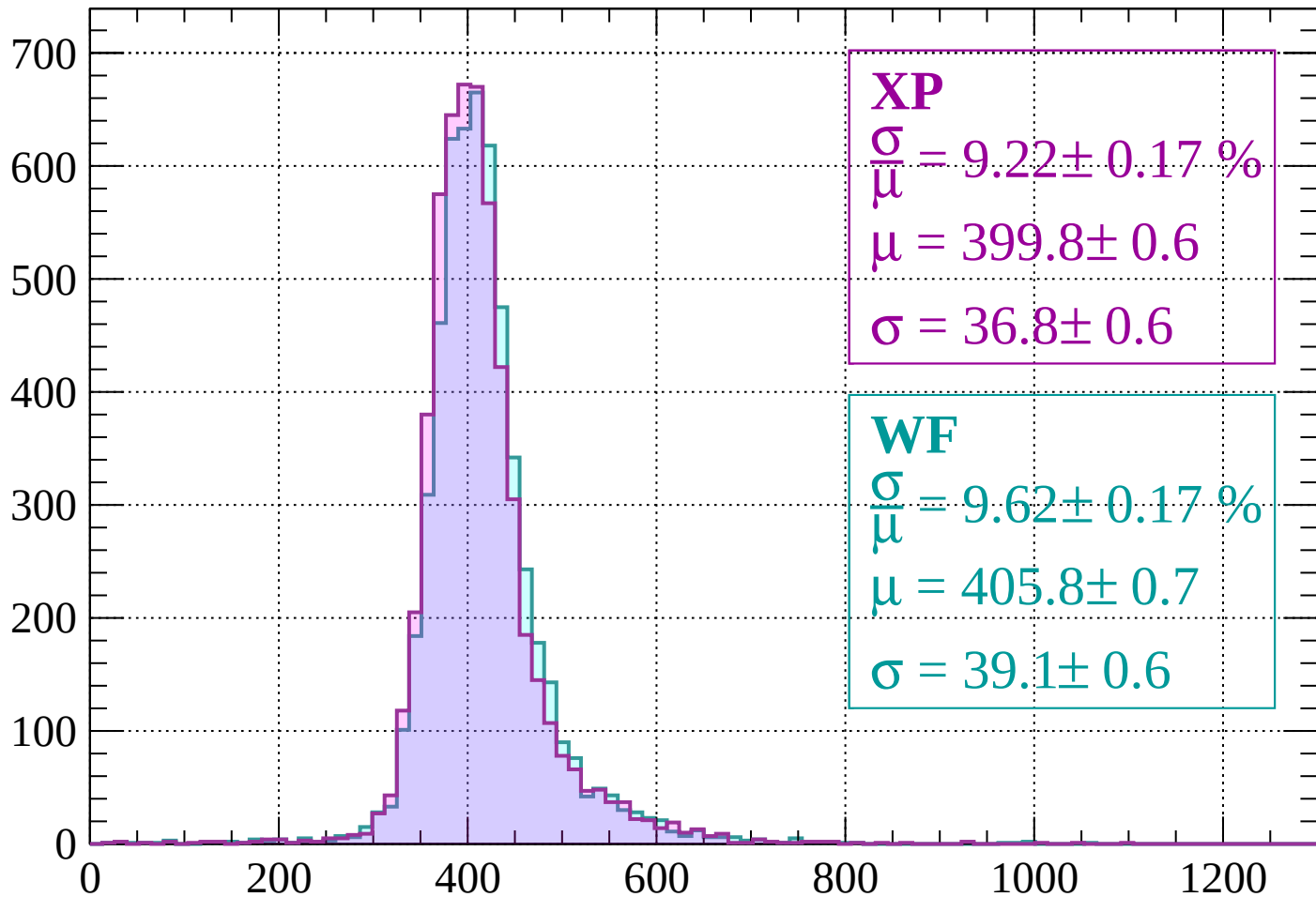
$$\mu = 408.6 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 9.22 \pm 0.17 \%$$

$$\mu = 399.8 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

WF

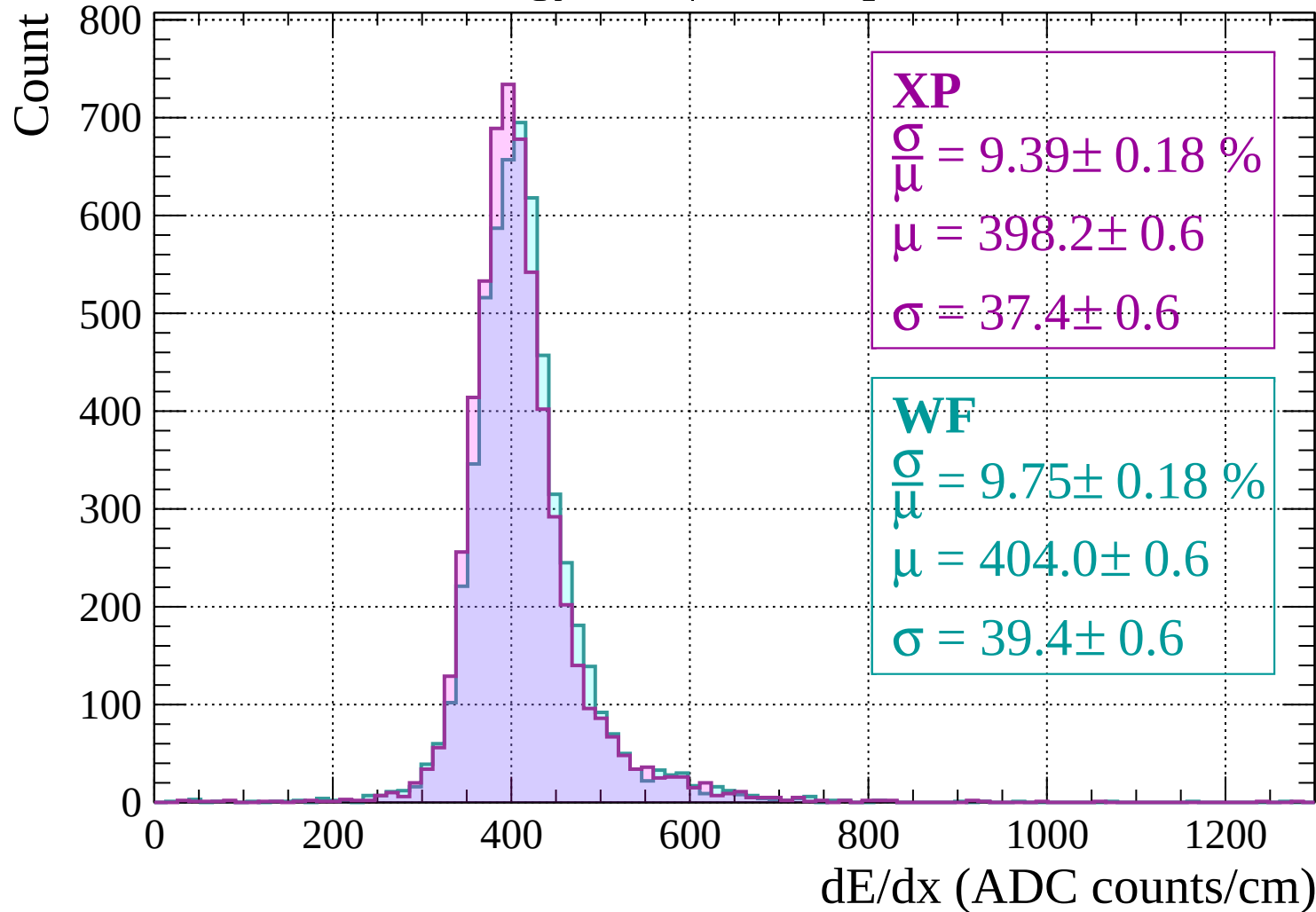
$$\frac{\sigma}{\mu} = 9.62 \pm 0.17 \%$$

$$\mu = 405.8 \pm 0.7$$

$$\sigma = 39.1 \pm 0.6$$

dE/dx (ADC counts/cm)

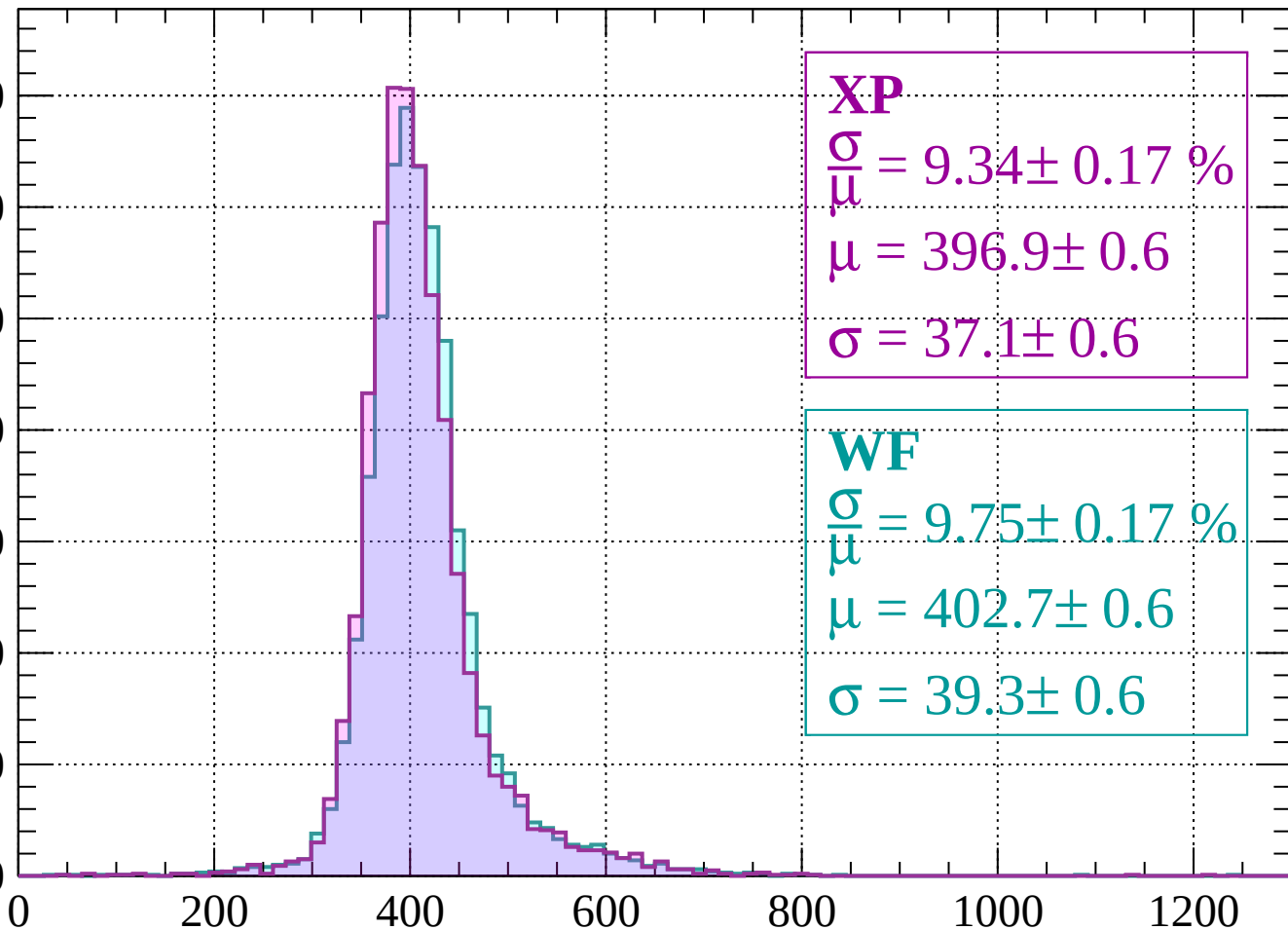
Energy loss | $-440 < p < -400$



Energy loss | $-400 < p < -360$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 9.34 \pm 0.17 \%$$

$$\mu = 396.9 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.17 \%$$

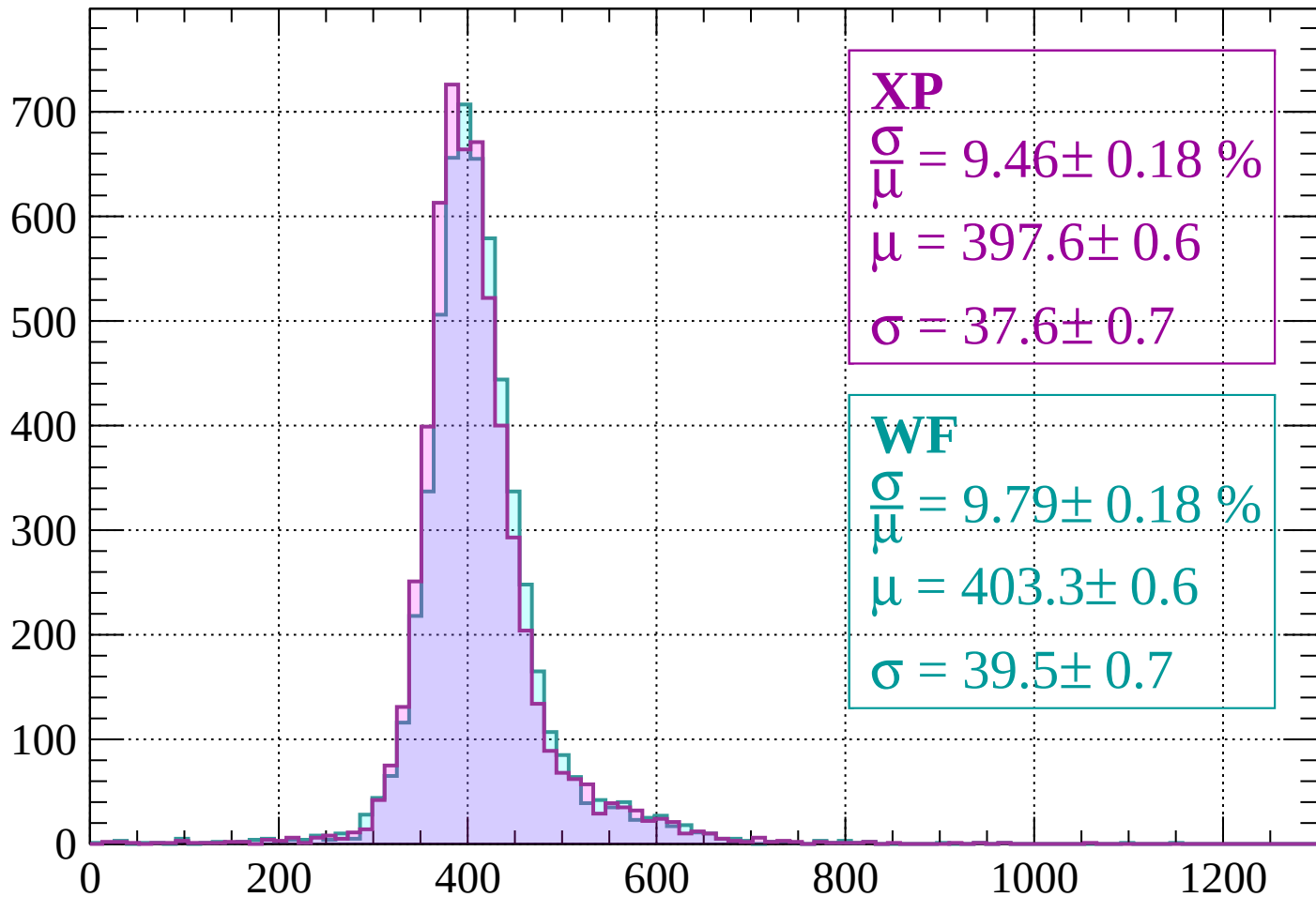
$$\mu = 402.7 \pm 0.6$$

$$\sigma = 39.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.18 \%$$

$$\mu = 397.6 \pm 0.6$$

$$\sigma = 37.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.18 \%$$

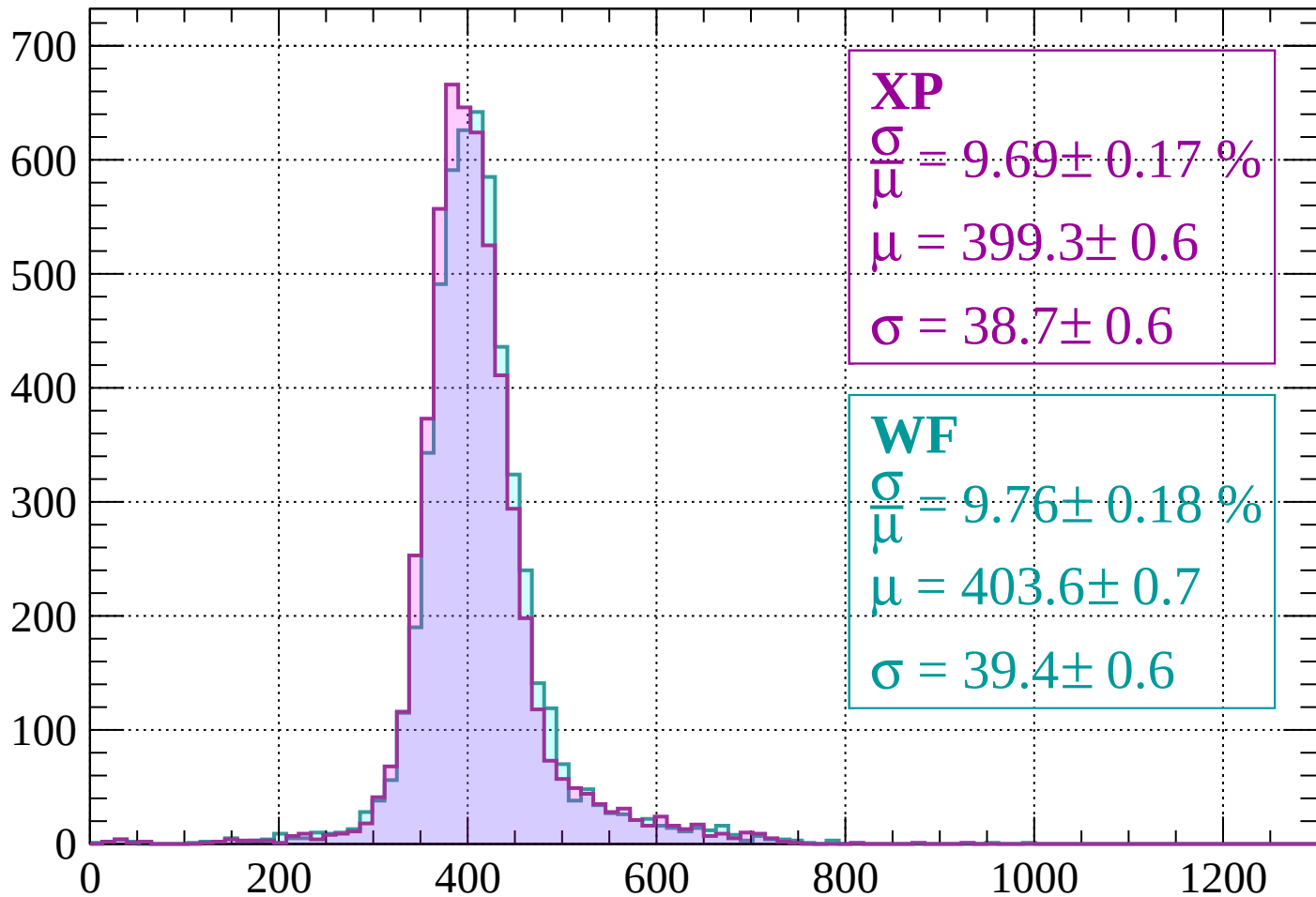
$$\mu = 403.3 \pm 0.6$$

$$\sigma = 39.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 9.69 \pm 0.17 \%$$

$$\mu = 399.3 \pm 0.6$$

$$\sigma = 38.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.76 \pm 0.18 \%$$

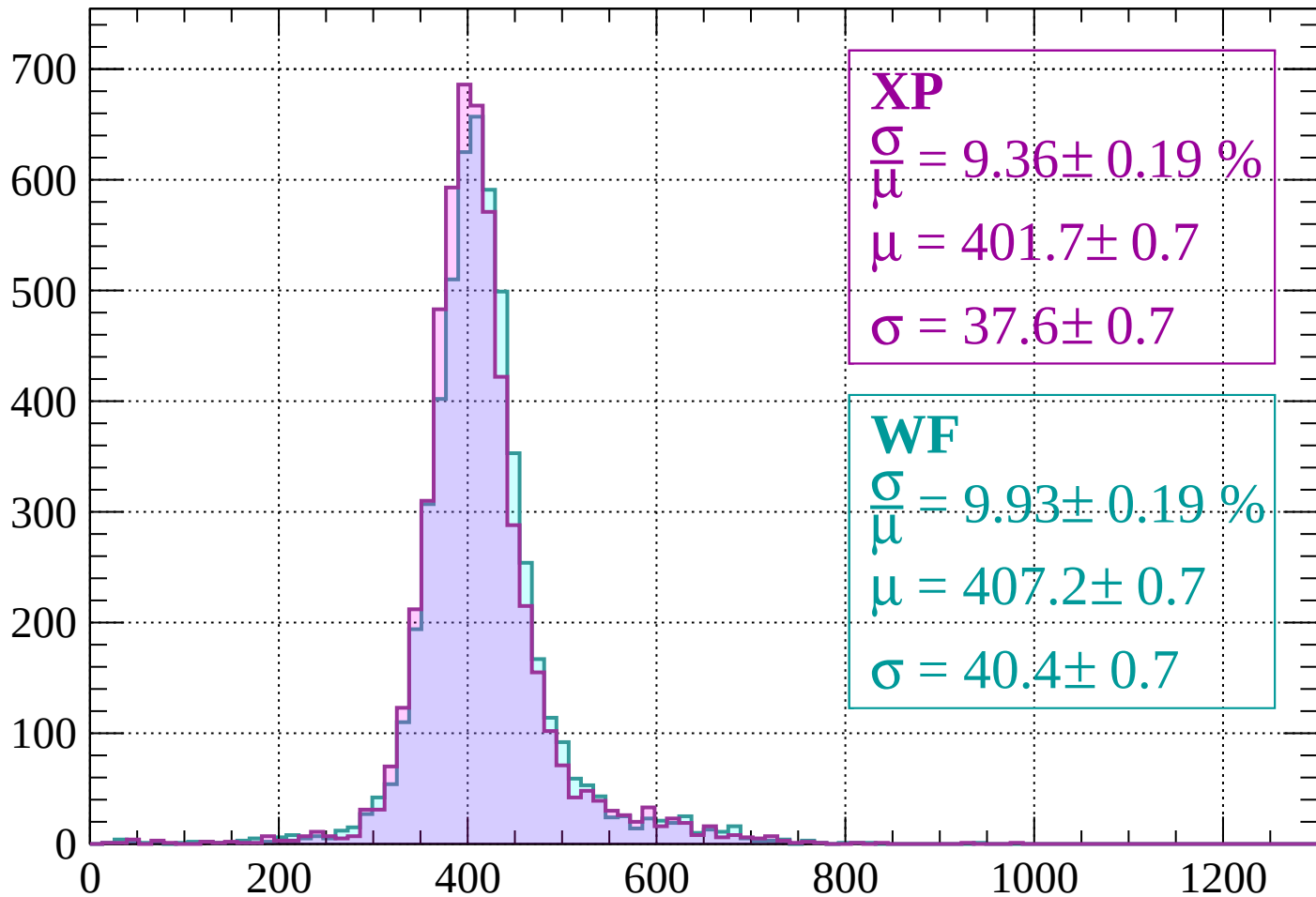
$$\mu = 403.6 \pm 0.7$$

$$\sigma = 39.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 9.36 \pm 0.19 \%$$

$$\mu = 401.7 \pm 0.7$$

$$\sigma = 37.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.93 \pm 0.19 \%$$

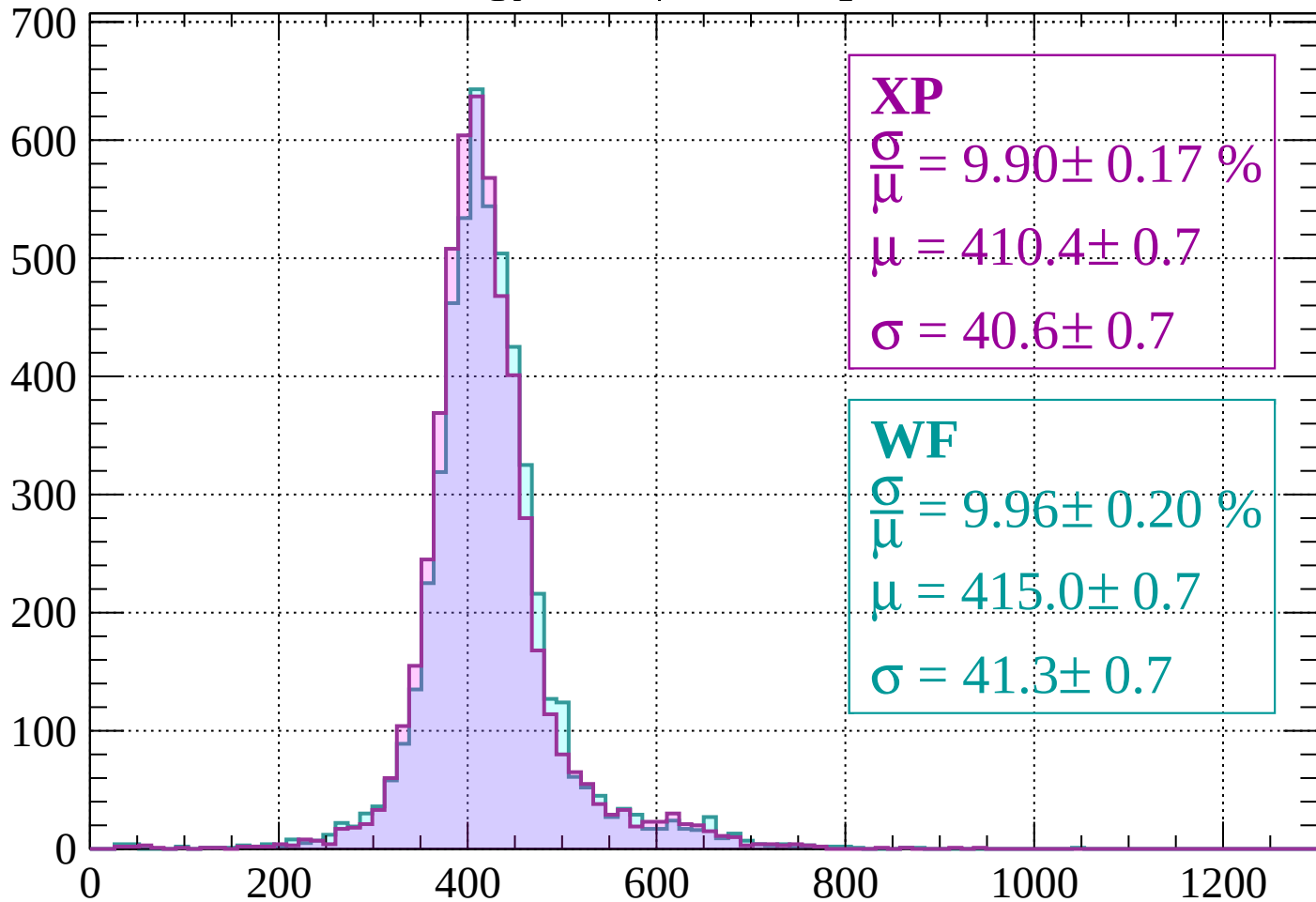
$$\mu = 407.2 \pm 0.7$$

$$\sigma = 40.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 9.90 \pm 0.17 \%$$

$$\mu = 410.4 \pm 0.7$$

$$\sigma = 40.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.96 \pm 0.20 \%$$

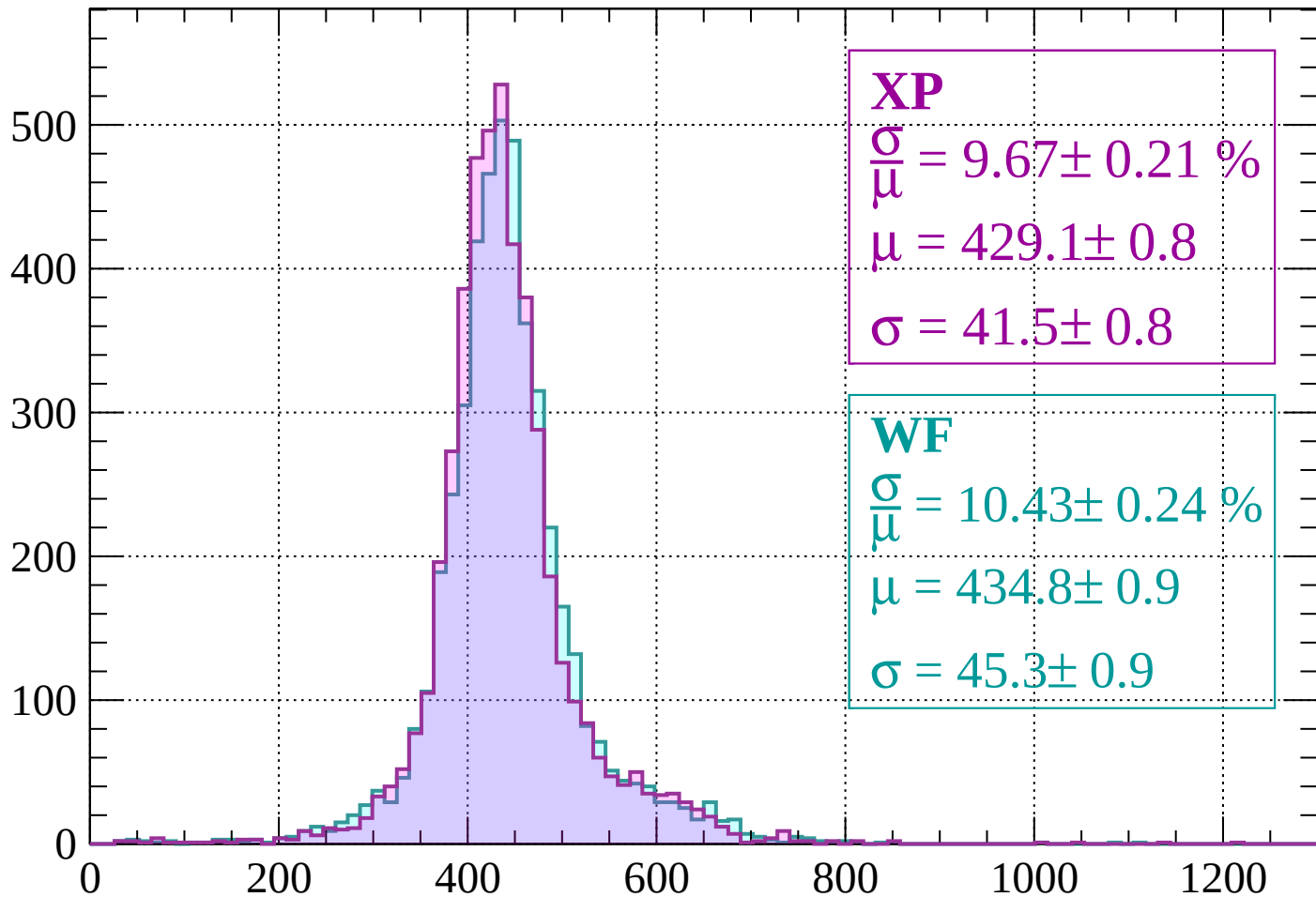
$$\mu = 415.0 \pm 0.7$$

$$\sigma = 41.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 9.67 \pm 0.21 \%$$

$$\mu = 429.1 \pm 0.8$$

$$\sigma = 41.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.43 \pm 0.24 \%$$

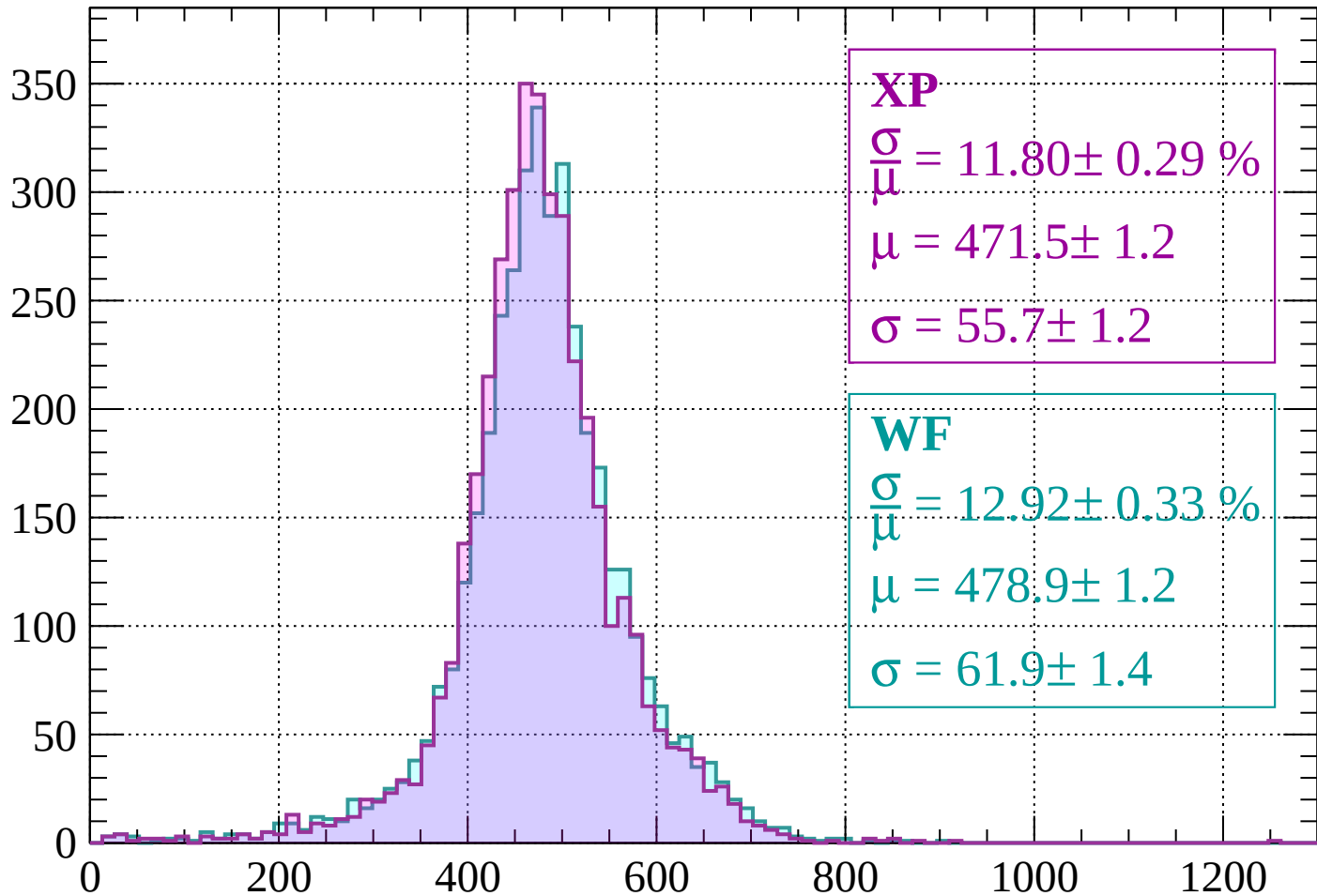
$$\mu = 434.8 \pm 0.9$$

$$\sigma = 45.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -160 < p < -120

Count



XP

$$\frac{\sigma}{\mu} = 11.80 \pm 0.29 \%$$

$$\mu = 471.5 \pm 1.2$$

$$\sigma = 55.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 12.92 \pm 0.33 \%$$

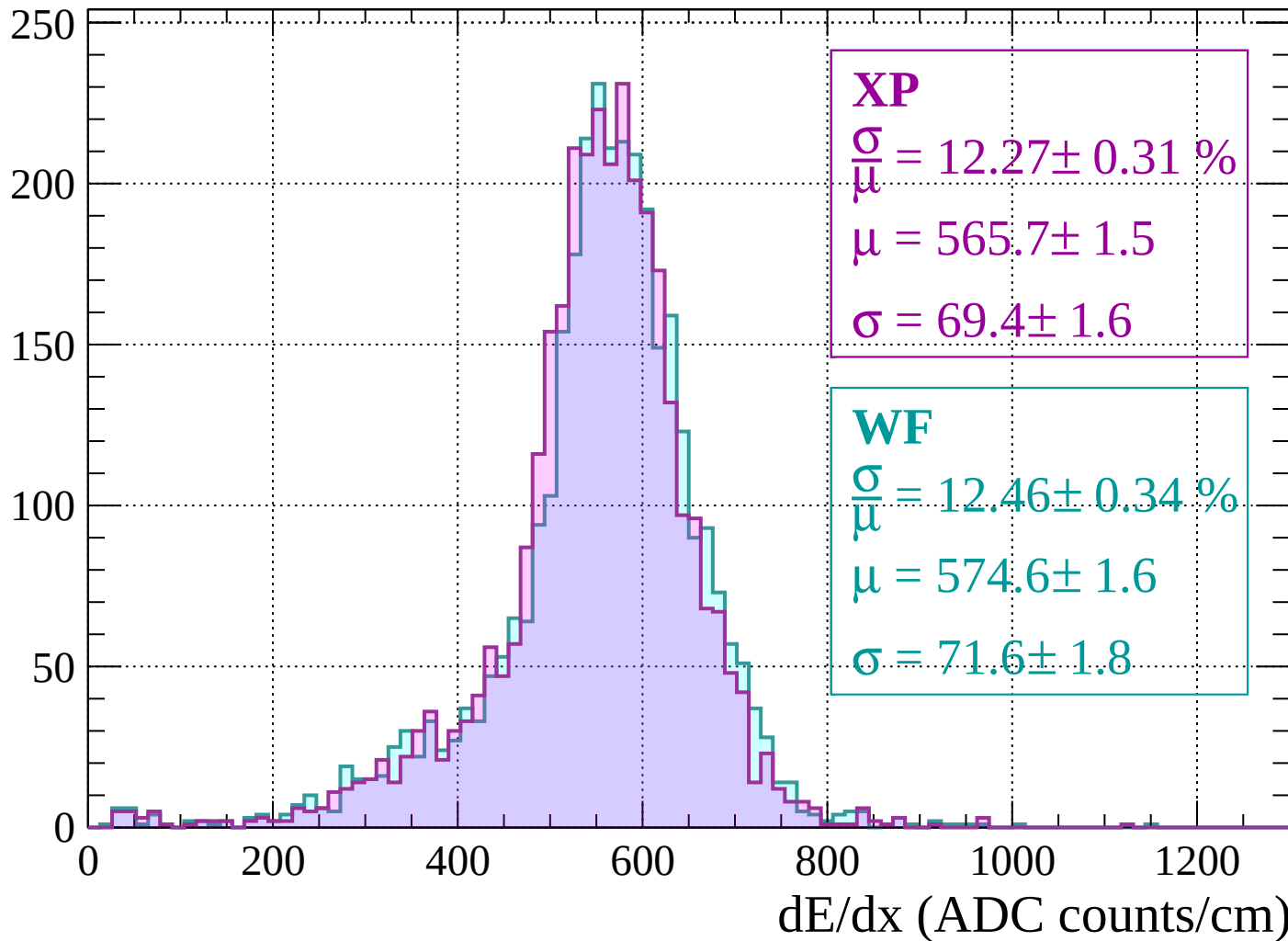
$$\mu = 478.9 \pm 1.2$$

$$\sigma = 61.9 \pm 1.4$$

dE/dx (ADC counts/cm)

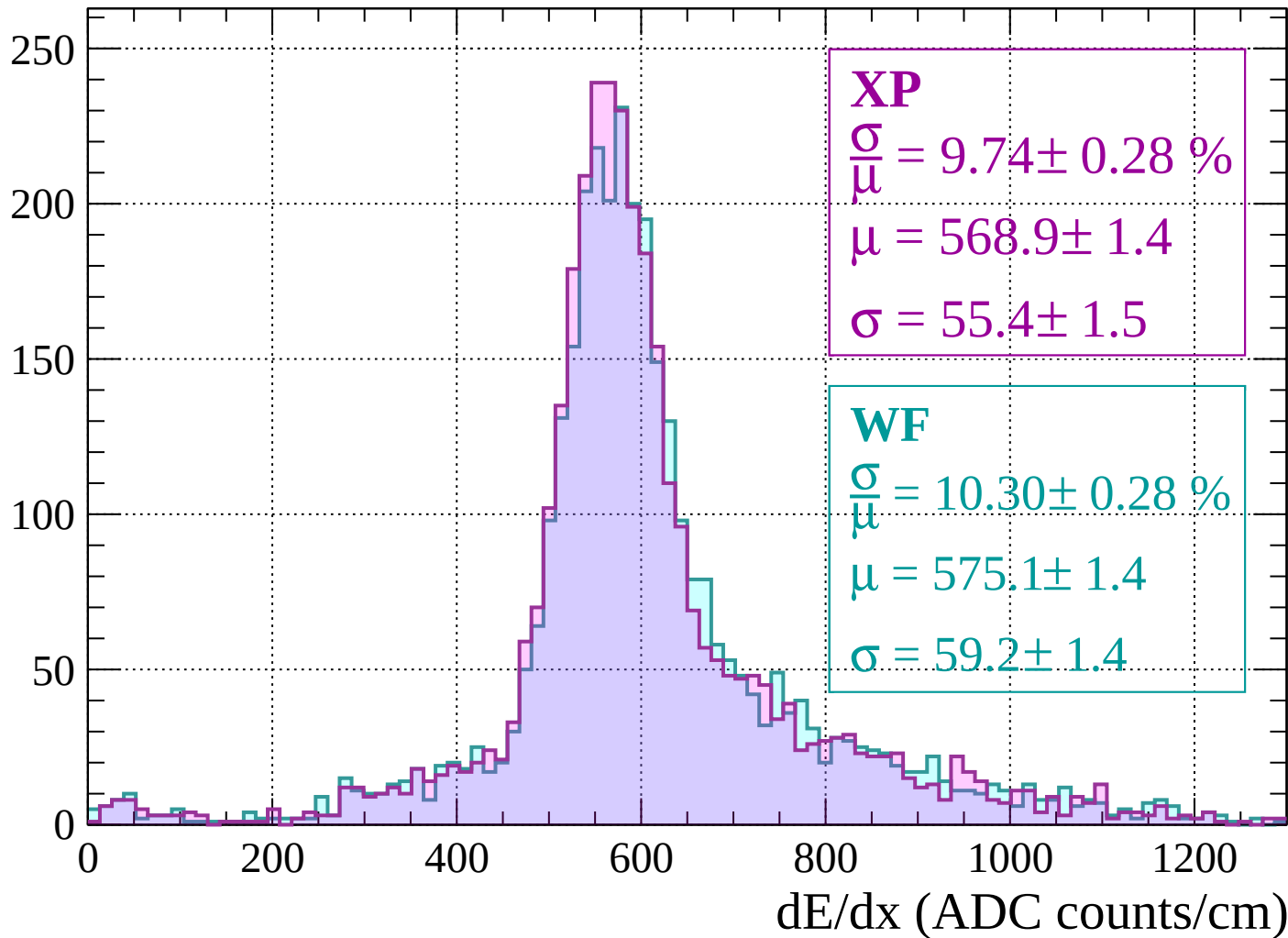
Energy loss | $-120 < p < -80$

Count

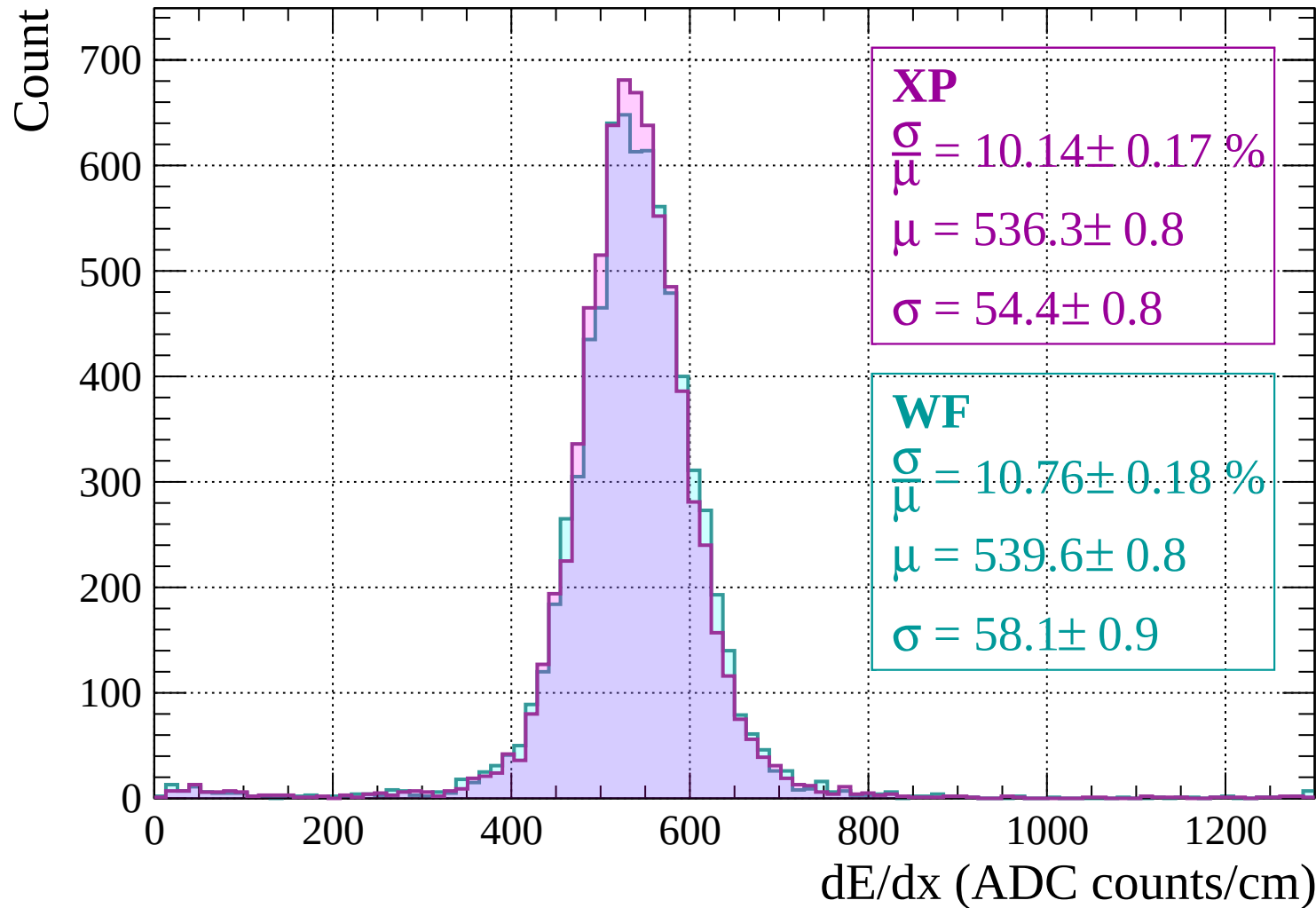


Energy loss | $-80 < p < -40$

Count

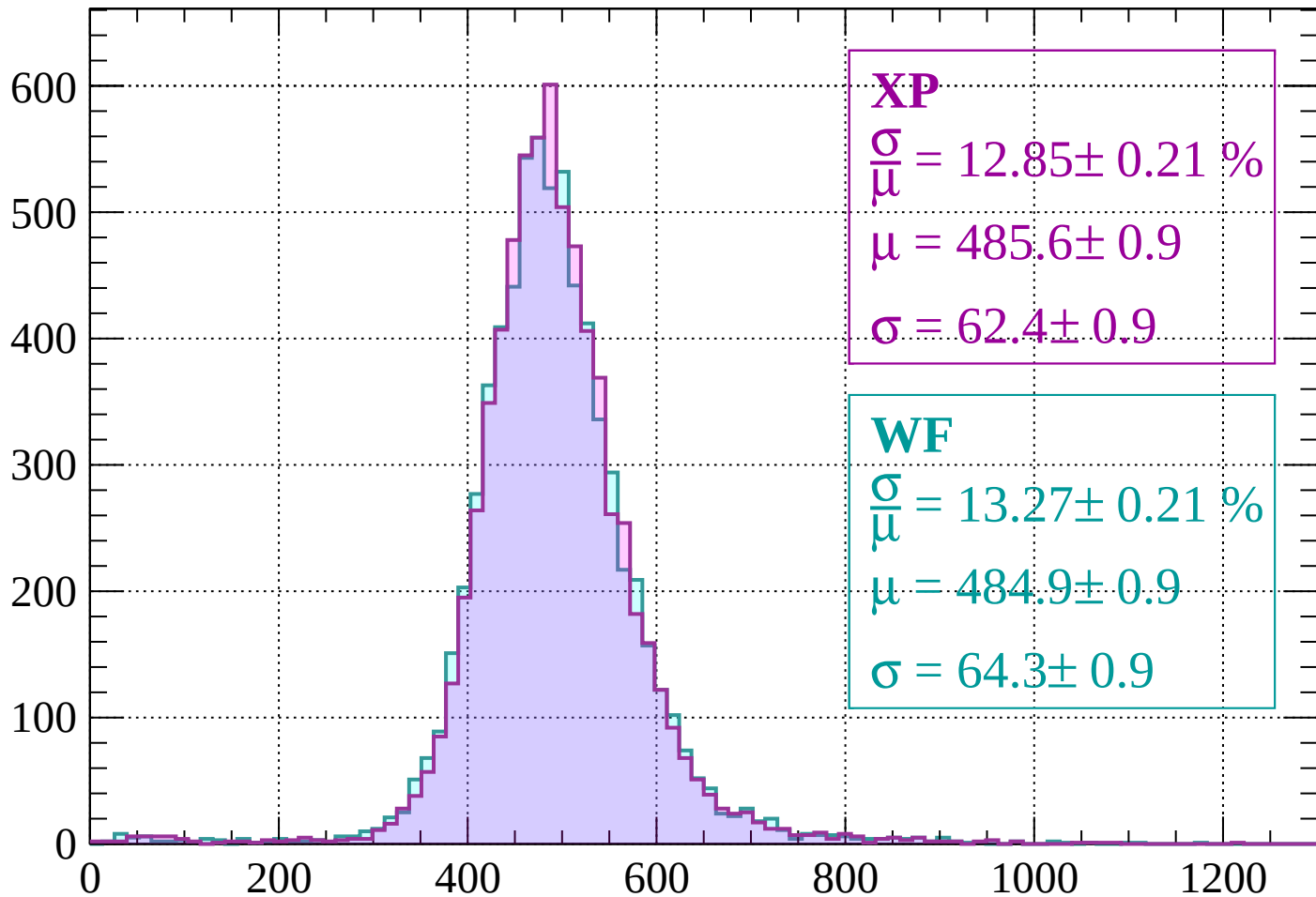


Energy loss | $-40 < p < 0$



Energy loss | $0 < p < 40$

Count



XP

$$\frac{\sigma}{\mu} = 12.85 \pm 0.21 \%$$

$$\mu = 485.6 \pm 0.9$$

$$\sigma = 62.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 13.27 \pm 0.21 \%$$

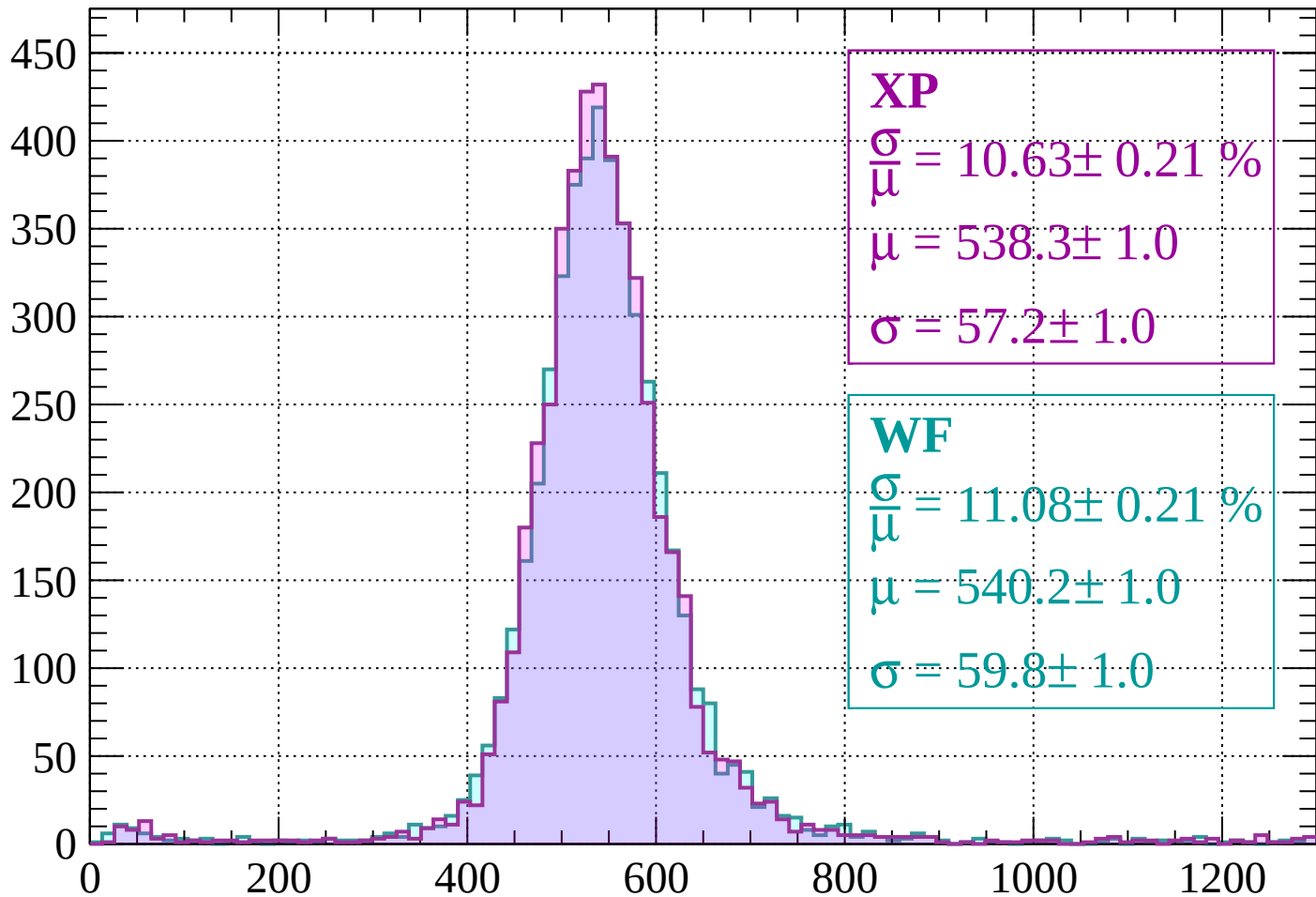
$$\mu = 484.9 \pm 0.9$$

$$\sigma = 64.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $40 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 10.63 \pm 0.21 \%$$

$$\mu = 538.3 \pm 1.0$$

$$\sigma = 57.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.08 \pm 0.21 \%$$

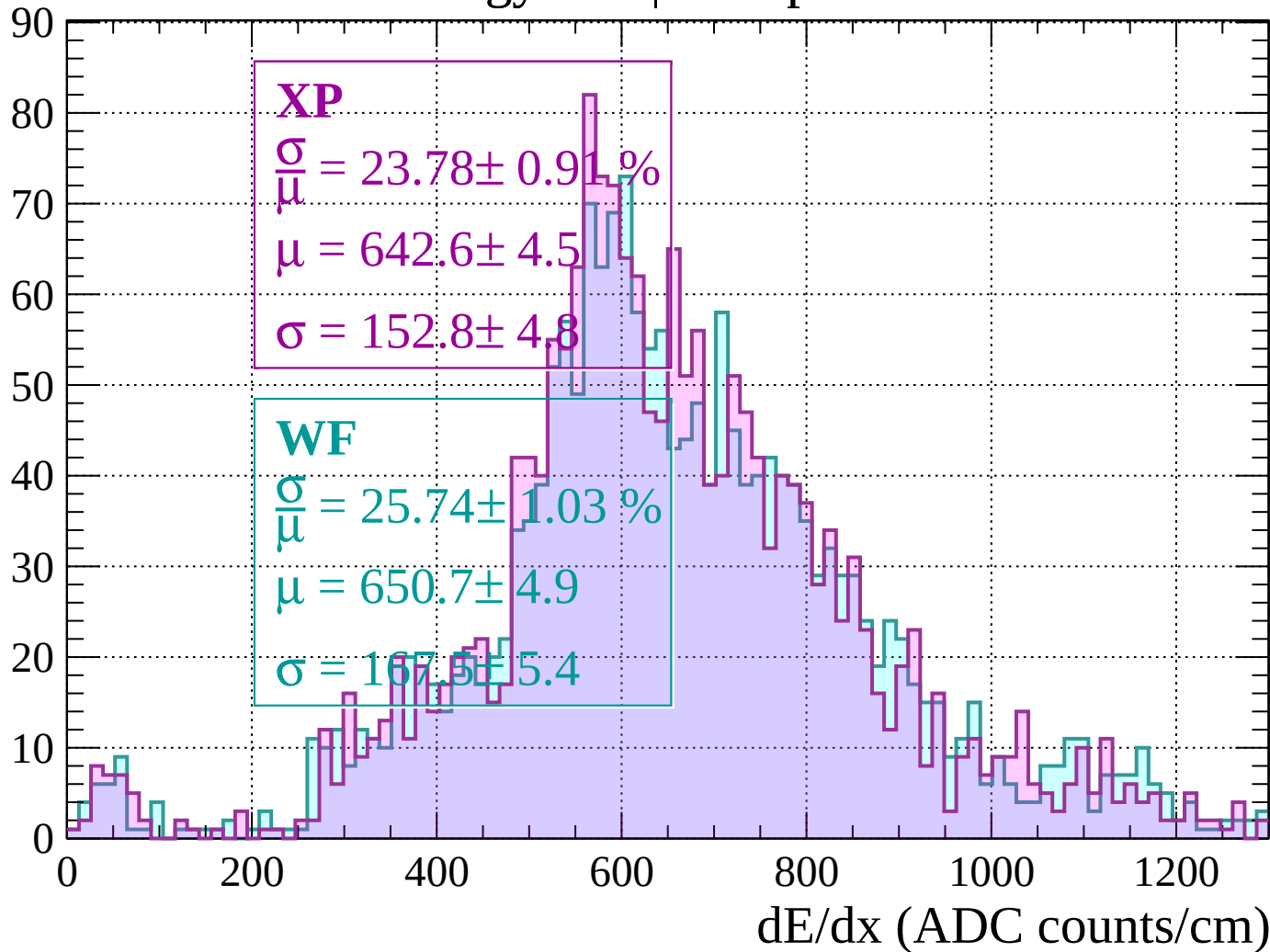
$$\mu = 540.2 \pm 1.0$$

$$\sigma = 59.8 \pm 1.0$$

dE/dx (ADC counts/cm)

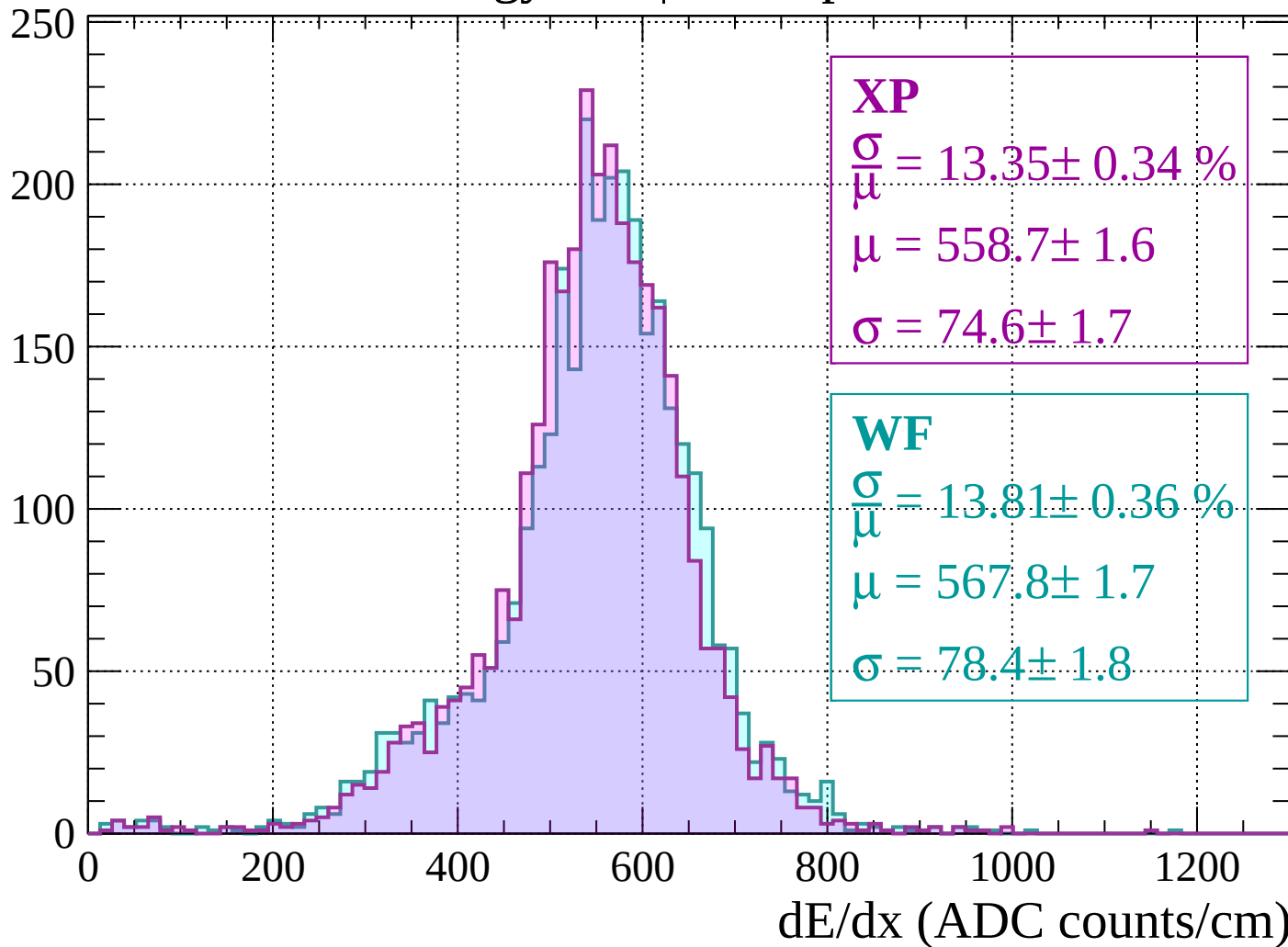
Energy loss | $80 < p < 120$

Count



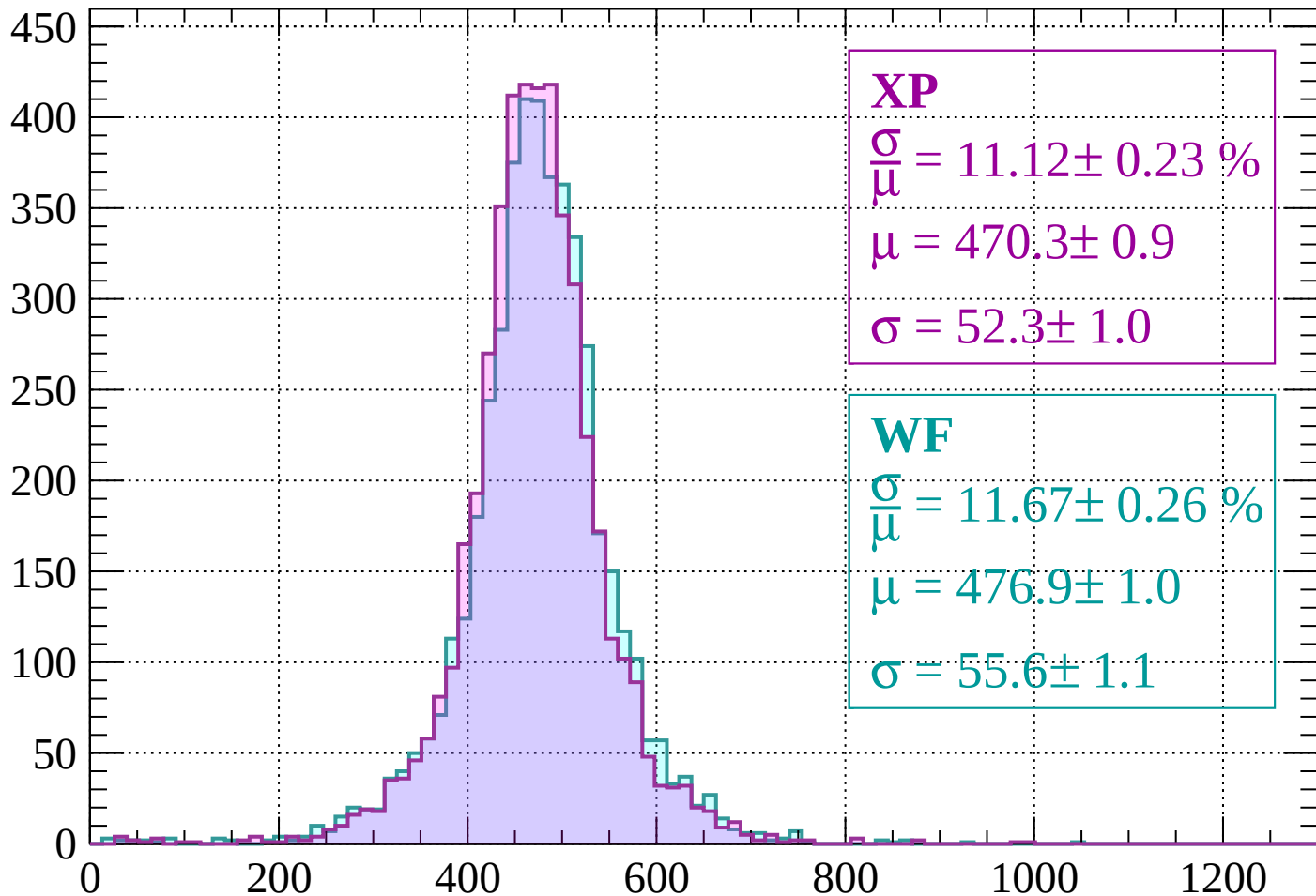
Energy loss | $120 < p < 160$

Count



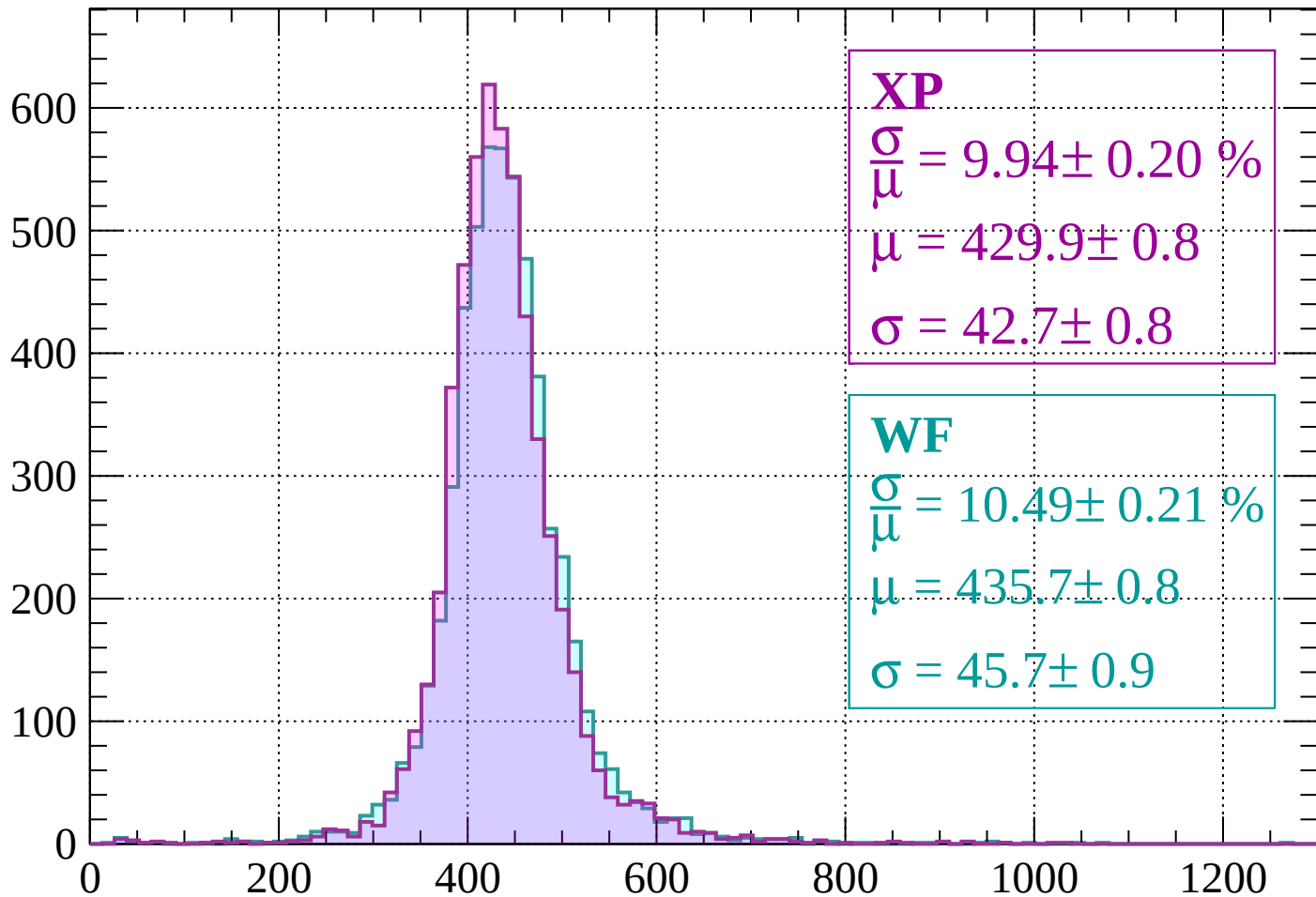
Energy loss | $160 < p < 200$

Count



Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 9.94 \pm 0.20 \%$$

$$\mu = 429.9 \pm 0.8$$

$$\sigma = 42.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.49 \pm 0.21 \%$$

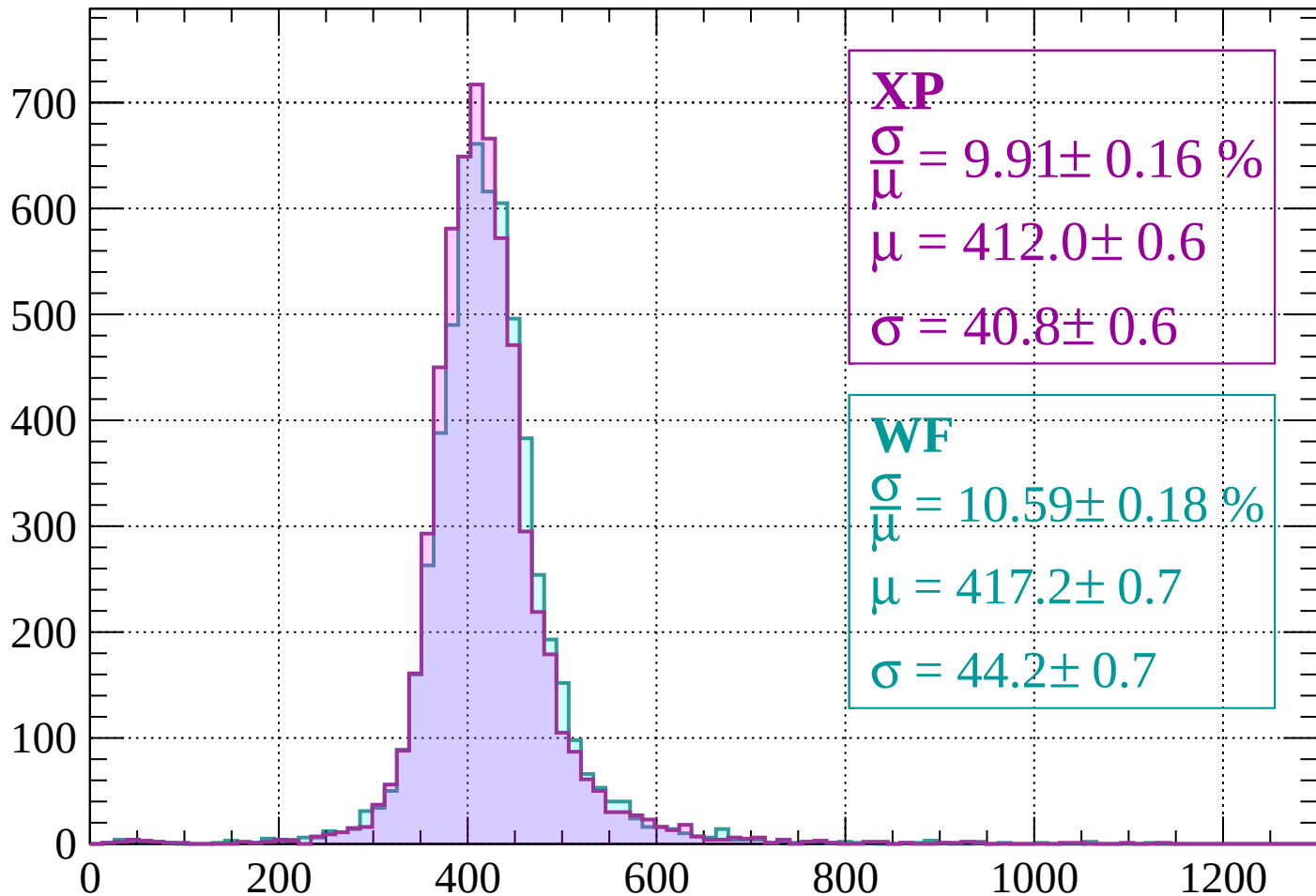
$$\mu = 435.7 \pm 0.8$$

$$\sigma = 45.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 9.91 \pm 0.16 \%$$

$$\mu = 412.0 \pm 0.6$$

$$\sigma = 40.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.59 \pm 0.18 \%$$

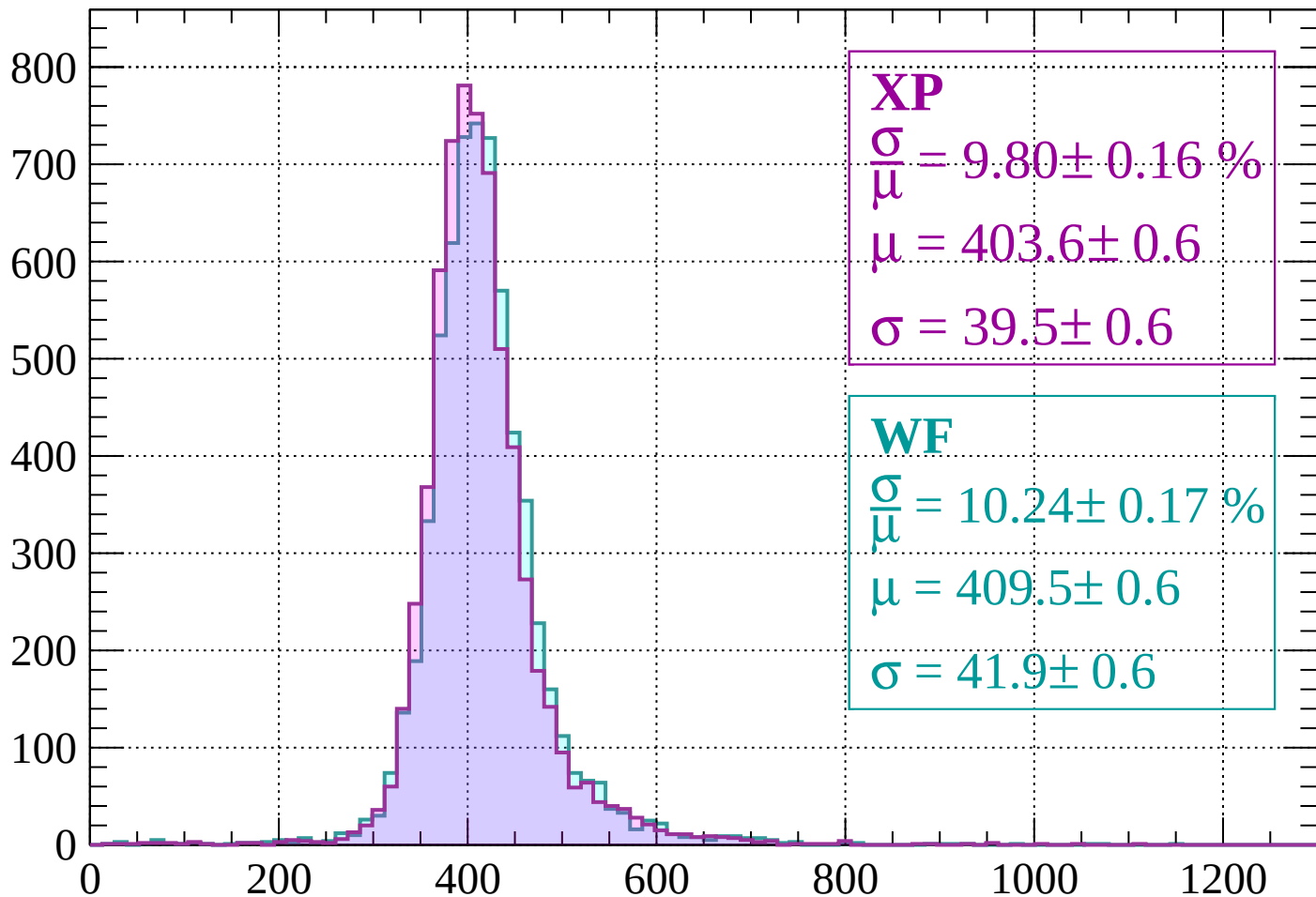
$$\mu = 417.2 \pm 0.7$$

$$\sigma = 44.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 9.80 \pm 0.16 \%$$

$$\mu = 403.6 \pm 0.6$$

$$\sigma = 39.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.24 \pm 0.17 \%$$

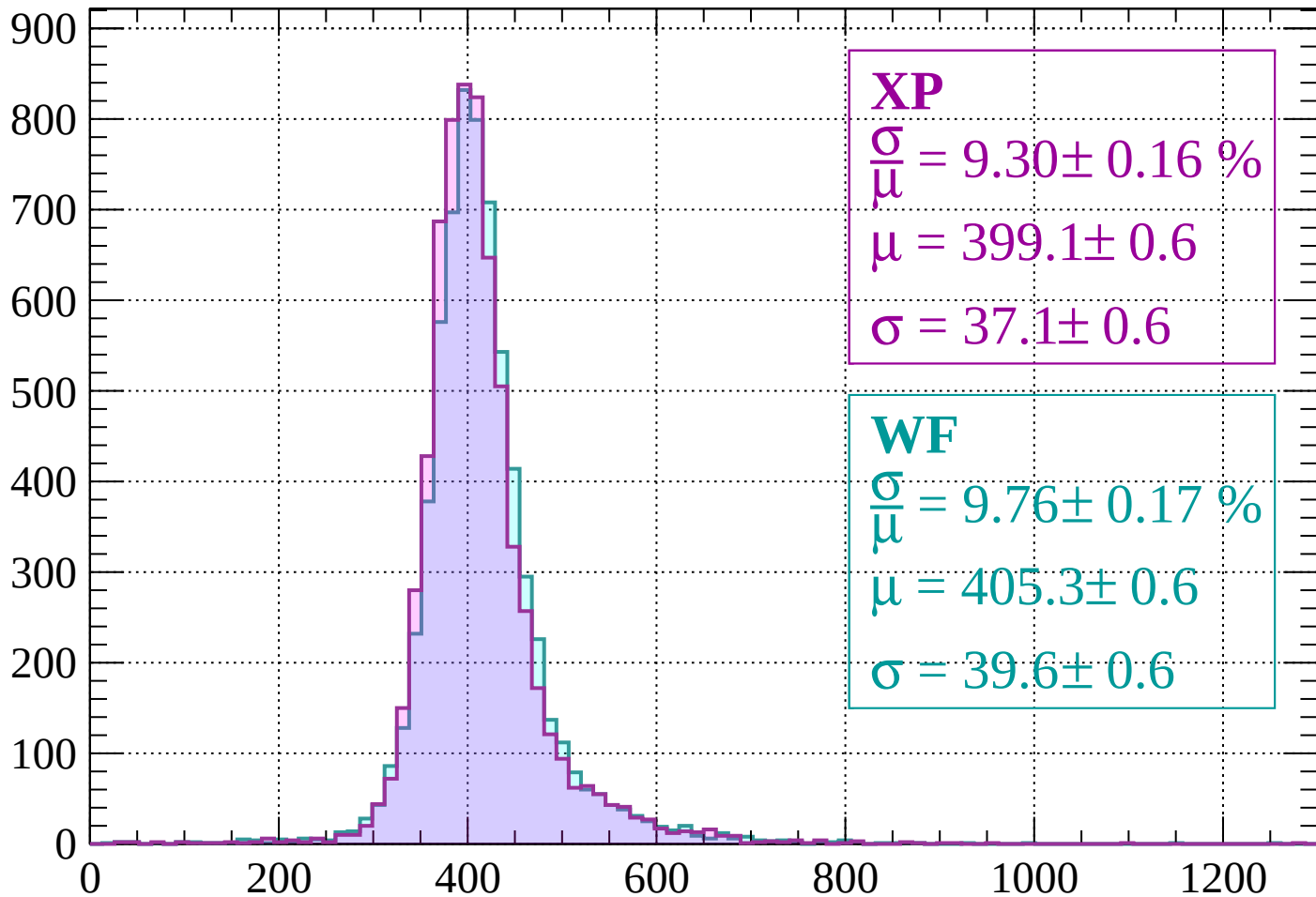
$$\mu = 409.5 \pm 0.6$$

$$\sigma = 41.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 9.30 \pm 0.16 \%$$

$$\mu = 399.1 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.76 \pm 0.17 \%$$

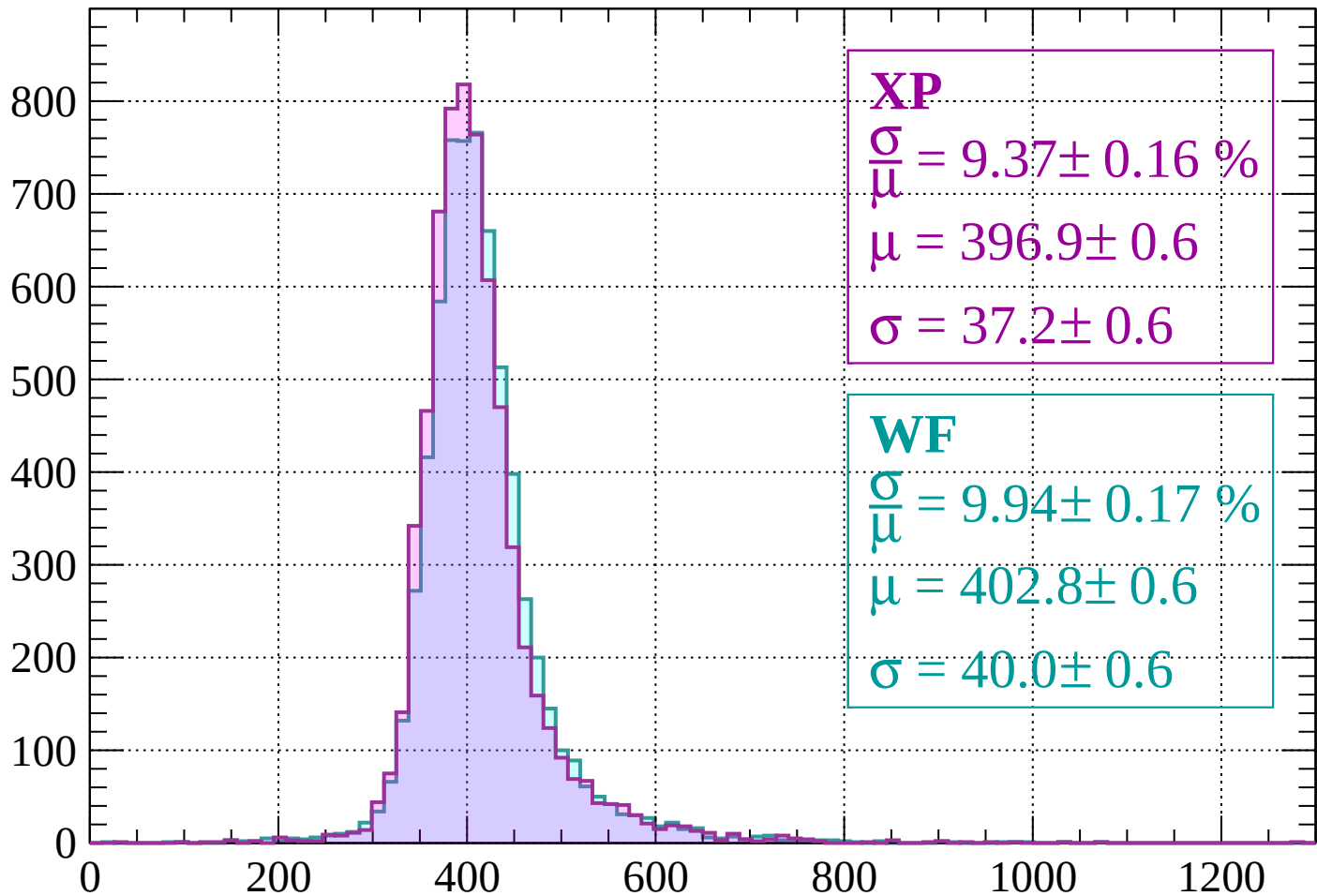
$$\mu = 405.3 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 9.37 \pm 0.16 \%$$

$$\mu = 396.9 \pm 0.6$$

$$\sigma = 37.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.94 \pm 0.17 \%$$

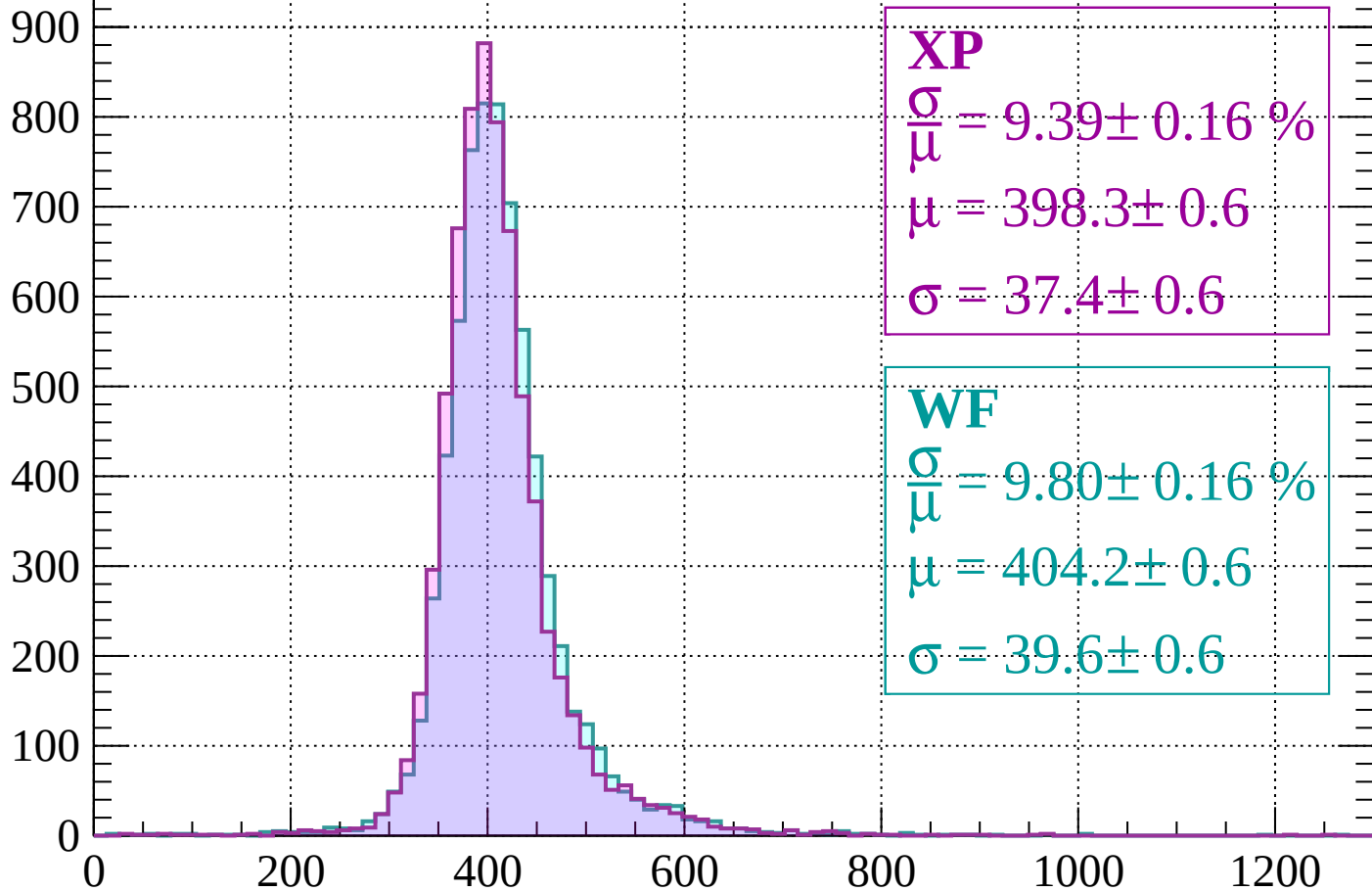
$$\mu = 402.8 \pm 0.6$$

$$\sigma = 40.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.16 \%$$

$$\mu = 398.3 \pm 0.6$$

$$\sigma = 37.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.80 \pm 0.16 \%$$

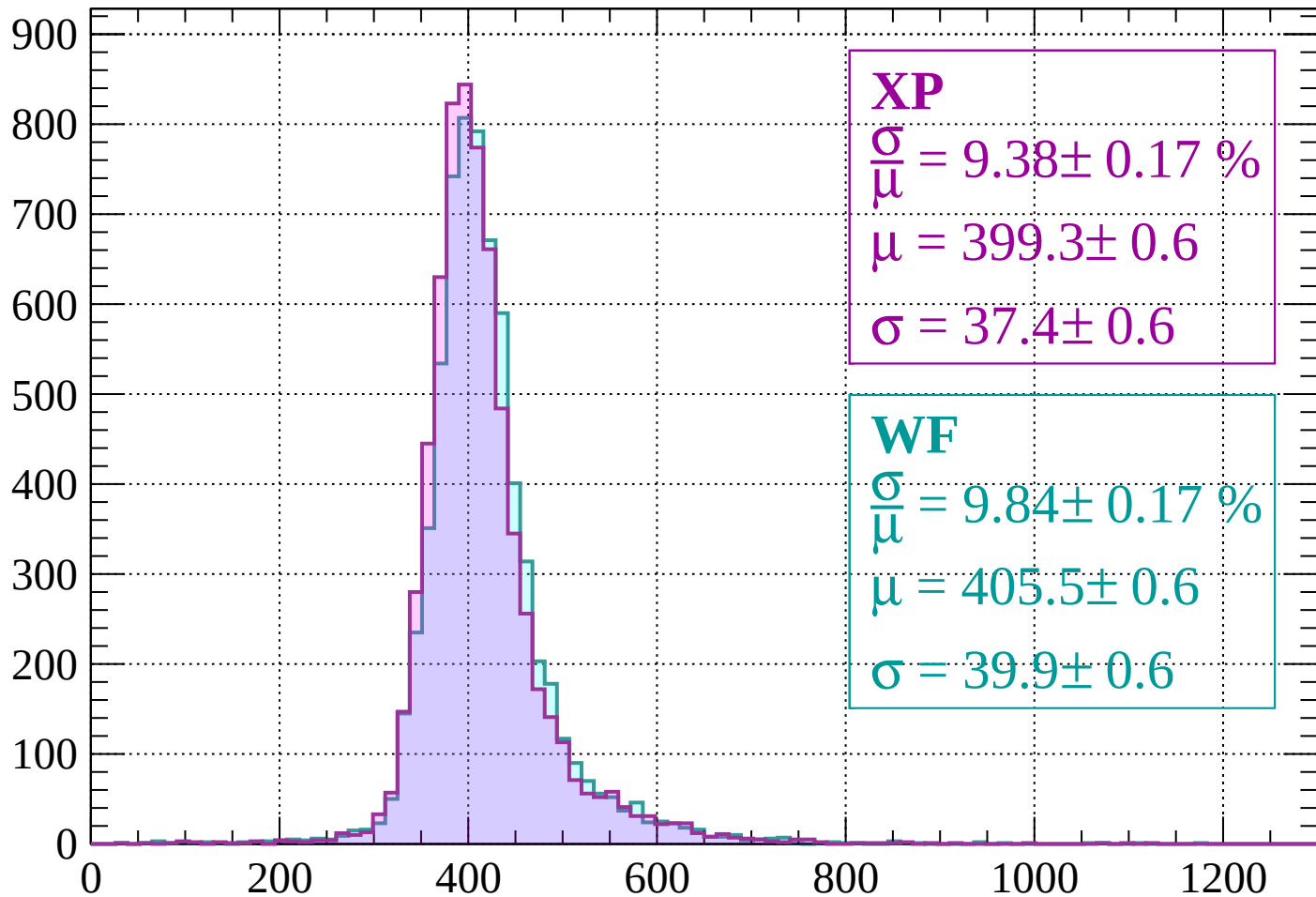
$$\mu = 404.2 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 9.38 \pm 0.17 \%$$

$$\mu = 399.3 \pm 0.6$$

$$\sigma = 37.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.84 \pm 0.17 \%$$

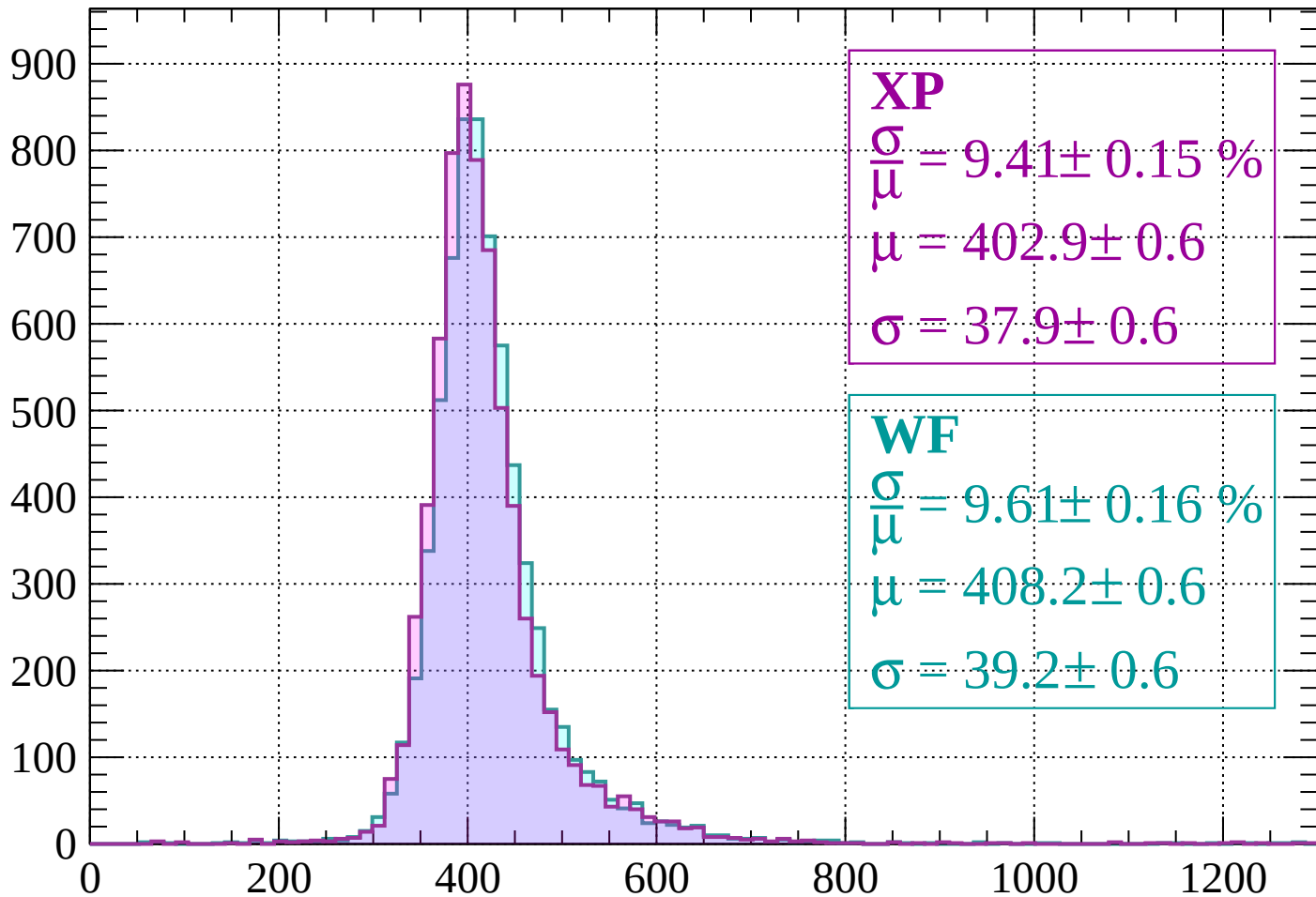
$$\mu = 405.5 \pm 0.6$$

$$\sigma = 39.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 9.41 \pm 0.15 \%$$

$$\mu = 402.9 \pm 0.6$$

$$\sigma = 37.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.61 \pm 0.16 \%$$

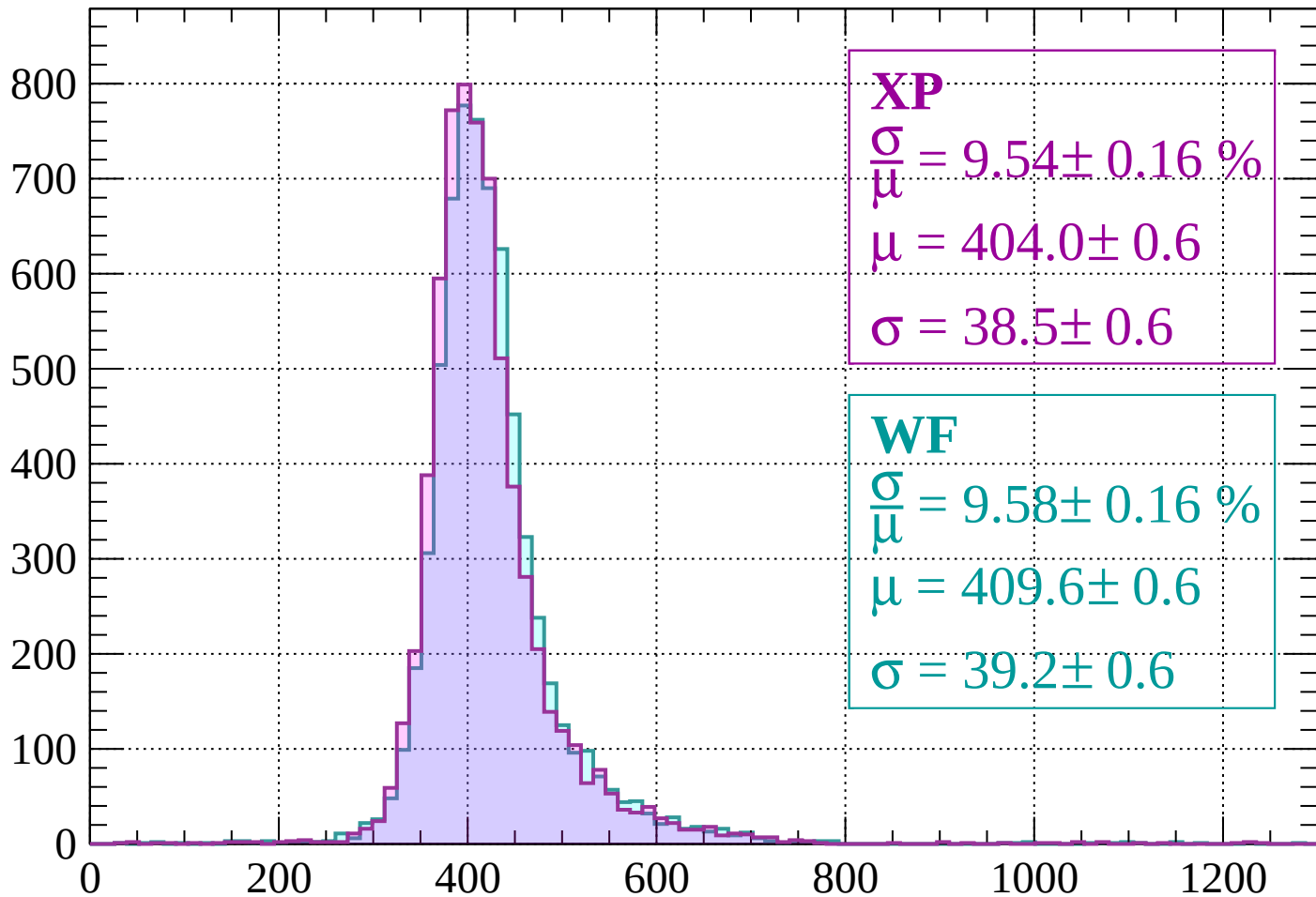
$$\mu = 408.2 \pm 0.6$$

$$\sigma = 39.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 9.54 \pm 0.16 \%$$

$$\mu = 404.0 \pm 0.6$$

$$\sigma = 38.5 \pm 0.6$$

WF

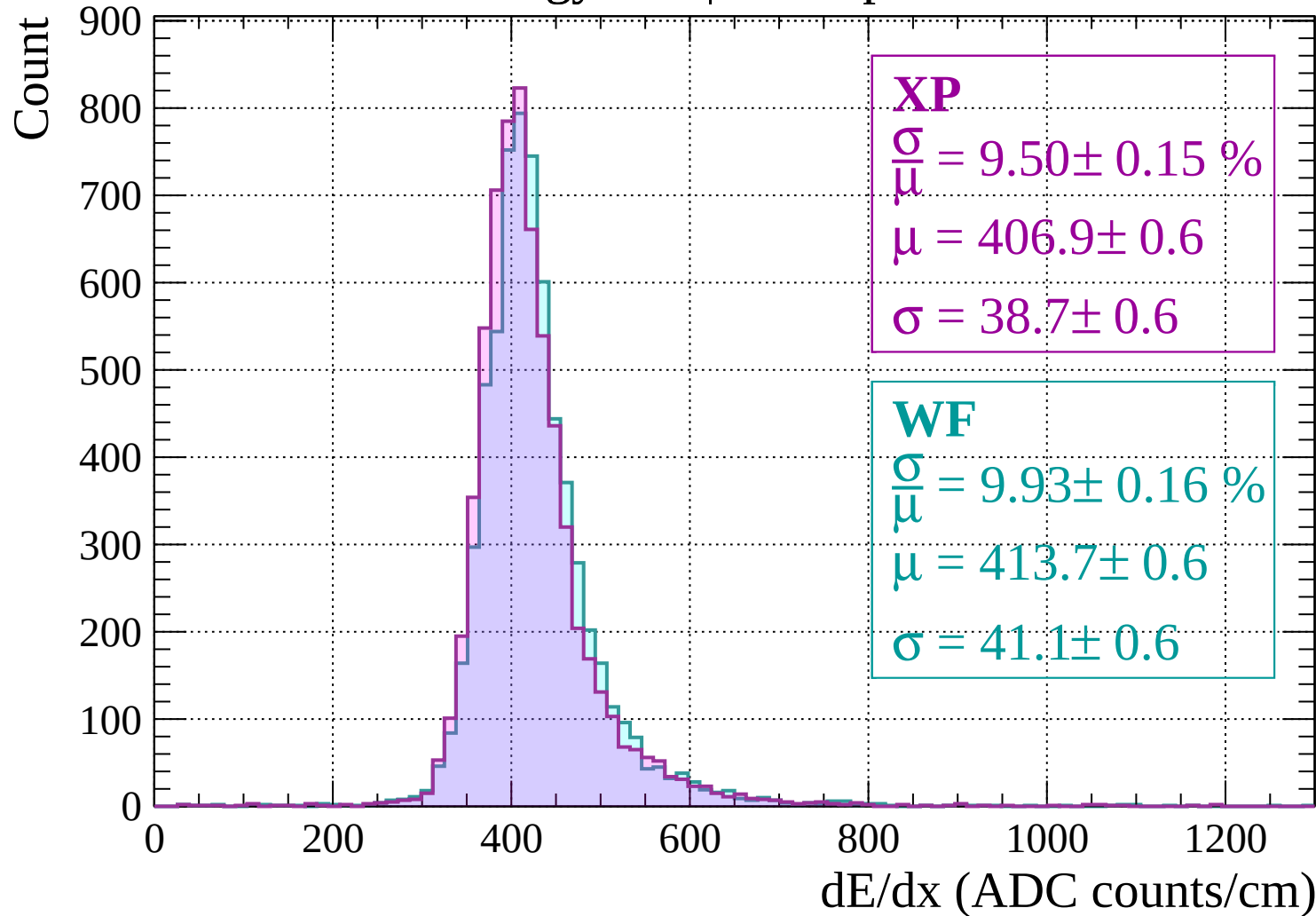
$$\frac{\sigma}{\mu} = 9.58 \pm 0.16 \%$$

$$\mu = 409.6 \pm 0.6$$

$$\sigma = 39.2 \pm 0.6$$

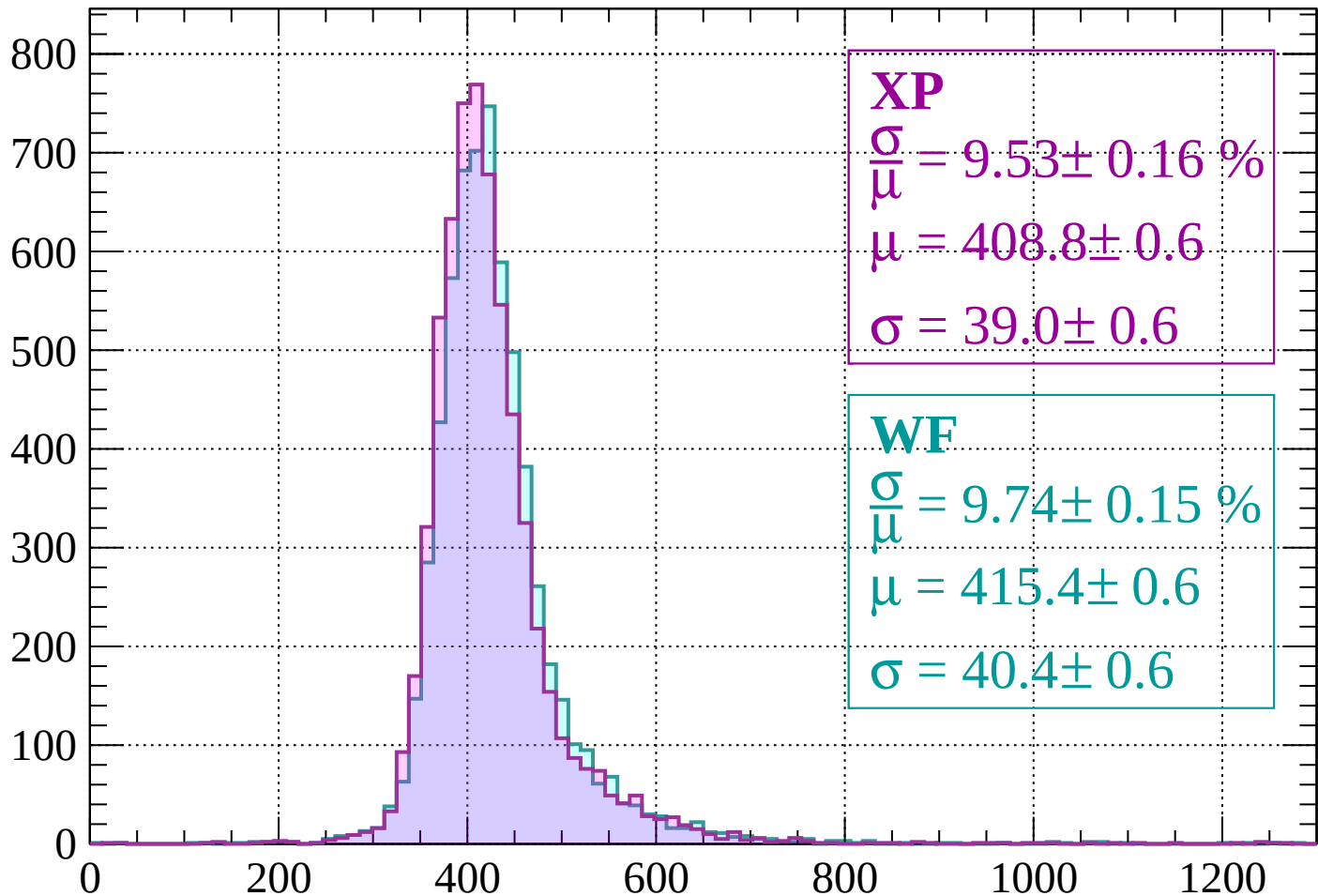
dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$



Energy loss | $600 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 9.53 \pm 0.16 \%$$

$$\mu = 408.8 \pm 0.6$$

$$\sigma = 39.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.74 \pm 0.15 \%$$

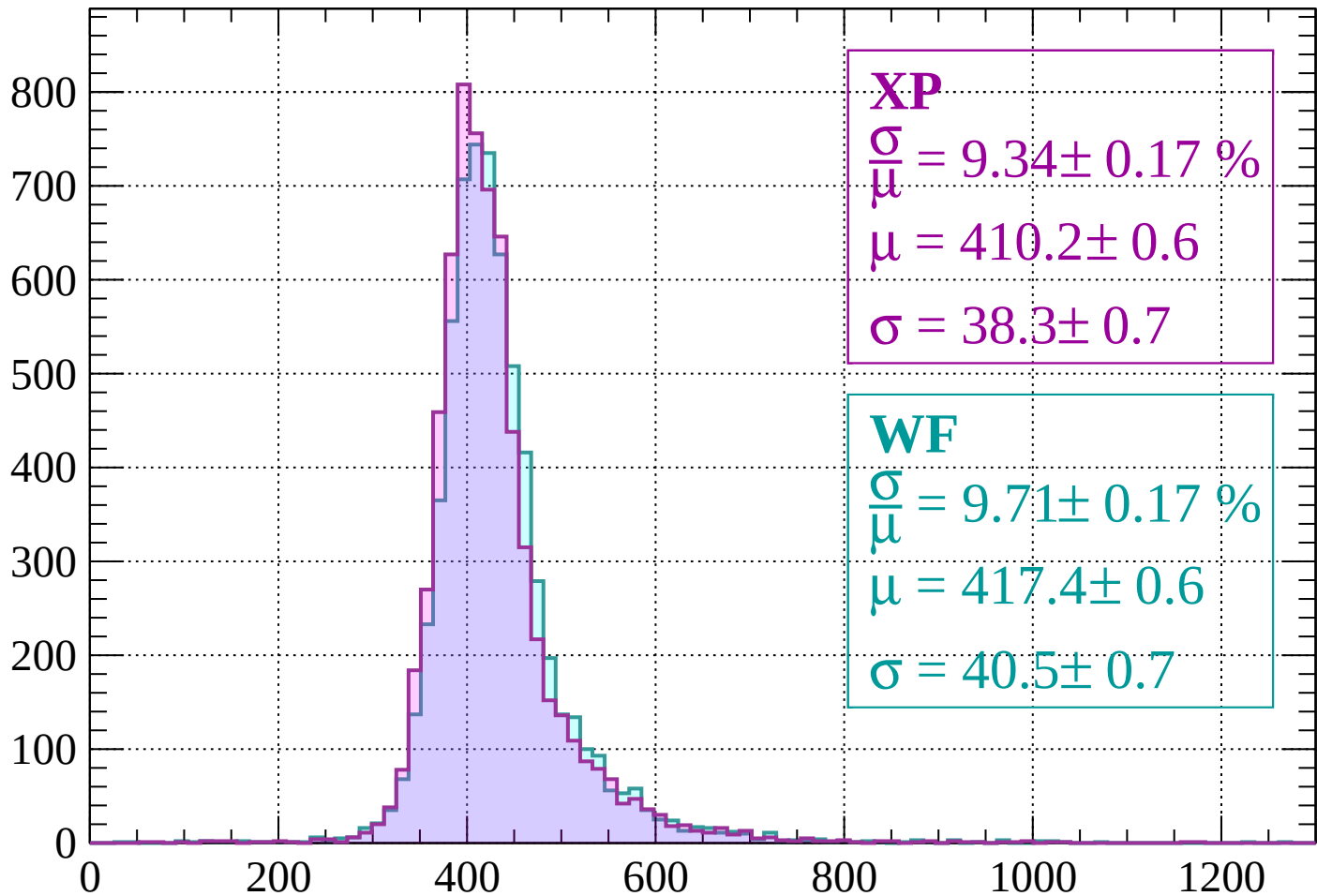
$$\mu = 415.4 \pm 0.6$$

$$\sigma = 40.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 9.34 \pm 0.17 \%$$

$$\mu = 410.2 \pm 0.6$$

$$\sigma = 38.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.71 \pm 0.17 \%$$

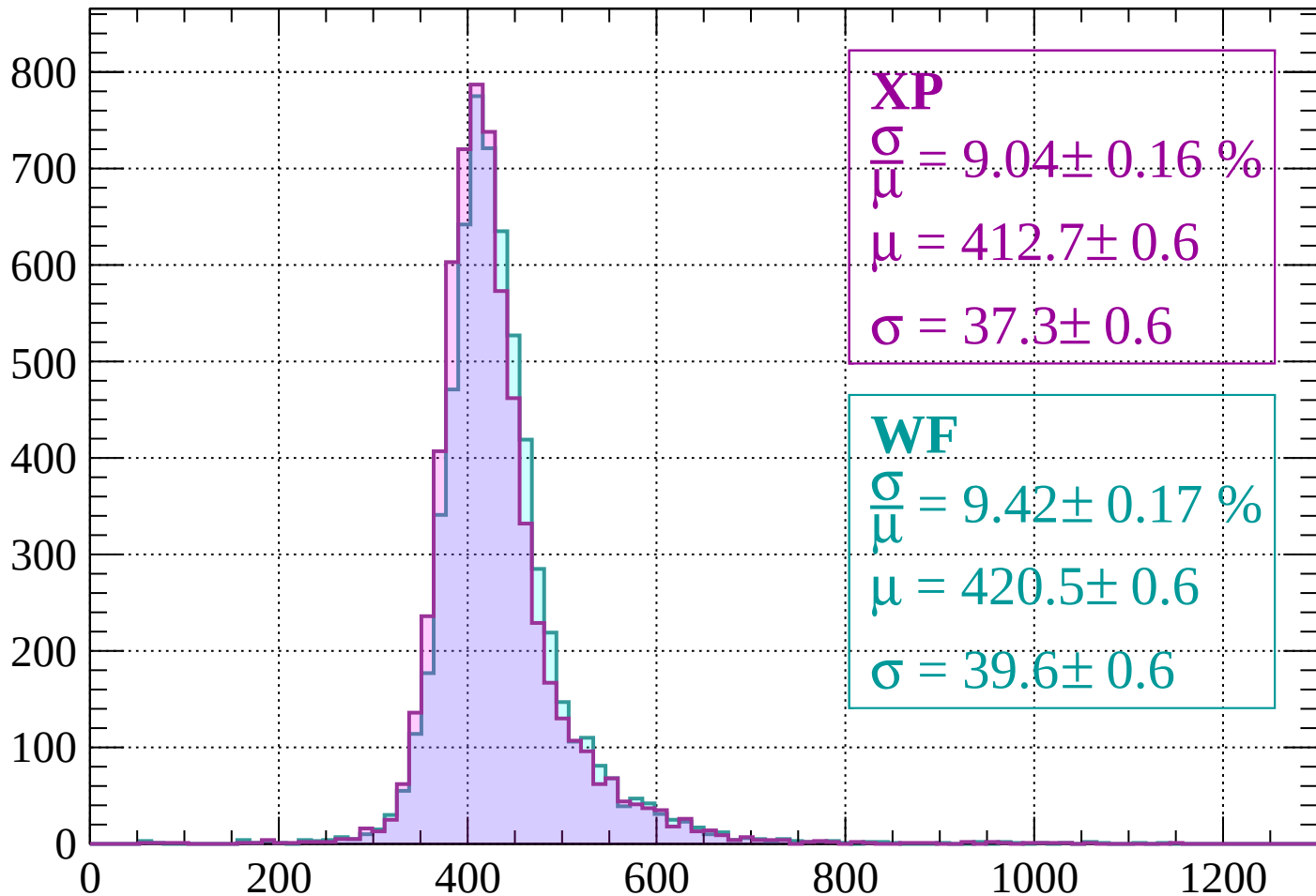
$$\mu = 417.4 \pm 0.6$$

$$\sigma = 40.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.16 \%$$

$$\mu = 412.7 \pm 0.6$$

$$\sigma = 37.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.17 \%$$

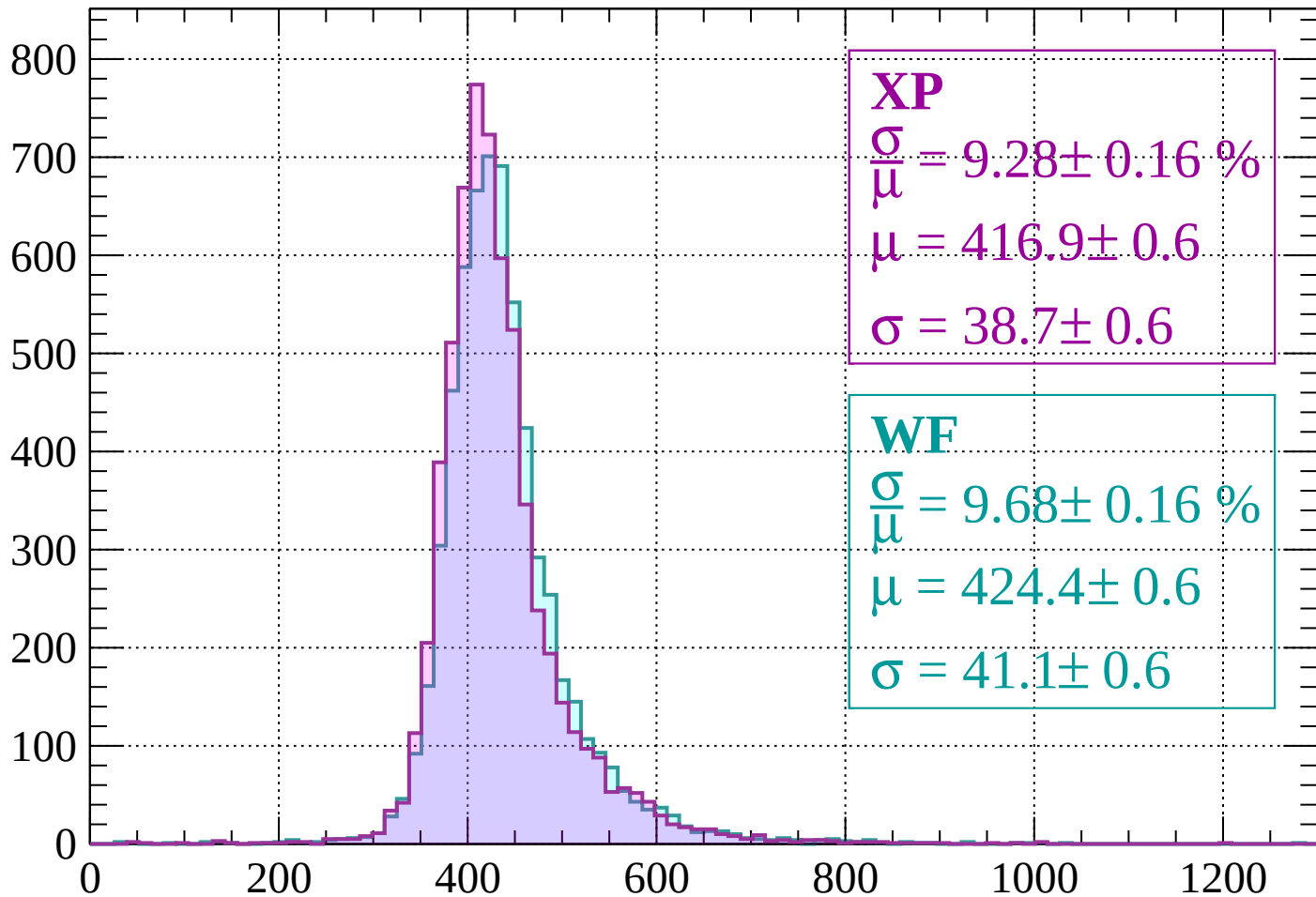
$$\mu = 420.5 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 9.28 \pm 0.16 \%$$

$$\mu = 416.9 \pm 0.6$$

$$\sigma = 38.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.68 \pm 0.16 \%$$

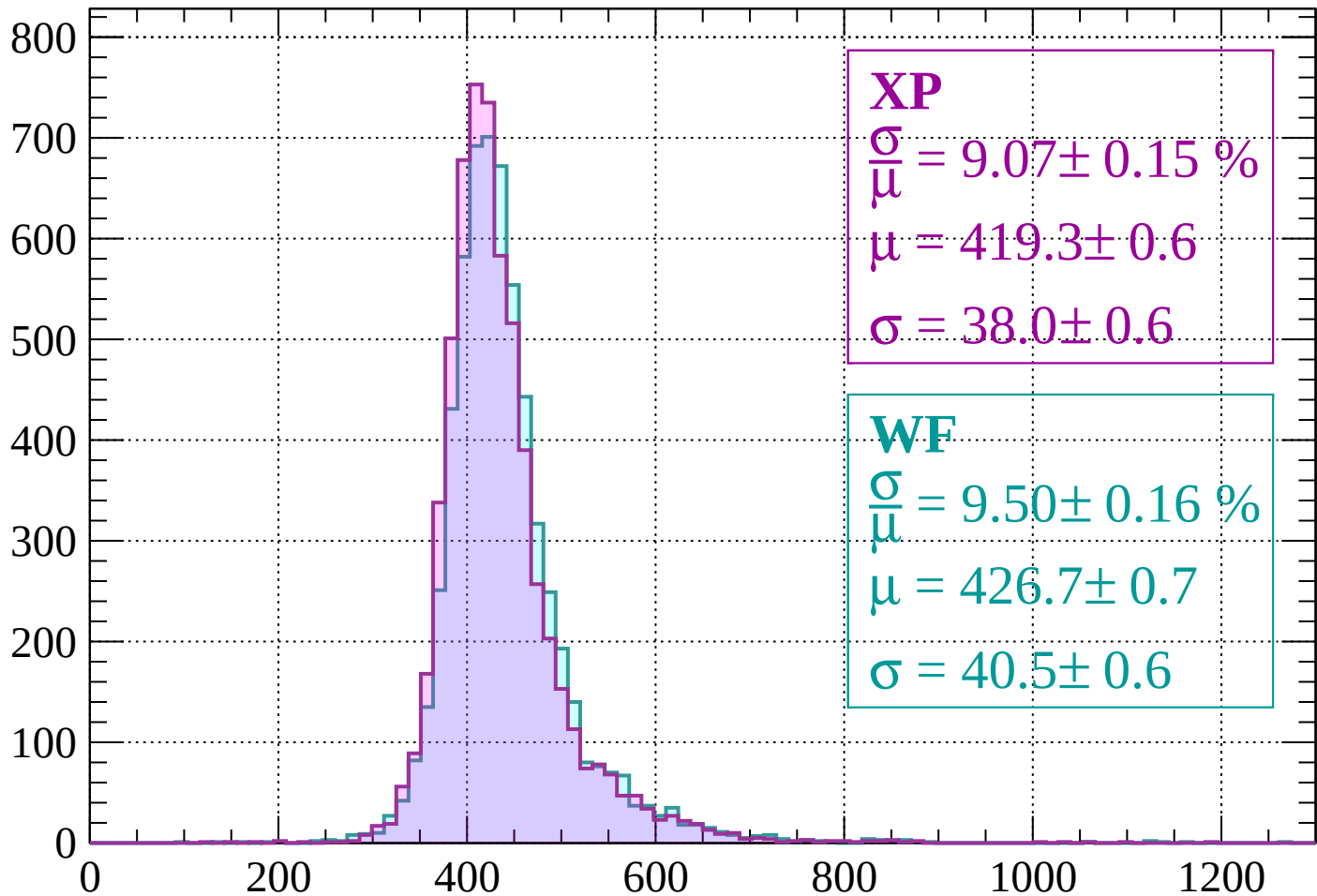
$$\mu = 424.4 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 9.07 \pm 0.15 \%$$

$$\mu = 419.3 \pm 0.6$$

$$\sigma = 38.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.50 \pm 0.16 \%$$

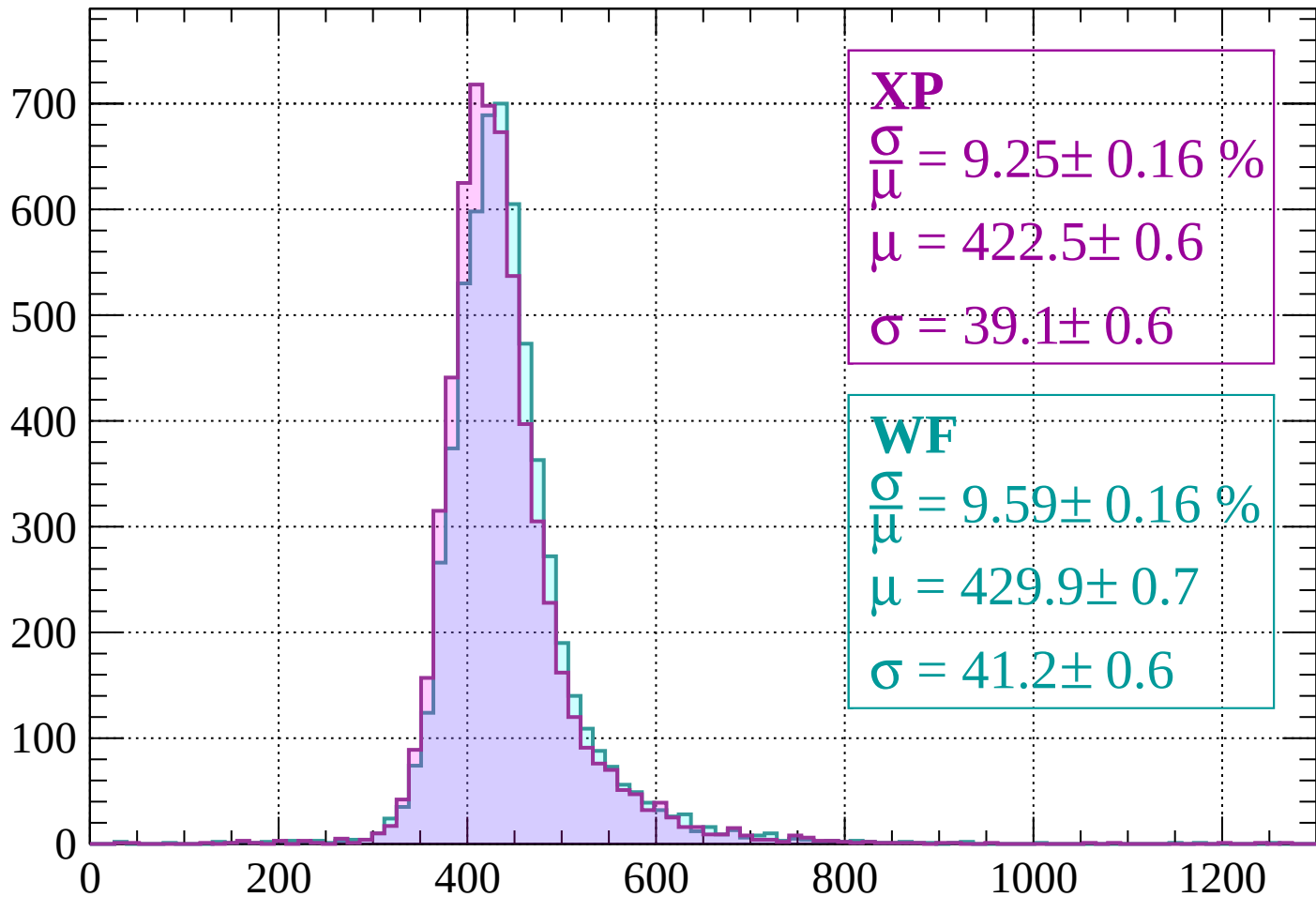
$$\mu = 426.7 \pm 0.7$$

$$\sigma = 40.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 9.25 \pm 0.16 \%$$

$$\mu = 422.5 \pm 0.6$$

$$\sigma = 39.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.16 \%$$

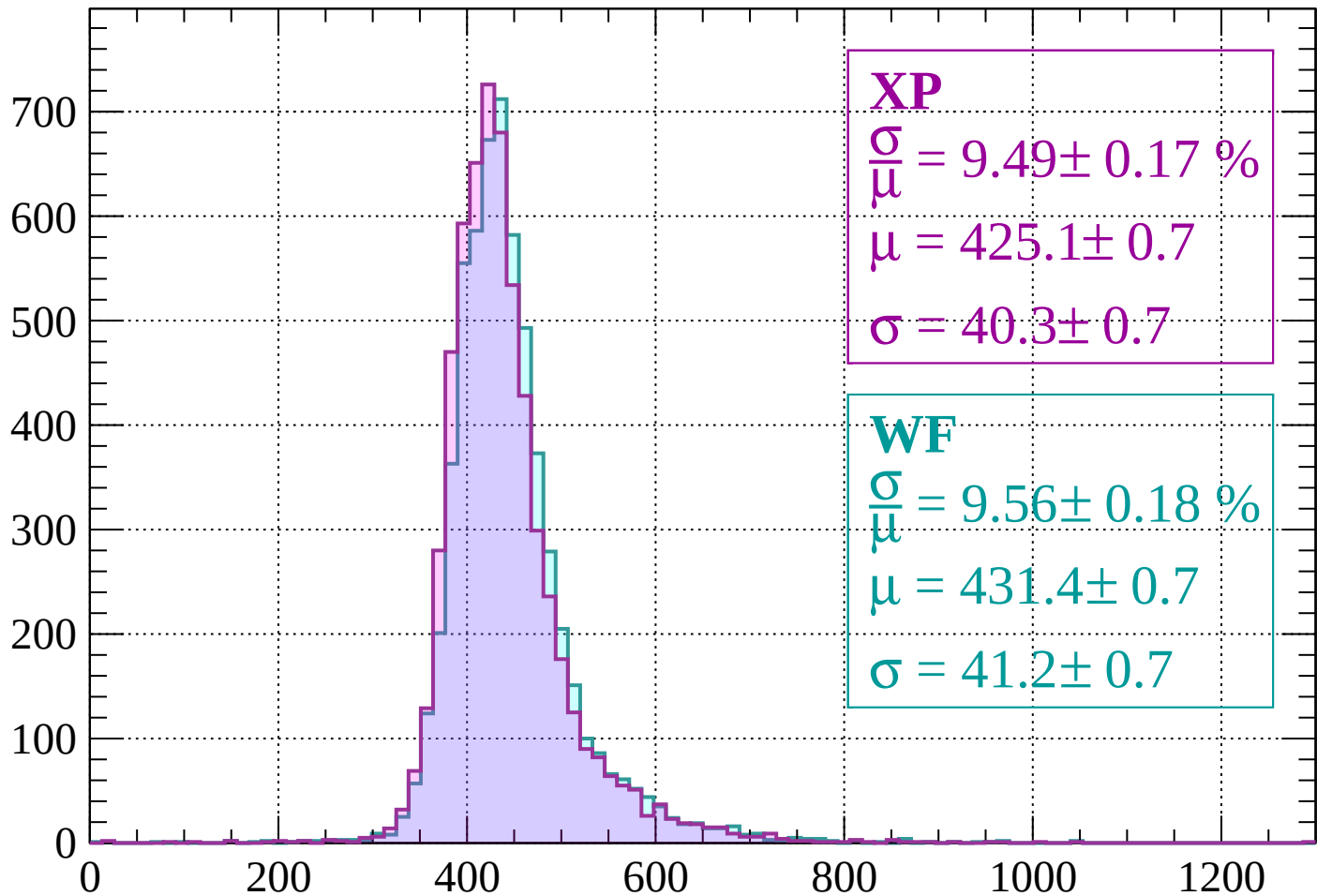
$$\mu = 429.9 \pm 0.7$$

$$\sigma = 41.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.17 \%$$

$$\mu = 425.1 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.18 \%$$

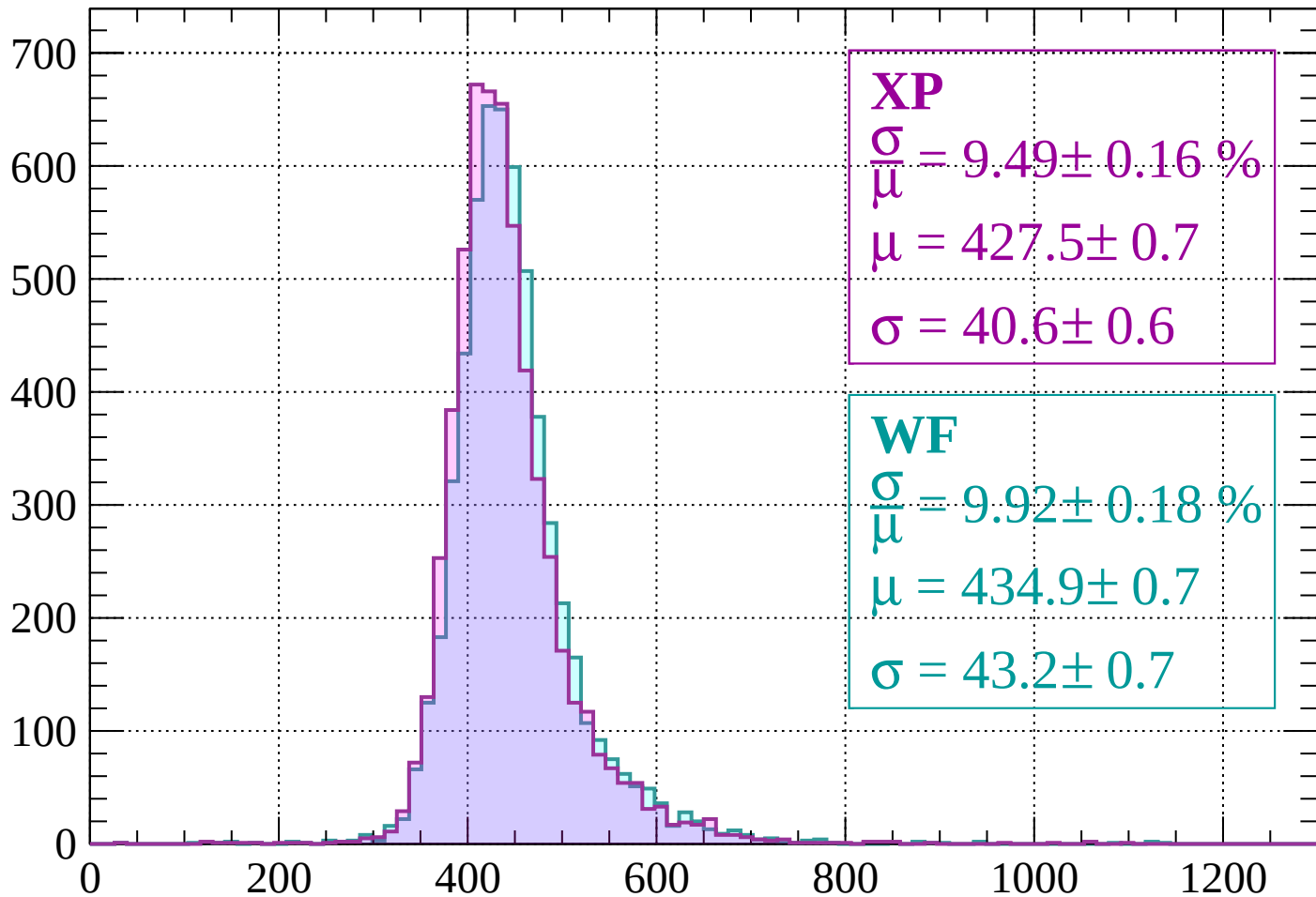
$$\mu = 431.4 \pm 0.7$$

$$\sigma = 41.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 920$

Count



XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.16 \%$$

$$\mu = 427.5 \pm 0.7$$

$$\sigma = 40.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.92 \pm 0.18 \%$$

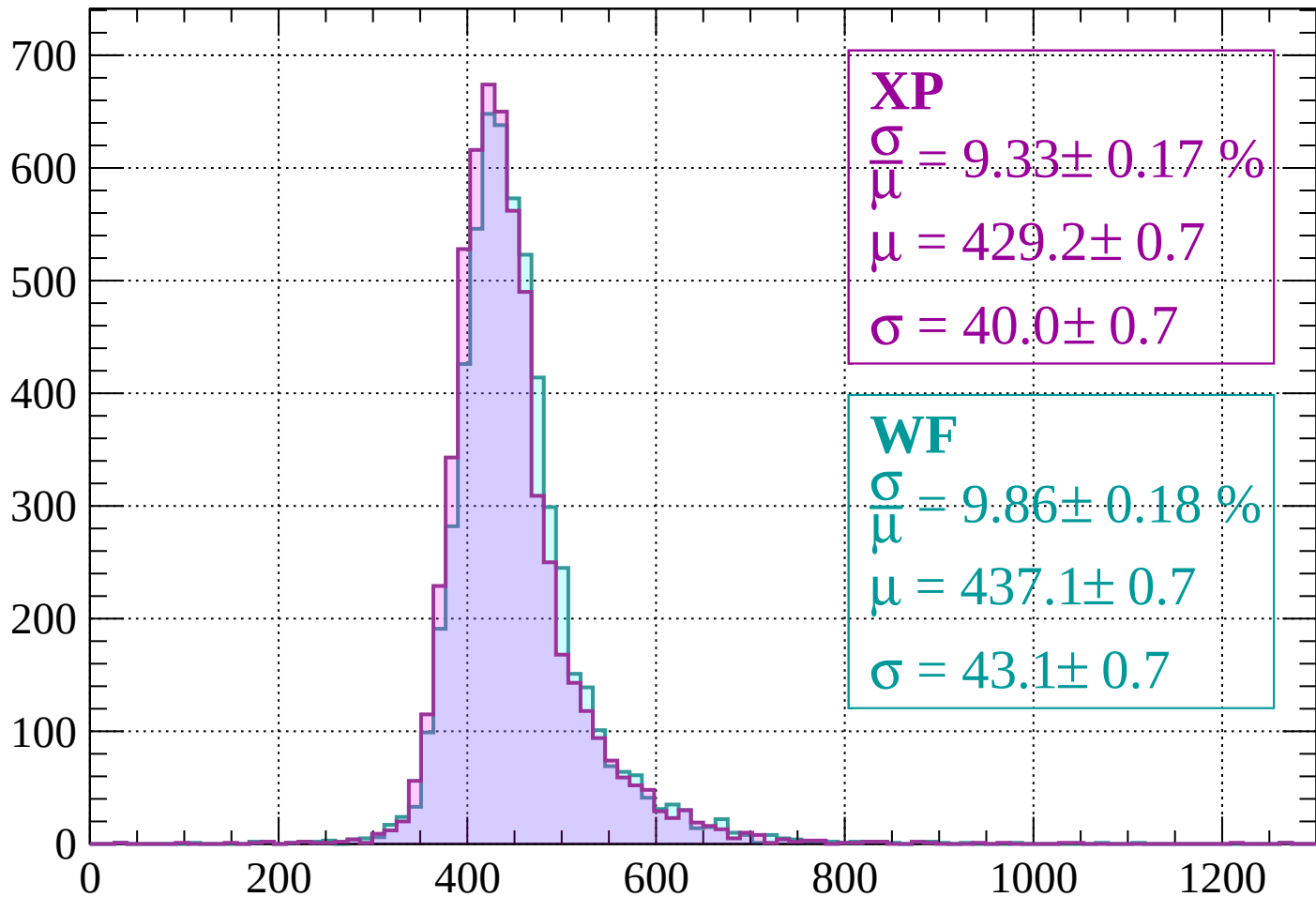
$$\mu = 434.9 \pm 0.7$$

$$\sigma = 43.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 9.33 \pm 0.17 \%$$

$$\mu = 429.2 \pm 0.7$$

$$\sigma = 40.0 \pm 0.7$$

WF

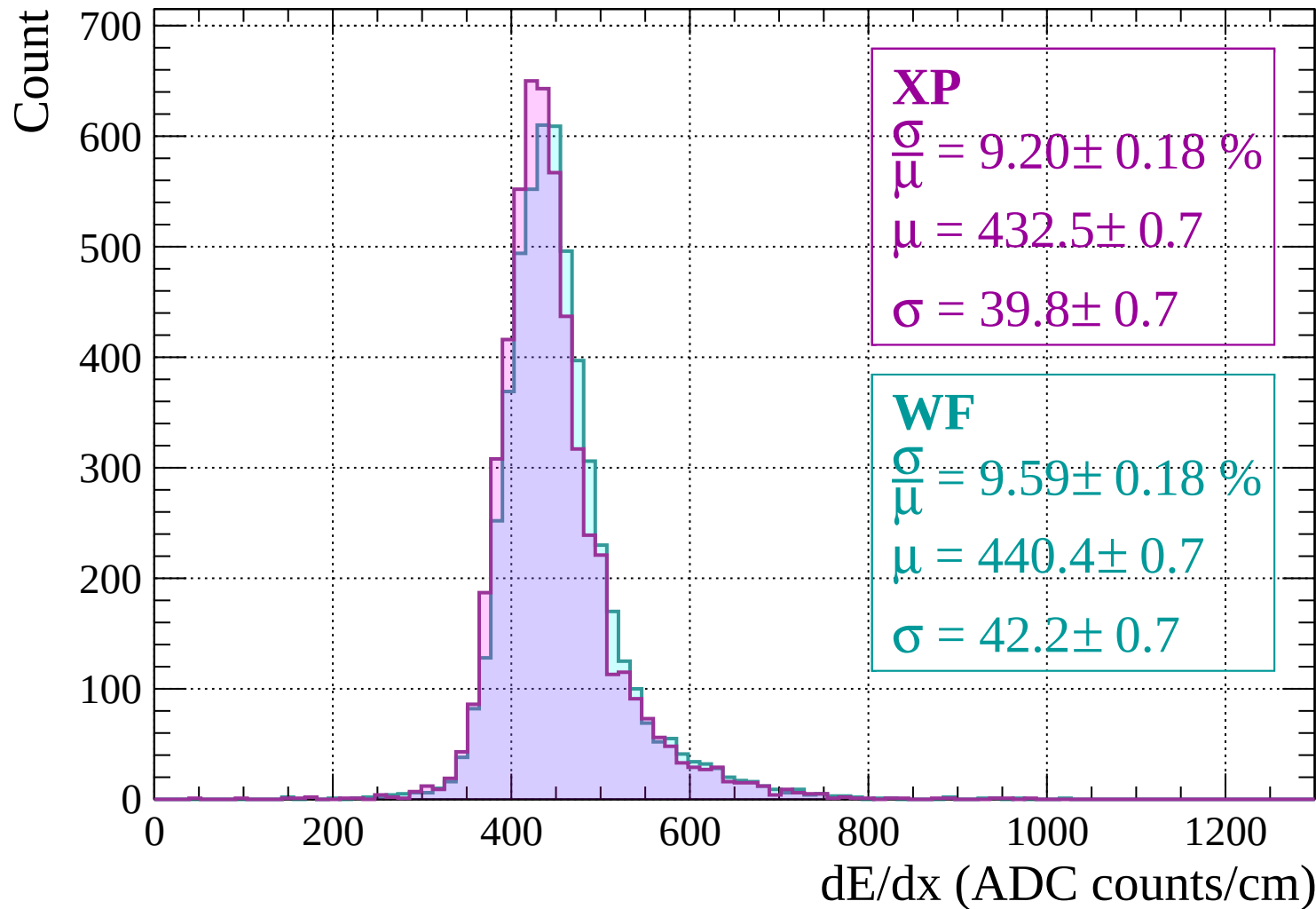
$$\frac{\sigma}{\mu} = 9.86 \pm 0.18 \%$$

$$\mu = 437.1 \pm 0.7$$

$$\sigma = 43.1 \pm 0.7$$

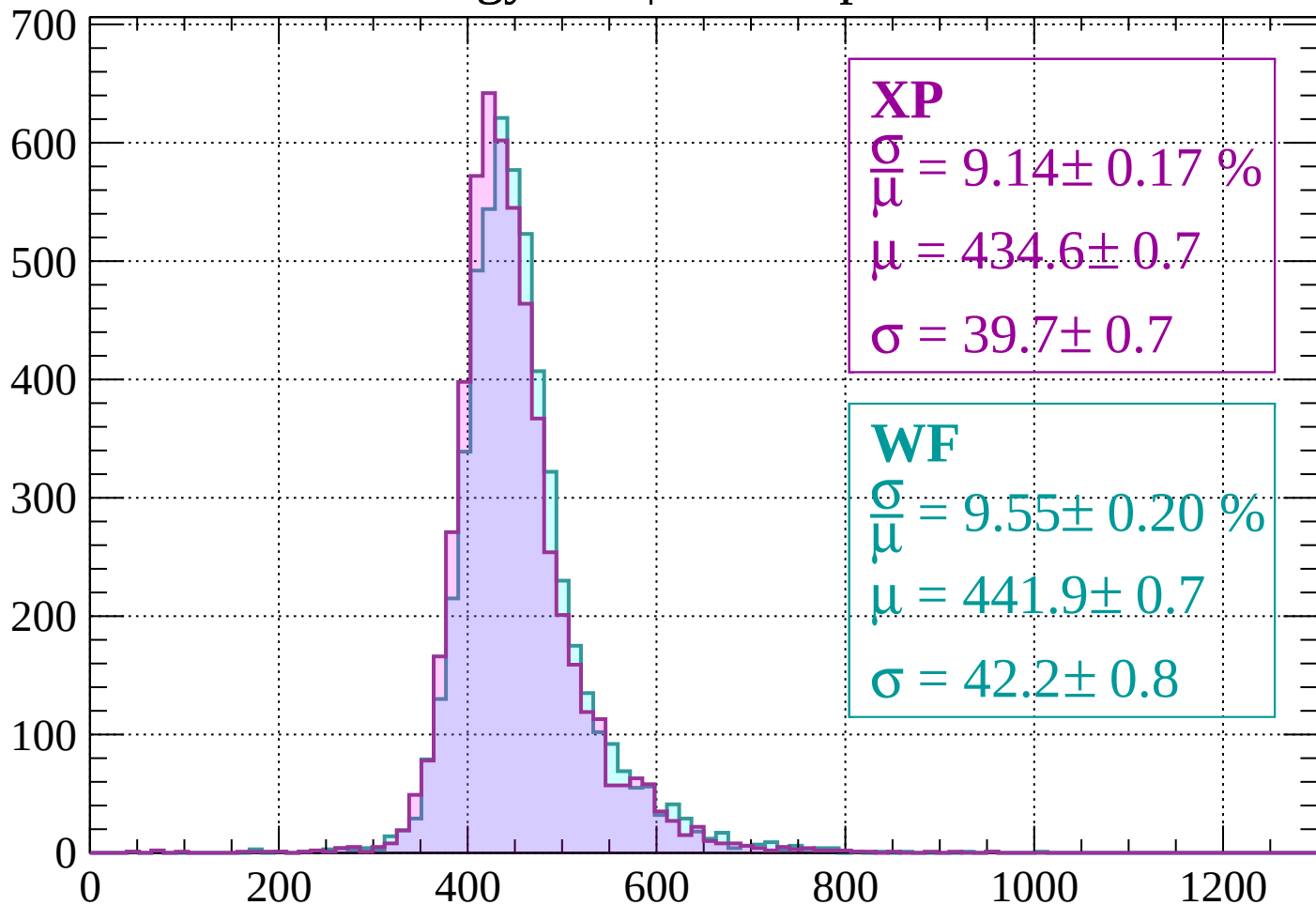
dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$



Energy loss | $1000 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 9.14 \pm 0.17 \%$$

$$\mu = 434.6 \pm 0.7$$

$$\sigma = 39.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.20 \%$$

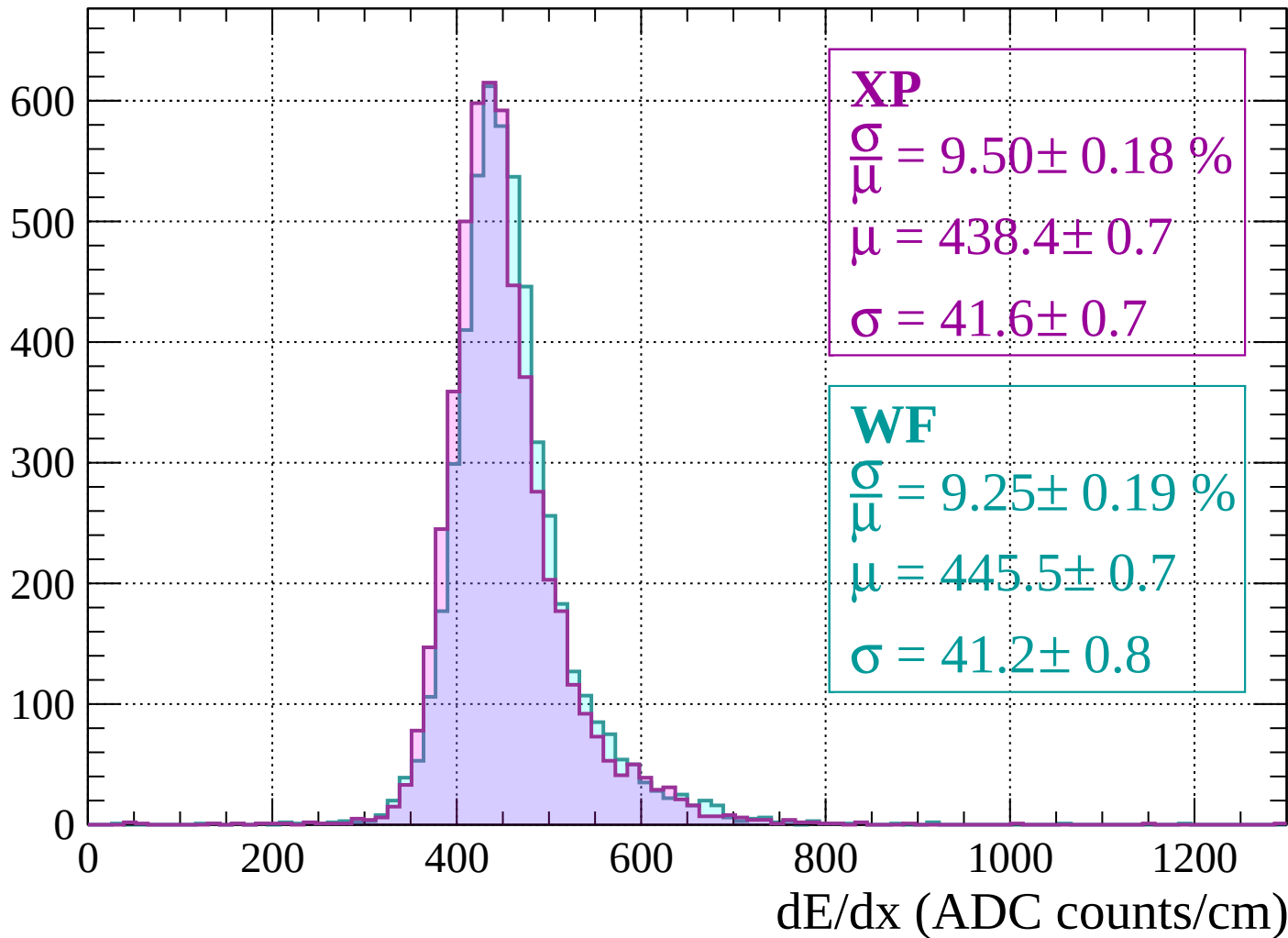
$$\mu = 441.9 \pm 0.7$$

$$\sigma = 42.2 \pm 0.8$$

dE/dx (ADC counts/cm)

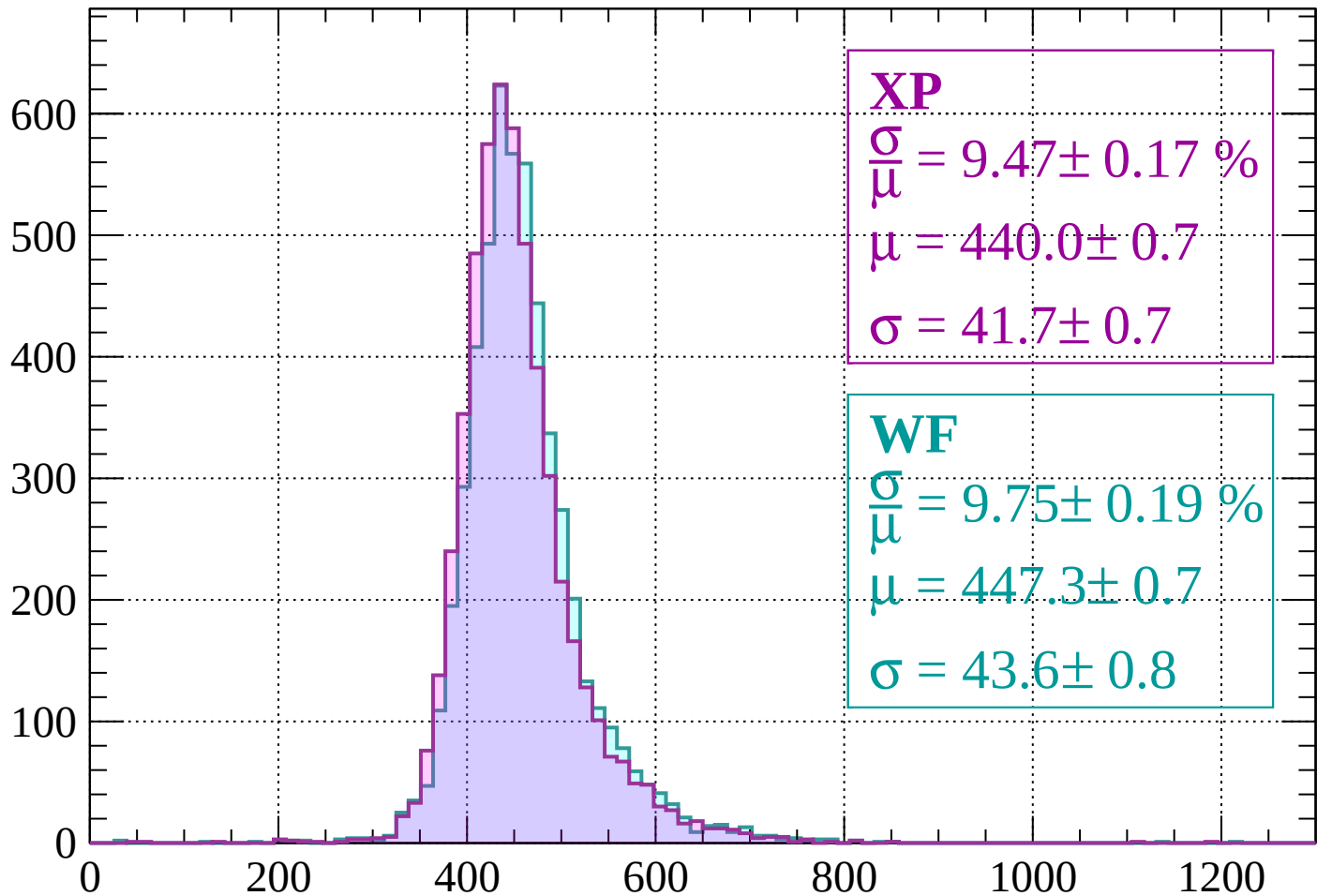
Energy loss | $1040 < p < 1080$

Count



Energy loss | $1080 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 9.47 \pm 0.17 \%$$

$$\mu = 440.0 \pm 0.7$$

$$\sigma = 41.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.19 \%$$

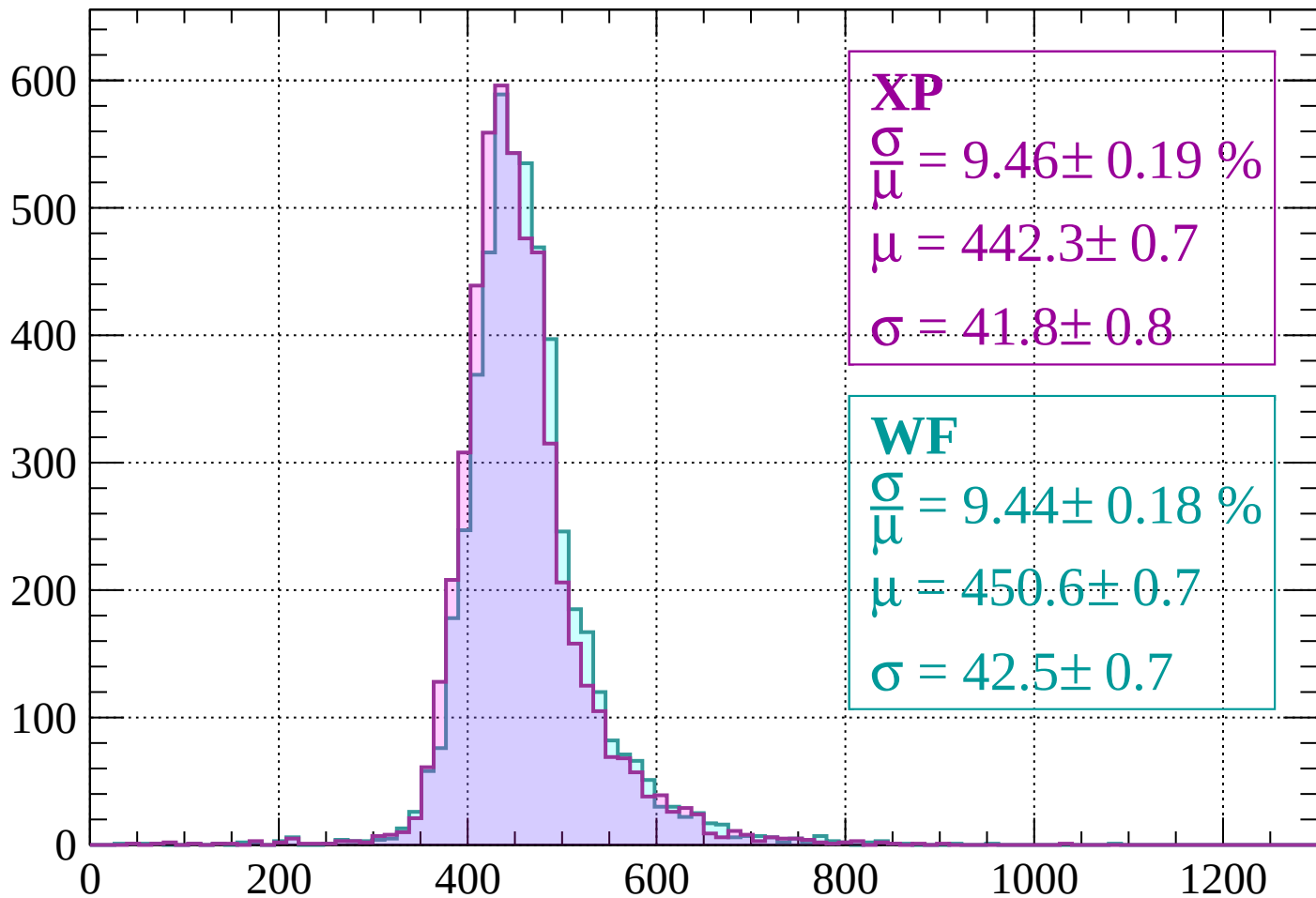
$$\mu = 447.3 \pm 0.7$$

$$\sigma = 43.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.19 \%$$

$$\mu = 442.3 \pm 0.7$$

$$\sigma = 41.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.44 \pm 0.18 \%$$

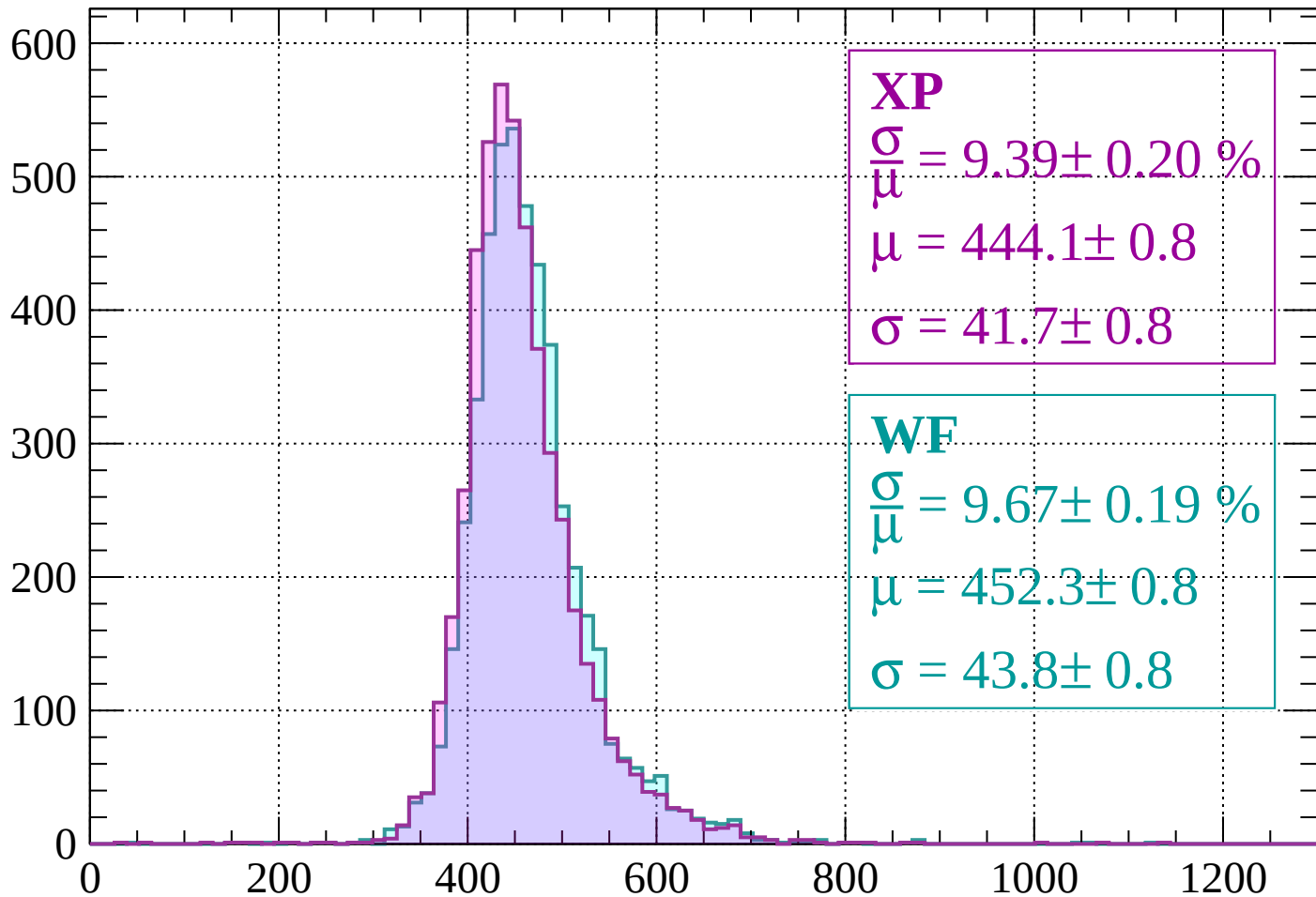
$$\mu = 450.6 \pm 0.7$$

$$\sigma = 42.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.39 \pm 0.20 \%$$

$$\mu = 444.1 \pm 0.8$$

$$\sigma = 41.7 \pm 0.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.67 \pm 0.19 \%$$

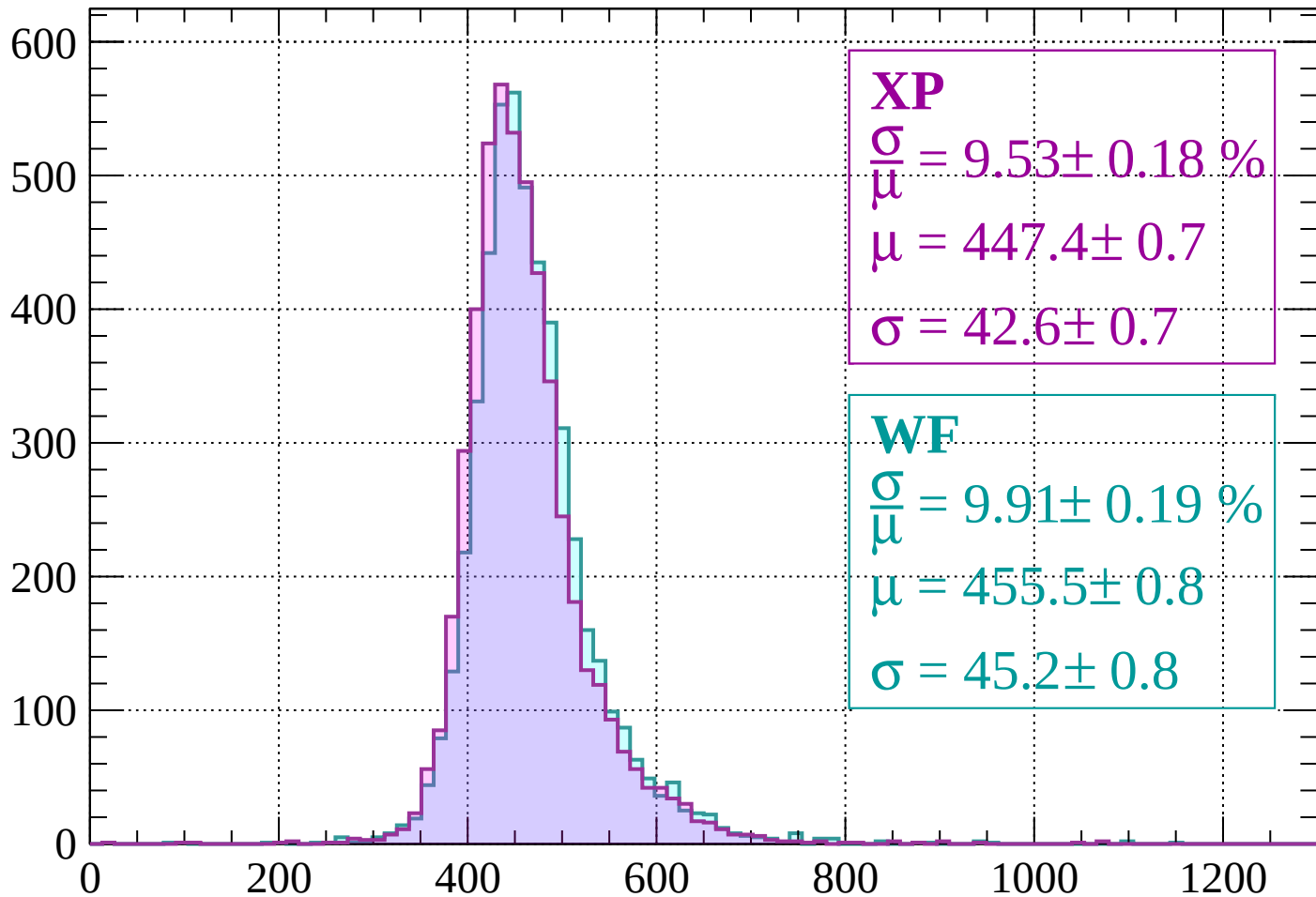
$$\mu = 452.3 \pm 0.8$$

$$\sigma = 43.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\mu} = 9.53 \pm 0.18 \%$$

$$\mu = 447.4 \pm 0.7$$

$$\sigma = 42.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.91 \pm 0.19 \%$$

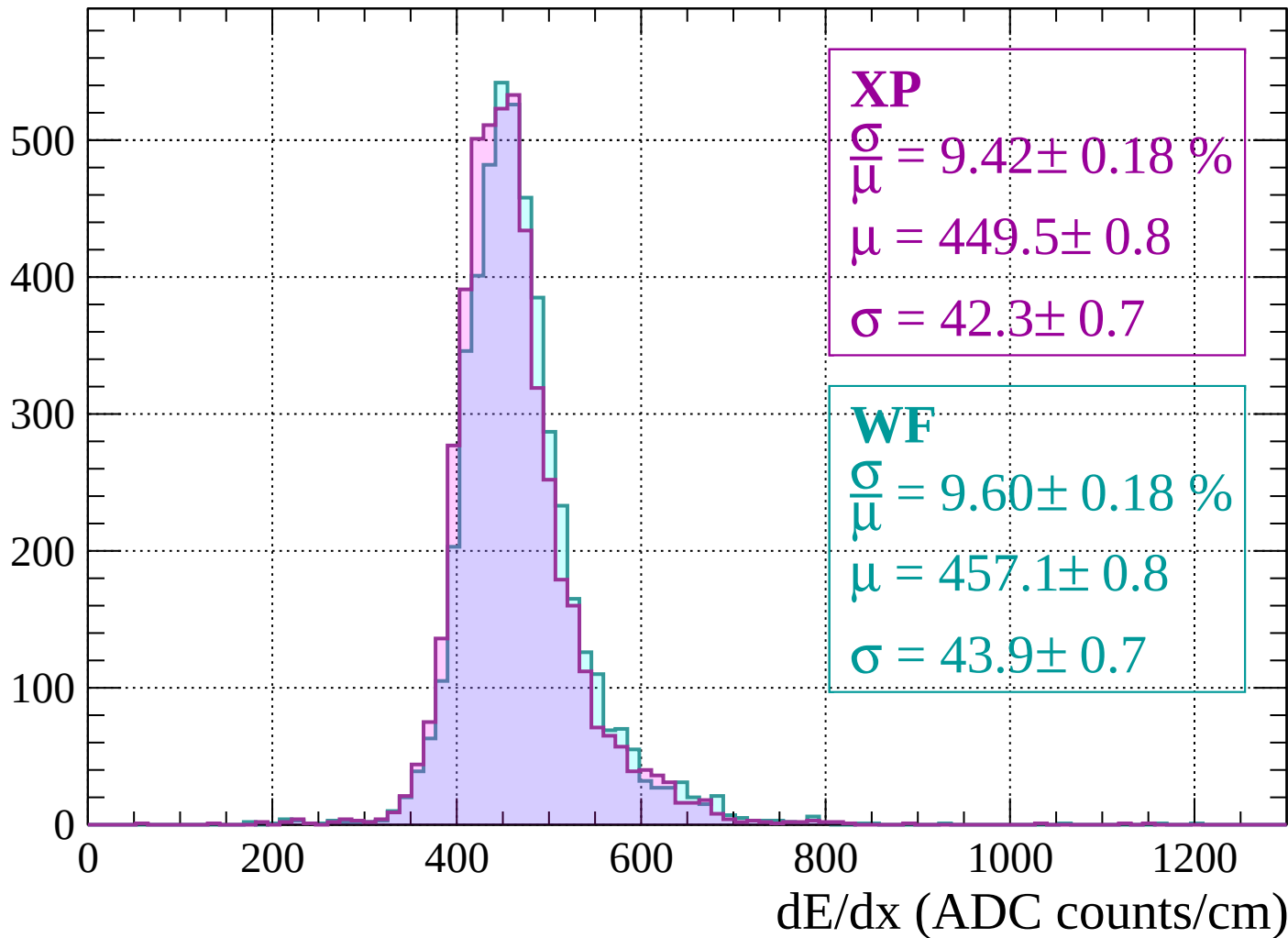
$$\mu = 455.5 \pm 0.8$$

$$\sigma = 45.2 \pm 0.8$$

dE/dx (ADC counts/cm)

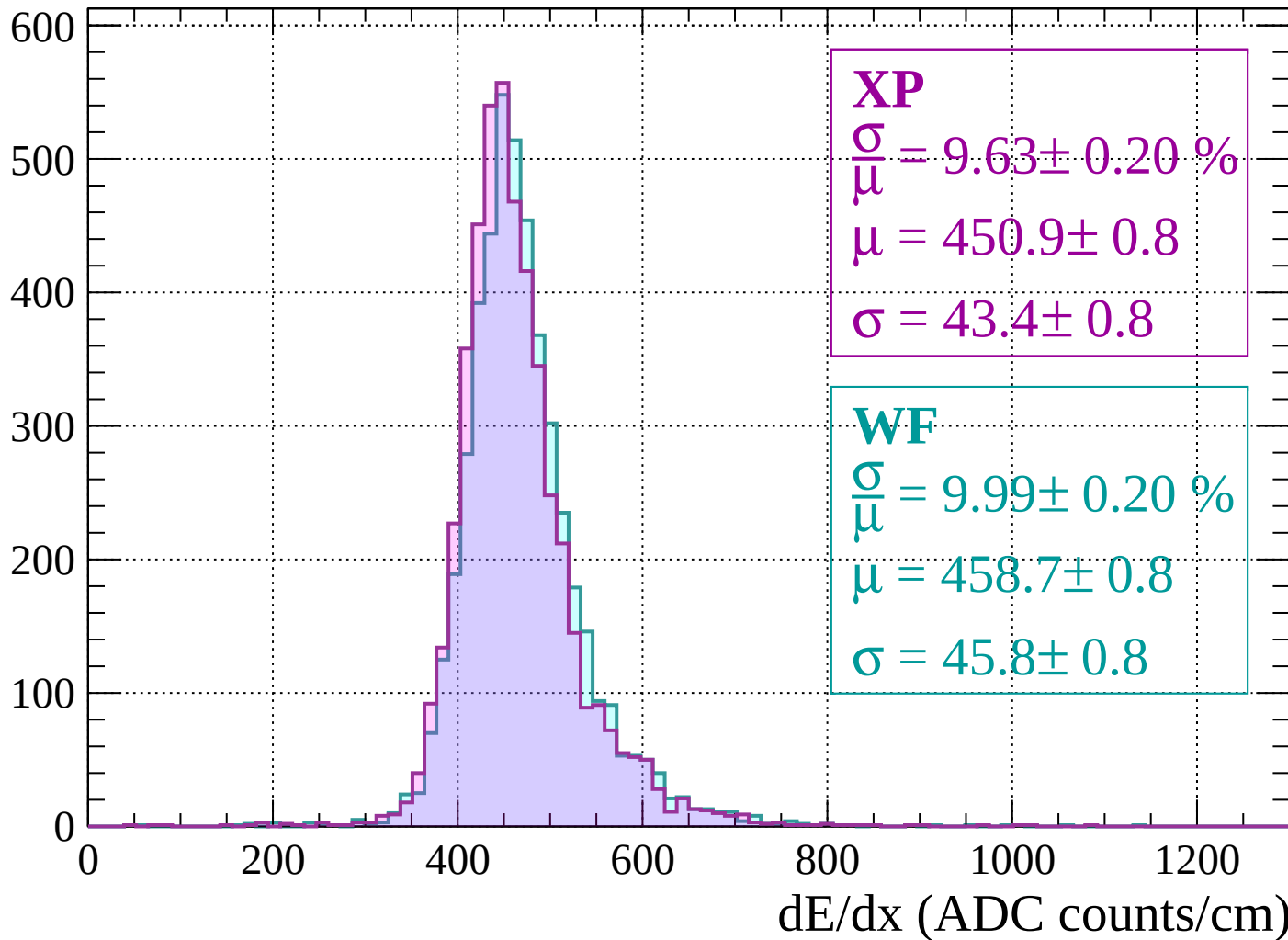
Energy loss | $1240 < p < 1280$

Count



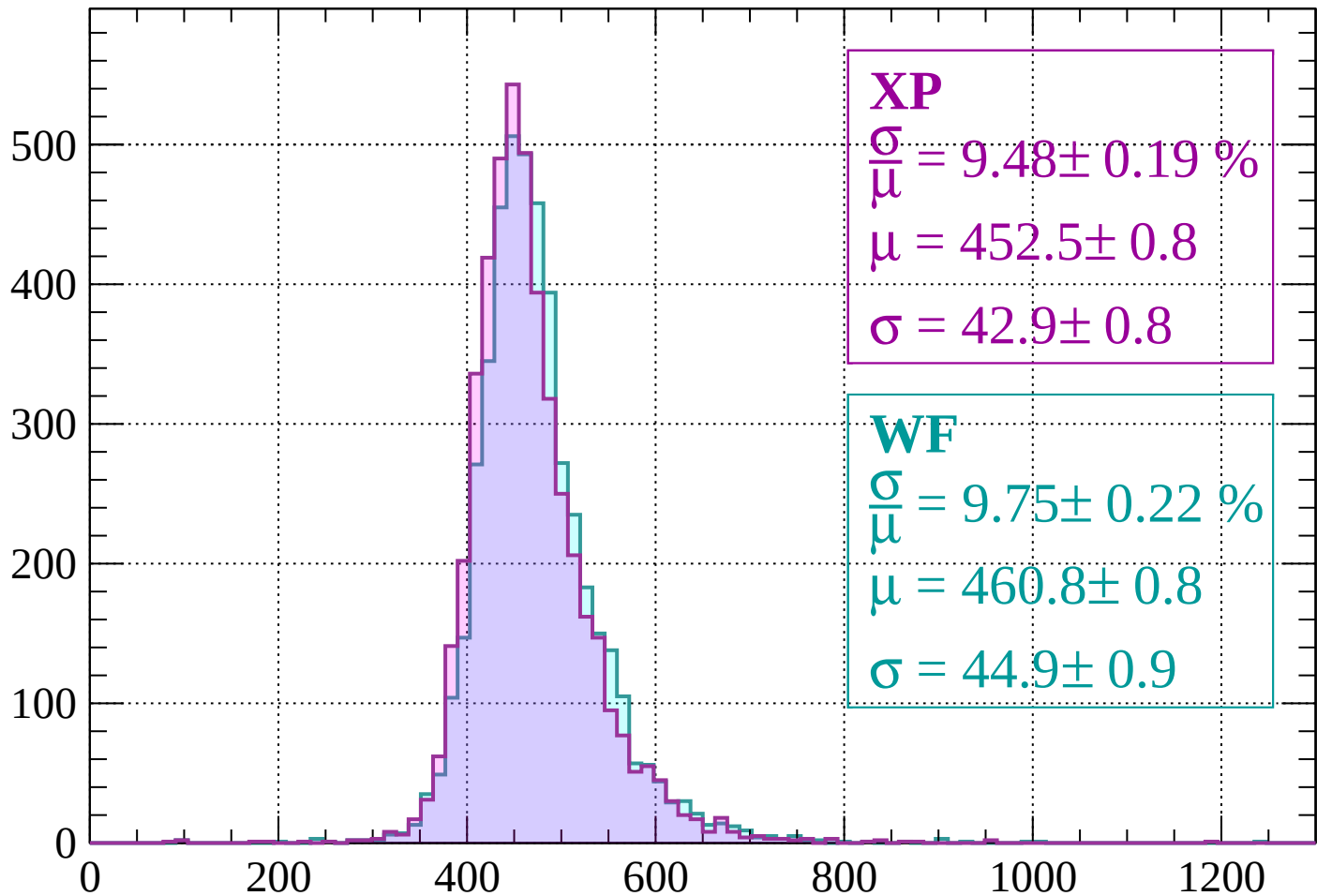
Energy loss | $1280 < p < 1320$

Count



Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.19 \%$$

$$\mu = 452.5 \pm 0.8$$

$$\sigma = 42.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.22 \%$$

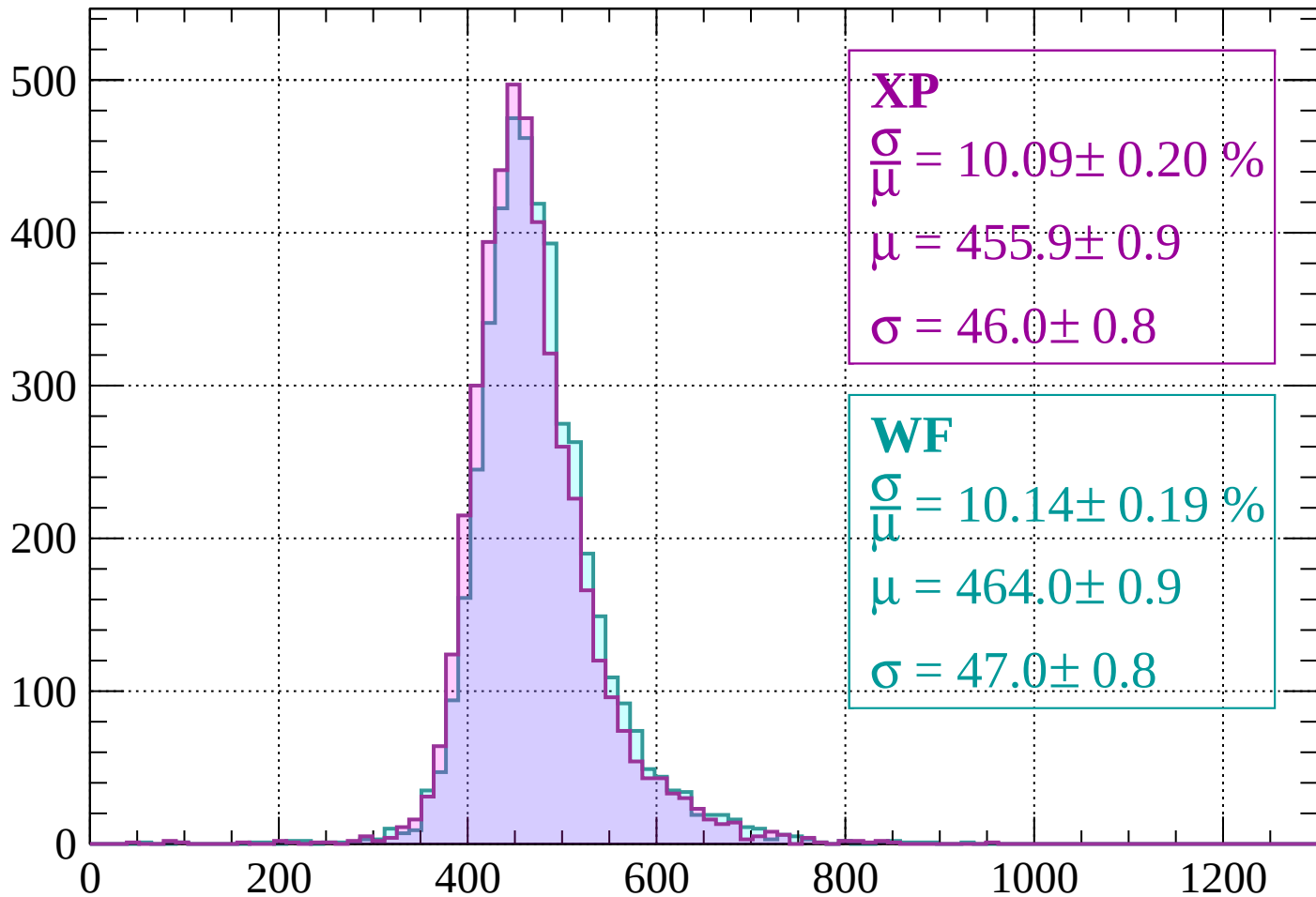
$$\mu = 460.8 \pm 0.8$$

$$\sigma = 44.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1400$

Count



XP

$$\frac{\sigma}{\mu} = 10.09 \pm 0.20 \%$$

$$\mu = 455.9 \pm 0.9$$

$$\sigma = 46.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.14 \pm 0.19 \%$$

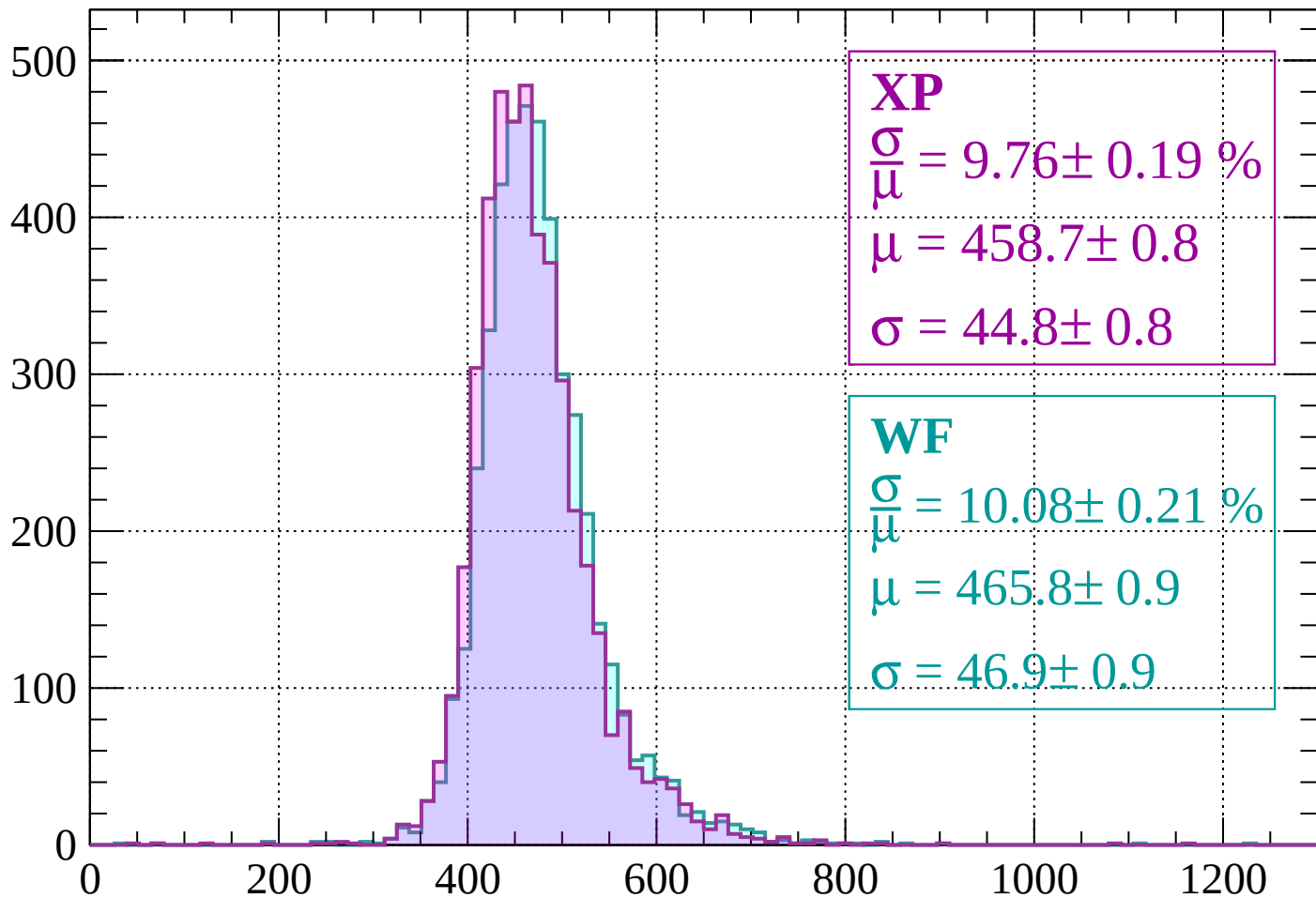
$$\mu = 464.0 \pm 0.9$$

$$\sigma = 47.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1400 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 9.76 \pm 0.19 \%$$

$$\mu = 458.7 \pm 0.8$$

$$\sigma = 44.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.08 \pm 0.21 \%$$

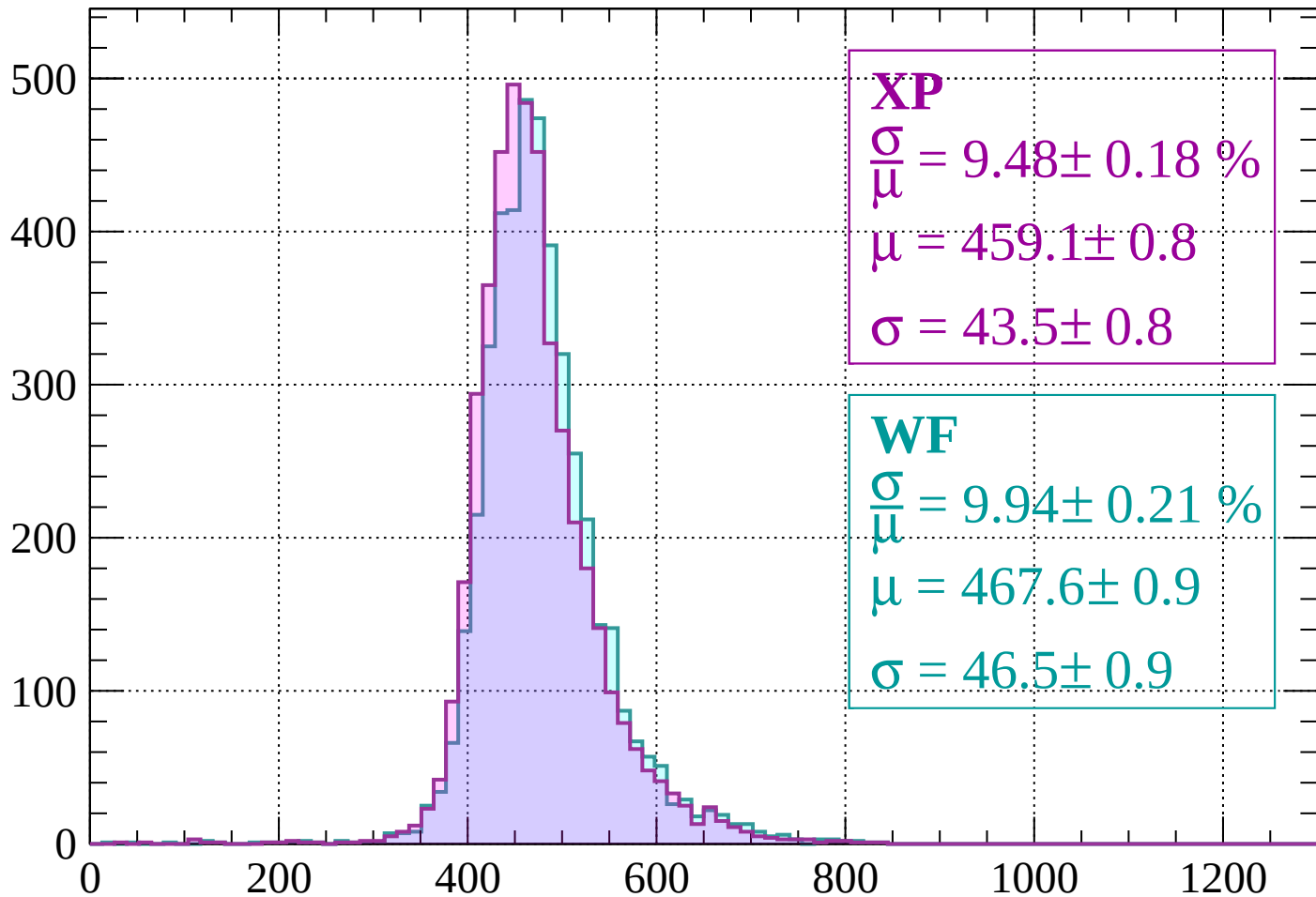
$$\mu = 465.8 \pm 0.9$$

$$\sigma = 46.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.18 \%$$

$$\mu = 459.1 \pm 0.8$$

$$\sigma = 43.5 \pm 0.8$$

WF

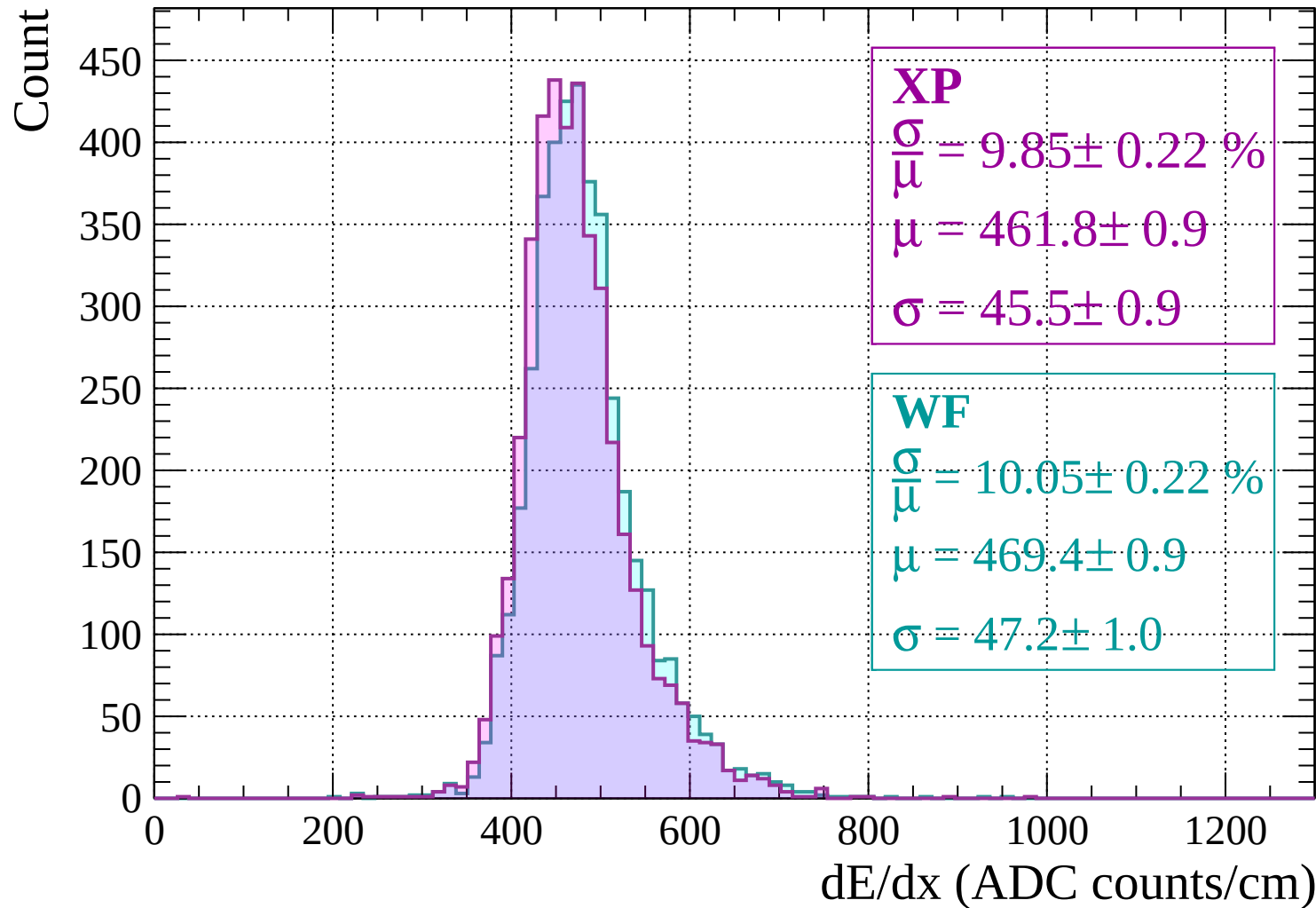
$$\frac{\sigma}{\mu} = 9.94 \pm 0.21 \%$$

$$\mu = 467.6 \pm 0.9$$

$$\sigma = 46.5 \pm 0.9$$

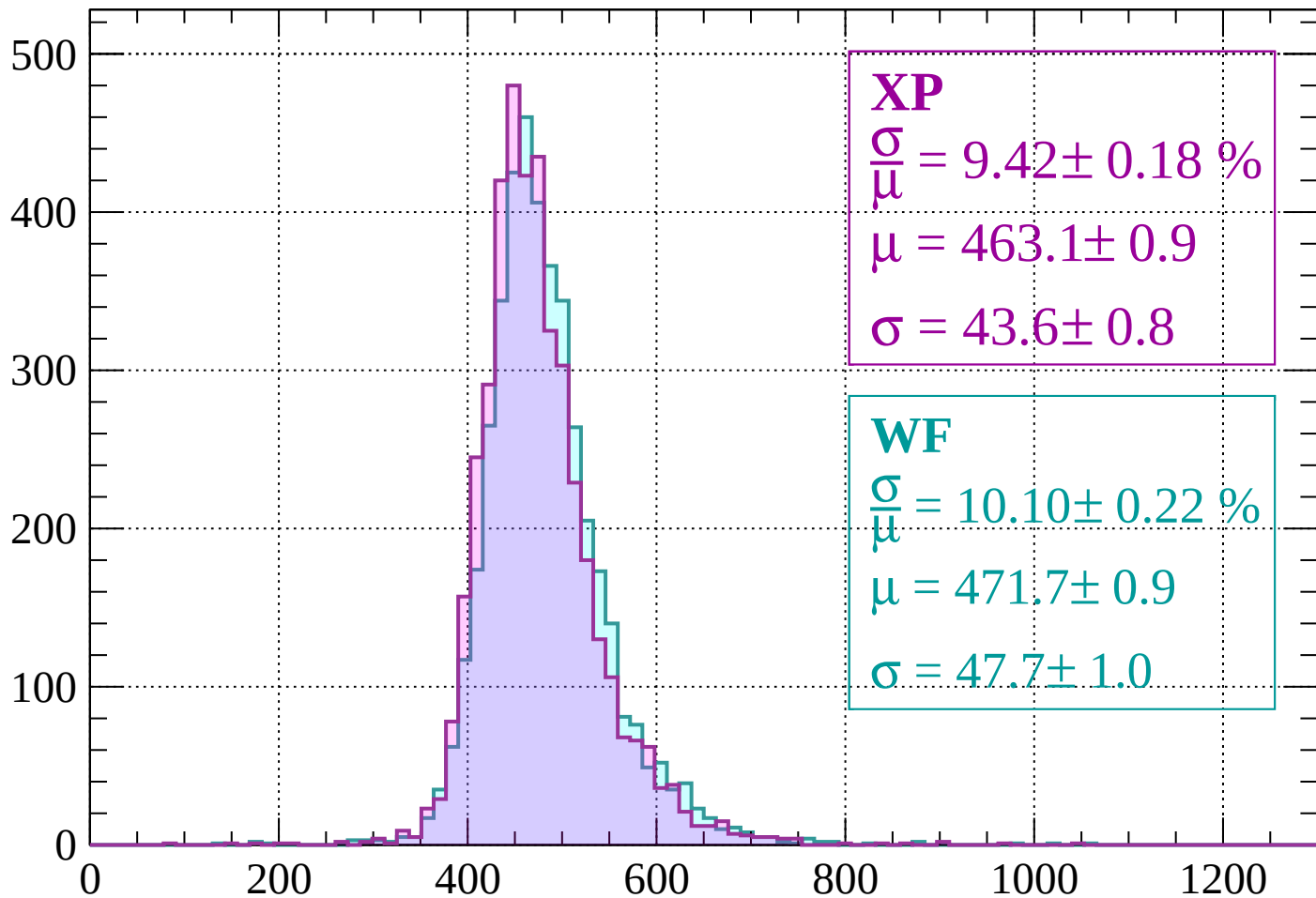
dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$



Energy loss | $1520 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 9.42 \pm 0.18 \%$$

$$\mu = 463.1 \pm 0.9$$

$$\sigma = 43.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.10 \pm 0.22 \%$$

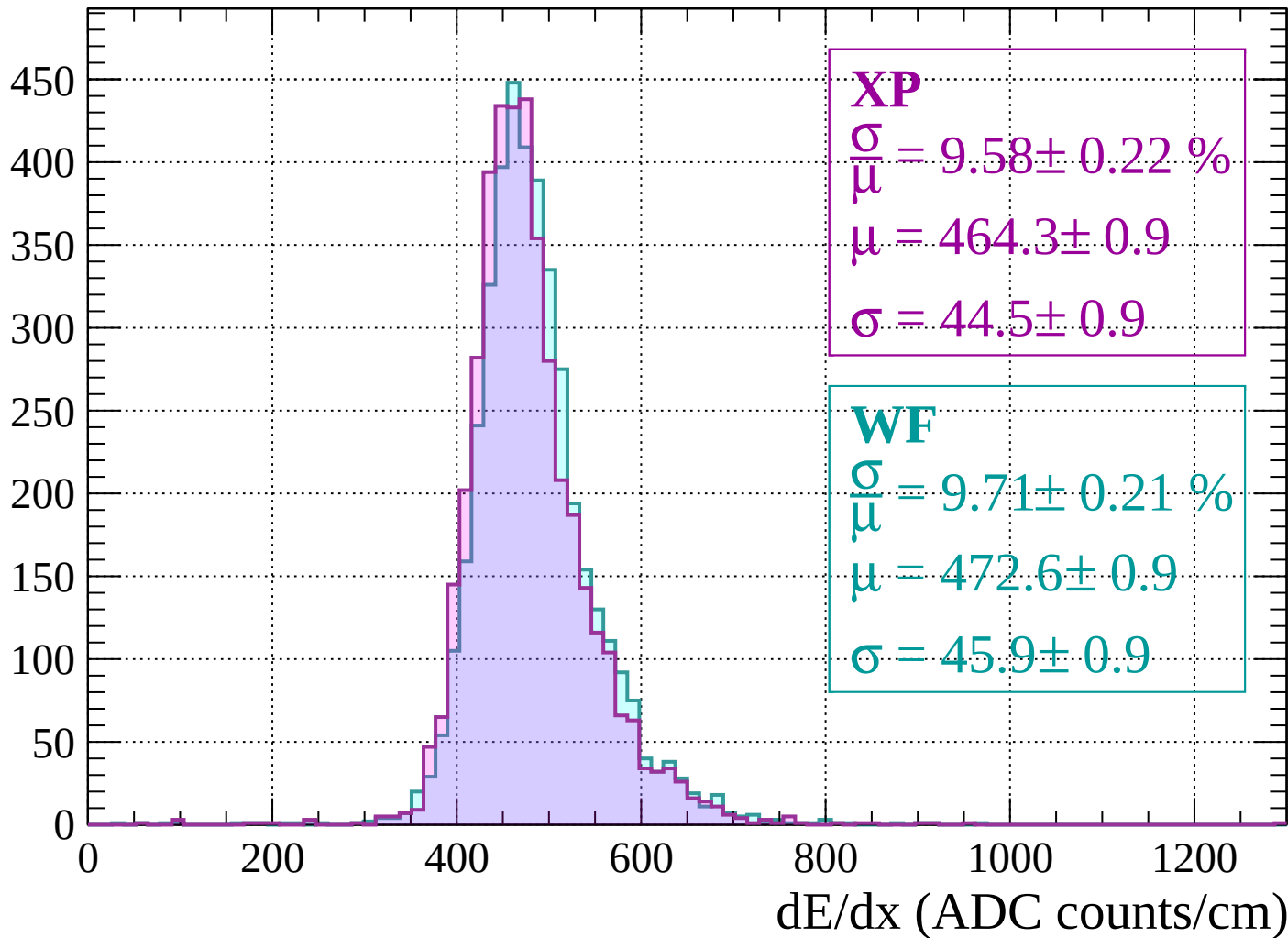
$$\mu = 471.7 \pm 0.9$$

$$\sigma = 47.7 \pm 1.0$$

dE/dx (ADC counts/cm)

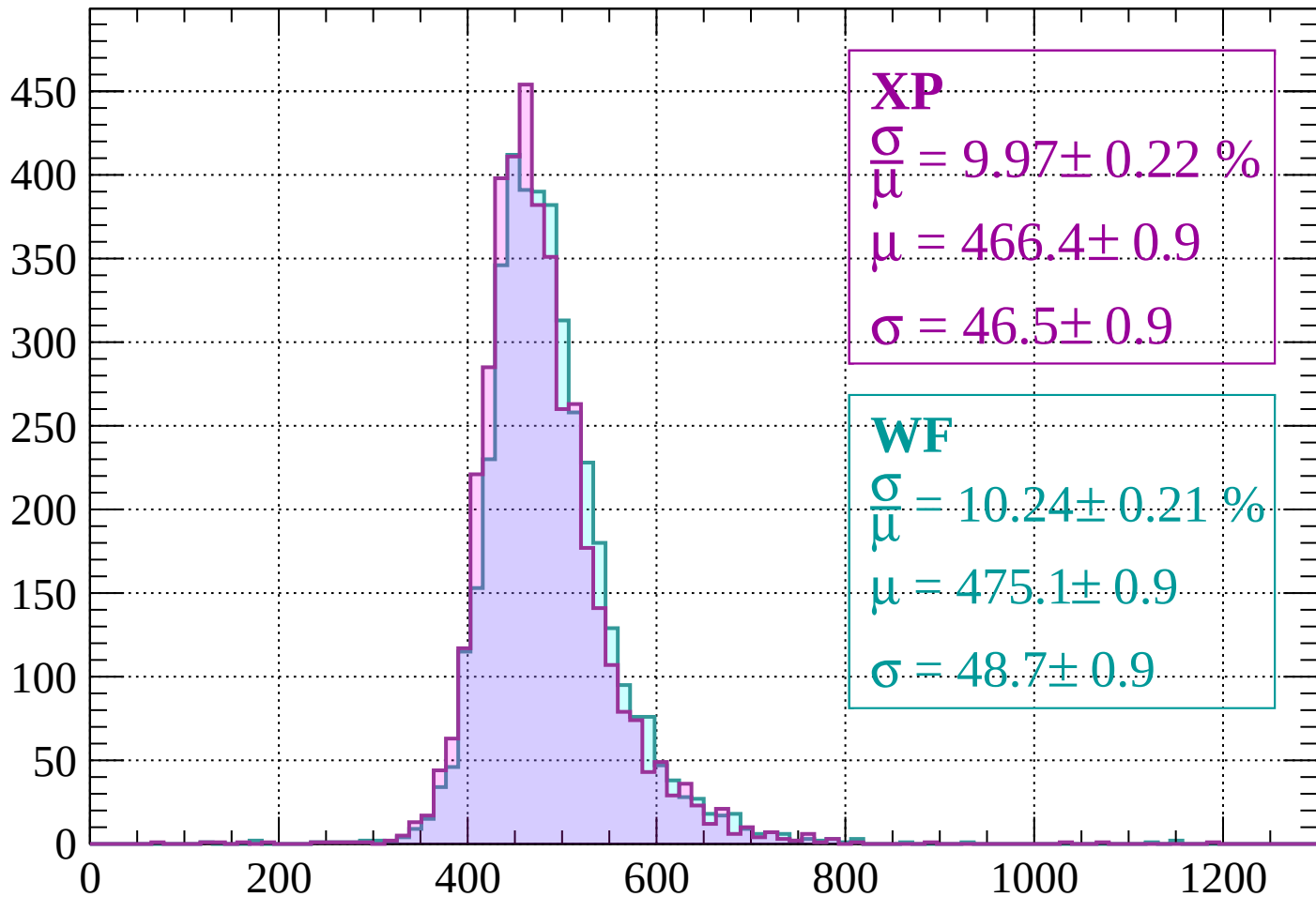
Energy loss | $1560 < p < 1600$

Count



Energy loss | $1600 < p < 1640$

Count



XP

$$\frac{\sigma}{\mu} = 9.97 \pm 0.22 \%$$

$$\mu = 466.4 \pm 0.9$$

$$\sigma = 46.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.24 \pm 0.21 \%$$

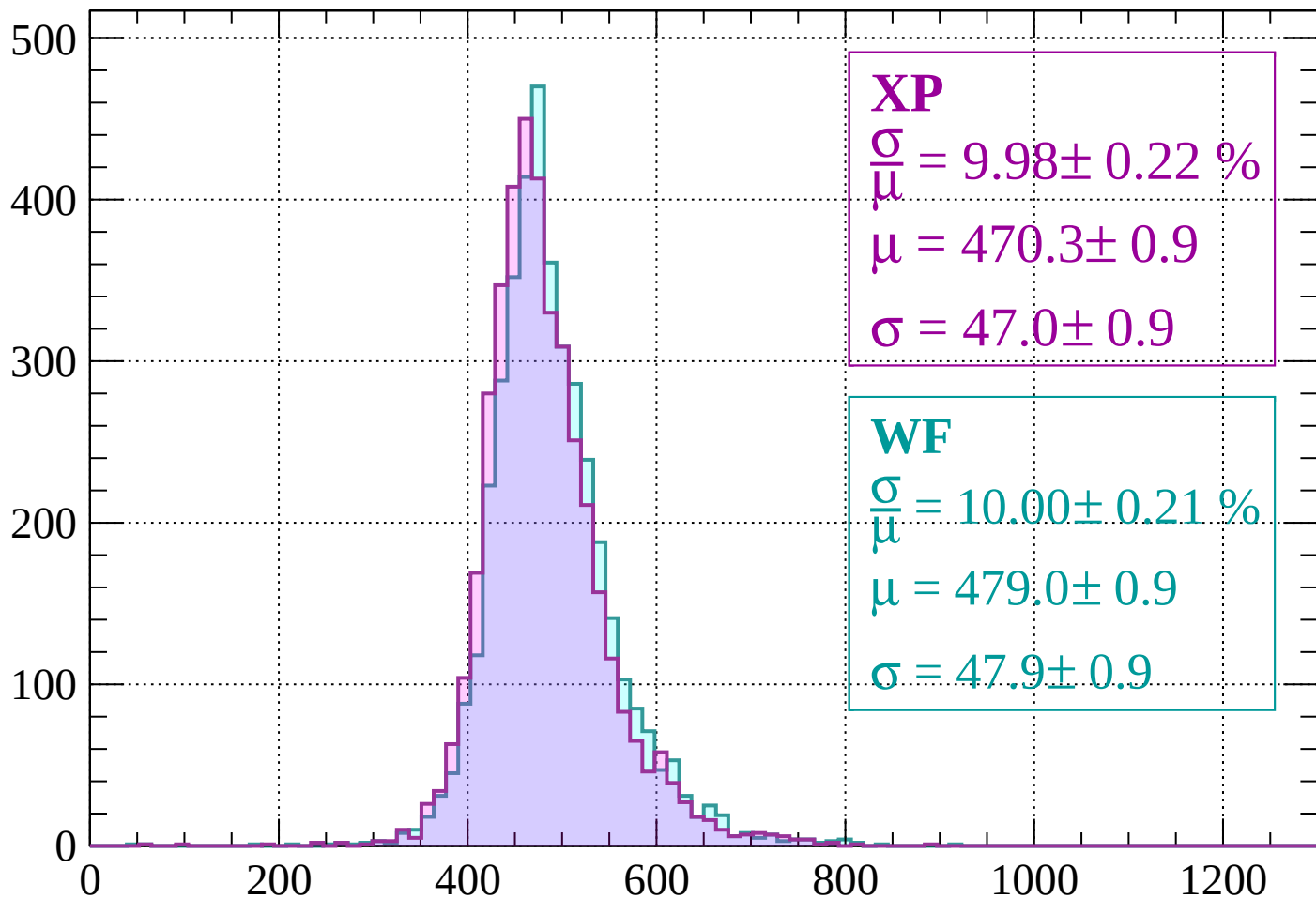
$$\mu = 475.1 \pm 0.9$$

$$\sigma = 48.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1640 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 9.98 \pm 0.22 \%$$

$$\mu = 470.3 \pm 0.9$$

$$\sigma = 47.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.00 \pm 0.21 \%$$

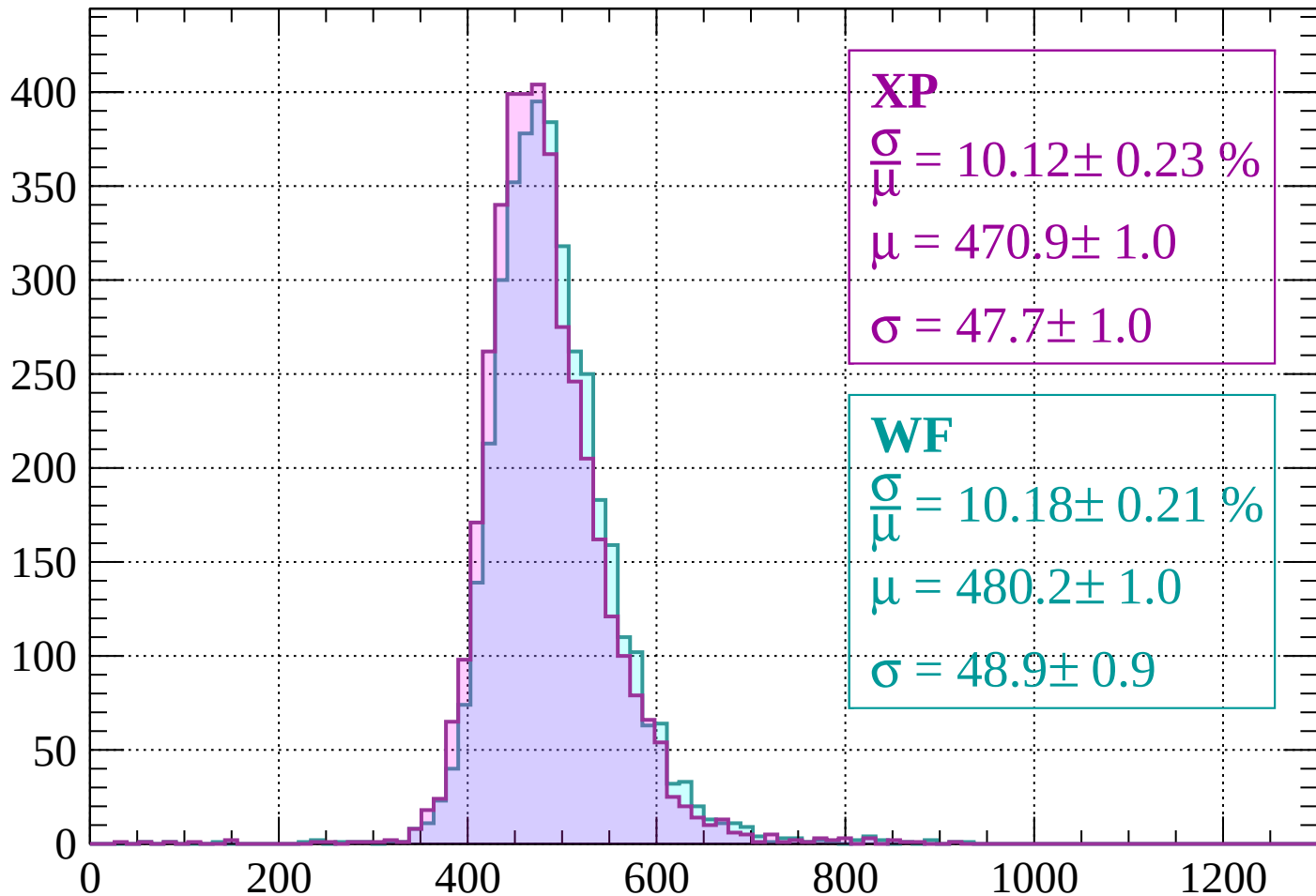
$$\mu = 479.0 \pm 0.9$$

$$\sigma = 47.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 10.12 \pm 0.23 \%$$

$$\mu = 470.9 \pm 1.0$$

$$\sigma = 47.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.18 \pm 0.21 \%$$

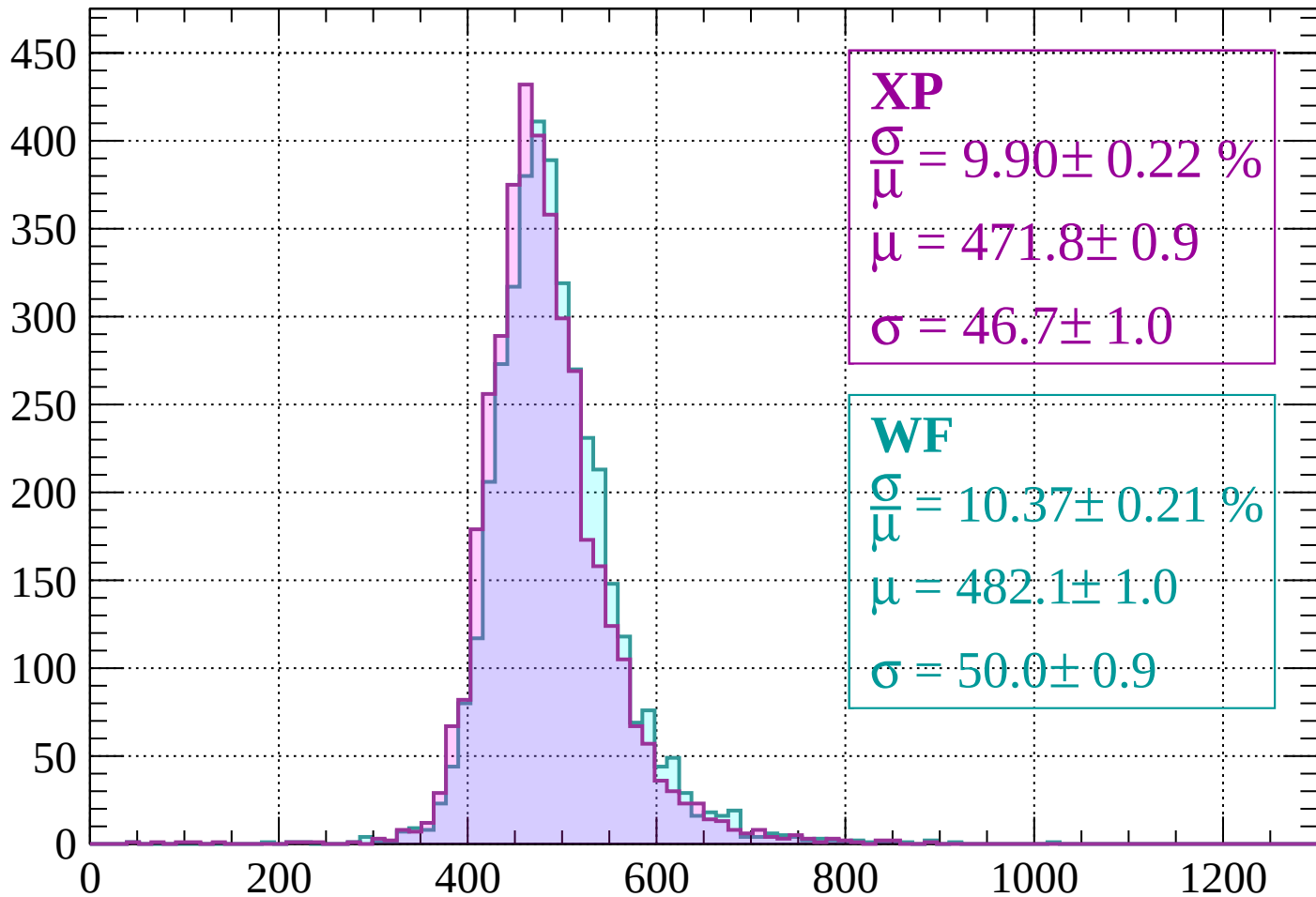
$$\mu = 480.2 \pm 1.0$$

$$\sigma = 48.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 9.90 \pm 0.22 \%$$

$$\mu = 471.8 \pm 0.9$$

$$\sigma = 46.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.37 \pm 0.21 \%$$

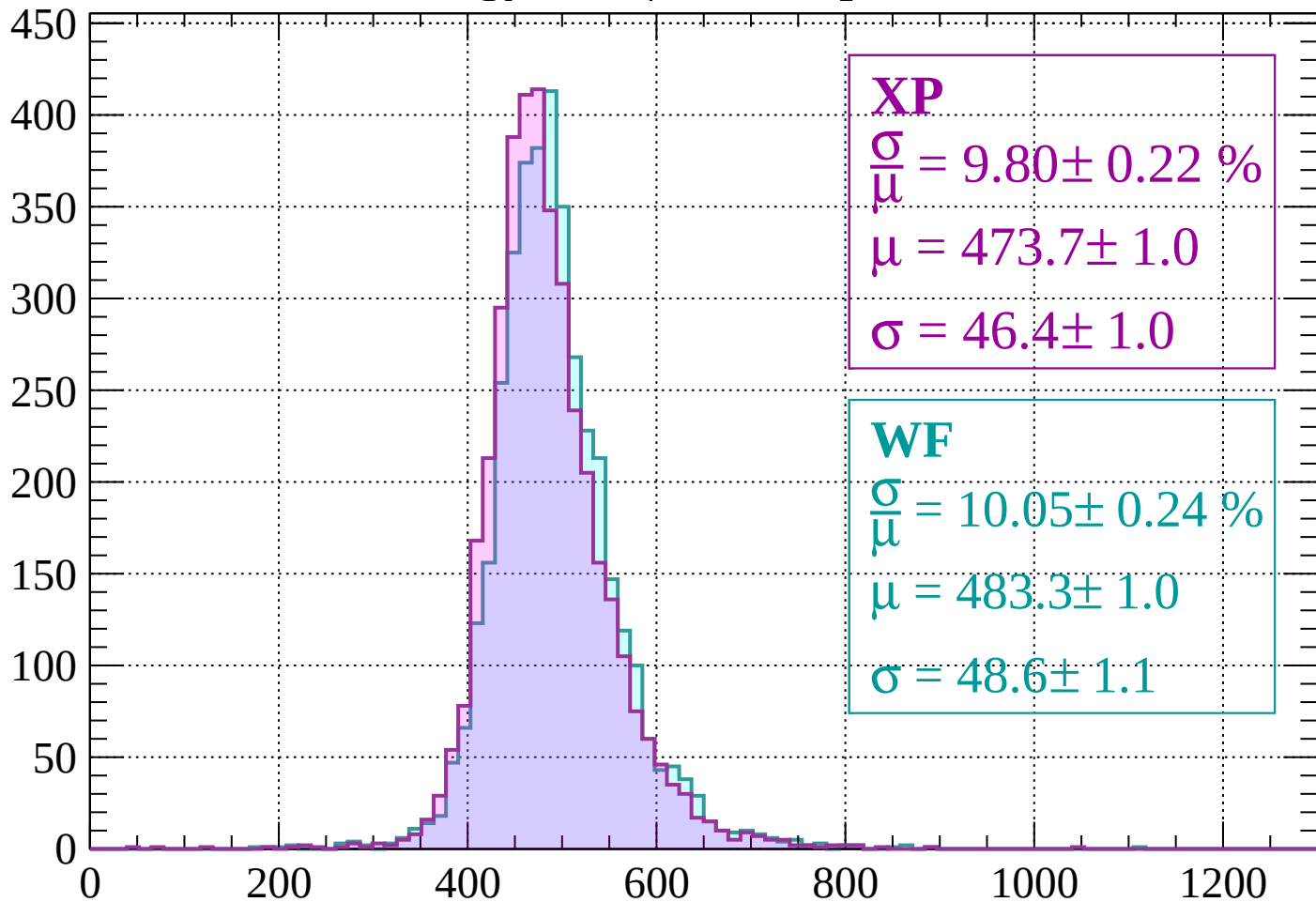
$$\mu = 482.1 \pm 1.0$$

$$\sigma = 50.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 9.80 \pm 0.22 \%$$

$$\mu = 473.7 \pm 1.0$$

$$\sigma = 46.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.05 \pm 0.24 \%$$

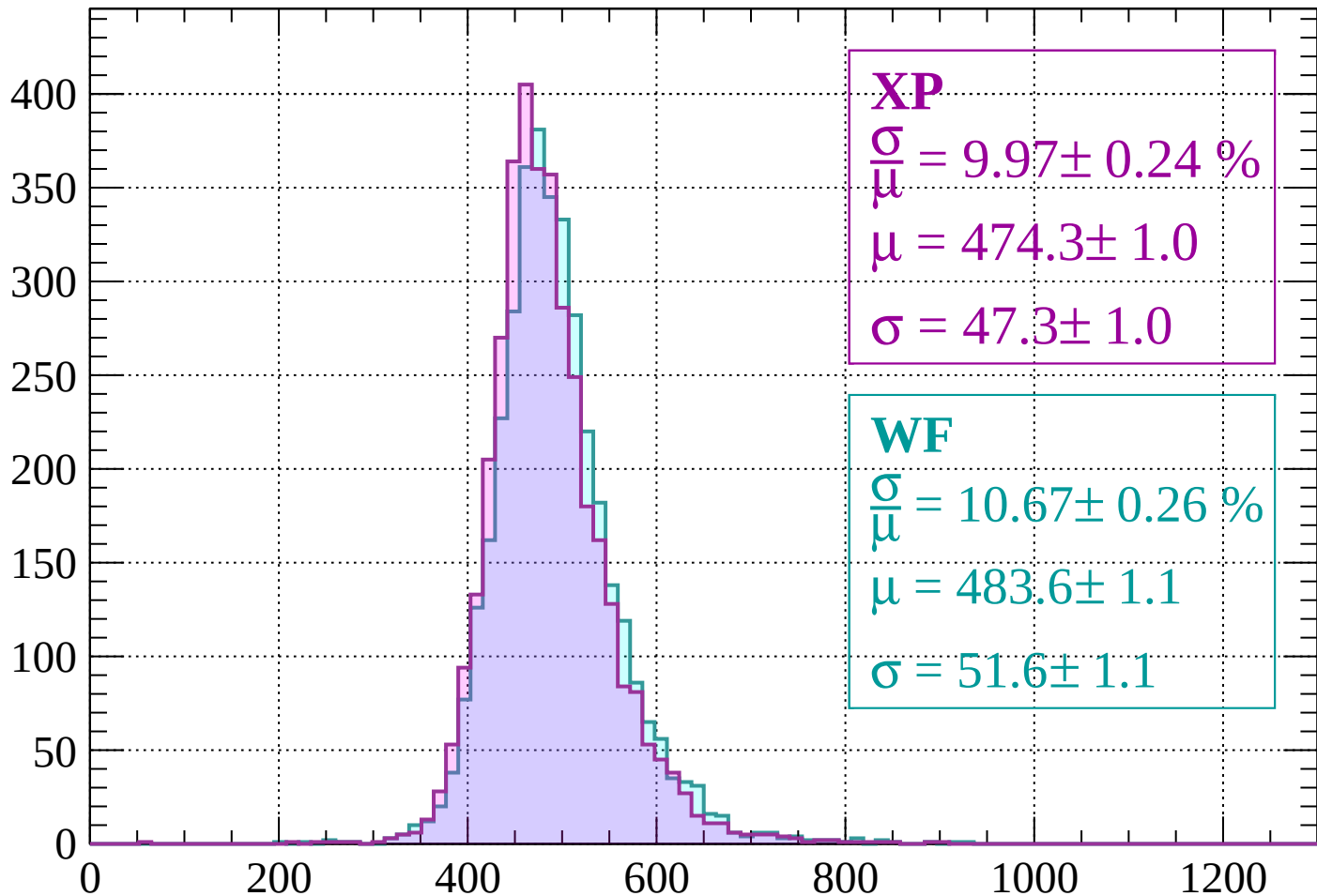
$$\mu = 483.3 \pm 1.0$$

$$\sigma = 48.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 9.97 \pm 0.24 \%$$

$$\mu = 474.3 \pm 1.0$$

$$\sigma = 47.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.67 \pm 0.26 \%$$

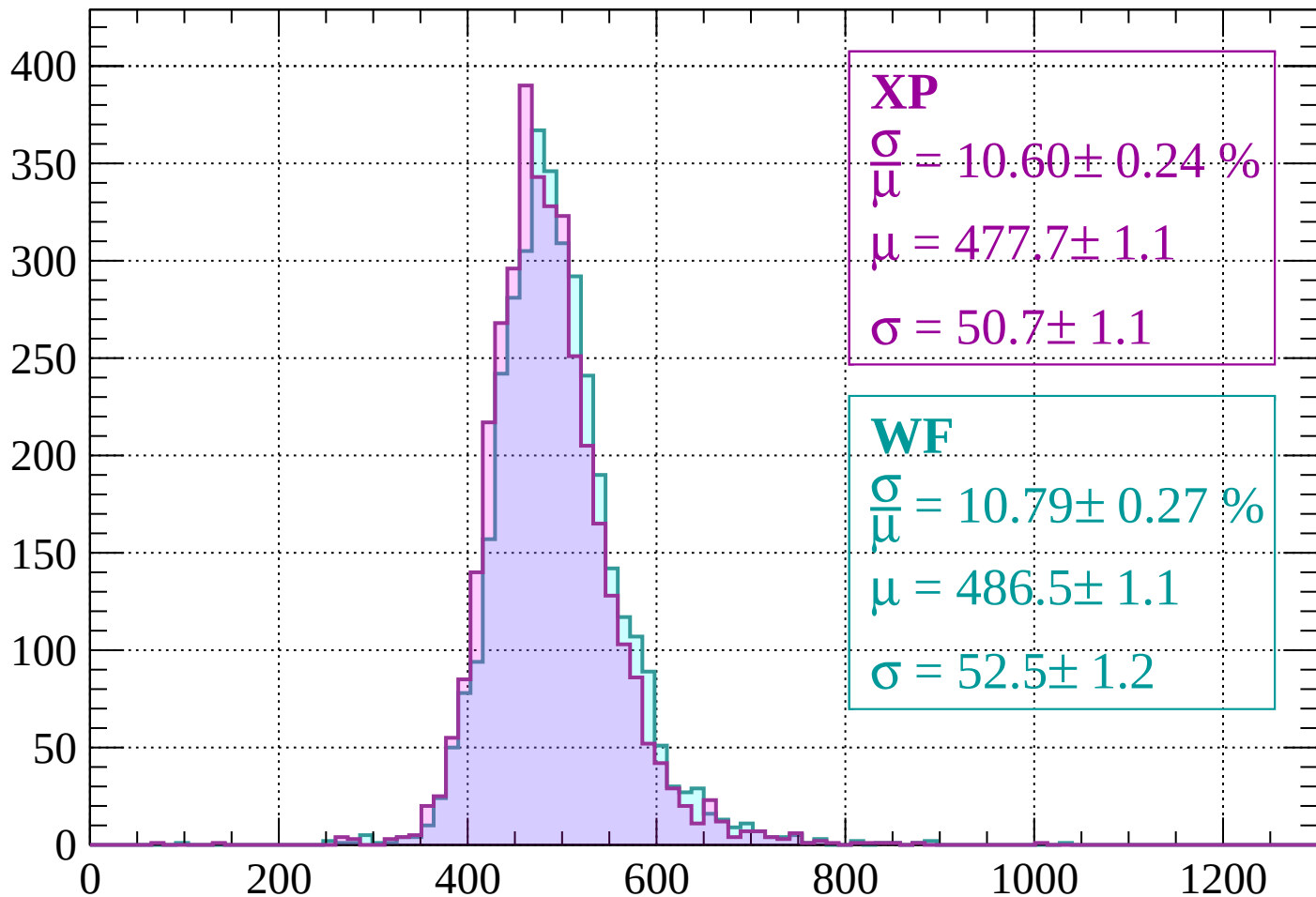
$$\mu = 483.6 \pm 1.1$$

$$\sigma = 51.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$

Count



XP

$$\frac{\sigma}{\mu} = 10.60 \pm 0.24 \%$$

$$\mu = 477.7 \pm 1.1$$

$$\sigma = 50.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.79 \pm 0.27 \%$$

$$\mu = 486.5 \pm 1.1$$

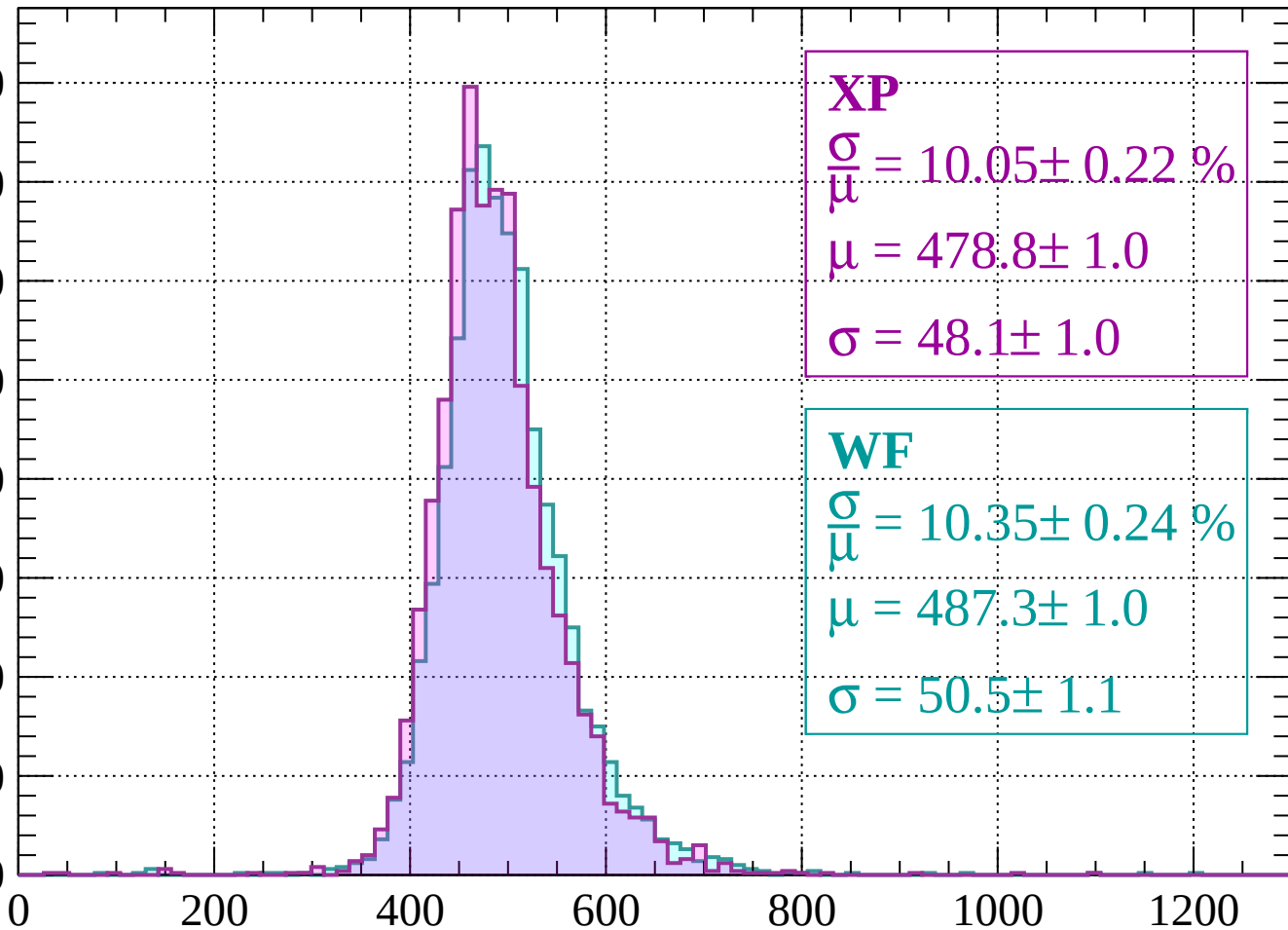
$$\sigma = 52.5 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 10.05 \pm 0.22 \%$$

$$\mu = 478.8 \pm 1.0$$

$$\sigma = 48.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.35 \pm 0.24 \%$$

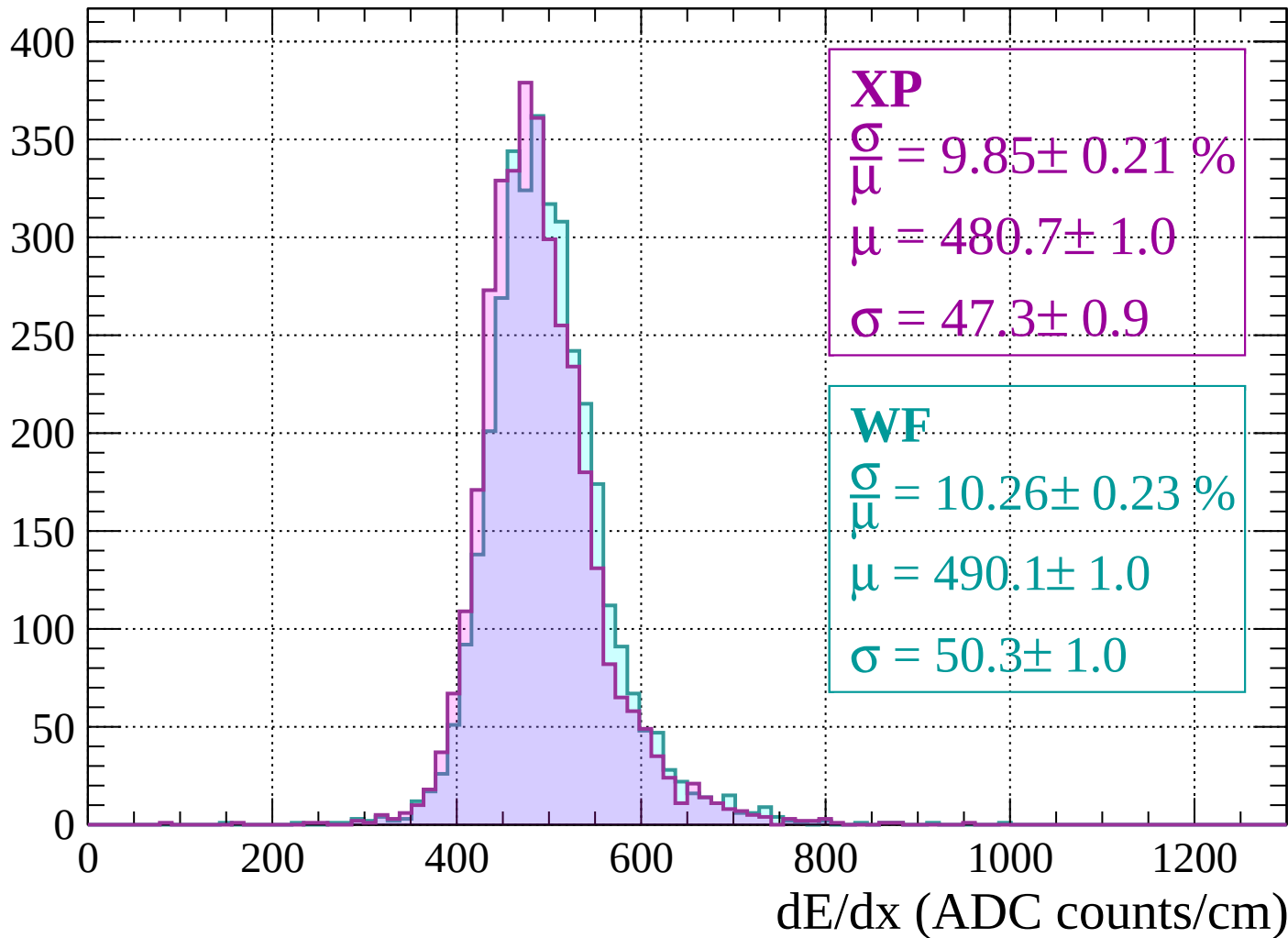
$$\mu = 487.3 \pm 1.0$$

$$\sigma = 50.5 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$

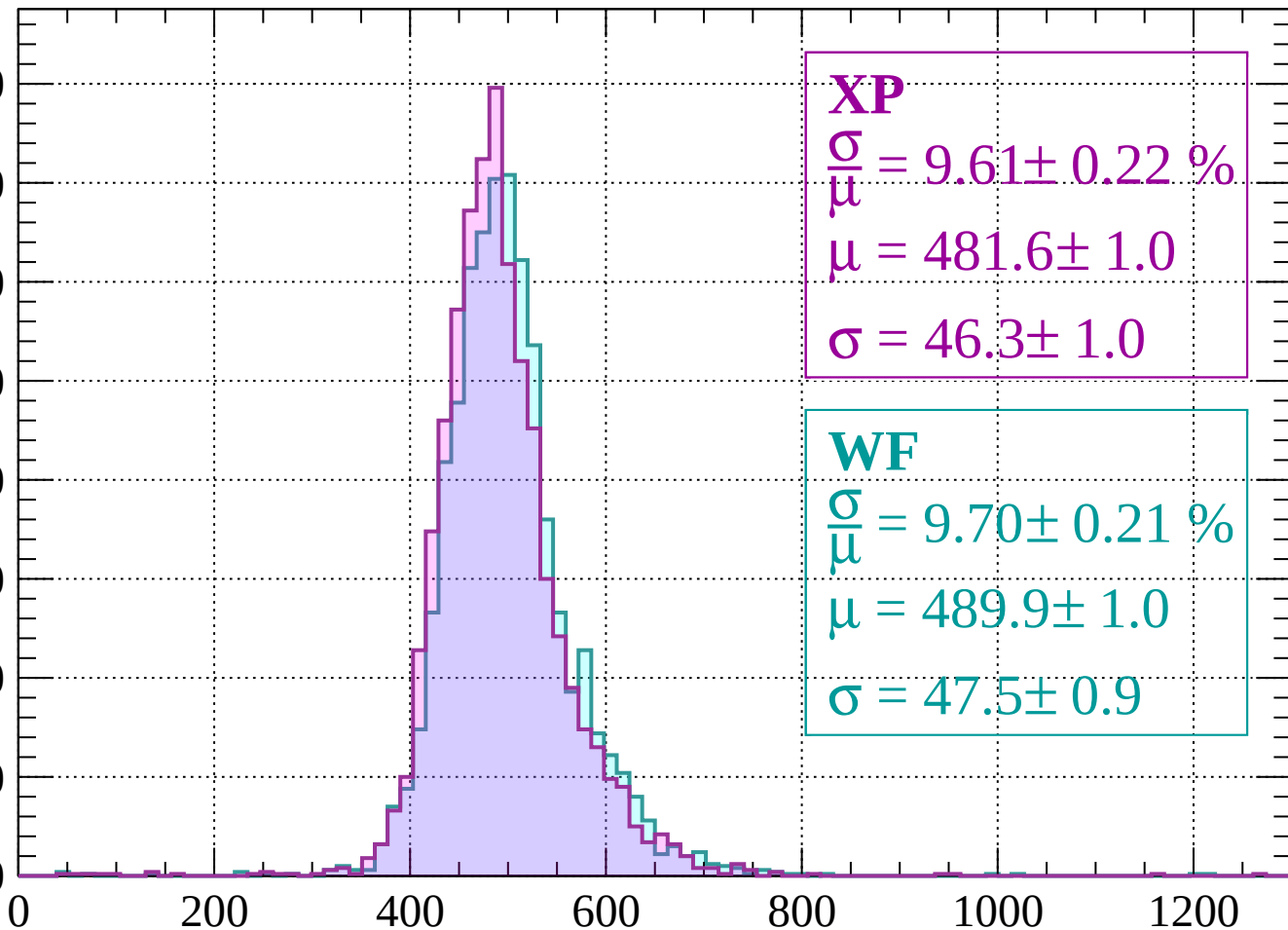
Count



Energy loss | $1960 < p < 2000$

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.61 \pm 0.22 \%$$

$$\mu = 481.6 \pm 1.0$$

$$\sigma = 46.3 \pm 1.0$$

WF

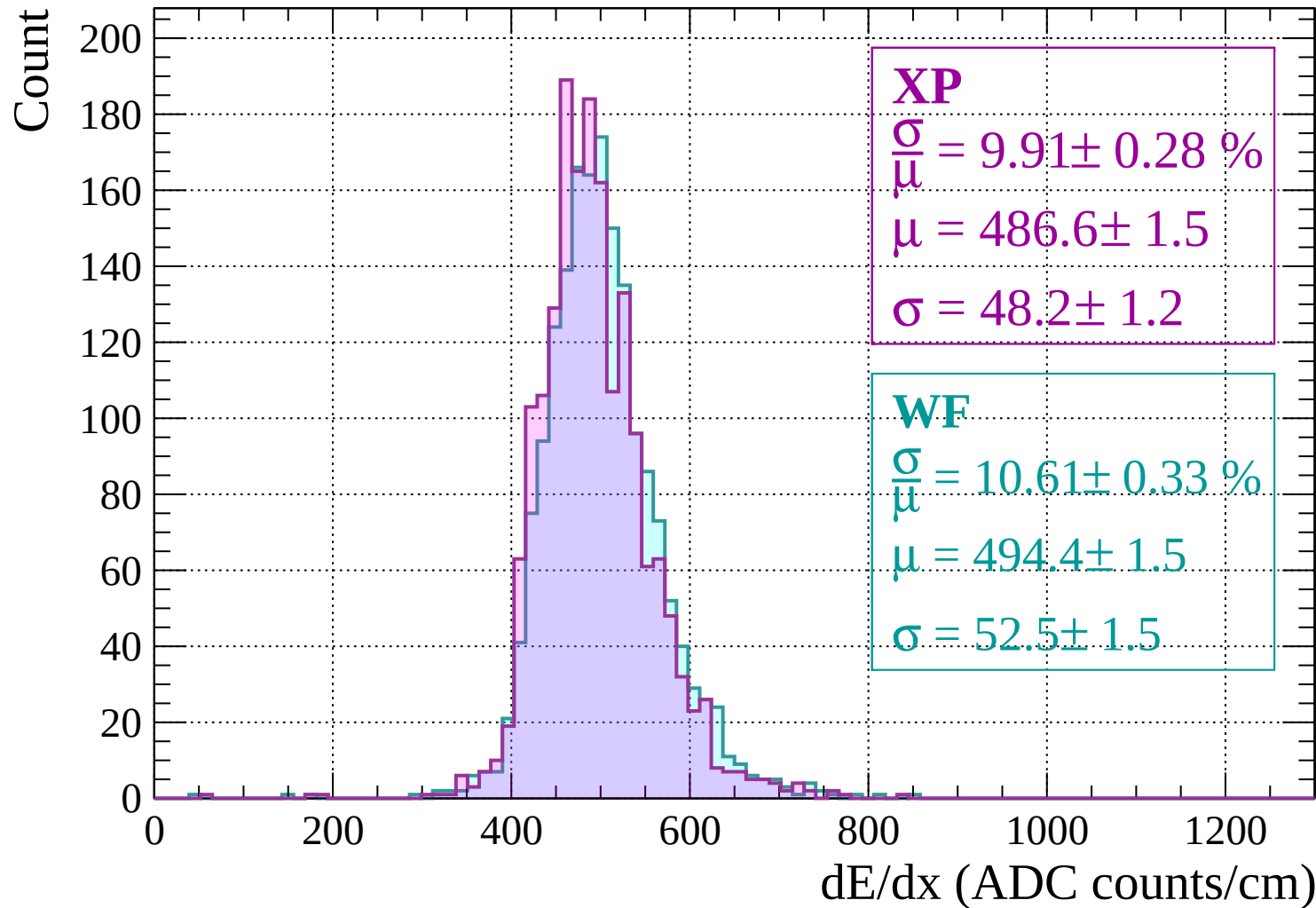
$$\frac{\sigma}{\mu} = 9.70 \pm 0.21 \%$$

$$\mu = 489.9 \pm 1.0$$

$$\sigma = 47.5 \pm 0.9$$

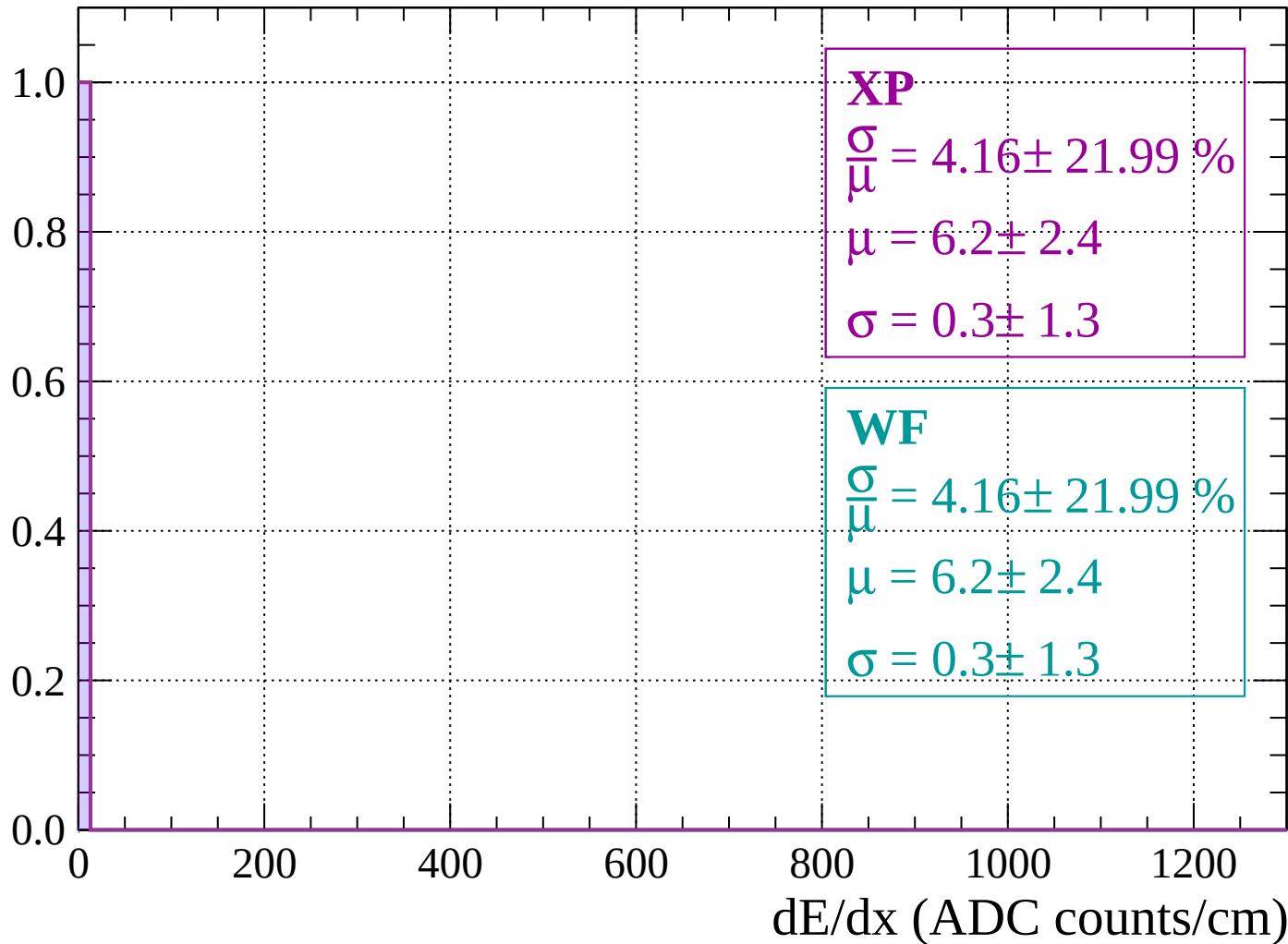
dE/dx (ADC counts/cm)

Energy loss | $2000 < p < 2040$



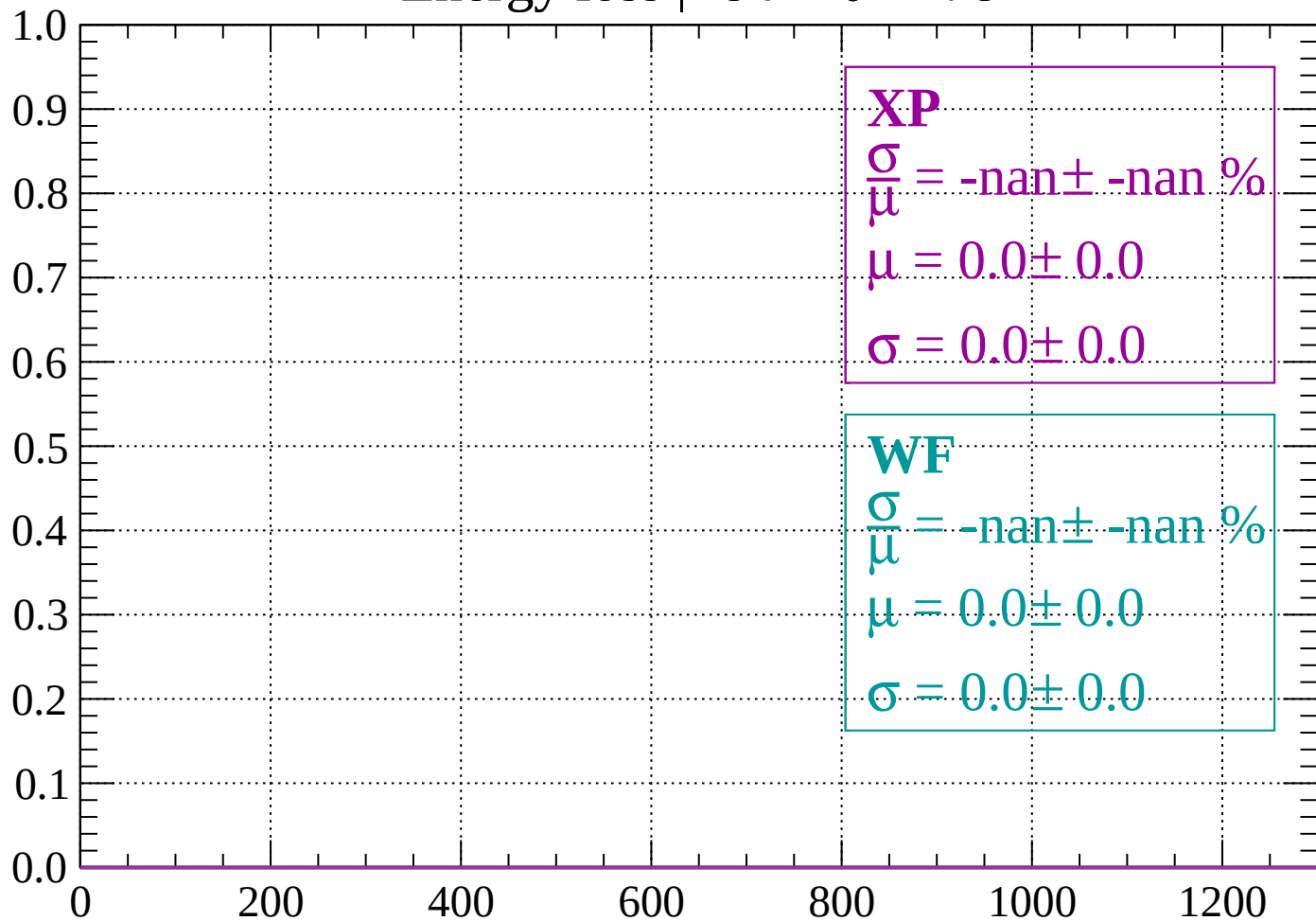
Energy loss | $-90 < \theta < -84$

Count



Energy loss | $-84 < \theta < -78$

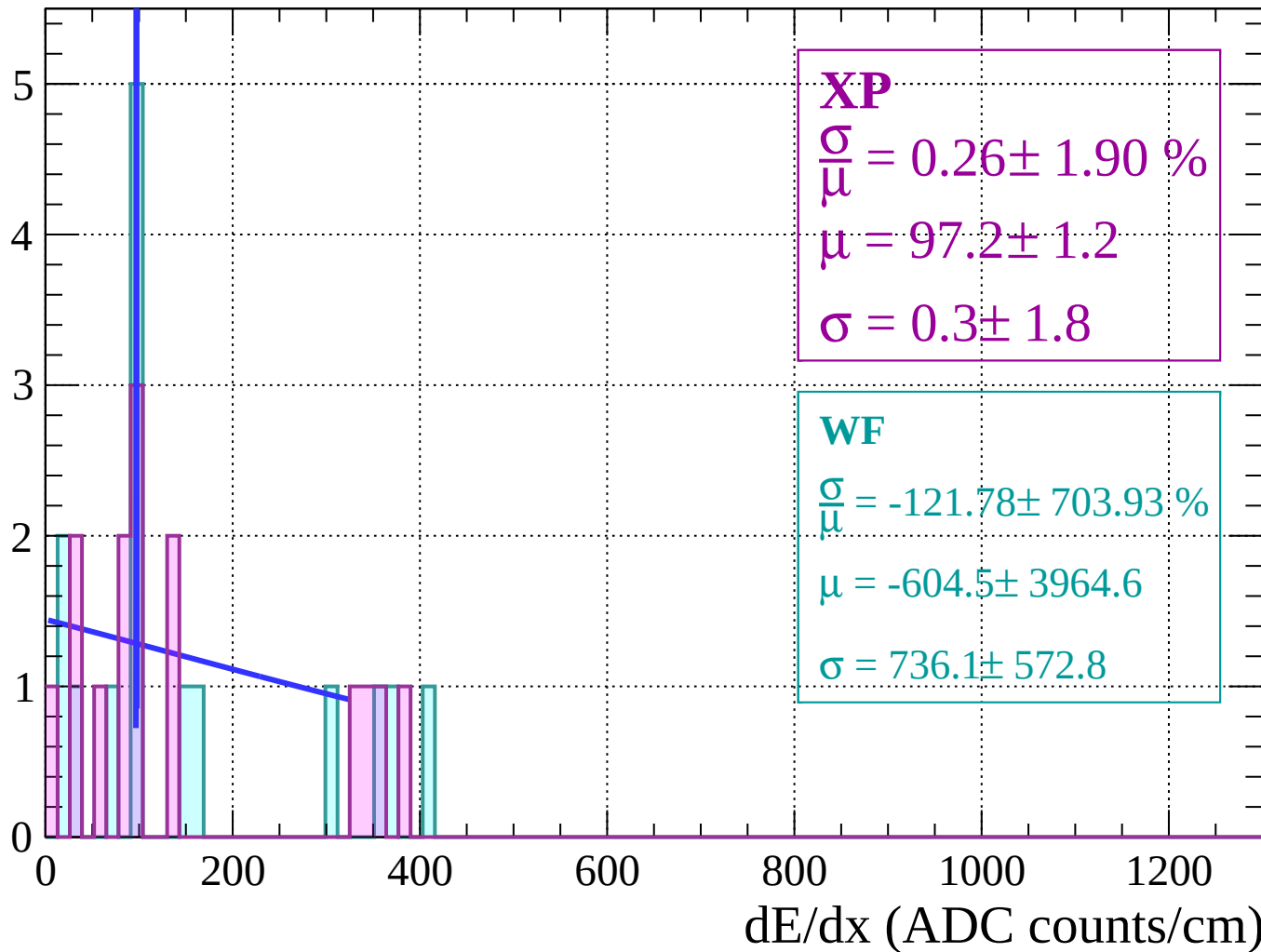
Count



dE/dx (ADC counts/cm)

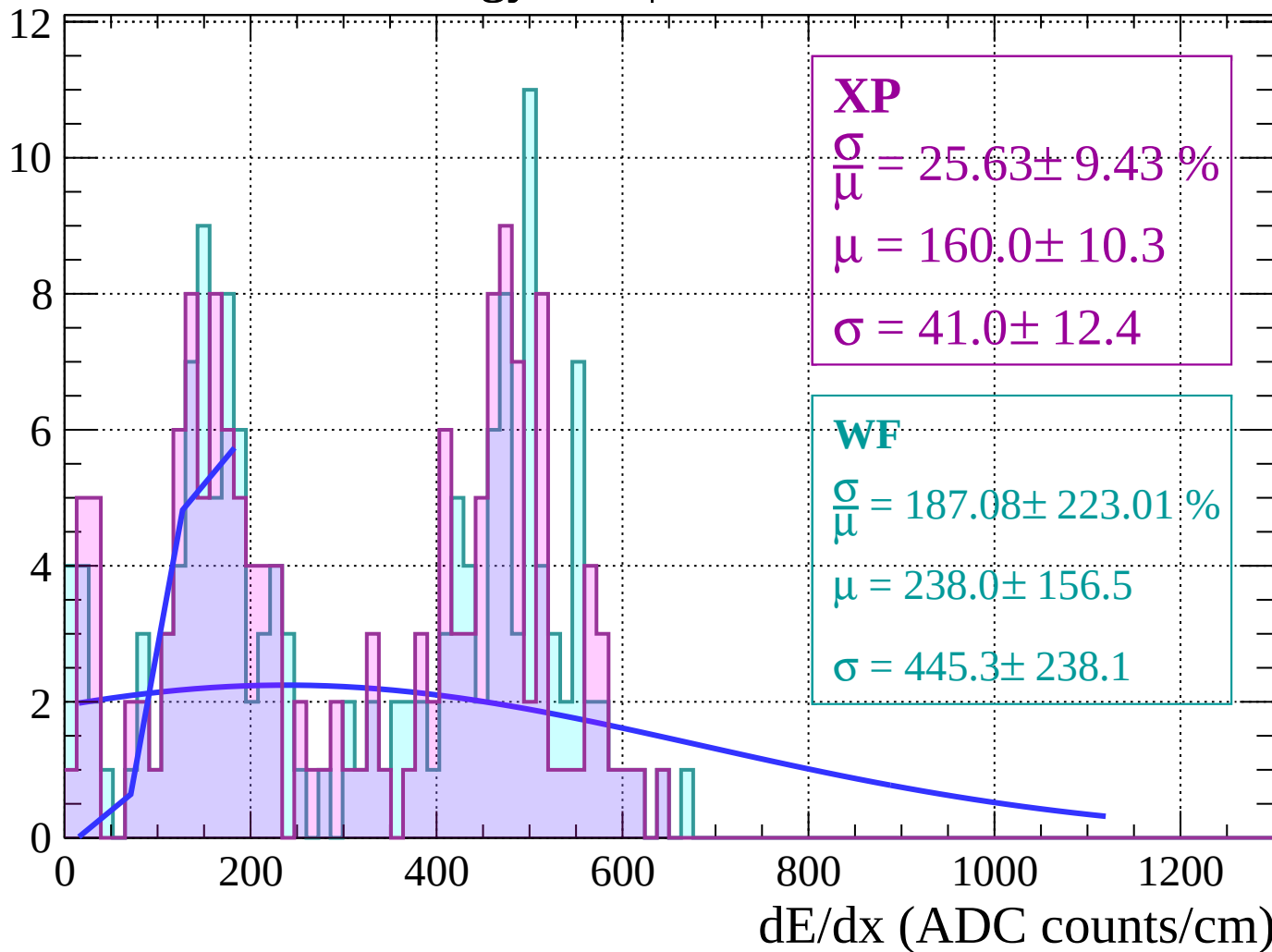
Energy loss | $-78 < \theta < -72$

Count



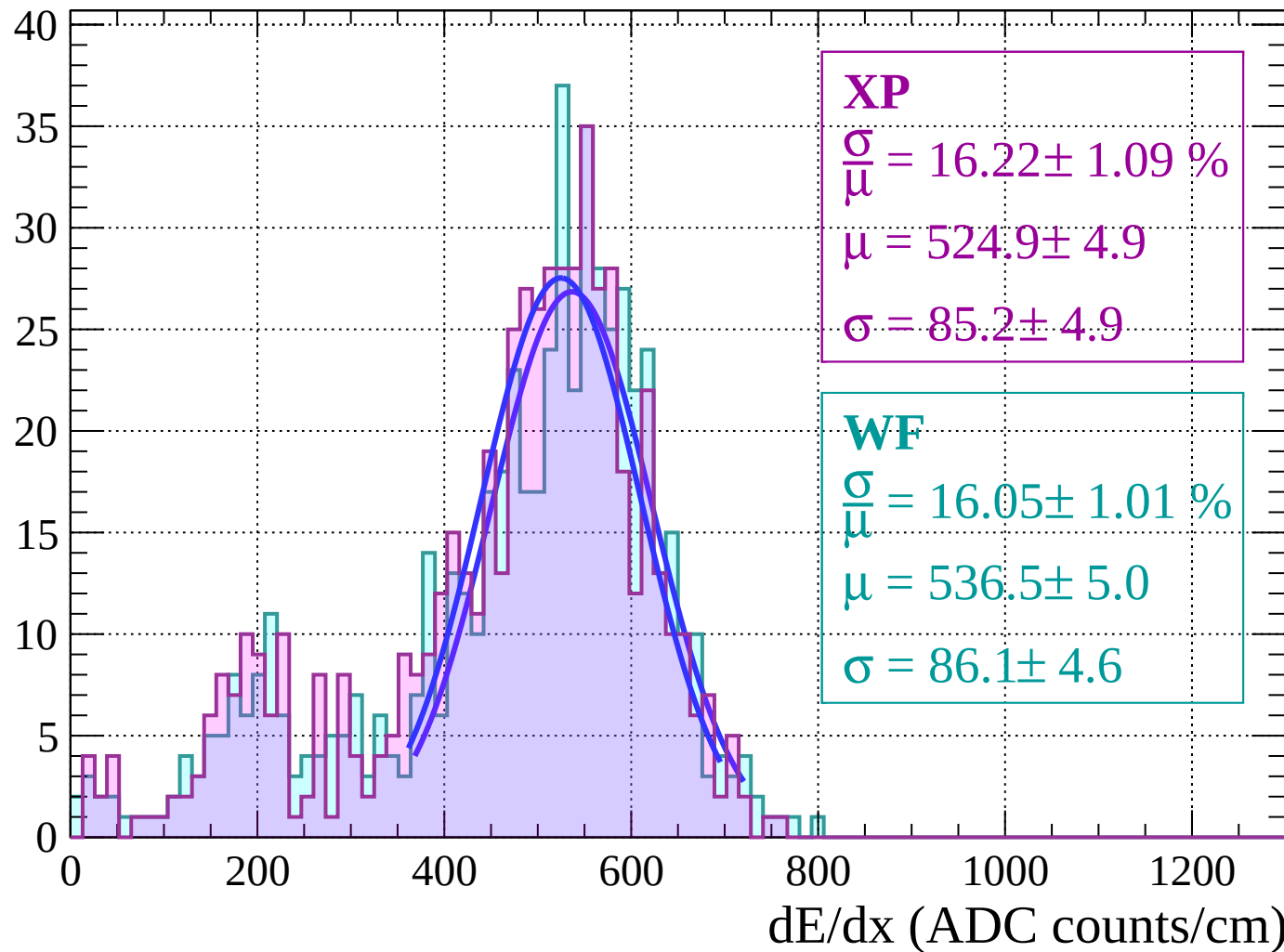
Energy loss | $-72 < \theta < -66$

Count



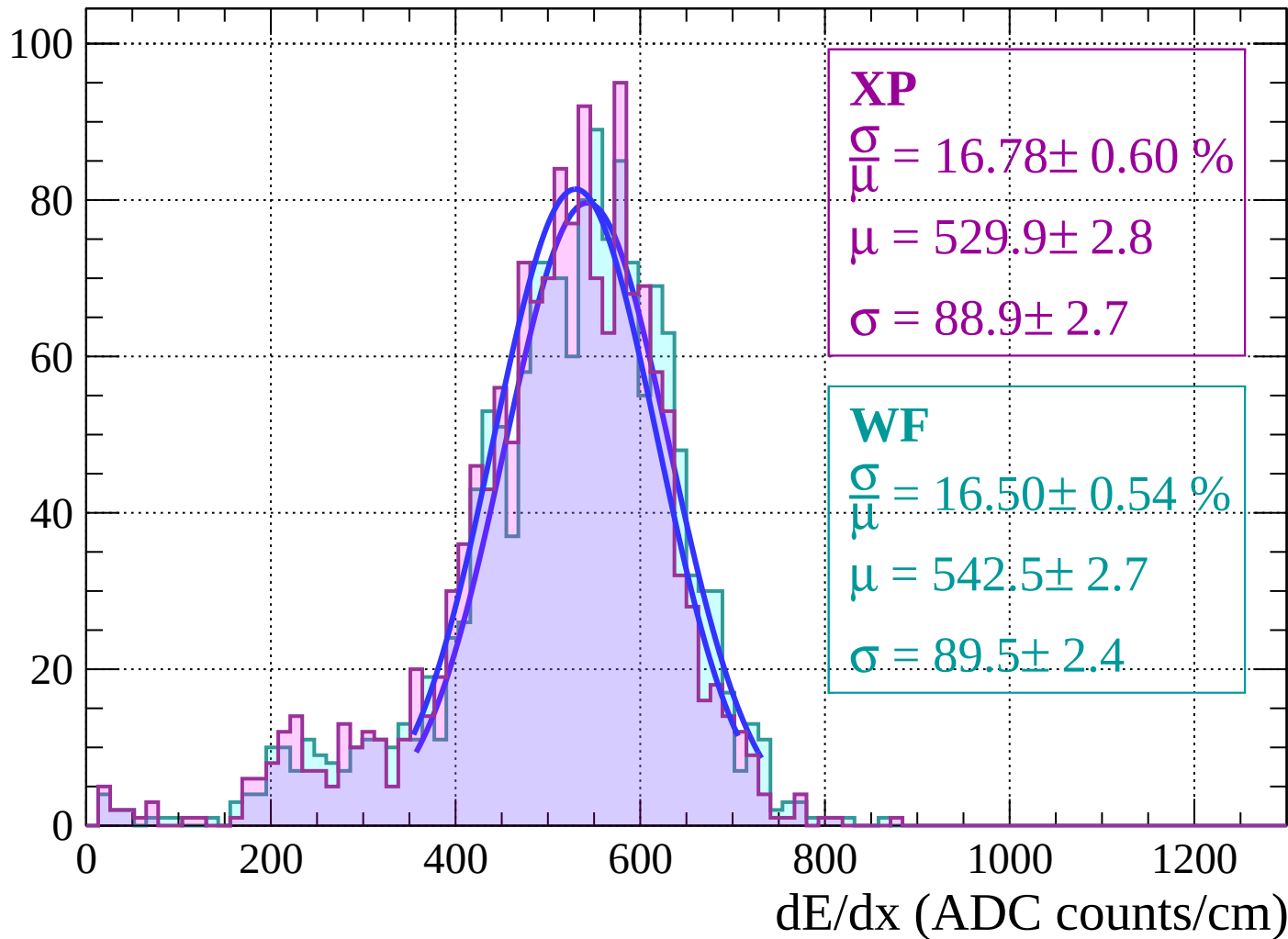
Energy loss | $-66 < \theta < -60$

Count



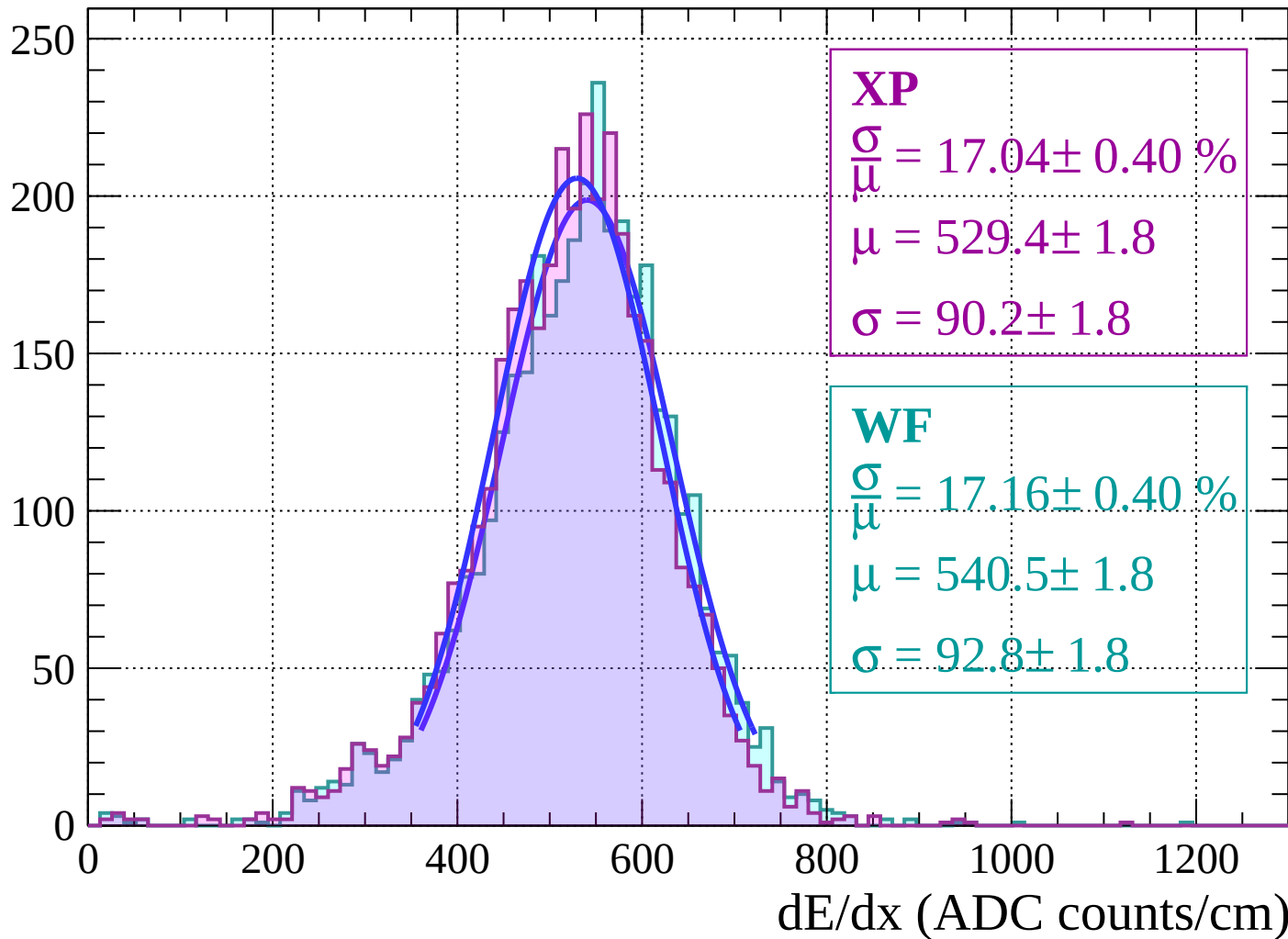
Energy loss | $-60 < \theta < -54$

Count

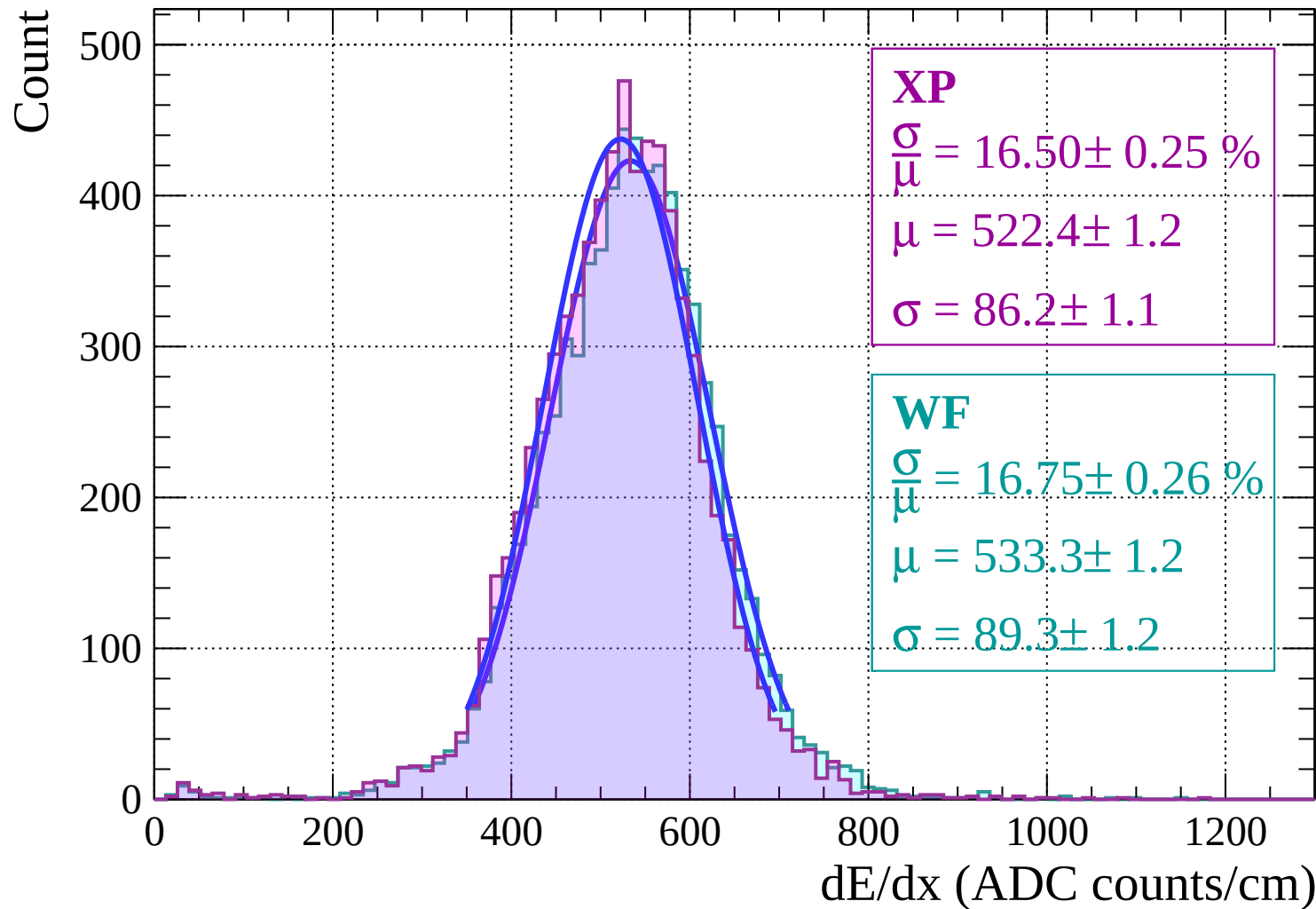


Energy loss | $-54 < \theta < -48$

Count

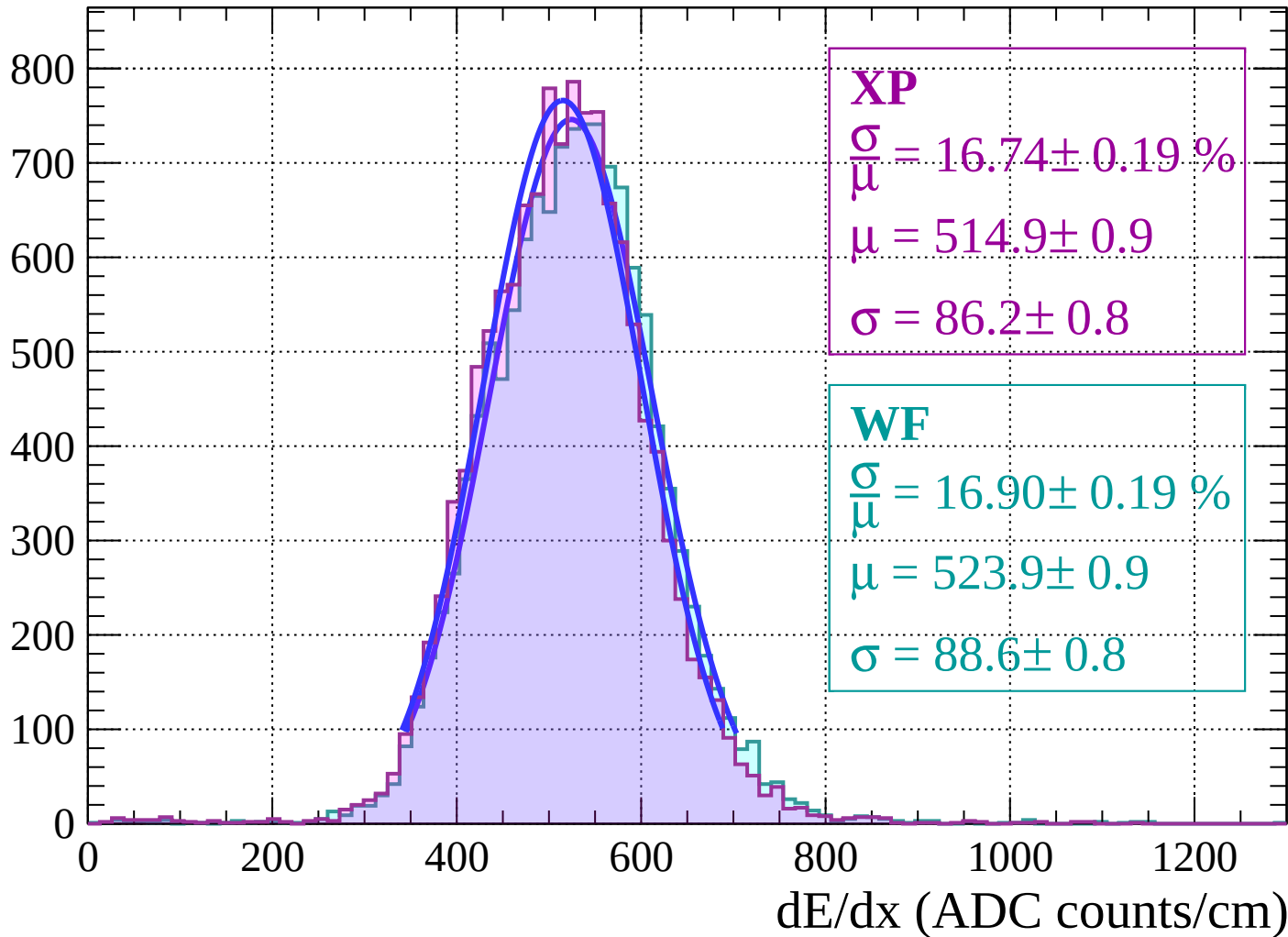


Energy loss | $-48 < \theta < -42$

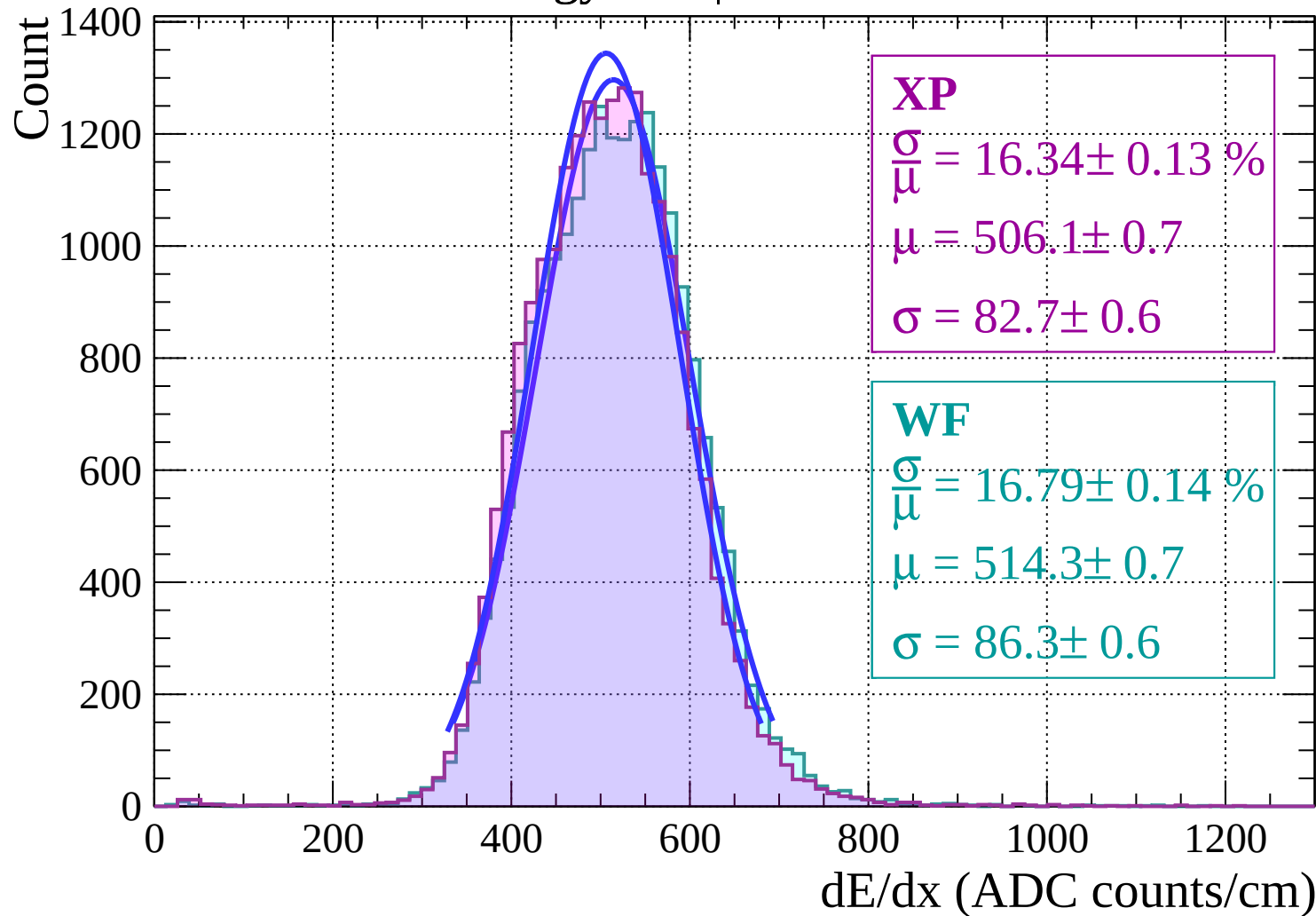


Energy loss $|-42 < \theta < -36$

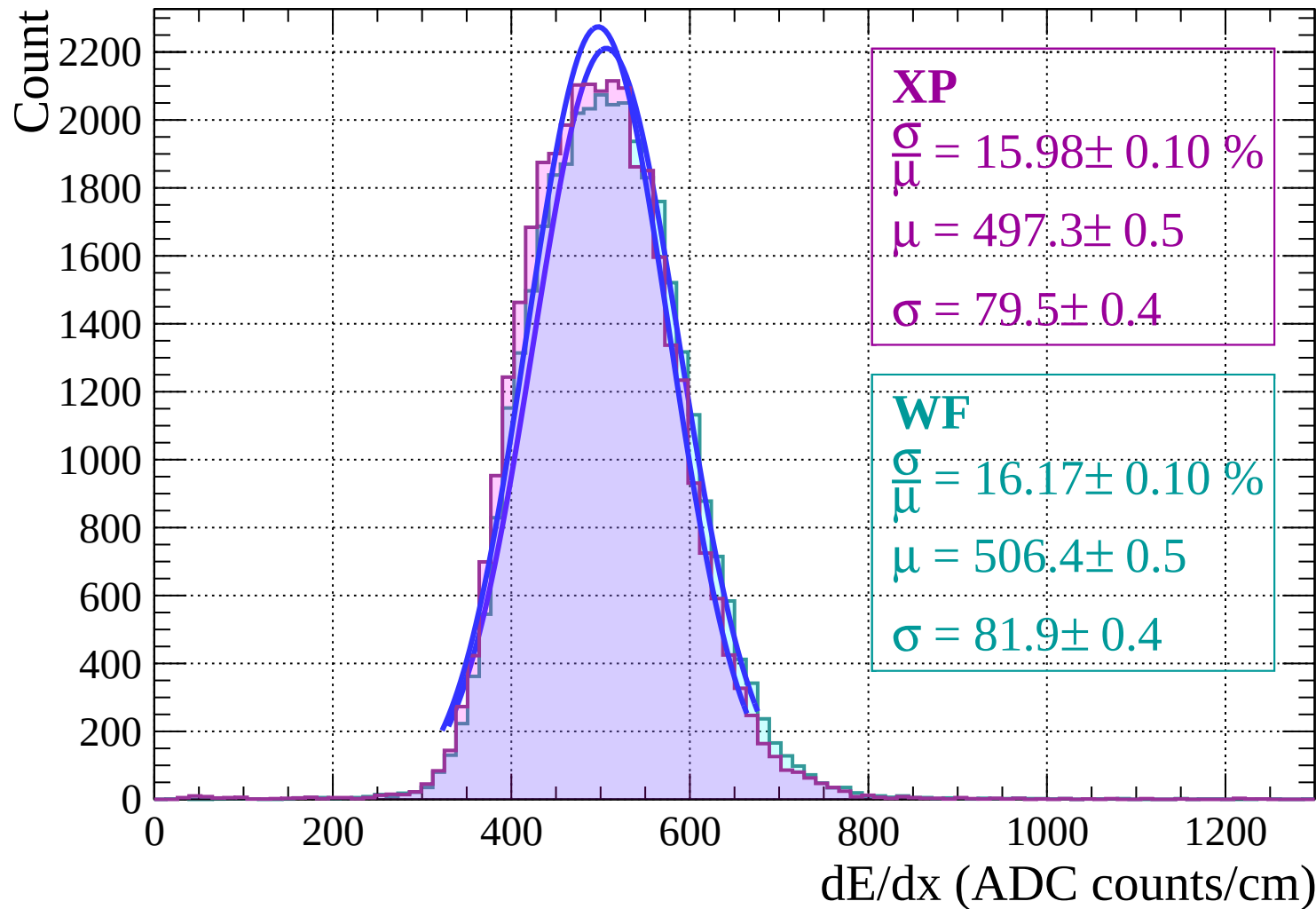
Count



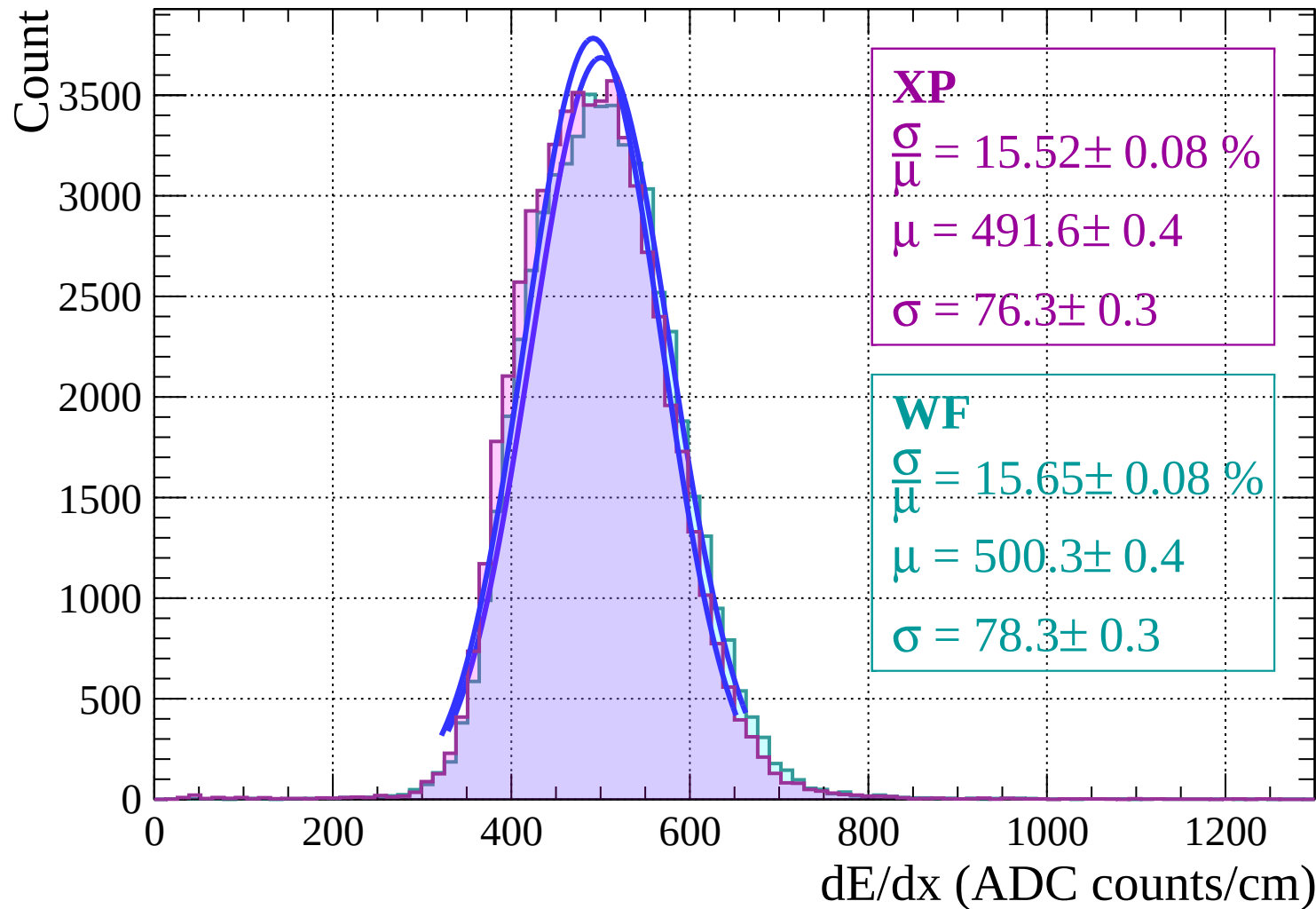
Energy loss | $-36 < \theta < -30$



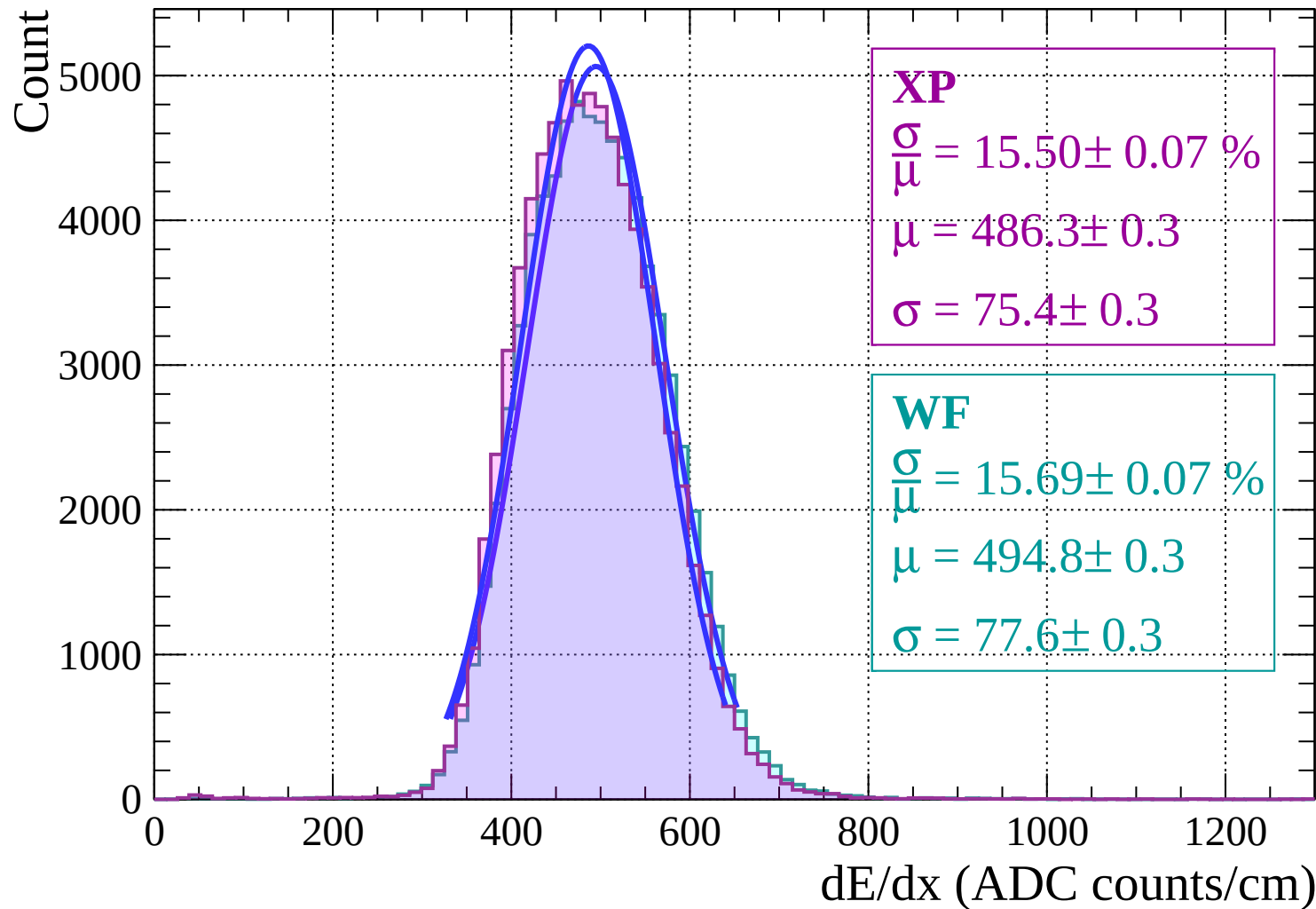
Energy loss | $-30 < \theta < -24$



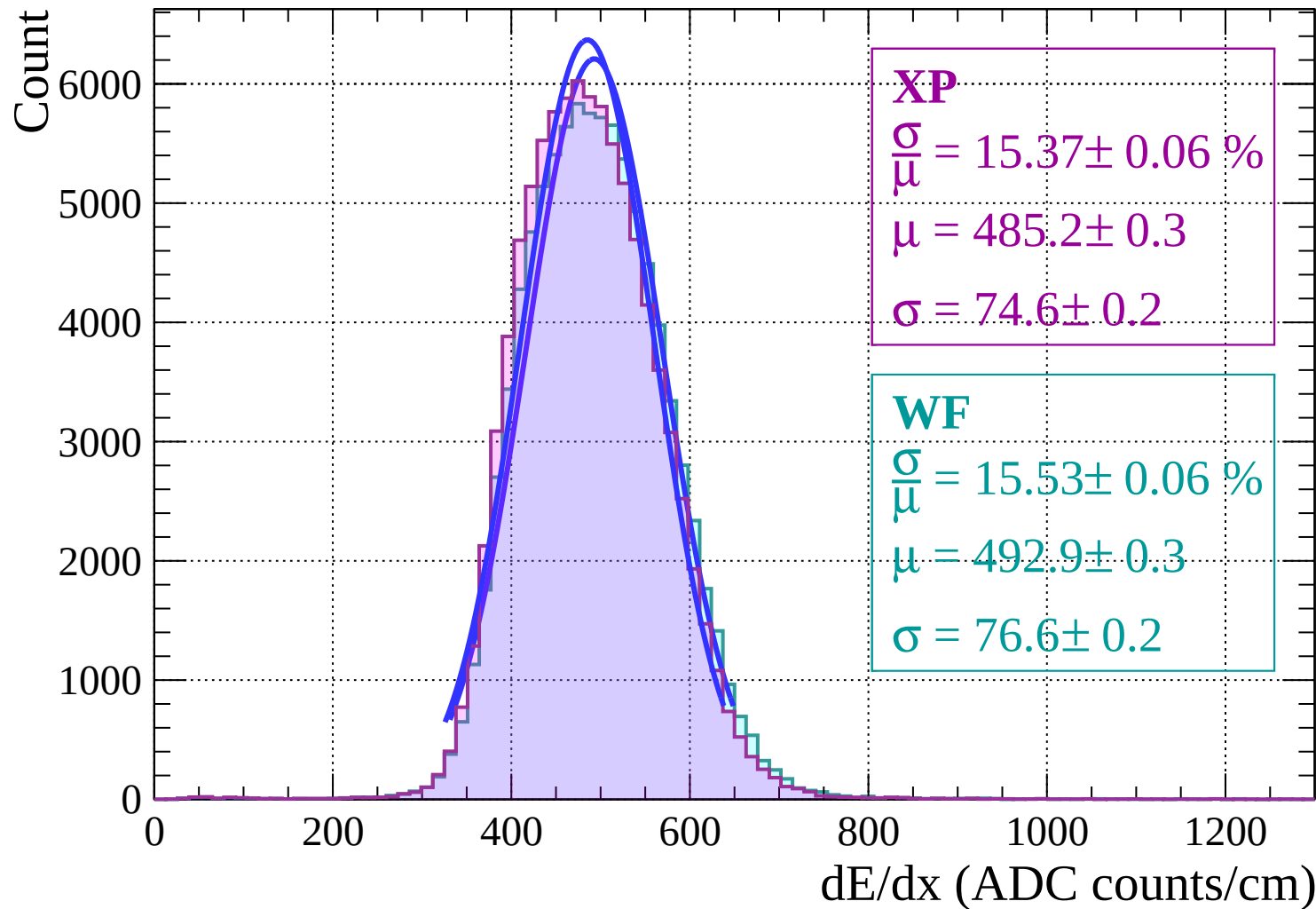
Energy loss | $-24 < \theta < -18$



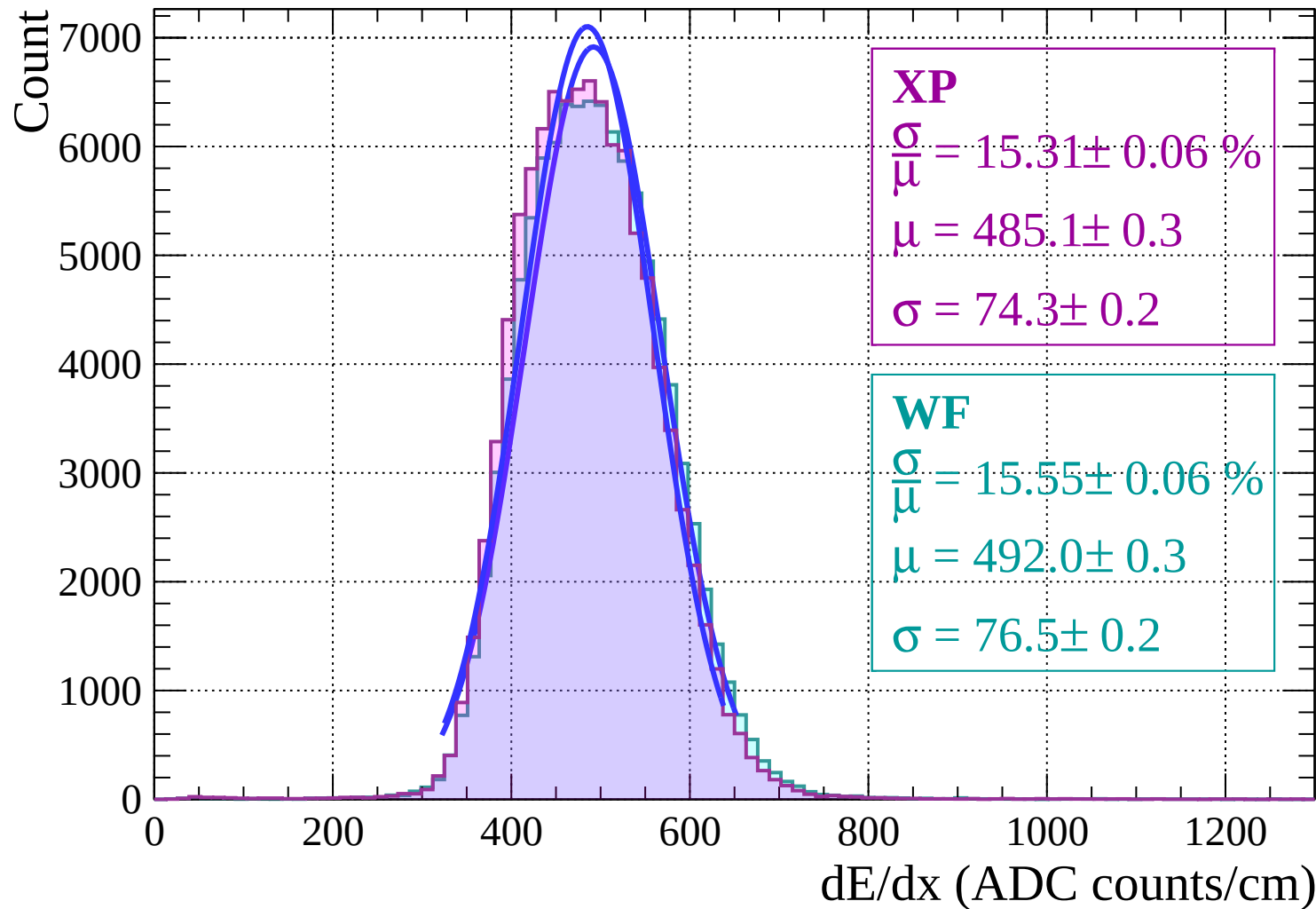
Energy loss | $-18 < \theta < -12$



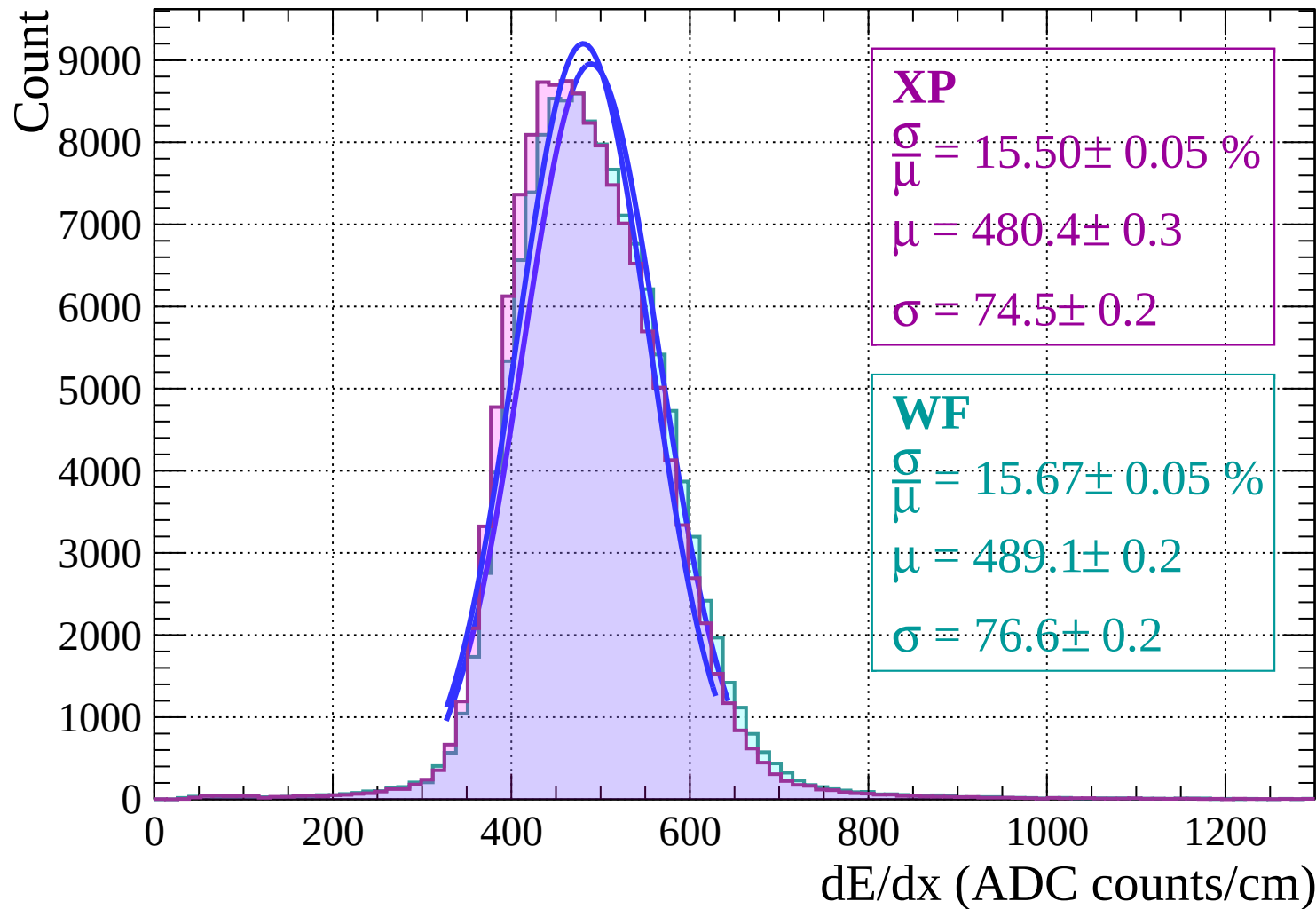
Energy loss | $-12 < \theta < -6$



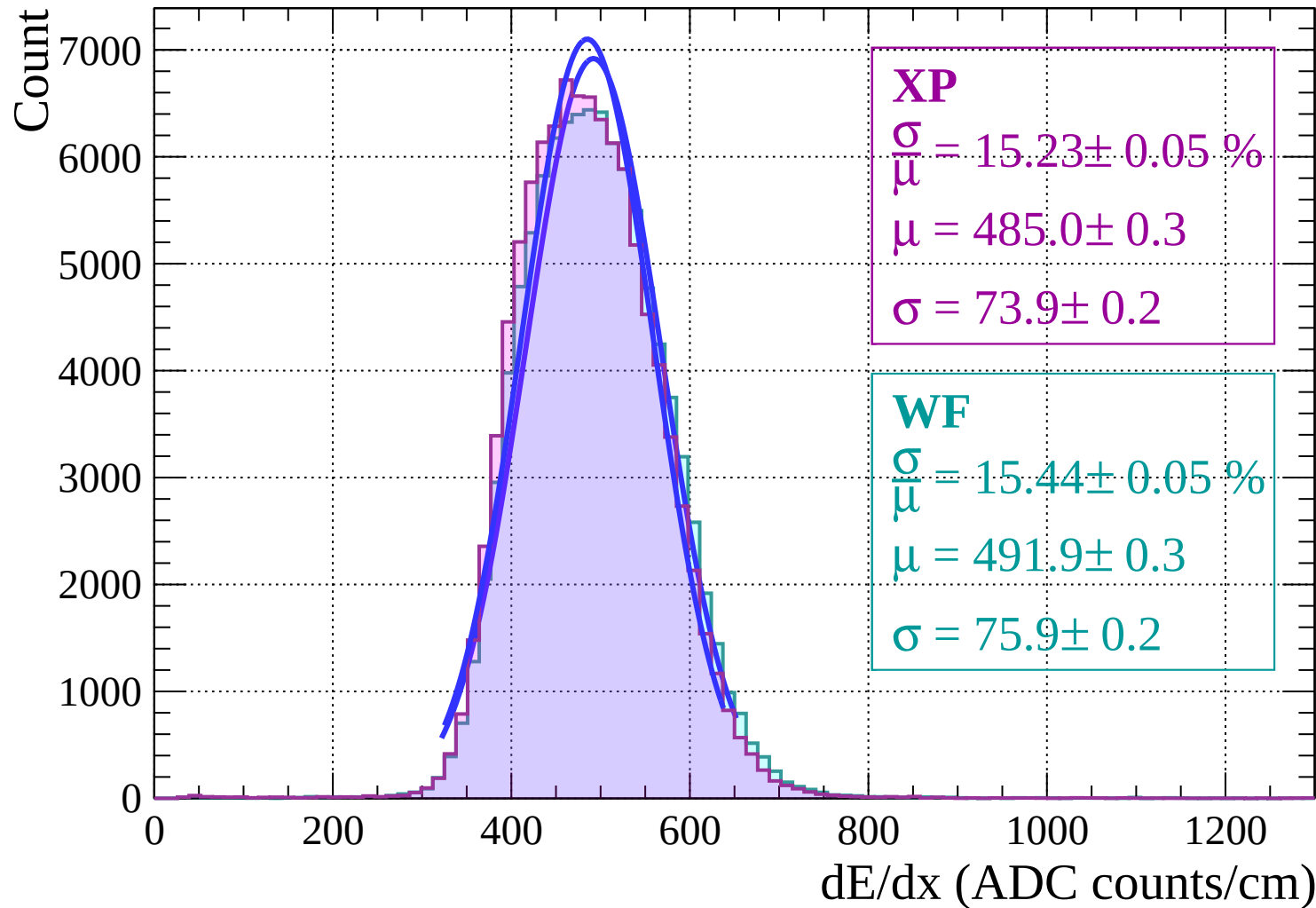
Energy loss | $-6 < \theta < 0$



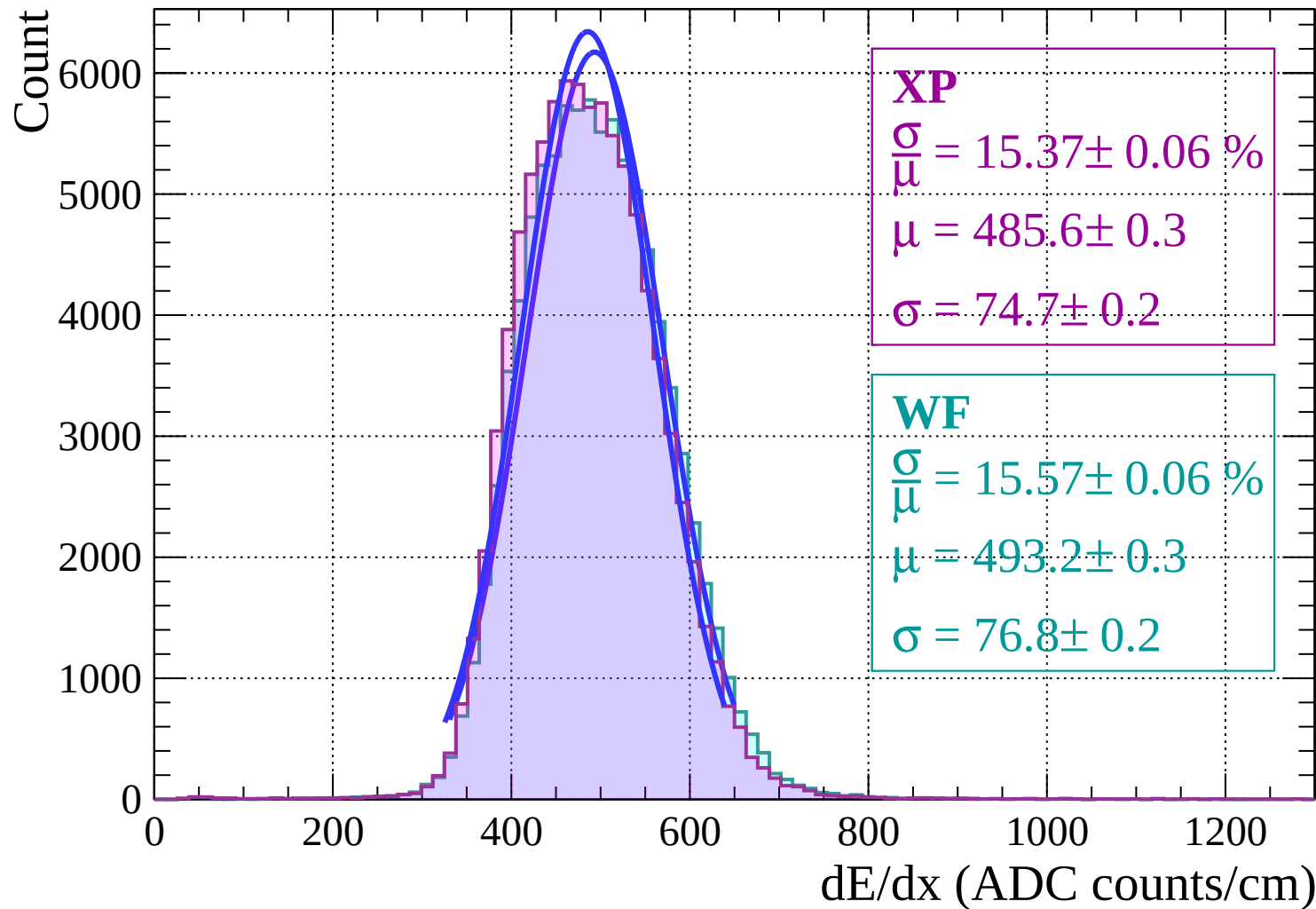
Energy loss | $0 < \theta < 6$



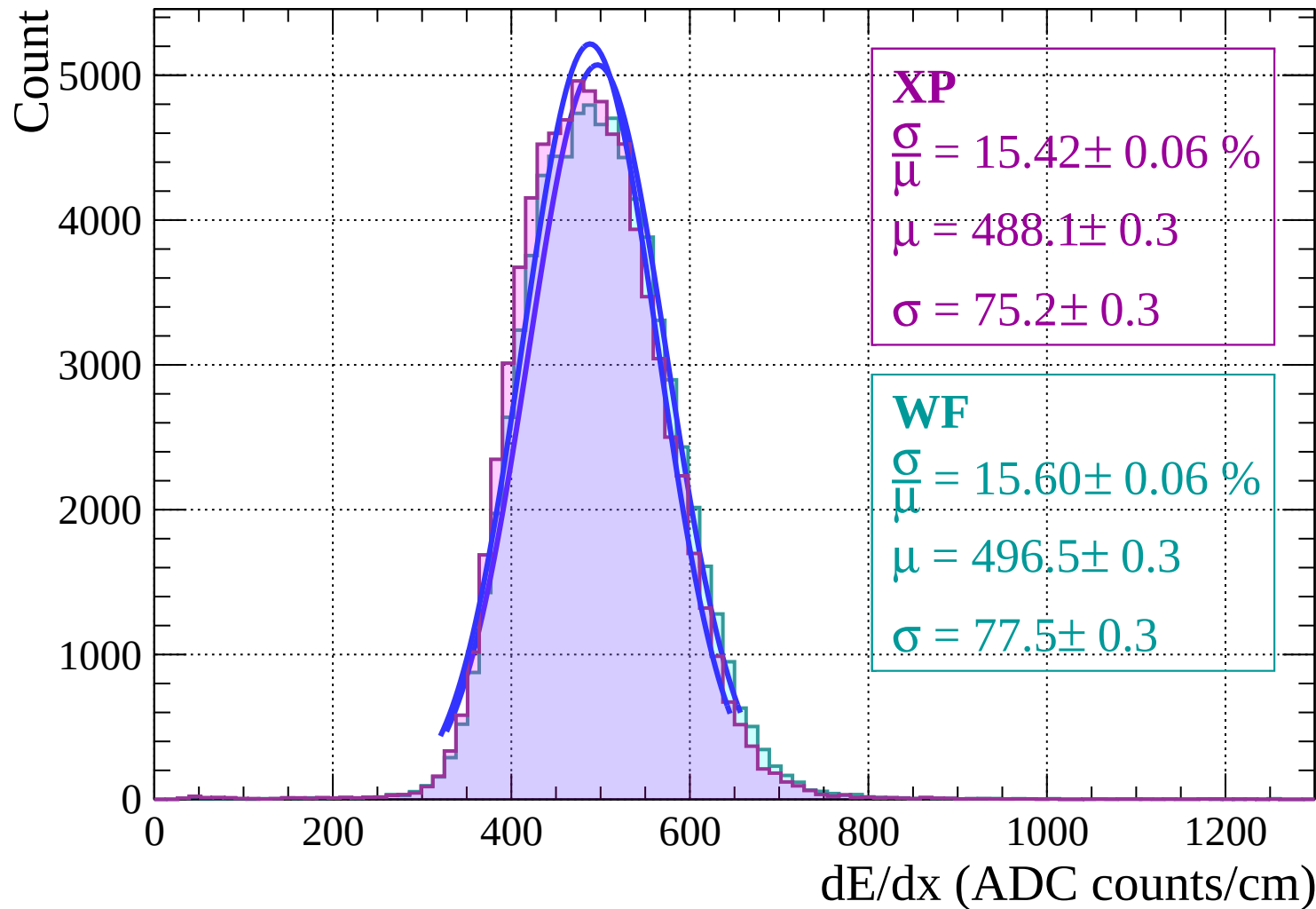
Energy loss | $6 < \theta < 12$



Energy loss | $12 < \theta < 18$

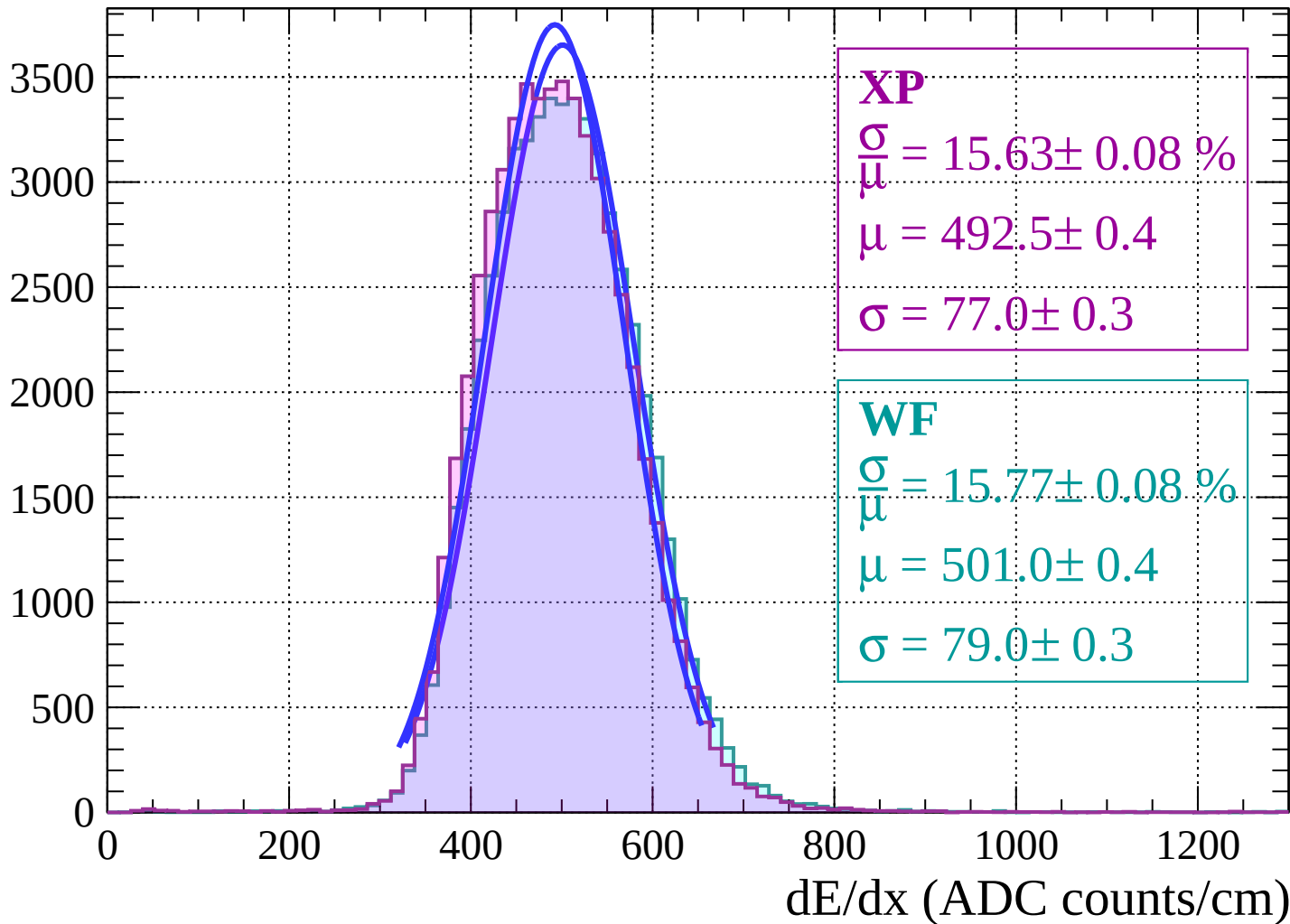


Energy loss | $18 < \theta < 24$

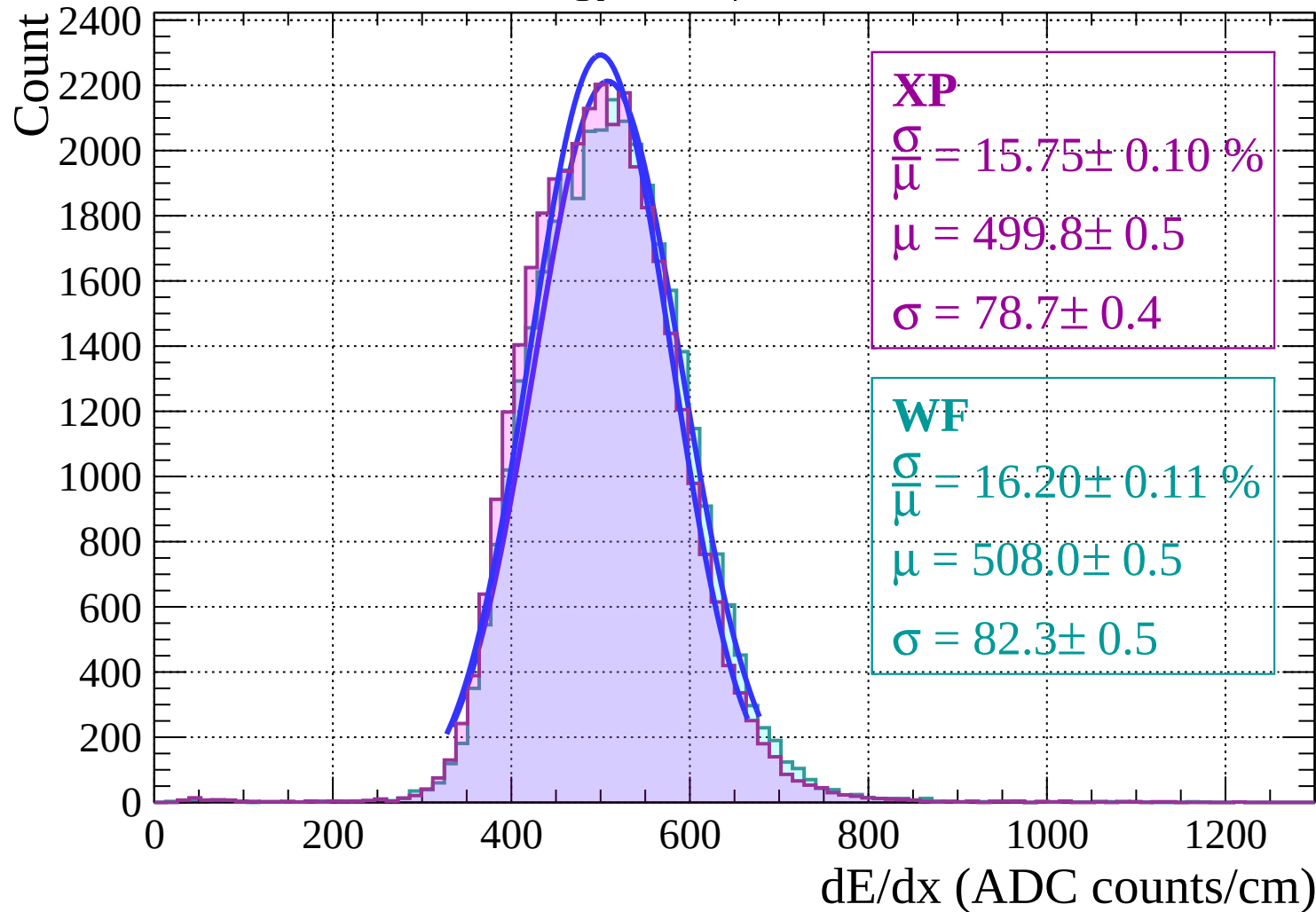


Energy loss | $24 < \theta < 30$

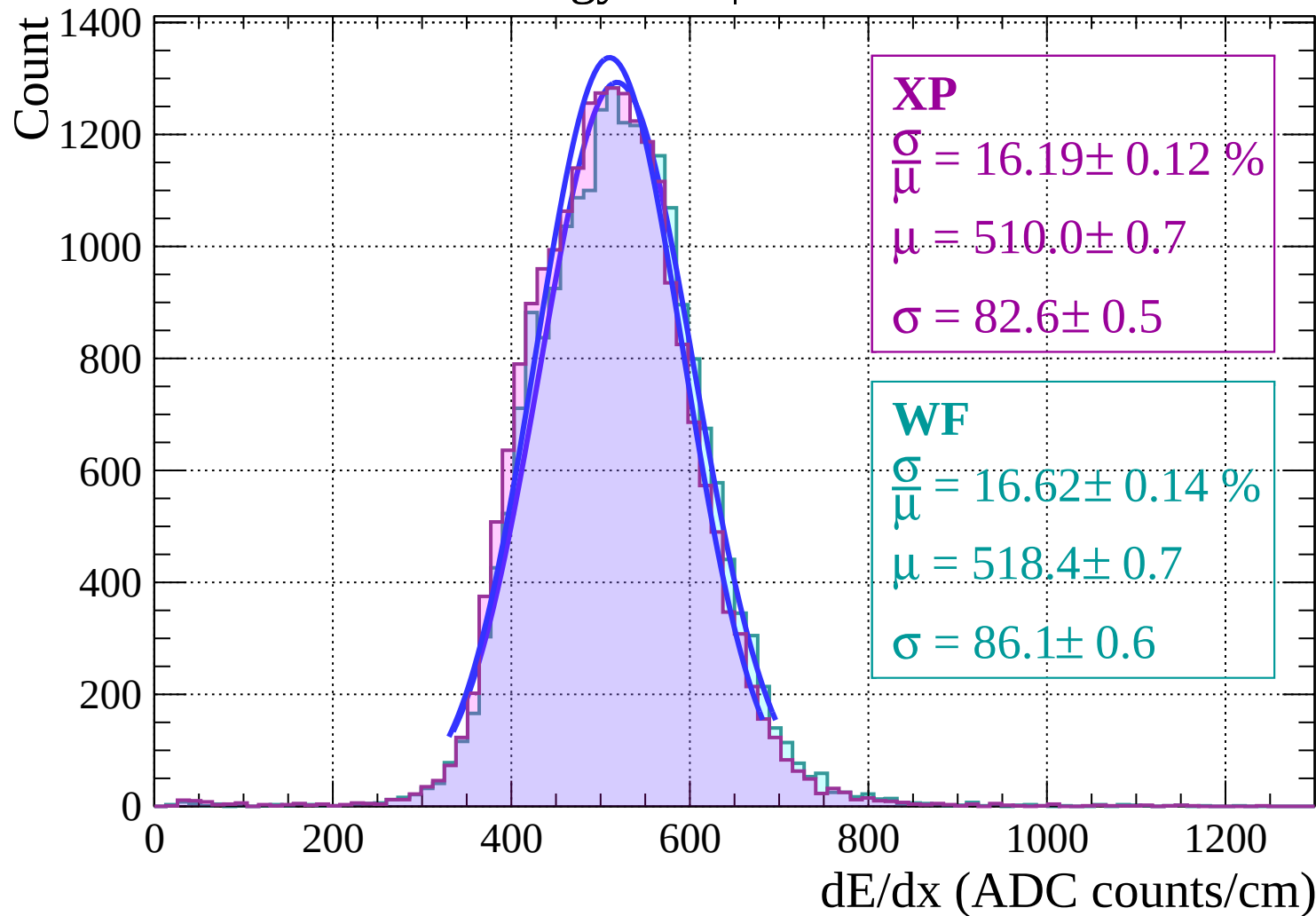
Count



Energy loss | $30 < \theta < 36$

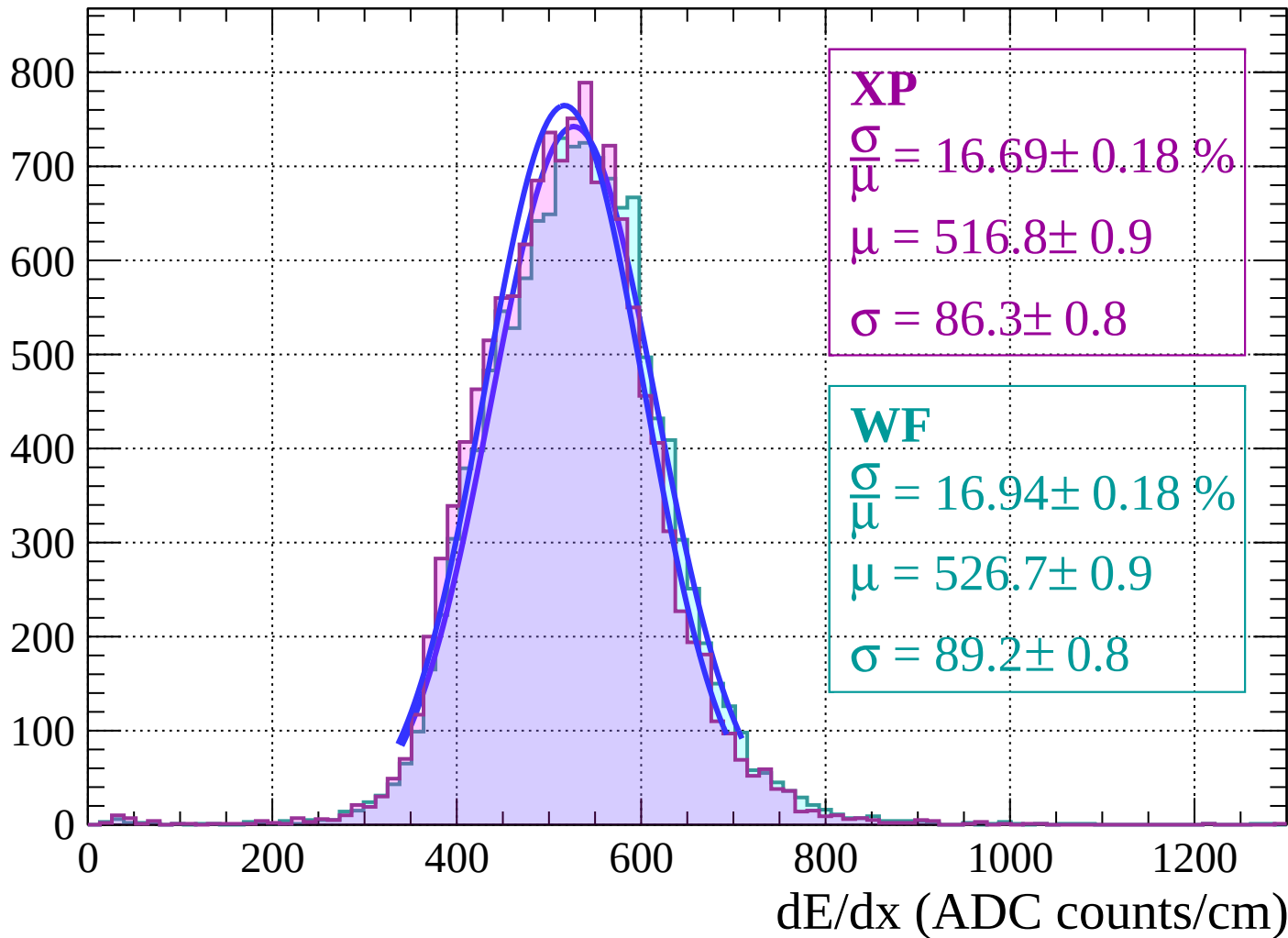


Energy loss | $36 < \theta < 42$

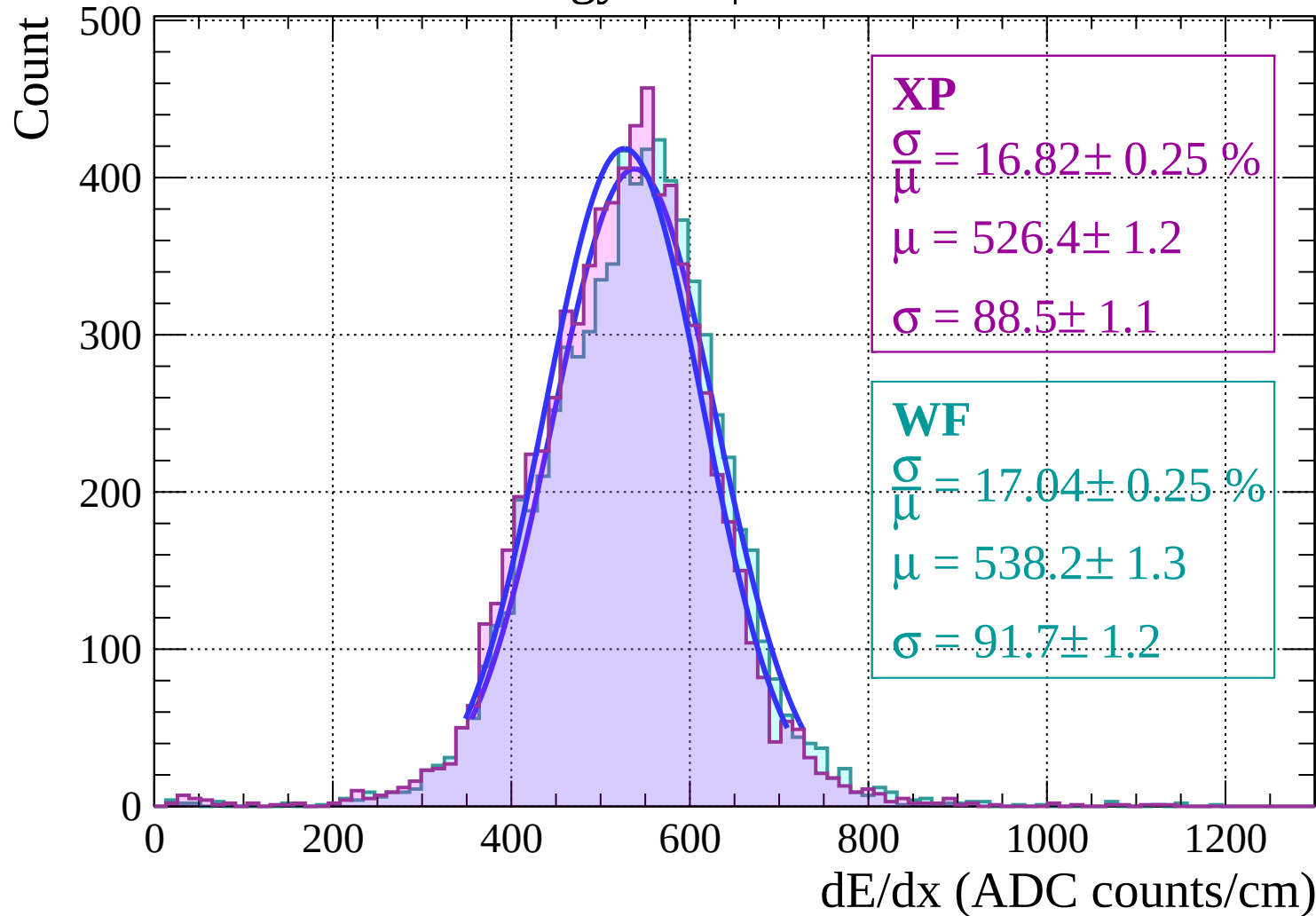


Energy loss | $42 < \theta < 48$

Count

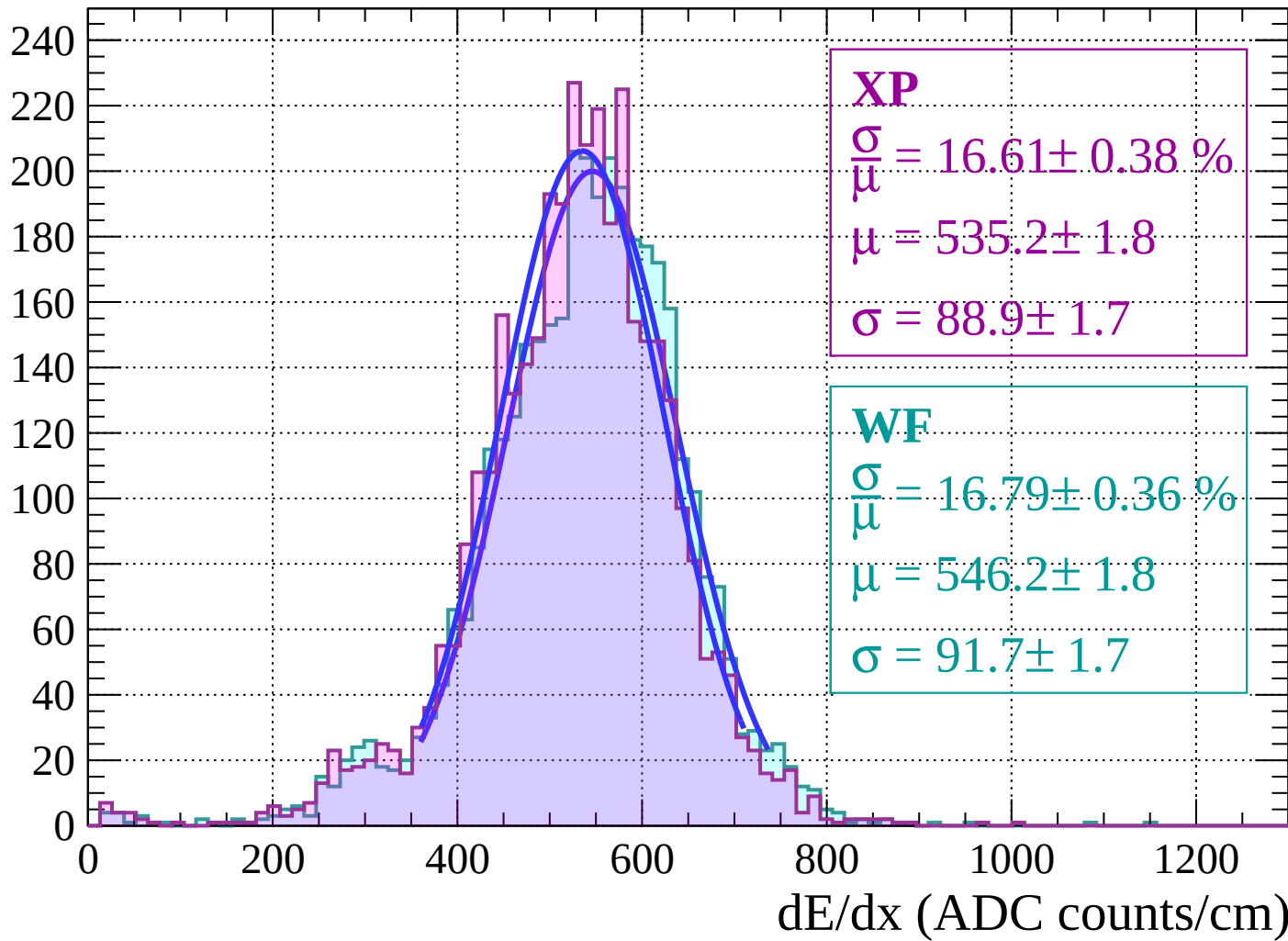


Energy loss | $48 < \theta < 54$



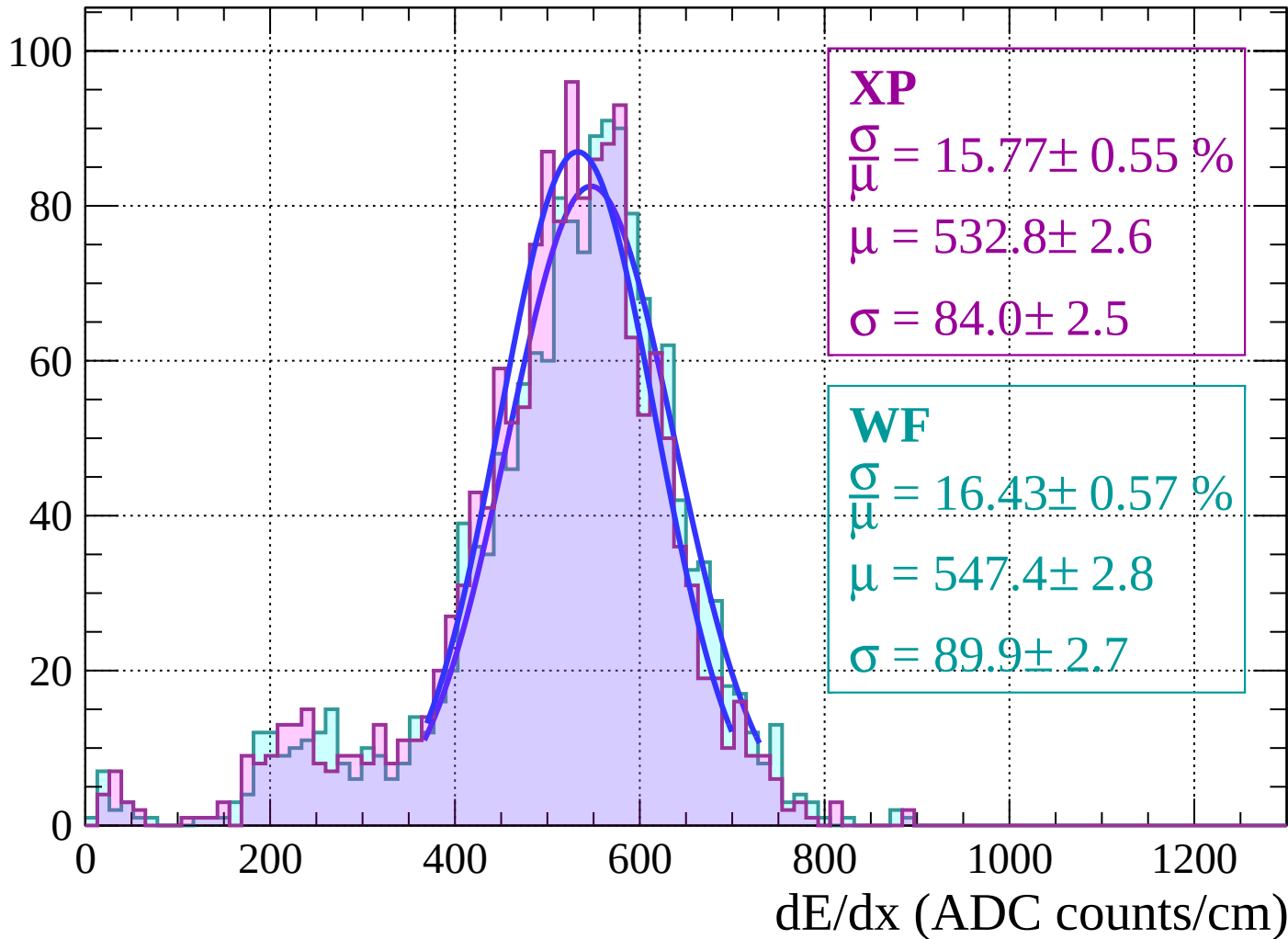
Energy loss | $54 < \theta < 60$

Count



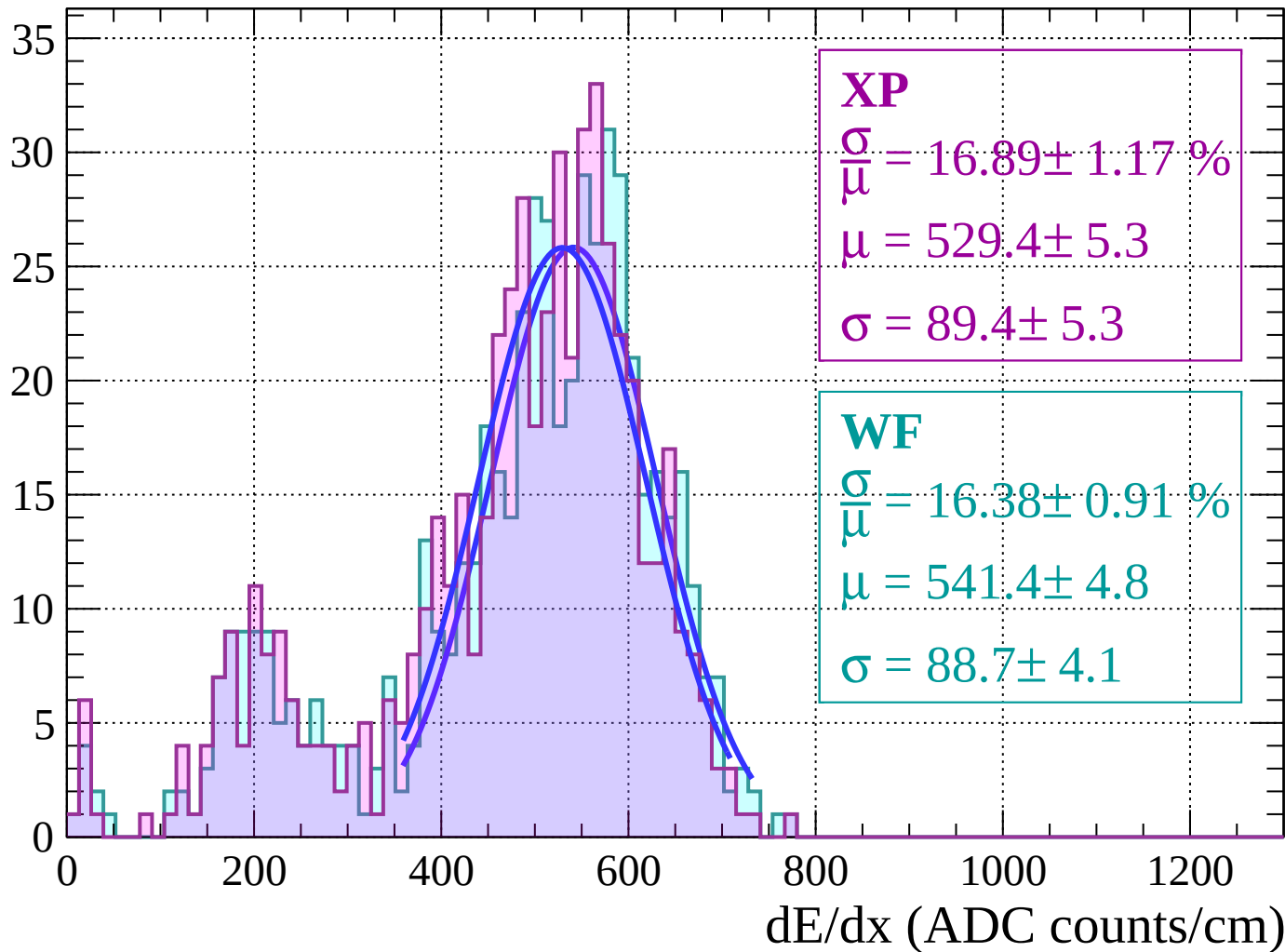
Energy loss | $60 < \theta < 66$

Count



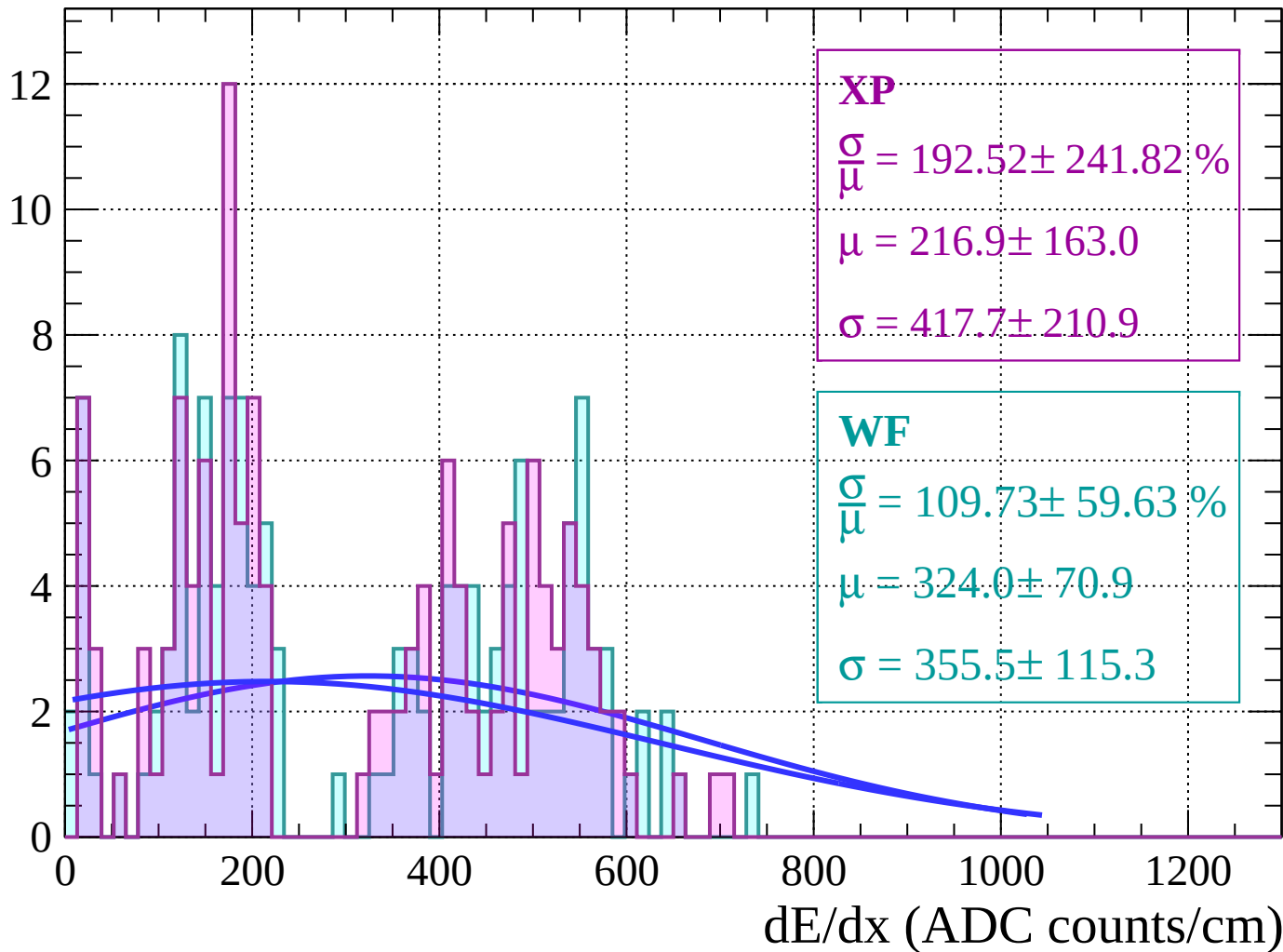
Energy loss | $66 < \theta < 72$

Count



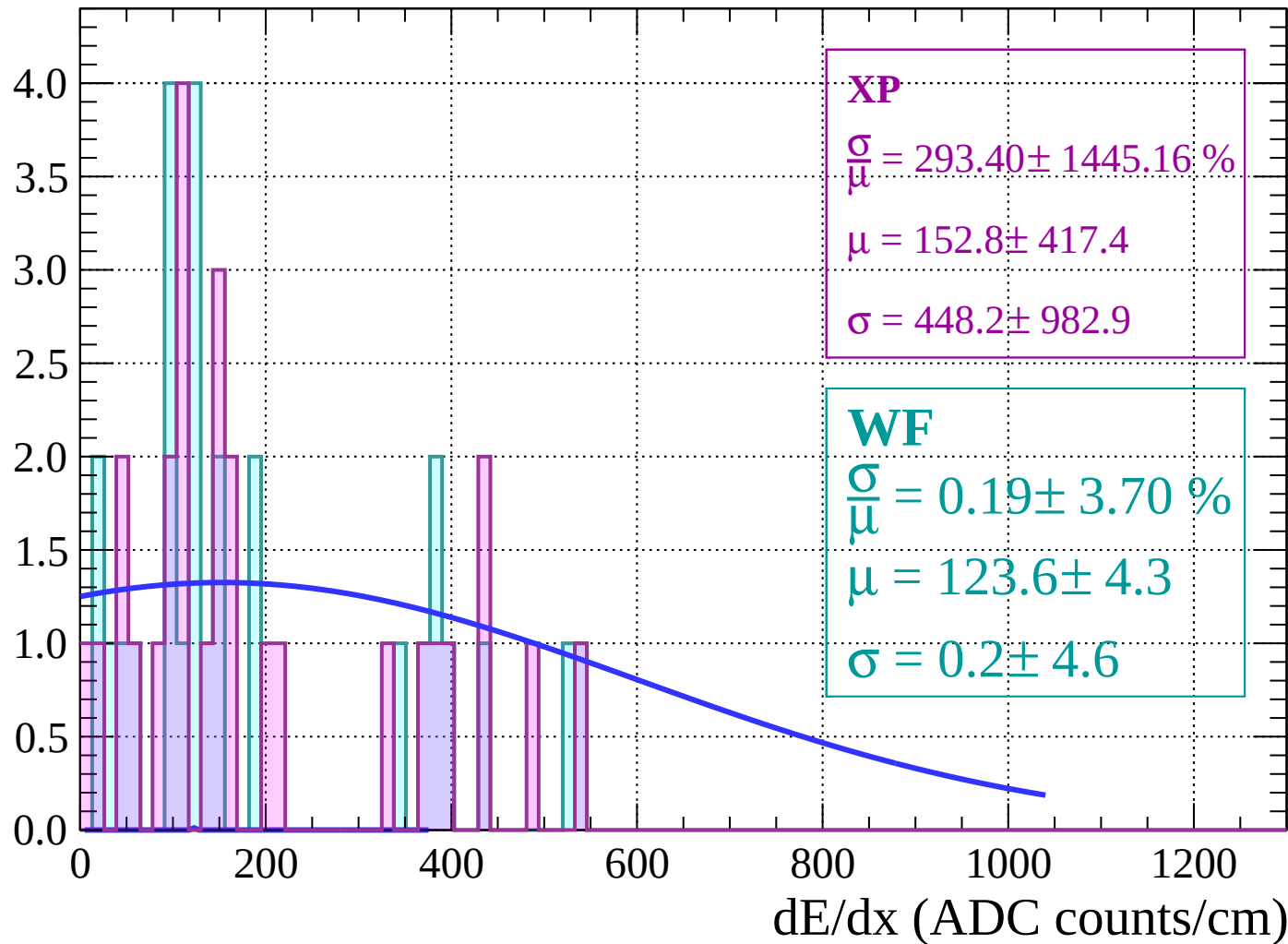
Energy loss | $72 < \theta < 78$

Count



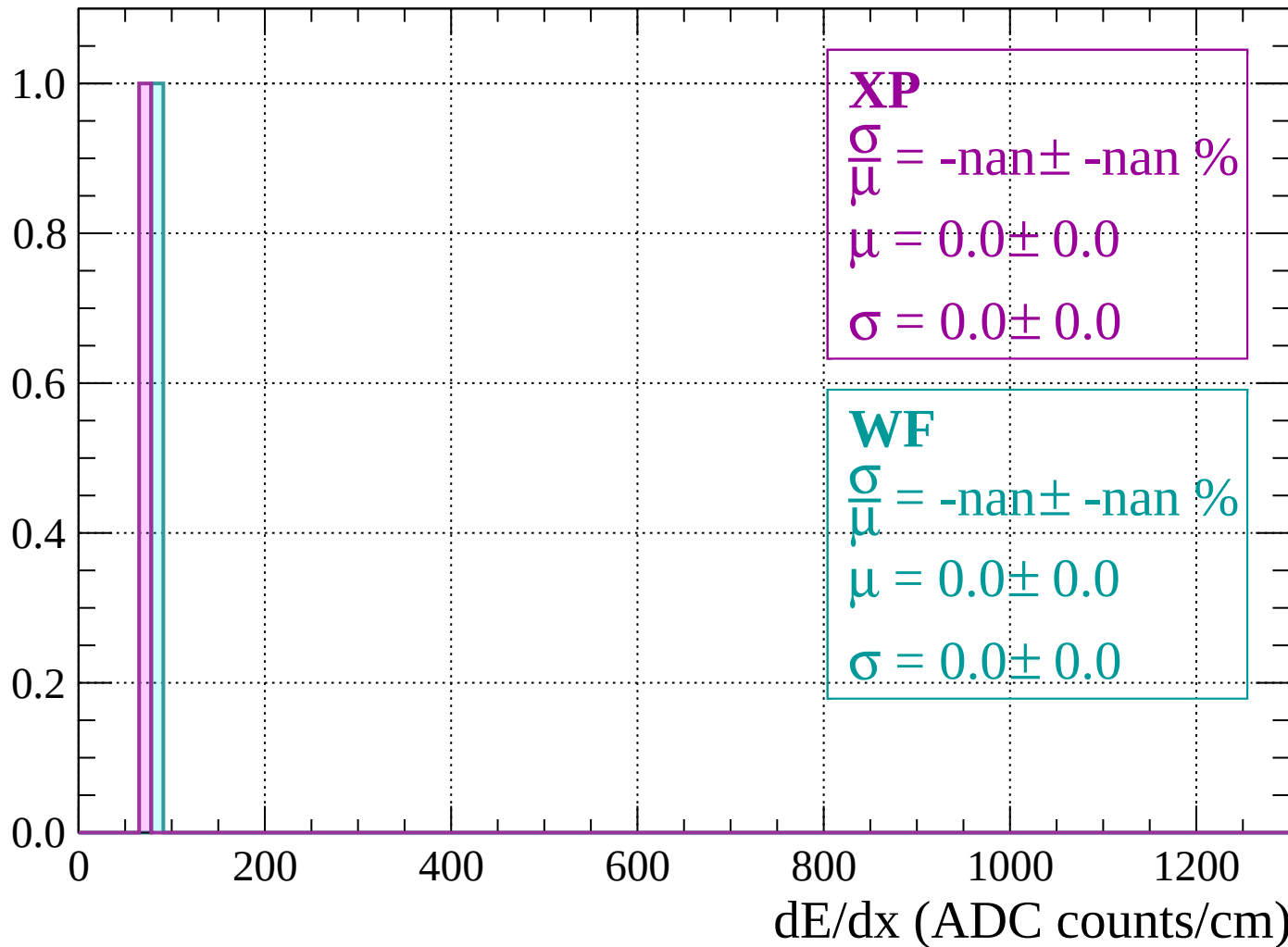
Energy loss | $78 < \theta < 84$

Count



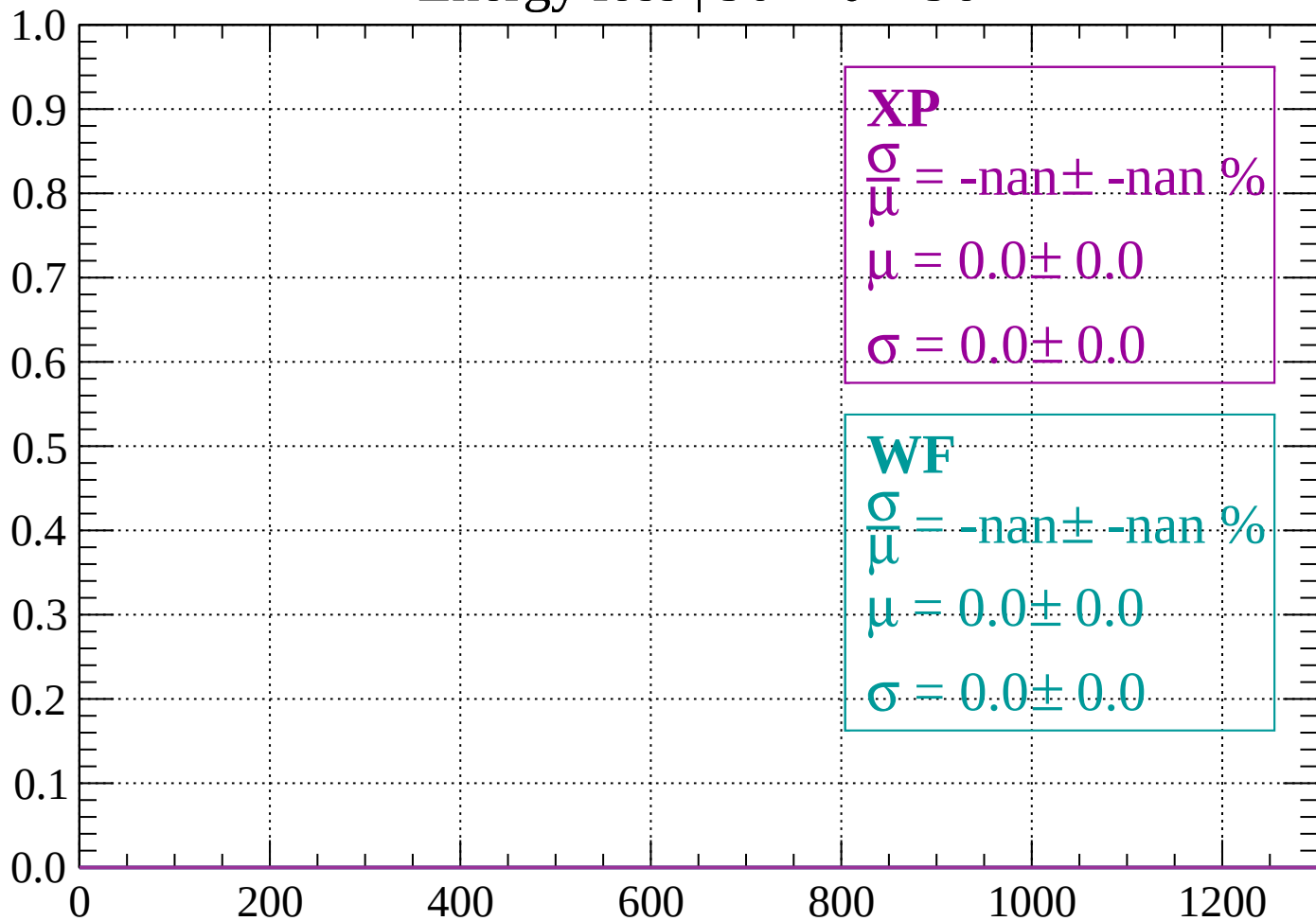
Energy loss | $84 < \theta < 90$

Count



Energy loss | $90 < \theta < 96$

Count



dE/dx (ADC counts/cm)

